Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

PART I

Theory

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

CHAPTER 1

Five Components of Tonality

The word "tonal" is contested territory. Some writers use it restrictively, to describe only the Western art music of the eighteenth and nineteenth centuries. For them, more recent music is "post-tonal"—a catch-all term including everything from Arvo Pärt's consonances to the organized sonic assaults of Varèse and Xenakis. This way of categorizing music makes it seem as if Pärt, Varèse, and Xenakis are clearly and obviously of a kind, resembling one another more than any of them resembles earlier composers.

"Tonal" can also be used expansively. Here, the term describes not just eighteenth-and nineteenth-century Western art music, but rock, folk, jazz, impressionism, minimalism, medieval and Renaissance music, and a good deal of non-Western music besides. "Tonality" in this sense is almost synonymous with "non-atonality"—a double negative, most naturally understood in contrast to music that was deliberately written to contrast with it. The expansive usage accords with the intuition that Schubert, the Beatles, and Pärt share musical preoccupations that are not shared by composers such as Varèse, Xenakis, and Cage. But it also raises awkward questions. What musical feature or features lead us to consider works to be tonal? Is "tonality" a single property, or does it have several components? And how does tonality manifest itself across the broad spectrum of Western and non-Western styles? Faced with these questions, contemporary music theory stares at its feet in awkward silence.[1]

The purpose of this book is to provide general categories for discussing music that is neither classically tonal nor completely atonal. This, in my view, includes some of the most fascinating music of the twentieth century, from impressionism to postminimalism. It also includes some of the most mysterious and alluring music of Chopin, Liszt, and Wagner—music that is as beloved by audiences as it is recalcitrant to analytical scrutiny. My goal is to try to develop a set of theoretical tools that will help us think about these sophisticated tonal styles, which are in some ways freer and less rule-bound than either eighteenth-century classical music or twentieth-century modernism.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

More specifically, I will argue that five features are present in a wide range of genres, Western and non-Western, past and present, and that they jointly contribute to a sense of tonality:

1. *Conjunct melodic motion*. Melodies tend to move by short distances from note to note.
2. *Acoustic consonance*. Consonant harmonies are preferred to dissonant harmonies, and tend to be used at points of musical stability.
3. *Harmonic consistency*. The harmonies in a passage of music, whatever they may be, tend to be structurally similar to one another.
4. *Limited macroharmony*. I use the term "macroharmony" to refer to the total collection of notes heard over moderate spans of musical time. Tonal music tends to use relatively small macroharmonies, often involving five to eight notes.
5. *Centricity*. Over moderate spans of musical time, one note is heard as being more prominent than the others, appearing more frequently and serving as a goal of musical motion.

The aim of this book is to investigate the ways composers can use these five features to produce interesting musical effects. This project has empirical, theoretical, and historical components. Empirically, we might ask how each of the five features contributes to listeners' perceptions of tonality: which is the most influential, and are there any interesting interactions between them? For instance, is harmonic consistency more important in the context of some scales than others? Theoretically, we might ask how the various features can *in principle* be combined. Is it the case, for example, that diatonic music necessarily involves a tonic? Conversely, is chromatic music necessarily non-centric? Finally, we can ask historical questions about how different Western styles have combined these five tonal ingredients—treating the features as determining a space of possibilities, and investigating the ways composers have explored that space.

This book is primarily concerned with the theoretical and historical questions. I am a musician, not a scientist, and although I will sometimes touch on perceptual issues, I will largely leave empirical psychology to the professionals. Instead, I will ask how composers *have combined* and *might combine* the five features. Part I of the book develops theoretical tools for thinking about the five features. Part II uses these tools to argue for a broader,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

more continuous conception of the Western musical tradition. Rather than focusing narrowly on the eighteenth and nineteenth centuries (the so-called "common practice period"), I attempt to identify an "extended common practice" stretching from the beginning of Western counterpoint to the music of recent decades. The point is to retell the history of Western music in such a way that the tonal styles of the last century—including jazz, rock, and minimalism—emerge as vibrant and interesting successors to the tonal music of earlier periods.

My central conclusion is that the five features impose much stronger constraints than we would intuitively expect. For example, if we want to combine conjunct melodic motion and harmonic consistency, then we have only a few options, the most important of which involve acoustically consonant sonorities. And if we want to combine harmonic consistency with limited macroharmony, then we are led to a collection of very familiar scales and modes. Thus the materials of tonal music are, in Richard Cohn's apt description, "overdetermined," in the sense that they are special or distinctive for multiple different reasons.[2] This suggests that when we look closely, we should find important similarities between different tonal styles: since there are only a few ways to combine the five features, different composers—from before Palestrina to after Bill Evans—will necessarily use the same basic techniques. In the second part of this book I make good on this claim, tracing common practices that connect the earliest examples of Western counterpoint to music of the very recent past.

1.1 THE FIVE FEATURES
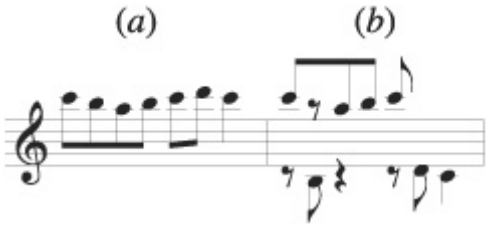
Let's consider the five features in more detail, with an eye toward understanding why they might be so widespread throughout Western and non-Western music. A preference for *conjunct melodic motion* likely derives from the features of the auditory system that create a three-dimensional "auditory scene."[3] An eardrum, in effect, is a one-dimensional system that can only move back and forth. From this meager input our brains create a vivid three-dimensional sonic space consisting of individually localized sounds: the phone ringing in front of you, the honk of a car horn outside your window, and the sound of a droning music theorist off to your right. To accomplish this dazzling transfiguration, the brain relies on a number of computational tricks, one of which is to group sonic events that are

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

nearby in pitch.[4] Thus, a sequence like Figure 1.1.1a tends to be heard as belonging to a single sound source, whereas Figure 1.1.1b creates the impression of multiple sources. In this sense, small melodic steps are intrinsic to the very notion of "melody."

Figure 1.1.1 Small movements sound melodic (*a*), while large registral leaps create the impression of multiple melodies (*b*).
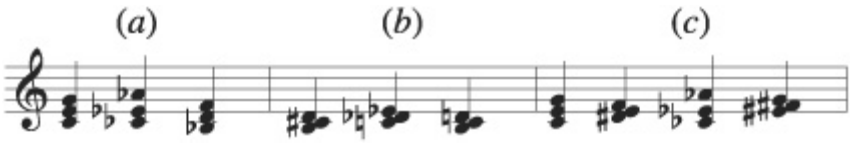


Acoustic consonance, or intrinsic sonic restfulness, is another very widespread musical feature.[5] Many styles make heavy use of consonant intervals such as the octave and perfect fifth, assigning them privileged melodic and harmonic roles. Scales containing a large number of consonant intervals are found in seemingly independent musical cultures, and there is evidence from infant psychology that the preference for consonance is innate.[6] At present, however, we do not know for certain how universal or innate this preference is. Fortunately, this issue is largely irrelevant in the present context: what matters is just that many listeners, both Western and non-Western, do have a fairly deep-seated preference for consonant sonorities.

Slightly more general than acoustic consonance is *harmonic consistency*, or the use of sonorities that resemble one another. For example, Figure 1.1.2a features a series of major and minor chords, all audibly similar. Their resemblance gives the passage a kind of smoothness, and we experience the chords as belonging together. Likewise, Figure 1.1.2b uses a series of very dissonant chromatic clusters that also seem to belong together. By contrast, Figure 1.1.2c uses very different-sounding harmonies, switching aimlessly between different sonic worlds. This sort of harmonic incongruity is quite unusual in Western music, and often provokes spontaneous laughter—suggesting that the expectation of harmonic consistency is very strong, even though it is rarely discussed.

Figure 1.1.2 Harmonic consistency using consonant sonorities (*a*) and dissonant sonorities (*b*). Sequence (*c*) does not exhibit harmonic consistency.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



In most Western and non-Western music, pitches are drawn from a relatively small reservoir of available notes—typically, between five and eight.[7] As a result, Western music has a two-tiered harmonic consistency: at the local (or instantaneous) level, a passage like Figure 1.1.3 presents a series of major and minor chords, which are audibly related, while over larger time spans it articulates a scale by using only seven different notes. The scale can thus be considered a kind of "large" or *macro* harmony that subsumes the individual chords.[8] Even though there is no one instant at which this larger harmony is presented, it nevertheless has a significant effect on our listening experience: scale-based melodies are easier to remember than nonscalar melodies, and notes outside the scale sound more pungent than notes in the scale. As we will see, macroharmonies can be relatively consonant, like the diatonic or pentatonic scale, or relatively dissonant, like the chromatic scale. They can also participate in larger-level voice leadings analogous to those connecting individual chords.

Finally, we often hear some pitches or notes as being more important (or "central") than others. These pitches tend to serve as points of musical arrival, to which others are heard as "leading" or "tending." Thus, one and the same sequence of notes—such as that in Figure 1.1.4—can be heard in a variety of ways. If we hear it in a musical context where C is the most stable pitch, then it sounds like a beginning that is in need of some sort of continuation, ending with a comma rather than a period. But in a context where F is stable, it sounds more complete, as if it ends with a period or exclamation point. Centricity is again a very widespread feature of human music, appearing in a large number of seemingly unrelated musical cultures. However, different styles can emphasize the tonic note to different extents: as Harold Powers once noted, there is a much stronger feeling of centricity in Indian music than in Renaissance polyphony.[9]

Figure 1.1.3 Major and minor triads belonging to the same diatonic scale.
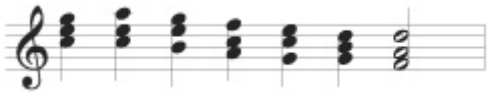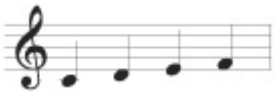


Figure 1.1.4 A single melody will sound different in different harmonic contexts.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Of these five features, harmonic consistency is clearly the most culturally specific. The idea that music should consist of rapidly changing chords is a deeply Western idea, in a double sense: it is deep, insofar as it characterizes much Western music since before the Renaissance; and it is Western, since there are many cultures in which the notion of a "chord progression" simply plays no role. (Many traditional non-Western styles are purely monophonic, or feature an unchanging "drone" harmony; however, there are now a large number of syncretistic styles that combine Western harmonies with non-Western melodic and rhythmic ideas.) Acoustic consonance is also somewhat culture-specific: although many cultures make some use of consonant intervals, and although some have recognizably Western conceptions of consonance, other non-Western styles sound quite dissonant to Western ears. By contrast, the three other features—conjunct melodic motion, limited macroharmony, and centricity—are common to virtually all human music. This near universality may be attributable, at least in part, to features of our biological inheritance.[10]

Now for an important disclaimer. While I think that typical Western listeners *prefer* music that exemplifies the five features, I do not mean to suggest that such music is intrinsically *better* than any other kind of music. "Tonal," for me, is not synonymous with "good." (Nor is "popular," for that matter: there is plenty of unpopular, nontonal music that I happen to like, from Nancarrow to Xenakis to Ligeti.) In particular, I have no interest in arguing that atonal composers are misguided, fighting against biology, or anything of the sort. Instead, the purpose of this book is an affirmative one: to develop new theoretical tools for thinking about tonality, and to provide new insights into the relations between various musical styles. My hope is that this investigation will be useful to composers and theorists of all varieties, as even the advocates of atonality will stand to gain from a deeper understanding of that which they are trying to avoid.

1.2 PERCEPTION AND THE FIVE FEATURES

This book is primarily concerned with what composers do, rather than what listeners hear: the goal is to describe various ways in which the five features have been or might be

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

combined. But it is not possible to avoid perceptual issues altogether. After all, readers have a right to wonder whether my five features do indeed contribute to the experience of tonality, and if so, whether they are the only factors that contribute.

The first question can be easily answered: all one needs to do is constrain randomly generated notes according to each of the five features. Insofar as the constraints cause random music to sound increasingly tonal, or at least ordered, then I am right about their psychological importance. Furthermore, the same experiment can be used not just to show *that* these features have important psychological effects, but also *what particular psychological effects they have*. This is useful because familiar styles tend either to combine many of the features, and hence be fully tonal in a traditional sense, or else to abandon most of them à la radical atonality. Consequently, existing musical works do not always help us to understand the specific contributions made by each of our five components individually.

I strongly urge you to try this experiment for yourself: the results are not subtle, and they demonstrate the powerful psychological effects that can be obtained with simple musical means. (In particular, I encourage you to use the book's companion website, which contains a number of illustrative examples.[11]) Unfortunately, this is a case where a musical experience is worth a thousand words: no amount of merely verbal description on my part will substitute for your own investigation. Nevertheless, it is worth trying to describe these effects, if only to persuade you to actually perform the experiment. Figure 1.2.1a presents a series of completely random three-note chords, with pitches chosen arbitrarily from the range C2 to C7. It provides a baseline against which subsequent examples can be judged. (Some people, myself included, find this sort of random texture to be oddly appealing.) Figure 1.2.1b constrains the randomness by requiring that the notes move just a few semitones from chord to chord. What results is very rudimentary sort of counterpoint, consisting of three independent melodic strands. Although it is considerably less random-sounding than the pointillistic texture of (*a*), the harmonic structure of the sequence still sounds somewhat indistinct, providing the ear with relatively little to grab onto. Figure 1.2.1c combines conjunct melodic motion with harmonic consistency, requiring that each chord be a "stack of fourths." The melodic lines, rather than wandering aimlessly, now seem to create chords with a distinctive harmonic identity, which in turn gives the passage a feeling of increased consistency.[12] (This example might seem

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

reminiscent of some Stravinsky or Hindemith.) Finally, Figure 1.2.1d restricts the chords to the same diatonic scale. To complete the transition from utter randomness to something recognizably tonal, I have replaced the "fourth chords" of (*c*) with more consonant diatonic triads. Although the result will not win any composition prizes, it does demonstrate that a kind of rudimentary tonalness is in fact generated by my five features. Indeed, the differences between Figures 1.2.1a–d are striking and unmistakable, even for a layperson with no specialized musical training.

Informal experiments like these suggest that, for typical listeners, the five features play an important role in determining the tonalness (or perhaps "orderedness") of musical stimuli. Furthermore, I strongly suspect that for many listeners, "tonalness," "orderedness," and "pleasantness" are correlated: all else being equal, music displaying many of the five features will be preferred to music that does not.[13] This constitutes the testable psychological theory lurking at the core of this book. Personally, I think it would be interesting to try to determine not just *that* the five features have important psychological effects, but also their relative importance. What makes a larger difference to listeners' perceptions, harmonic consistency, acoustic consonance, or conjunct melodic motion? We lack even the most rudimentary data that would allow us to answer this question. One of my hopes is that some psychologist readers will be motivated to undertake the obvious experiments suggested by the ideas I will be discussing.

Figure 1.2.1 Four randomly generated sequences. Sequence (*a*) is completely random; (*b*) exhibits efficient voice leading; (*c*) exhibits harmonic consistency and efficient voice leading; (*d*) exhibits harmonic consistency, efficient voice leading, and limited macroharmony.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
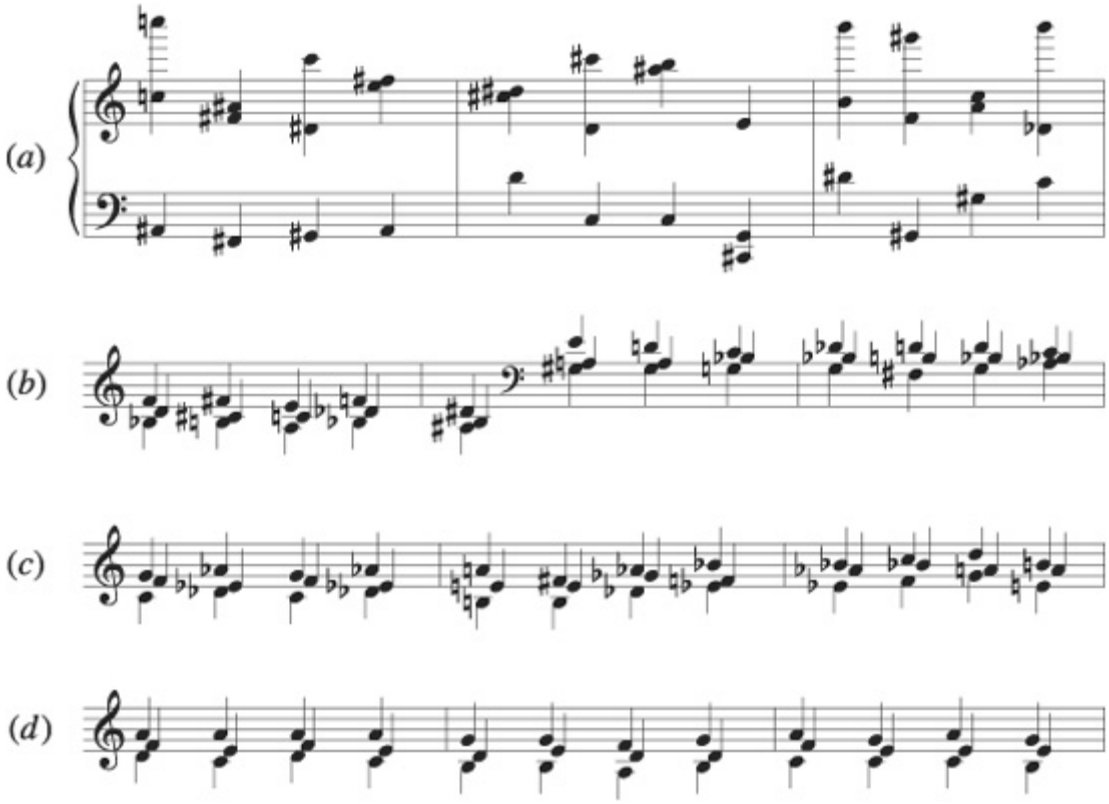


Figure 1.2.1 shows that the five features *can* contribute fairly dramatically to the sense of tonality. But are they *necessary* for creating tonal effects? And if so, are they the only such features or are there others?

In some ways, I think the question is misguided. The point here is not to police the use of the word "tonality" by setting strict limits on what may or may not be described with the term, but rather to replace the crude opposition "tonal/atonal" with a more nuanced set of distinctions. You can decide for yourself whether to use "tonal" to describe (say) diatonic music without a tonal center, or chromatic music with a strong sense of harmonic consistency; my job is just to show that the term "tonality" typically applies to music that exhibits my five basic features. I should also point out that I am neglecting important issues such as rhythm, motivic variation, timbre, form, performance, and rubato, all of which can contribute to the sense of tonality. (My goal is not to provide a complete theory of all music, but rather to discuss a few general features whose musical importance is for the most part unquestioned.) That said, however, I confess that it is difficult for me to imagine that I would ever want to use the term "tonal" to describe music in which acoustic consonance plays no role, in which there is no conjunct melodic motion or harmonic consistency, in which no tone is heard as central, and which does not limit itself to a

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

relatively small number of pitch classes over short stretches of time. In this sense, it seems that at least some of the five features are necessary for tonality, at least as I personally understand it.
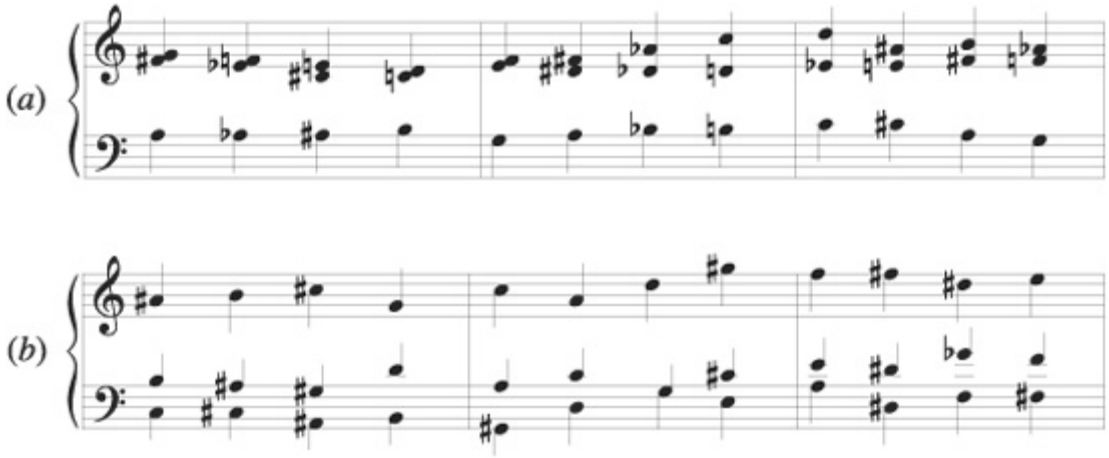
Throughout the twentieth century, composers devised new musical languages—some idiosyncratic, some very widely used—intended to replace traditional tonality. It is instructive to subject these alternative systems to the experimental test described above: that is, to constrain random musical notes according to their basic principles, and to listen for the perceptual differences that result.[14] To that end, Figure 1.2.2 uses "constrained randomness" to investigate one of the most prominent alternatives to tonality, the twelve-tone system.[15] On a first (or even tenth) listening, I do not find the sequences in Figure 1.2.2 to be dramatically different from randomly generated sequences such as Figure 1.2.1b. Indeed, if someone were to present the three passages in a psychology experiment, I doubt I would notice that one was random while the other two were not. This is not to say that twelve-tone structure makes *no* audible difference, or that one cannot learn to hear the coherence in Figure 1.2.2, but rather that the psychological effects here are relatively subtle—considerably less dramatic than those produced by our five features.

Does this show that twelve-tone music is aesthetically problematic? Or that the twentieth-century quest for alternatives to traditional tonality is fruitless? Not at all.[16] But it does suggest that twelve-tone rows produce less powerful psychological consequences than harmonic consistency, conjunct melodic motion, acoustic consonance, macroharmony, and centricity.[17] And while twelve-tone music is just one twentieth-century musical system, similar comments might be made about other approaches. To my mind, this suggests that the five features are unusually powerful tools for creating musical coherence. To say this is not to deny that alternative tools may in principle exist, but simply to reiterate the basic point that tonality constitutes a fairly unique solution to some elementary compositional problems. If this is right, then the task of providing an alternative to tonality is much more difficult than one might intuitively have imagined.

Figure 1.2.2 Randomly generated twelve-tone music.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



1.3 FOUR CLAIMS

The argument of this book revolves around four basic claims, each of which concerns ways in which the five features can interact with or constrain one another. In this section I'll briefly outline these claims as a way of foreshadowing some of my central preoccupations.

1.3.1 Harmony and Counterpoint Constrain One Another

Imagine a composer, Lyrico, who would like to combine conjunct melodic motion with harmonic consistency; that is, he would like to write melodies that move by short distances while using harmonies that are structurally similar to one another. Intuitively, it might seem that there are innumerable ways to satisfy these two constraints—that there is an entire universe of syntaxes consistent with these fundamental principles. But in fact, there are just a few ways in which they can be combined.

Consider the simplest possible situation. Suppose Lyrico decides to combine an unchanging "drone" harmony with a moving melodic voice. Harmonic consistency is thus obtained trivially: the chords in the passage will all be similar because *there is only one chord*. Let us further imagine that Lyrico chooses to use a C major chord as his drone. If he were to confine the melody to the notes of this chord, the result would be a series of unmelodic leaps (Figure 1.3.1a). This is because its notes are all reasonably far apart. To obtain conjunct melodic motion, Lyrico can therefore introduce "passing tones" that connect the chord tones by short melodic steps. The result, shown in Figure 1.3.1b, is a *scale* covering an entire octave, in which successive notes are connected by relatively small distances, and in which chord tones alternate with nonchordal "passing tones." Lyrico can

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

now write melodies that move freely along this scale, alternating between stable and unstable notes.

Figure 1.3.1 (*a*) Confining a melody to the notes of the C major chord produces large leaps, so it is necessary to add "passing tones" (*b*).

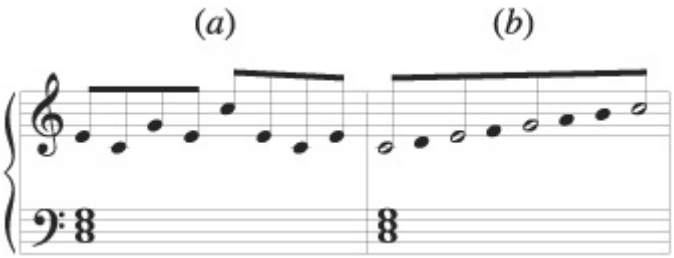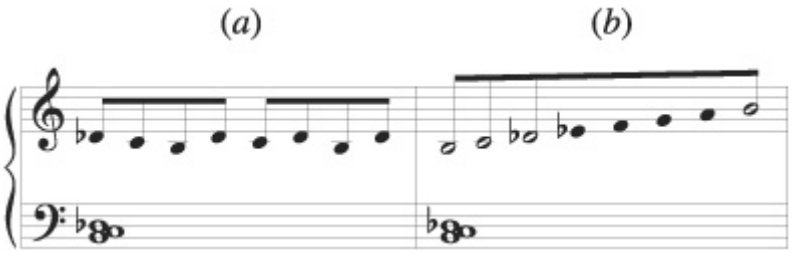


Figure 1.3.2 (*a*) Confining a melody to the notes of the cluster {B, C, D♭} produces conjunct melodic motion, but changing octaves requires a large number of passing tones (*b*).



But suppose Lyrico wakes up one day in a more ornery mood and decides to use the dissonant chromatic cluster {B, C, D♭} as his drone. Here the compositional situation is reversed: where the major chord is consonant, and has its notes spread relatively far apart, this chord is very dissonant and has all its notes close together. Consequently, by confining himself to the notes of the chord, Lyrico can obtain conjunct melodies, as in Figure 1.3.2. However, it takes a large number of passing tones to connect the D♭ in one octave to the B in the next. Since the resulting scale does not exhibit a regular alternation of stable "chord tones" and unstable "passing tones," it is difficult to hear the nonharmonic tones as connective devices that simply decorate an underlying harmony.

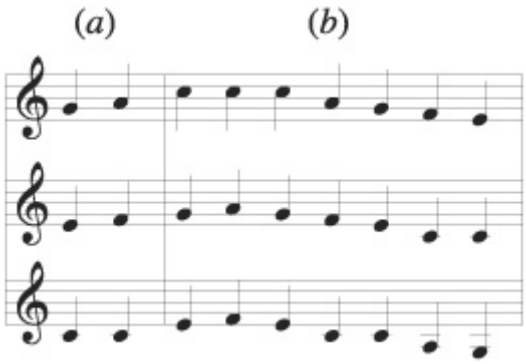
These two examples suggest a general moral: *harmony and melody constrain one another*. Different types of chords suggest different musical uses. In particular there is a fundamental difference between chords like {C, E, G}, whose notes are all far away from each other, and chords like {B, C, D♭}, whose notes are clustered close together. When a chord's notes are clustered close together, it is possible to create conjunct melodies that

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

use only the chord's notes, but it is not possible to create scales that have a regular alternation between chord tones and passing tones. When a chord's notes are relatively spread out, it is not possible to create conjunct melodies by using only the notes of the chord, but it *is* possible to create scales with a nice arrangement of chord and non-chord tones.

Figure 1.3.3 (*a*) Every note of the C major triad is near some note of the F major triad. (*b*) It is possible to use a series of C and F major triads to construct three simultaneous melodies.
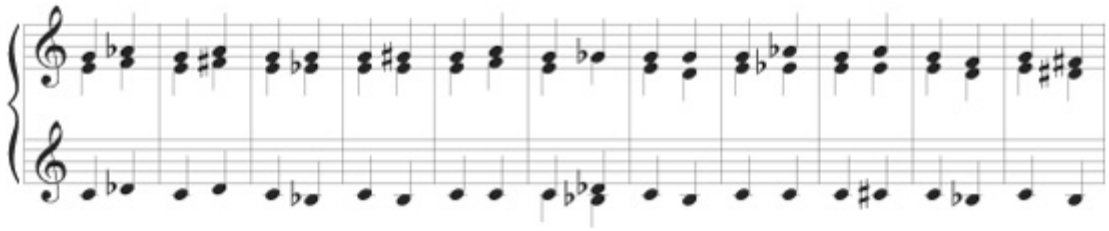


Now consider a more sophisticated problem. Suppose Lyrico decides to write a C major chord followed by an F major chord, its transposition by ascending perfect fourth. As shown in Figure 1.3.3, *every* note in the C major chord is near some note in an F major chord: C is common to both, E is one semitone from F, and G is two semitones from both F and A. This means that Lyrico can write a sequence of C and F major chords that articulates *three separate melodies*, each moving by small distances. This is *counterpoint*—a group of simultaneous melodies, or *voices*, articulating mappings, or *voice leadings*, between successive chords. For example, the first voice leading in Figure 1.3.3b maps G to A, E to F, and C to C. This voice leading is *efficient* because all the voices move by short distances. Clearly, efficient voice leading is simply conjunct melodic motion in all parts of a contrapuntal texture.

Figure 1.3.4 Any two major triads can be linked by stepwise voice leading; in the case of the tritone, this requires four voices.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



As it happens, the major chord is particularly well suited for contrapuntal music. Figure 1.3.4 shows that any two major chords can be connected by stepwise voice leading in which no voice moves by more than two semitones. This means that Lyrico can write a harmonic progression *without worrying about melody*; that is, for any sequence of major triads, there is always some way to connect the notes so as to form step-wise melodies. Conversely, Lyrico can write any melody whatsoever without worrying about harmony, as there will always be some way to harmonize it with a sequence of efficient voice leadings between major chords.

But what if Lyrico writes the chromatic cluster {B, C, D♭} followed by {E, F, G♭}, its transposition by ascending fourth? Here, none of the notes of the first chord are within two semitones of any note in the second, and hence there is no way to combine a sequence of these chords so as to produce conjunct melodies (Figure 1.3.5). At the same time, however, the chromatic cluster can do things that the C major chord can't: Figure 1.3.6 shows that it is possible to write contrapuntal music in which individual melodic lines move by short distances *within* a single, unchanging harmony. Clearly, this is possible only because the chord's notes are all clustered together, ensuring that there is always a short path between any two of them. The resulting music produces a feeling of burbling within stasis, a kind of melodic activity within overall harmonic restfulness.[18]

Once again, we see that different kinds of chords are useful for different purposes. Chords that divide the octave very unevenly, such as {B, C, D♭}, are ideally suited for static music in which harmonies do not change. By contrast, chords that divide the octave relatively evenly, like C major, can be connected to their transpositions by efficient voice leading, and are therefore suited for contrapuntal music in which harmonies change quickly. Remarkably, the most nearly even chords in twelve-tone equal temperament are the familiar chords of Western music: perfect fifths, triads, seventh chords, ninth chords, and familiar scales. Besides being well suited for contrapuntal music, these chords are all *acoustically consonant*, or restful and stable-sounding. This, then, is an example of Cohn's "overdetermination," a situation in which a familiar musical object is remarkable for

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

multiple reasons: nearly even chords are interesting not just because they permit the combination of harmonic consistency and conjunct melodic motion, but also because they can be acoustically consonant.[19]

Figure 1.3.5 Chromatic clusters cannot always be linked by efficient voice leading.
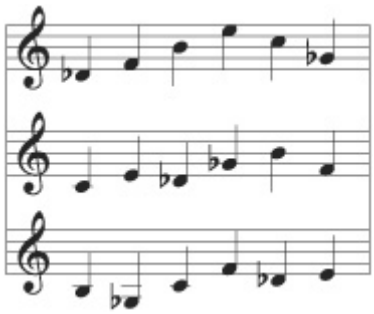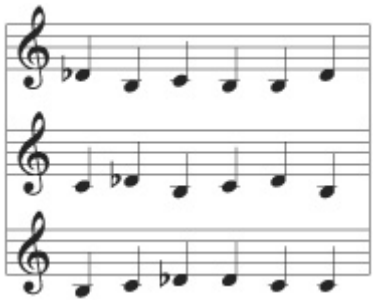


Figure 1.3.6 Chromatic clusters allow conjunct melodic motion to be combined with harmonic stasis.



More generally, it would seem that composers who wish to combine harmonic consistency and conjunct melodic motion have relatively few options: they can either use familiar sonorities in more or less familiar ways, or they can use chromatic chords whose notes are clustered together. The next few chapters will elaborate on this point, using the notion of *near symmetry* to explain exactly how chord structure constrains contrapuntal function. Later chapters will trace the practical consequences of this interdependence, considering a range of superficially different styles, from Renaissance polyphony to contemporary jazz, and showing that they all utilize fundamentally similar procedures. Our theoretical work will demonstrate that these similarities are not simply the byproduct of historical influence, but also of the more fundamental ways in which the five features constrain one another. In other words, they testify to the fact that there are only a few ways to combine harmonic consistency and stepwise melodies.

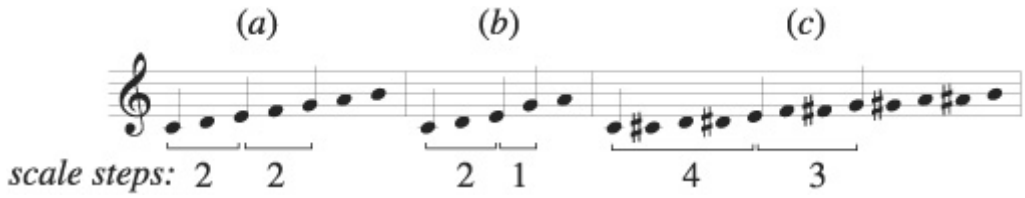
1.3.2 Scale, Macroharmony, and Centricity are Independent

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

My second claim is that we need to distinguish the closely related phenomena of scale, macroharmony, and centricity. A *scale*, as I use the term, is a means of measuring musical distance—a kind of musical ruler whose unit is the "scale step." Relative to the C diatonic scale, the notes E and G are two scale steps apart, since there is precisely one white note between them (Figure 1.3.7).[20] These same notes are one step apart relative to the pentatonic scale C-D-E-G-A and three steps apart relative to the chromatic scale. Similarly, C and E are two steps apart relative to the diatonic and pentatonic scales, and four apart relative to the chromatic scale. The three scales therefore give us three different ways of measuring musical distance, and three different estimates of the relative sizes of the intervals C-E and E-G. In principle, we should not ask whether the intervals C-E and E-G are "the same size" unless we also specify a particular musical "ruler." As we will see, the richness of tonal music lies partly in the way it exploits these various conceptions of musical distance.

Figure 1.3.7 Scale steps provide a means of measuring musical distance. The intervals C-E and E-G are two steps large relative to the diatonic scale (*a*), two and one steps large relative to the pentatonic scale (*b*), and four and three steps large relative to the chromatic scale (*c*).
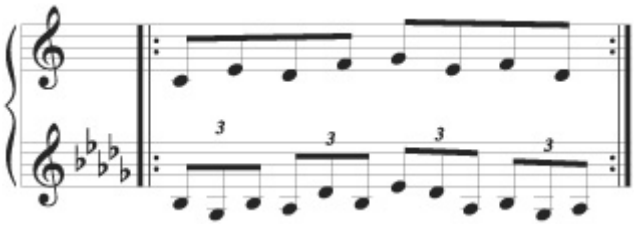


If a scale is a musical ruler, then a *macroharmony* is the total collection of notes used over small stretches of musical time. Typically, macroharmonies are also scales: a composer might (for example) use only the white notes on the piano keyboard, while also exploiting the unit of distance defined by adjacent notes (the "scale step"). But in principle it is possible to separate the two phenomena. Figure 1.3.8 shows a passage of "polytonal" music in which the upper staff moves systematically along the C diatonic scale, while the lower staff moves along G♭ pentatonic. To explain how this music works, we need to postulate two different scales, one for each staff. (Of course it would be possible to combine the notes in both staves into a single chromatic scale, but the resulting collection would be useless for explaining why the individual voices move as they do.) Here, then, is

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

a passage of music that uses pentatonic and diatonic scales to create a chromatic macroharmony. The concept of "scale" allows us to describe the structure within each voice, while the concept "macroharmony" allows us to describe the global harmony they produce.

Figure 1.3.8 The music makes use of two scales to create a chromatic macroharmony.



A second fundamental distinction is between *macroharmony* and *centricity*. "Centricity" refers to the phenomenon whereby a particular pitch is felt as being more stable or important than the others. (In traditional tonality this is the tonic note; in modal theory, it is the "final.") Chapters 4 and 5 will suggest that macroharmony and centricity are completely independent: it is entirely possible, for example, to write diatonic music in which no note is heard as a tonal center, just as one can write chromatic music with a very clear center. There is, however, a strong historical association between the two, with diatonic music often being centric and chromatic music centerless. The explanation for this connection lies in the fact that some prominent musicians believed that the diatonic scale had a unique "natural" tonic, and hence that a centerless chromaticism was the main alternative to traditional tonality. Other musicians, believing that centricity and macroharmony were independent, felt free to explore a much wider range of scales and modes. Chapter 5 will thus propose that the cleavage between the "scalar" and "chromatic" traditions, exemplified by composers such as Debussy and Schoenberg, was exacerbated by a fundamental disagreement about the relationship between macroharmony and centricity.
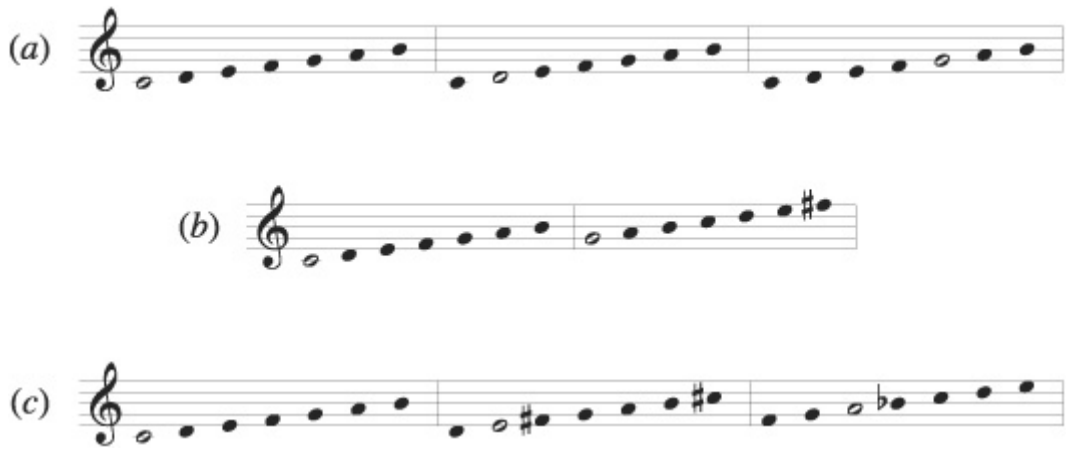
Together, scale, macroharmony, and centricity are the three principal components of what I think of as the "general theory of keys"—a set of tools for describing music that is tonal in the broad sense, even though it may not conform to the specific conventions of eighteenth-century tonality. One of this book's goals is to consider various sorts of "generalized keys" in Western music. Particularly interesting here is the gradual emergence of a musical language that combines a wide variety of macroharmonies and

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

tonal centers. The earliest Western music explored the tonal centers contained within a relatively static and largely diatonic macroharmony (Figure 1.3.9).[21] Classical music, which reduced the available modes to just major and minor, created large-scale harmonic contrasts by juxtaposing different scalar collections (e.g. G major and C major). It was only in the last decades of the nineteenth century that these two procedures were combined, as composers began to feel free to use *any* of the seven modes of *any* of the twelve diatonic collections. This scalar vocabulary was further extended with the use of additional nondiatonic scales—including the pentatonic, whole tone, melodic minor, harmonic minor, and octatonic—creating hundreds of possible combinations of macroharmony and tonal center. What results is a dazzling proliferation of "generalized keys" providing a wealth of alternatives to traditional major and minor modes.

Figure 1.3.9 (*a*) In earlier music, composers emphasized different centers within a single fixed diatonic collection. (*b*) In classical music, composers restricted the available modes to two, creating long-term harmonic change by emphasizing *different* major or minor scales. (*c*) Only in the twentieth-century were these two techniques systematically combined, creating a much wider range of tonal areas to choose from.



1.3.3 Modulation Involves Voice Leading
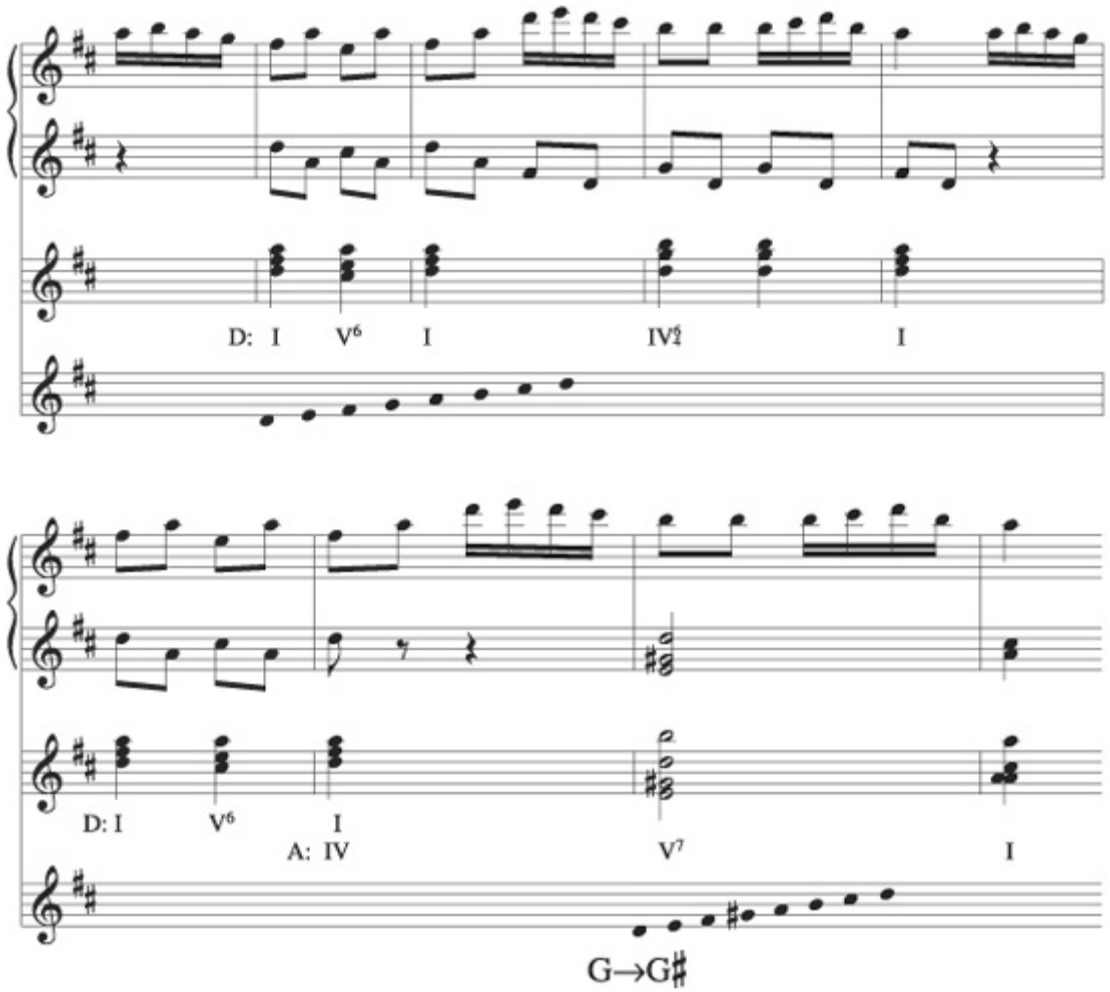
My third claim is that tonal music makes use of the same voice-leading techniques on two different temporal levels: *chord progressions* use efficient voice leading to link structurally similar chords, and *modulations* use efficient voice leading to link structurally similar scales.[22] As a result, tonal music is both self-similar and hierarchical, exploiting the same procedures at two different time scales.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 1.3.10 shows the opening of the last movement of Clementi's D major Piano Sonata, Op. 25 No. 6. The two parallel phrases present a series of three-voice chord progressions: I–V^6–I followed by I–IV–I and then V^7/V–V. The bottom staff shows that we can find a higher-level harmonic motion relating two diatonic collections: the first six measures limit themselves to the seven notes of D major, while the rest of the phrase abandons the G♮ in favor of the G♯. As we will see in Chapter 4, this modulation, or motion between macroharmonies, can be represented as a voice leading in which the G♮ *moves by semitone* to G♯. This means that the music exhibits two sorts of efficient voice leading: on the level of the half measure, there is a sequence of eight efficient voice leadings between triads; while on a larger temporal level there is a single efficient voice leading between D major and A major scales, occurring somewhere near the seventh measure of the example.

Figure 1.3.10 Two levels of voice leading in Clementi's Op. 25 No. 6.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
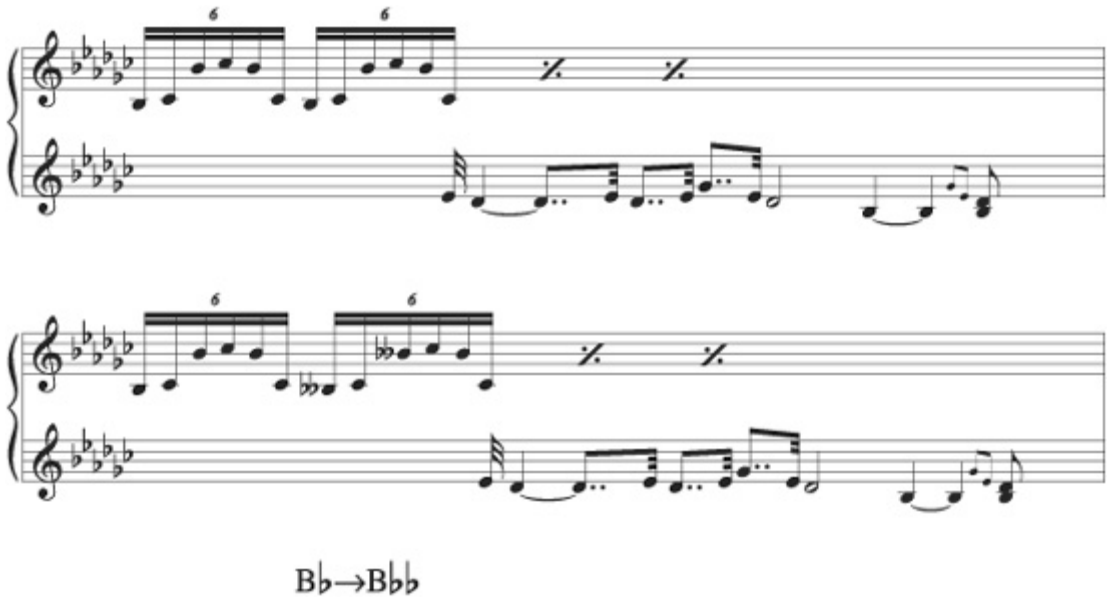
Figure 1.3.11 shows that similar processes occur in twentieth-century music. The top system depicts the main theme of Debussy's prelude "Le vent dans la plaine," which uses the pitches of E♭ natural minor. When the theme returns, B♭ moves by semitone to B♭♭, producing a collection that is enharmonically equivalent to F♯ melodic minor ascending. Debussy's "modulation" is thus analogous to Clementi's, although it involves modes that Clementi himself would never have used. Here, then, we have a familiar tonal technique appearing in the context of a significantly expanded modal vocabulary. As we will see in Chapter 9, this same technique has been used by a number of twentieth-century composers, including Stravinsky, Shostakovich, and the minimalists. There are also a few non-Western styles that use voice leading to link closely related scales.[23]

The idea that tonal music is hierarchically self-similar is central to the work of Heinrich Schenker, who claimed that tonal pieces consisted of recursively embedded patterns. The theory I have described is similar to Schenker's insofar as I consider tonal music to utilize efficient voice leading at two temporal levels. Unlike Schenker, however, I view macroharmonies and scales (rather than chords or melodic lines) as the primary vehicles of long-range harmonic progression: for me, the long-term voice leading in Figure 1.3.10 connects the D and A major *scales* rather than D and A major tonic triads. By contrast, a Schenkerian would likely interpret the passage—and indeed the entire piece—as involving chord progressions at all hierarchical levels. In Chapter 7 I will return to these issues, suggesting that my approach, though perhaps less unified than Schenker's, more closely reflects the cognitive processes involved in composition.

Figure 1.3.11 Voice leading between macroharmonies in Debussy's "Le vent dans la plaine."

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



1.3.4 Music Can Be Understood Geometrically

My fourth claim is that geometry provides a powerful tool for modeling musical structure. This is because there exists a family of geometrical spaces that depict the voice-leading relationships among virtually any chords we might care to imagine. Some of these (such as the familiar "circle of fifths") are relatively simple, but others (such as the Möbius strip containing two-note chords) are considerably more complex. One goal of the book is to provide a user-friendly introduction to these musico-geometrical spaces, explaining how they work, and showing how they allow us to visualize a wealth of musical possibilities at a glance.

Suppose, by way of illustration, that our friend Lyrico decides to write music using only the seven triads in the C diatonic collection. After a little exploration, he finds that some of these are closer together than others: for example, he can turn a C major triad into an A minor triad by moving only one note by one diatonic step, whereas he must move each voice to turn C major into D minor (Figure 1.3.12). Pondering this a little further, Lyrico eventually realizes that the diatonic triads can be linked in a "circle of thirds" (Figure 1.3.13), where each chord can be connected to its neighbors by a single-step motion. This circle allows him to define a kind of "distance" according to which C major and A minor are one step apart (since they are adjacent on the circle) while C major and D minor are three steps apart (since there are two chords between them).

Figure 1.3.12 Voice leading between diatonic triads.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Now suppose that Lyrico's rival Avanta becomes frustrated with all this conservatism. "Why limit yourself to just the triads in that one seven-note scale?" she asks, stamping her foot. "There's a whole world out there beyond the white notes, you know!" Avanta then proceeds to demonstrate some of the musical possibilities not represented in Lyrico's simple circular model: she shows that in the familiar chromatic scale, the C major chord can be linked to *four* separate triads by single-semitone voice leading, and to *seven* triads by a pair of semitone steps (Figure 1.3.14). What is the analogue, in Avanta's expanded musical world, to Lyrico's circular map? How can she depict the voice-leading possibilities between *all* the triads in the chromatic scale?

The answer turns out to be surprisingly complicated: instead of a simple seven-chord circular model, Avanta needs the three-dimensional, 40-chord lattice shown in Figure 1.3.15. This figure provides a map of all the contrapuntal possibilities available to a composer who wants to use traditional triads, but is willing to step outside the confines of a single diatonic scale. This complex-looking construction provides the first hint that ordinary musical questions might sometimes lead to nontrivial geometrical answers. In fact, Chapter 3 shows that Avanta's lattice lives in what mathematicians would call "the interior of a twisted triangular two-torus," otherwise known as a triangular doughnut. This space contains all possible three-note chords in any conceivable scale and any conceivable tuning system. Analogous spaces depict voice-leading relations among four-note chords, five-note chords, and so on. We will use them throughout this book, both for modeling chords and scales and for analyzing specific pieces.

Figure 1.3.13 Single-step voice-leading between diatonic triads can be modeled with a circle.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

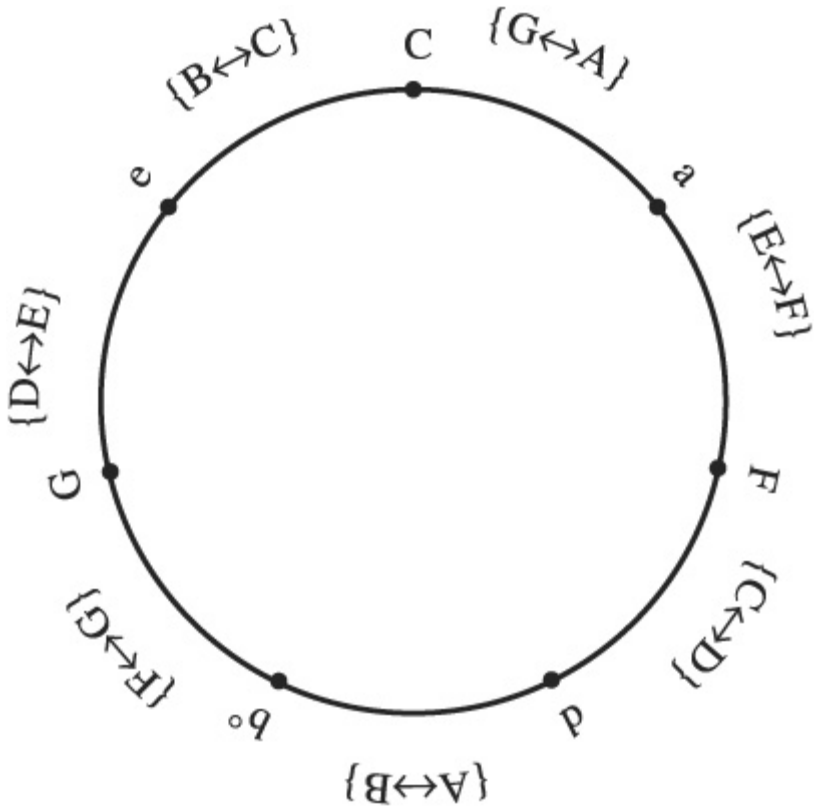


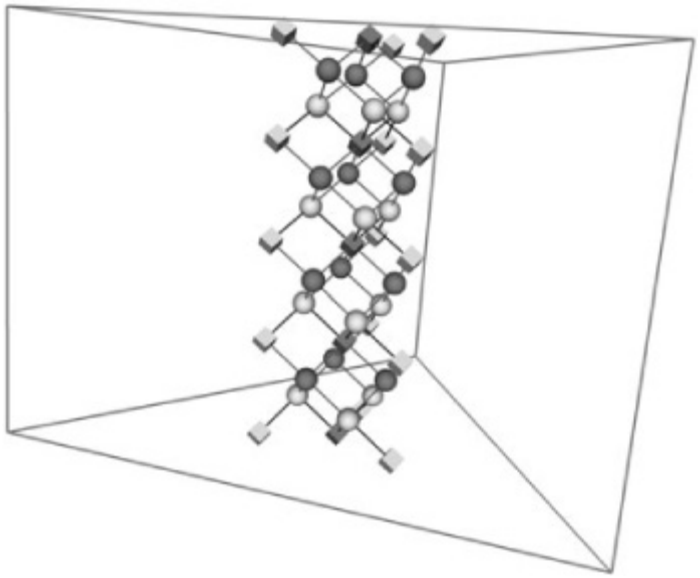
Figure 1.3.14 (*a*) One-semitone voice leading and (*b*) two-semitone voice leading among triads.
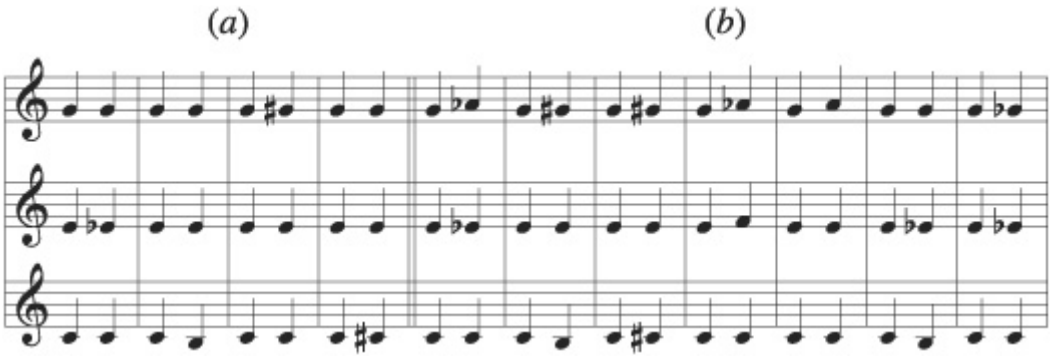


Figure 1.3.15 This three-dimensional graph represents single-semitone voice leading between major, minor, augmented, and diminished triads—represented by dark spheres, light spheres, dark cubes, and light cubes respectively. Chords on the same horizontal cross-section are related by major-third transposition; vertical motion corresponds to semitonal transposition, and the top face is glued to the bottom with a 120° twist. We will explore this figure more thoroughly in Chapter 3.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



For now, it is enough to note that the world of chromatic voice leading, as represented by Figure 1.3.15, is significantly more complicated than the simple circle representing diatonic triads. Because of this complexity, the fundamental logic animating nineteenth-century music has not always been clearly understood: theorists have sometimes depicted chromaticism as involving whimsical aberrations, departures from compositional good sense, rather than as the systematic exploration of a complex but coherent terrain. This has in turn led historians and composers to depict nineteenth-century chromaticism as pushing tonal logic to its breaking point, such that the step to complete atonality became all but inevitable. I will argue against this point of view, using new geometrical tools to demonstrate that the music of Chopin and Wagner can be just as rigorous as the music that preceded it.

1.4 MUSIC, MAGIC, AND LANGUAGE

Having sketched some major themes, let me now step back to make a few remarks about my approach to music theory. In this book I am primarily interested in the idealized composer's point of view: my goal is to describe conceptual structures that can be used to *create* musical works, rather than those involved in perceiving music. I stress the adjective "idealized," as my goal is not to undertake a historical investigation of the way past composers *actually* conceived of their music, but rather to describe concepts contemporary composers might find useful—in other words, to answer the question "what concepts would be helpful if I wanted to compose music like this?" Music theory, understood in this

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

way, helps composers steal from one another in a sophisticated fashion, allowing us to appropriate general procedures and techniques rather than particular chords or melodies. Of course it can also help performers and analysts understand music "from the inside," by showing how composers make use of the options available to them.

Of course, composer-based music theory cannot ignore listeners entirely: the point is to write music that other people want to listen to, and this can only occur if composers are dealing with musical features that listeners care about. But though composers and listeners need to be synchronized in general, there is room for considerable divergence when it comes to the details. For example, classical composers evidently considered it important to conclude a piece in the tonic key, even though listeners are relatively insensitive to this feature of musical organization.[24] This means that theorists should not assume that the cognitive structures involved in *making* music are the same as those involved in *perceiving* it: ideas that are central to the composer's craft, such as the principle that a classical sonata should recapitulate the second theme in the tonic key, may have only a glancing relevance to ordinary listeners.

I find it useful here to consider the analogy with magic. A stage magician uses various tricks to *cause* the audience to have extraordinary experiences—bunnies seem to disappear, beautiful assistants seem to be sawed in half, and so on. Enjoying a magician's performance does not require you to understand how the tricks are done; in fact, understanding may actually diminish your astonishment. Nor is the magician's "ideal audience" composed of professional magicians: the point is to perform the trick for people who will genuinely be *fooled*. In much the same way, I understand composition to be a process of using technical musical tools to ensure that audiences have certain kinds of extraordinary experiences. When composing, I make various choices about chords, scales, rhythm, and instrumentation to create feelings of tension, relaxation, terror, and ecstasy, to recall earlier moments in the piece or anticipate later events. But I do *not* in general expect listeners to be consciously tracking these choices. Listeners who do ("ooh, a dissonant ♯9 chord in the trombones, in polyrhythm against the flutes and inverting the opening notes of the piece!") are like professional magicians watching each others' routines—at best, engaged in a different sort of appreciation, and, at worst too intellectually engaged to enjoy the music as deeply as they might.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

One might contrast this approach with an alternative, on which the composer's and listener's perspectives are thought to be more closely aligned. Here the relevant analogy is not to magic, but to language: the idea is that composers write music that contains various sorts of patterns, including familiar sequences of chords and keys, thematic recurrences, and so on, while the listener's job is to recover these patterns. Expert listeners recover more patterns than inexpert listeners, and are consequently more qualified to pass aesthetic judgment on particular pieces of music. Insofar as some composers place patterns in their music that cannot be decoded aurally by expert listeners, their compositions are thereby aesthetically flawed.[25] On this view, music essentially involves a language-like transmission of syntactic patterns from producer to receiver.

The linguistic model is attractive, and there is no doubt some truth to it. But I think it understates the extraordinary distinctiveness of human language. Linguistic abilities are remarkable both for their accuracy and for their homogeneity: if you say "there is a tiger nearby," I have no trouble repeating the sentence word for word, writing it down, or explaining what it means. Furthermore, almost any English speaker can effortlessly distinguish grammatical utterances such as "there is a tiger nearby" from nongrammatical ones like "nearby tiger is." By contrast, musical abilities are both *heterogeneous* and fairly *in*accurate. First, there is an enormous spectrum ranging from congenital amusics to gifted listeners with extraordinarily accurate absolute pitch. These individuals have radically different musical capabilities as both producers and consumers of music. Second, while competent language-listeners are typically also competent language-speakers, this is not the case for music: most people enjoy listening to music, while only a few enjoy creating it. Third, even expert listeners lose a large amount of the "signal" in even moderately complicated music. I have had more than three decades of musical training, and yet I—like most other "expert" listeners—would have trouble notating, or recreating at the piano, the notes and rhythms in a ten-second excerpt from an unfamiliar four-voice Baroque fugue. Musically, in other words, I am analogous to a person who cannot reliably understand all the words in the sentence "there is a tiger nearby."[26]

These differences no doubt reflect the fact that music largely lacks semantic content. A listener who does not understand "be quiet, there's a tiger nearby" is a serious danger to himself and others; consequently, there is tremendous pressure for speakers and listeners to converge on the same interpretation of sentences. By contrast, a listener who does not

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

follow the detailed syntax of a Beethoven symphony, but who nevertheless enjoys it, creates no problems whatsoever. In some important sense *it simply does not matter* whether you follow all the details of a piece's syntax; what matters is that you follow the piece well enough to enjoy what you hear. For in the end, the composer's well-being depends on your willingness to listen, not on whether you interpret music in the same way that he or she does. Or to revert to our earlier analogy: what is important is not that you understand the magic trick, but simply that you feel the force of the illusion.

I should pause here to mention that I have a personal perspective on these issues, since I sometimes create music with the aid of algorithms and computers. In these cases I can be fairly removed from the underlying syntactical structure of the music I create, whether that be some complex melodic process, a gradually shifting probability distribution over the twelve chromatic notes, or a harmonic structure derived from a mathematical analysis of Mozart's music. Here my compositional role is that of a gatekeeper or judge, selecting the computer-generated passages that strike me as intuitively compelling, and arranging them, in a collage-like fashion, so as to produce the best musical effect. (Typically, I augment these algorithmic passages with music composed intuitively, producing a final product that blends the human and the inhuman.) The fact that some people seem to *like* this music only serves to highlight the oddness of the linguistic approach. For here the composer is only vaguely aware of the structures that, on the linguistic model, the listener is supposed to be recovering. Given that I am not at all sure that *I* am hearing these structures accurately, it seems presumptuous for me to demand much more from my listeners. Instead, I am hoping that listeners will be *directly affected* by the music in the same way I am—that is, tickled, amused, impressed, or awed.

Faced with this situation, I draw several morals.

First, there is a potential for real divergence between what we might call "composer's grammar" and "listener's grammar." Listeners may potentially grasp only a fraction of the underlying syntax of a Bach fugue, a Beethoven symphony, or a John Coltrane solo. (Some authors have claimed that this is particularly true when we consider the artificial syntaxes of twentieth-century atonality, and this may be true; but the more important point is that there will be significant gaps whenever we consider music of any complexity.[27]) In this sense, listening is like trying to catch up to a train that is forever just beyond your reach; indeed, the very fact that we miss so much structure is no doubt part of what leads us to

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

study scores, or listen repeatedly to the same pieces. This means that there are at least two separate projects that music theorists can engage in: modeling what composers actually do, and modeling what listeners actually experience. We should be careful not to conflate these by acting as if listening is "composing in reverse."

Second, we should be careful not to assume that there is any one thing that "listening" is. It is possible that some listeners are adept at recovering a large amount of the structure in muscial pieces, but many listeners are not. The heterogeneity of musical abilities makes it difficult to abstract away from individual variation in favor of some idealized "competent" listener. Can we describe as "competent" the listener who loves Beethoven, but who performs poorly on standard ear-training tests? What about the listener who hates music but perceives it very accurately? Does the "ideal" musical listener have absolute pitch? A perfect memory for every musical detail? Is the point of listening to music to experience aesthetic enjoyment, or is it to recover a kind of musical "syntax" that the composer placed in his or her music? Personally, I suspect there is no uncontroversial answer to these questions: there simply is no "competent" or "ideal" listener that is analogous to the "idealized speaker" of contemporary linguistics.

Third, listeners' perceptions may in some respects be more crude and statistical than we would initially think. We do not determine the meaning of sentences by estimating the proportion of nouns to verbs, but we do respond very strongly to relatively crude global features of the musical stimulus. Does the piece use consonant or dissonant harmonies? Does it restrict itself, over moderate spans of musical time, to a small set of notes, and do these notes themselves change over larger time spans? Do melodies in general move by short distances? The answers to these questions tell us an enormous amount about how untrained listeners will respond to a piece. Hence my five features might be compared to a set of basic tools which composers can use to perform their musical magic.

Fourth, musical heterogeneity poses special problems for composers, who confront an audience of widely varying interests and abilities. Traditional tonal composers dealt with this by writing music that was immediately attractive, largely by virtue of exploiting the five features. Many composers also built into their music layers of additional, more complicated structure—complex thematic and formal interrelationships, intricate rhythmic devices, sophisticated contrapuntal tricks, and so on. The result was a kind of music that listeners could engage with in multiple ways: laypersons could simply enjoy a piece for the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

gross statistical features it shared with many other compositions in the same style, while cognoscenti could become more involved in the subtleties particular to that work.

I find it fascinating that so many twentieth-century musicians chose to abandon this strategy. Composers such as Schoenberg, Webern, Berg, and Varèse—and later Babbitt, Boulez, Cage, Xenakis, and Stockhausen—wrote music that listeners often found quite unpleasant. In many cases, this was because these composers rejected not just acoustic consonance, but also harmonic consistency, conjunct melodic motion, limited macroharmony, and centricity. Some early modernist composers may have hoped that ordinary listeners would adapt to this new musical style, adjusting their ears to the absence of familiar musical structures and coming to enjoy atonality as deeply and directly as traditional music. But after several decades of avant-garde exploration, composers began to realize that this might never occur. The *locus classicus* of this new perspective is Milton Babbitt's 1958 manifesto "Who Cares if You Listen?"—in which he cheerfully acknowledged that his music was not enjoyed by laypersons, but only by a specialist musical community analogous to the specialist community of professional mathematicians. [28]

Rather than criticizing this point of view, let me just say that I am interested in music that does not make this choice: I like music that brings people together, rather than dividing them, and I think the traditional strategy—writing immediately attractive music that also contains deeper levels of structure—is as potent as it ever was. (Indeed, it may be all the more necessary in an economic and cultural environment in which notated music is somewhat marginal.) In this book, therefore, I will be primarily concerned with effects that can be achieved *within* a broadly attractive sound-world. From a technical standpoint, this restriction is relatively unconstraining, as a large number of avant-garde techniques are easily applied within a consonant harmonic context: it is perfectly possible, for example, to write music that is diatonic, or more generally macroharmonically consonant, while also being serial, aleatoric, indeterminate, wildly polyrhythmic, and so on.[29] Thus, the choice between superficial accessibility and off-putting dissonance is not forced on us by our interest in particular musical techniques. Instead, it is a relatively independent reflection of our own aesthetic preferences.

1.5 OUTLINE OF THE BOOK, AND A SUGGESTION FOR IMPATIENT READERS

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

The book is divided into two halves, with the theoretical material front-loaded into the first five chapters. Chapter 2 reviews basic theoretical concepts and introduces simple geometrical models of musical structure, representing pitches as points on a line and pitch classes as points on a circle. These models are then used to investigate the relations between conjunct melodic motion, harmonic consistency, and acoustic consonance. Chapter 3 introduces higher-dimensional "maps" of musical space, providing powerful tools for visualizing the interactions between harmony and counterpoint. Chapter 4 introduces scales, describing them as musical "rulers" that allow musicians to measure the distance between notes; it then identifies a set of familiar scales that are interesting for a number of distinct reasons. Finally, Chapter 5 describes the components of the generalized theory of keys. It proposes various tools for representing macroharmony and centricity, and contrasts two traditions in twentieth-century music: the *chromatic tradition*, which largely abandons scales in favor of highly chromatic textures, and the *scalar tradition*, which makes use of an expanded range of scales and modes. Since Chapters 2–4 are the most technically demanding portion of the book, Appendix F provides a series of study questions to help reinforce the material. These questions make good homework assignments when the book is used in a classroom.

The second part uses these ideas to reinterpret the history of Western music. Chapter 6 proposes that there is an "extended common practice" stretching from the beginning of Western counterpoint to the tonal music of the twentieth century. What links these different styles is the combination of harmonic consistency and conjunct melodic motion: the idea that music should have a two-dimensional coherence, both harmonic (or vertical) and melodic (or horizontal). Chapter 7 uses geometrical models to investigate the functional harmony of the classical period, briefly considering the relation between traditional harmonic theory and the views of Heinrich Schenker. Chapter 8 explores the ways in which nineteenth-century composers exploited efficient voice leading in chromatic space, suggesting that there is more structure to chromatic music than we might expect. Chapter 9 argues that twentieth-century tonal composers used scales to counteract the trend toward a saturated chromaticism, fusing chromatic voice-leading techniques with the limited macroharmony of earlier periods. Finally, Chapter 10 treats what I call the "modern jazz synthesis," a contemporary common practice that unites impressionist chords and scales, chromatic voice leading, and the functional harmony of the classical era.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

The design of the book means that readers will need to absorb a considerable amount of theoretical material before reaching the analytical payoff in the second half. Readers who are less interested in theory for its own sake may therefore want to read the book out of order. Chapter 8 can largely be read directly after Chapters 2 and 3, although the discussion of *Tristan* will be enhanced by familiarity with the material in §§4.8–10. Chapter 9 can be read directly after Chapter 4. This abbreviated path hits the main theoretical highlights (new tools for understanding voice leading and scales), as well as the most important analytical applications: to the chromatic tonality of Schubert, Chopin and Wagner, and to the twentieth-century scale-based tonality of Debussy, Stravinsky and Reich. When using the book in an advanced undergraduate theory class, I have assigned pieces from Chapters 8–9 as homework assignments early in the semester, while students are still reading the introductory chapters. (For instance, I assign Chopin's E minor prelude and F minor mazurka in the week when students are reading Chapter 3.) That way, students have an opportunity to work with the music on their own before being introduced to my own particular perspective on these remarkable works.

Finally, a word of encouragement: it is possible to understand the gist of later chapters even while remaining somewhat fuzzy about the technical material in Chapters 2–4. So don't be afraid to forge ahead, returning to earlier sections as the need arises.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

CHAPTER 2

Harmony and Voice Leading

The enterprise of "musical set theory" aspires to catalogue all the chords available to contemporary composers. Unfortunately, this project turns out to be more complicated than one might imagine. This is, first, because chords have an intrinsic and sophisticated geometry, with distance being determined by voice-leading size. Second, basic musical concepts such as "transposition," "inversion," "triad," and "chord type" can all be *relativized* to scales; thus two chords may belong to the same category relative to one scale but not another. (To make matters worse, we can also do set theory in an unquantized space that admits a continuous infinity of notes between B and C. This perspective can even teach us useful lessons about the discrete world of ordinary musical experience.) Third, chord tones themselves can be differentiated in terms of their importance, leading to phenomena such as rootedness and centricity. Finally, in the most sophisticated versions of set theory, the objects of comparison may themselves be chord progressions, with musical categorization proceeding flexibly and according to an ever-changing variety of symmetry operations (to be discussed shortly).

The next four chapters will try to address these issues, describing a new approach to chords and rebuilding "musical set theory" from the ground up. This chapter introduces the basic concepts and definitions. We begin with elementary geometrical models of musical structure, representing notes as points on a line (*pitch space*) and on a circle (*pitch-class space*). We then turn to higher-order objects—chord progressions and voice leadings—that describe motion through time. (Because this formalism sets the stage for much of the rest of the book, I encourage you to work through the study questions in Appendix F.) Next we consider the importance of symmetry in music theory, asking under what conditions harmonic consistency and conjunct melodic motion can be combined. This leads to our first significant theoretical result: a general understanding of the interdependence between acoustic consonance, efficient voice leading, and harmonic consistency.

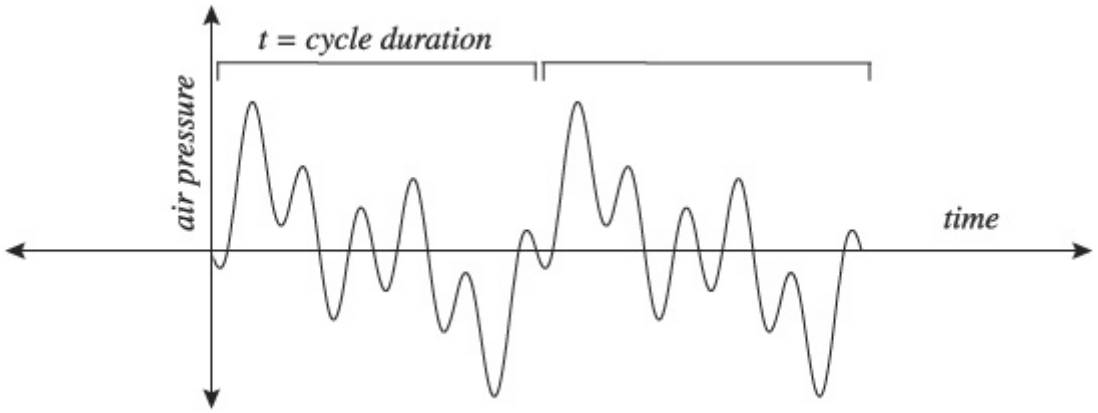
2.1 LINEAR PITCH SPACE

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Sound consists of small fluctuations in air pressure, akin to changes in barometric pressure. These fluctuations are heard as having a definite pitch when they repeat themselves (at least approximately) after some period of time t (Figure 2.1.1). The reciprocal of the period, $1/t$, is the *fundamental frequency* of the sound—a number that measures how many repetitions occur per unit of time. Musicians without absolute pitch are sensitive not to fundamental frequencies as such, but rather to the *ratios* between them. Suppose, on Tuesday, a typical person whistles a tune whose pitches have frequencies $f, g, h, \ldots$. On Wednesday, if asked to reproduce the tune, she is likely to whistle the frequencies $cf, cg, ch, \ldots$, where c is some number close to one. Musicians say that the whistler has *transposed* the notes, changing their fundamental frequencies so as to preserve the ratios between them. This suggests that frequency ratios represent a kind of musical distance, and that listeners are more attuned to the distances between notes than to their absolute positions in frequency space.[1]

Figure 2.1.1 Pitched sounds are periodic. The fundamental frequency is equal to $1/t$, where t is the duration of the cycle.



Musicians do not like to work with fundamental frequencies, since ratios are awkward and division is hard; furthermore, we typically measure numerical distances using subtraction rather than division. It is useful, therefore, to relabel pitches by mapping every fundamental frequency f onto a number p according to an equation of the form

$$p = c_1 + c_2 \log_2 (f/440)$$

Don't be put off by the logarithm: the important point is that in the new system, the whistler whistles the pitches $p, q, r, \ldots$ on Tuesday, and $p + x, q + x, r + x, \ldots$ on

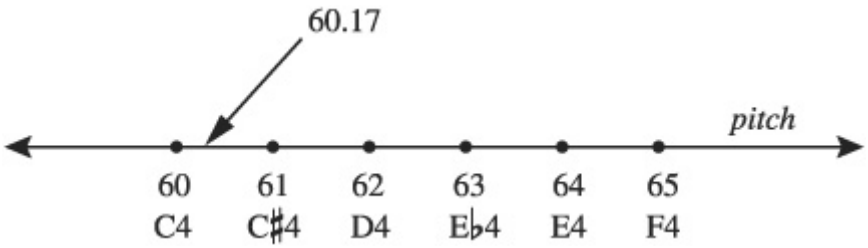
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Wednesday. The distance between two pitches p and q is now calculated by subtraction ($|p - q|$) rather than division (f/g).

The equation in the preceding paragraph has two constants, c_1 and c_2, the first of which determines the number corresponding to the frequency A440, and the second of which determines the size of the octave. In this book, I will always choose $c_1 = 69$ and $c_2 = 12$. This creates a linear *pitch space* in which the unit of distance is the *semitone* and middle C is (arbitrarily) assigned the number 60. (Note that the term "semitone" is defined in continuous space, without presupposing any particular temperament or chromatic scale.) As shown in Figure 2.1.2, labels for familiar equal-tempered pitches can be determined by counting keys on an ordinary piano, so that the C♯ above middle C is 61, the D above that is 62, and so on. Since the space is continuous, we can use fractional numbers, labeling the pitch 17 hundredths of a semitone (or *cents*) above middle C as 60.17. In principle, we could eschew letter-names for the remainder of this book, conducting the discussion entirely in numerical terms. For clarity, however, I will use familiar letter names whenever possible. These should be taken as shorthand for numerical pitch labels: "C♯4" and "61," like "Bob Dylan" and "Robert Zimmerman," are different ways of referring to the same thing.

Figure 2.1.2 Linear pitch space is a continuous line, containing a point for every conceivable pitch.



2.2 CIRCULAR PITCH-CLASS SPACE

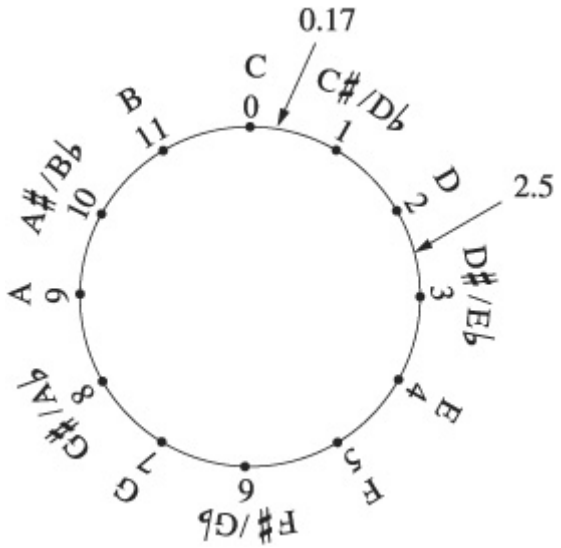
In linear pitch space octaves are not special, and the distance 12 is no different from any other. However, human beings hear octave-related pitches as having the same quality, color, or—as psychologists call it—*chroma*.[2] (As Maria puts it in *The Sound of Music*, the note "Ti" brings us *back* to "Do.") Music theorists express this by saying that two pitches

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

an octave apart belong to the same *pitch class*. Geometrically, pitch classes can be represented using a circle (Figure 2.2.1). A single point in this space corresponds to the quality that is common to all the pitches sharing the same chroma: thus, the point "2" represents the quality ("D-ishness") that is shared by the pitches D0, D1, D2, D3, and so on. [3] Alternatively, we can take points on the circle to represent *categories* of notes all sharing the same chroma.

Figure 2.2.1 Circular pitch class space. The number 0.17 refers to the pitch class seventeen cents above the pitch class C, while the number 2.5 refers to the pitch class D♯, or D quarter-tone sharp, halfway between D and E♭.



It is important to understand that pitch and pitch-class space are not separate and independent: pitch-class space is formed out of pitch space when we choose to ignore, or abstract away from, octave information.[4] As a result, many properties of linear pitch space are transferred to circular pitch-class space. For example, it is not necessary to devise a new system of naming pitch classes, as we can simply use pitch names in the range $0 \leq x < 12$ to refer to the chromas they possess (Figure 2.2.1). The terms "above" and "below" also inherit their meaning from pitch space: the phrase "the pitch class a quarter tone above pitch class D" simply refers to the chroma possessed by any pitch a quarter tone above any pitch with chroma D. (Of course, since pitch-class space is circular, it is not meaningful to say that one pitch class is absolutely above or below another; the pitch class E is both two semitones above D and ten semitones below it.) Finally, the *distance* between two pitch classes can be defined as the shortest distance between any two pitches

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

belonging to those pitch classes. Thus, when a musician says "pitch class E is four semitones away from C," this means that for every pitch with chroma C, the nearest pitch with chroma E is precisely four semitones away.

Ultimately, pitch classes are important because they provide a language for making generalizations about pitches. The statement "pitch class E is four semitones above C" implies the statements "E4 is four semitones above C4," "E5 is four semitones above C5," and so on. Since it would take too long to list these statements individually, musicians instead use the convenient shorthand that pitch classes provide. Readers may find it useful here to consider the analogy between pitch class and time of day. The question "what's the distance from C to C♯?" is exactly analogous to "how long is it from 8 AM to 9 AM?" The answer "there is one hour between 8 AM and 9 AM," means that on any given day, 9 AM occurs precisely one hour after 8 AM. (Similarly, we can say "9 AM is one hour after 8 AM" or even "9 AM is 23 hours before 8 AM," much as we might observe that pitch class A is both one semitone above and eleven semitones below G♯.) Here we use time of day to summarize infinitely many facts about times: the statement "there is one hour between 8 AM and 9 AM" implies that there is one hour between 8 AM and 9 AM on Monday, August 8, 2006, one hour between 8 AM and 9 AM on Tuesday, August 9, 2006, and so on *ad infinitum*.

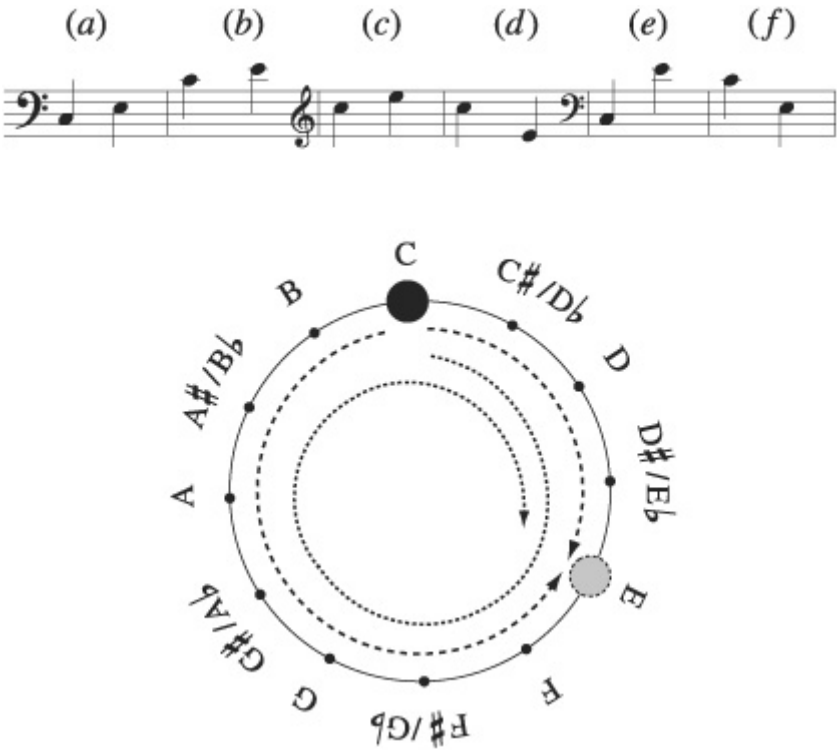
One unusual feature of this book is that I use *paths in pitch-class space* to represent intervals, or particular ways of moving from one pitch class to another.[5] The motivation here is that progressions like the first three in Figure 2.2.2 are closely related: in each case a C moves up by four semitones to an E, with the only difference being the octave in which the motion occurs. In the final three cases, C also moves to E, but differently—by eight descending semitones or 16 ascending semitones. We can capture what is similar about the first three cases, and their difference from the others, by modeling intervals using phrases such as "start at C and move up by four semitones." These combine an initial pitch class (C), a direction (up), and a distance (4), and can be notated $\mathrm{C} \xrightarrow{+4} \mathrm{E}$. Geometrically, these phrases correspond to ways of moving around the pitch-class circle: $\mathrm{C} \xrightarrow{+4} \mathrm{E}$ is represented by a four-semitone clockwise (ascending) path, whereas $\mathrm{C} \xrightarrow{-8} \mathrm{E}$ is represented by an eight-semitone *counterclockwise* (descending) path. (Motions can even wrap around the circle one or more times, as shown on the figure.) By contrast, music theorists have traditionally represented motion in pitch-class space using phrases like

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

"start at C and move to E however you want." This gives us no way to formalize the elementary musical observation that the first three progressions in Figure 2.2.2 are particularly closely related.[6]

Figure 2.2.2 The passages in (*a*)–(*c*) are similar in that they move C to E by four ascending semitones. The passages in (*d*)–(*f*) also move C to E, but differently—by eight descending semitones or by sixteen ascending semitones. We can capture what is similar about (*a*)–(*c*) by modeling these progressions as paths in pitch-class space. The progressions in (*a*)–(*c*) move C to E clockwise by four semitones along the pitch-class circle; (*d*) and (*f*) move eight semitones counterclockwise, and (*e*) moves sixteen clockwise semitones.



2.3 TRANSPOSITION AND INVERSION AS DISTANCE-PRESERVING FUNCTIONS

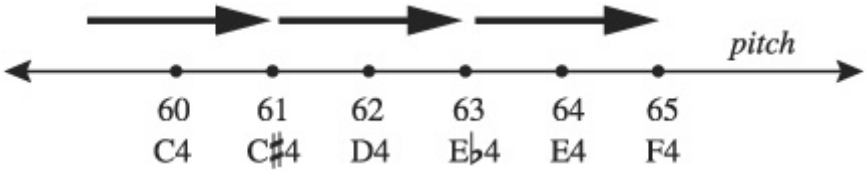
Since musicians are primarily sensitive to the distances between notes, we have reason to be interested in the distance-preserving transformations of musical space. It turns out that there are only two of these—*transposition* and *inversion*, corresponding to the geometrical operations of *translation* and *reflection*. These transformations play an important role in many different musical styles, and are central to contemporary music theory.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Transposition, discussed in §2.1, moves every pitch in the same direction by the same amount: the transposition of pitch p by x semitones, written $\mathbf{T}_x(p)$, is equal to $p + x$ (Figure 2.3.1). The distance moved represents the *size* of the transposition: virtually any listener can immediately distinguish a passage of music from its transposition by five octaves, while even the most subtle absolute-pitch listeners, extensively trained at the best conservatories, cannot discern the effects of transposition by 0.00001 of a semitone.

Figure 2.3.1 Transposition moves every point in the same direction by the same amount. Here, the arrows indicate ascending transposition by two semitones.



The second type of distance-preserving transformation, inversion, turns musical space upside-down. (Warning: music theorists use the term "inversion" in two unrelated senses, which go by the names of "registral inversion" and "pitch-space inversion"; I am talking about the second of these.[7]) Figure 2.3.2 depicts inversion as it operates on the theme of Bach's A minor prelude from Book II of the *Well-Tempered Clavier*. Where the left hand of (*a*) begins with an ascending octave leap, followed by a series of descending semitones, the left hand of (*b*) begins with a *descending* octave leap followed by a series of ascending semitones. In other words, the direction of motion has changed while the distances remain the same. (The passages in the upper staff are also related by inversion, although the first and last notes are altered slightly.) Because they are *direction reversing*, inversions change the character of musical passages more dramatically than transpositions: nobody would ever mistake the inverted form of Bach's theme for the original. Nevertheless, inversionally related chords often sound reasonably similar, at least in the grand scheme of things: the chords in Figure 2.3.3a contain a minor third, a minor seventh, and a perfect fifth, and sound reasonably restful, while those in (*b*) contain a perfect fifth, a tritone, and a minor second, and have a distinctive, dissonant "bite."[8] For this reason, many twentieth-century composers consider inversionally related chords to be similar.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.3.2 (*a–b*) Inversionally related passages in Bach's A minor prelude, WTC II. (*c*) Inversion as reflection in pitch space. Here, the note A3 is unaltered by the inversion, so the inversion can be written I^{A3}_{A3}. All other notes move by twice their distance from A3.
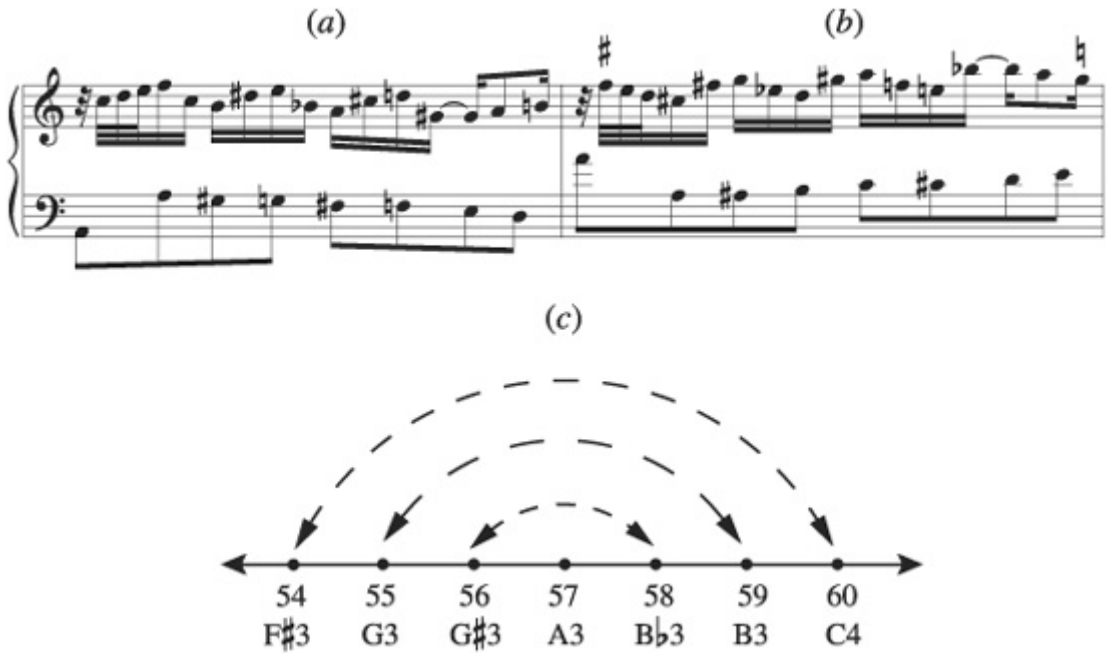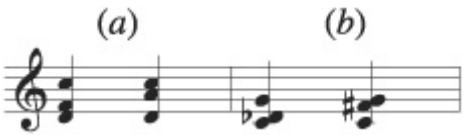


Figure 2.3.3 The chords in each measure are inversionally related, and sound more similar to each other than to either of the chords in the other measure.

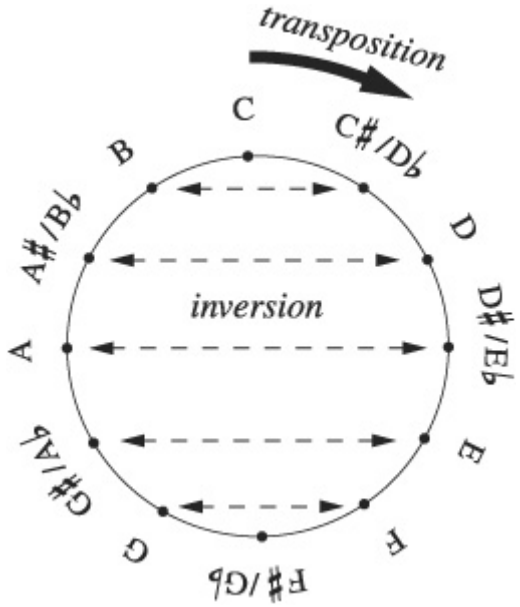


Inversion can be represented mathematically by subtraction from a constant value: the inversion that maps pitch x to pitch y, written $I^x_y(p)$, is $(x + y) - p$.[9] Geometrically, inversion corresponds to *reflection:* if we were to place a two-sided mirror at point A3 on Figure 2.3.2c, then every pitch would be sent to the place where its reflection would be. The point A3 is a *fixed point* of the inversion, since it is unaltered by the reflection. Every inversion has a fixed point, although it may lie between piano keys: for instance, the inversion that maps E4 to F4 has a fixed point at E♯4—the E quarter-tone sharp halfway between E4 and F4.[10] (Note that inversions move each pitch by twice its distance to this fixed point.) Inversions, unlike transpositions, cannot be distinguished by their size, since for any distance x, an inversion will move some pair of pitches by that amount.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.3.4 In circular pitch class space, transposition corresponds to rotation, while inversion corresponds to reflection.



Transposition and inversion can also be defined in pitch-class space. Figure 2.3.4 shows that transposition is represented by *rotation* while inversion is represented by reflection: the inversion around C sends every pitch class to the point where its image would appear, if there were a pointlike mirror at C. Note that every pitch-class inversion fixes *two* antipodal points on the circle, whereas pitch inversion fixes only a single point.[11] Our earlier mathematical formulas continue to apply: if p is a pitch class, then its transposition by x semitones is $p + x$, while the inversion that sends x to y is $(x + y) - p$. However, when operating with pitch classes we should always add or subtract 12 until the result lies in the range $0 \leq x < 12$. This can lead to odd-looking equations like $6 + 3 + 1.5 + 1.5 = 0$, since 12 and 0 are equivalent in pitch-class arithmetic.[12]

2.4 MUSICAL OBJECTS
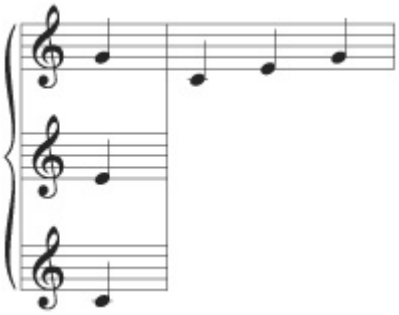
Understanding music is a matter of ignoring, or abstracting away from, information: we interpret the violinist's C4 and the cellist's C4 to be two instances of the same pitch, ignoring subtle differences in timbre and instrument. Similarly, we conceptualize the violinist's note, a little sharp and ever-so-slightly before the beat, as an in-tune note played on the beat. We can model this process using the mathematical concept of symmetry: to

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

abstract away from musical information is to define a collection of *symmetry operations* that leave the object's "essential identity" unchanged. Mathematicians say that objects related by a symmetry operation belong to the same *equivalence class*—the group of objects that are mutually "equivalent" under the effects of the symmetry operation. These equivalence classes can in turn be taken to represent the *properties* or *attributes* the objects share, much as the group of all white things can be taken to represent the property of whiteness. As we will see, many different music-theoretical concepts can be understood in this way.

Figure 2.4.1 The basic musical object (C4, E4, G4) can appear either harmonically or melodically.



We can begin by defining a *basic musical object* as an ordered series of pitches, uncategorized and uninterpreted.[13] Basic musical objects can be ordered in time or by instrumental voice: thus the object (C4, E4, G4) could represent an ascending C major arpeggio or a simultaneous chord in which the first instrument plays C4, the second E4, and the third G4 (Figure 2.4.1). (Instruments can be labeled arbitrarily; what matters is simply that we distinguish them somehow.) Basic musical objects are so particular as to be uninteresting. Until we have defined some symmetry operations, some way of grouping objects into larger categories, the object (C4, E4, G4) is absolutely different from and absolutely unrelated to (E4, C4, G4). Of course, the fact that we instinctively consider them to be similar shows that we categorize musical objects unthinkingly; indeed, it is difficult to conceive of music without grouping objects together in *some* way.

During the early eighteenth century, Jean-Philippe Rameau articulated the modern notion of a chord, classifying basic musical objects based on their pitch-class content rather than their order or registral arrangement.[14] Rameau implicitly suggested that three basic operations preserve the "chordal" or "harmonic" identity of a musical object: octave shifts, permutation (or reordering), and cardinality change (or note duplication). For

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

instance, one can transform (C4, E4, G4) by reordering its notes to produce (E4, G4, C4), transposing the second note up an octave to produce (C4, E5, G4), or duplicating the third note to produce (C4, E4, G4, G4)—all without changing its right to be called a "C major chord." Furthermore, as shown in Figure 2.4.2, these transformations can be combined to produce an endless collection of objects, all representing the same chord: (E4, G4, C5), (G3, G4, C5, E4), (E2, G3, C4, E4, E5), and so on. To be a C major chord is simply to belong to this equivalence class—or in other words, to contain all and only the three pitch classes C, E, and G. We can therefore represent the C major chord as the unordered set of pitch classes {C, E, G}. (Note that I use curly braces to represent unordered collections, reserving parentheses for ordered collections.) Geometrically, chords correspond to particular collections of points on a circle, as in Figure 2.4.2.

Traditional theory uses terms like "major chord" and "minor chord" to represent *chord types*, or collections of transpositionally related chords. We can interpret these terms as referring to equivalence classes formed by *four* symmetry operations: octave shifts, permutations, cardinality changes, and transpositions (Figure 2.4.3).[15] (Note that contemporary music theorists sometimes prefer the term *transpositional set class* to the more colloquial "chord type.") Geometrically, two chords belong to the same type if one can be rotated into the other in circular pitch-class space (Figure 2.4.3b). Such chords will share the same sequence of distances between their adjacent notes: for example, major chords divide the pitch-class circle into arcs of four, three, and five semitones, reading clockwise from the root.

Figure 2.4.2 (*a*) All of these musical objects represent the C major chord, and are related by some sequence of octave shifts, permutations (or rearrangement of voices), and note duplication. (*b*) The C major chord can be represented by an unordered set of points in pitch-class space.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

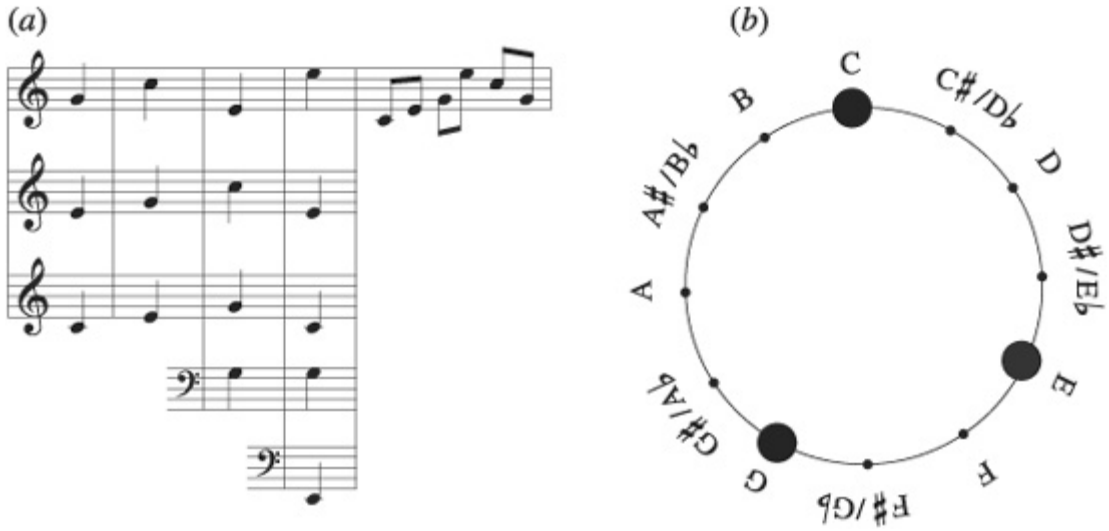
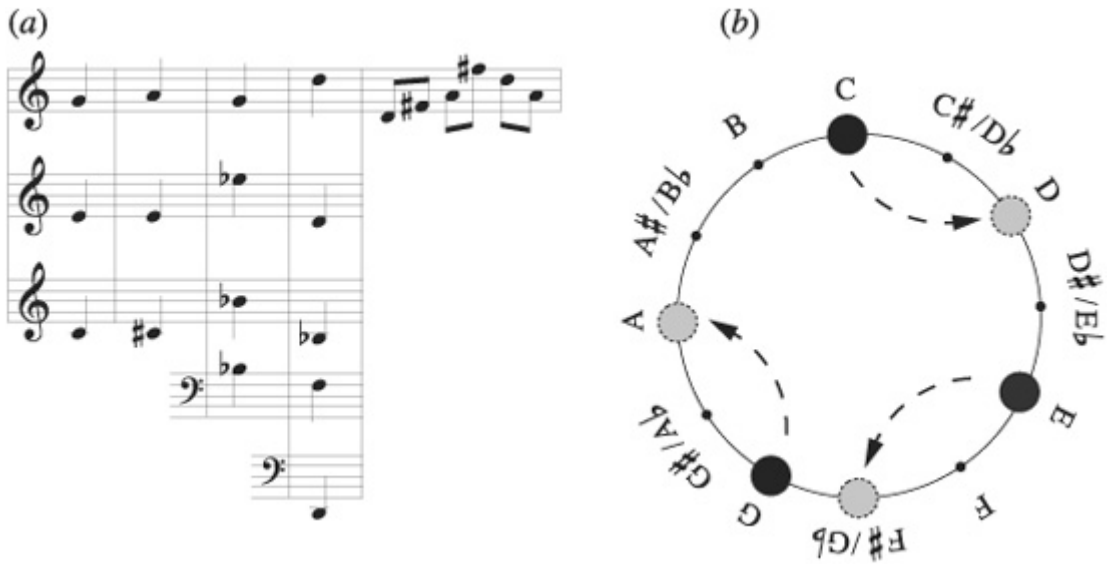


Figure 2.4.3 (*a*) All of these musical objects represent major chords, and are related by some combination of octave shifts, reordering of voices, note duplication, and transposition. (*b*) Any two major chords relate by rotation on the pitch-class circle and divide the pitch class circle into arcs that are 4, 3, and 5 semitones large, moving clockwise from the root. The figure shows the C and D major triads.



Since inversionally related chords share many properties (and sound reasonably similar), contemporary theorists often group them together as well. Music theorists say that two objects belong to the same *set class* if they are related by any combination of *five* symmetry operations (the "OPTIC" symmetries): octave shifts (O), permutations (P), transpositions (T), inversions (I), and cardinality changes (C) (Figure 2.4.4).[16] Figure 2.4.5 shows that the C major chord divides pitch-class space into arcs that are four, three, and five semitones large, reading clockwise from C, while its inversion, the C minor chord, divides the pitch-class circle into arcs that are four, three, and five semitones large reading

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
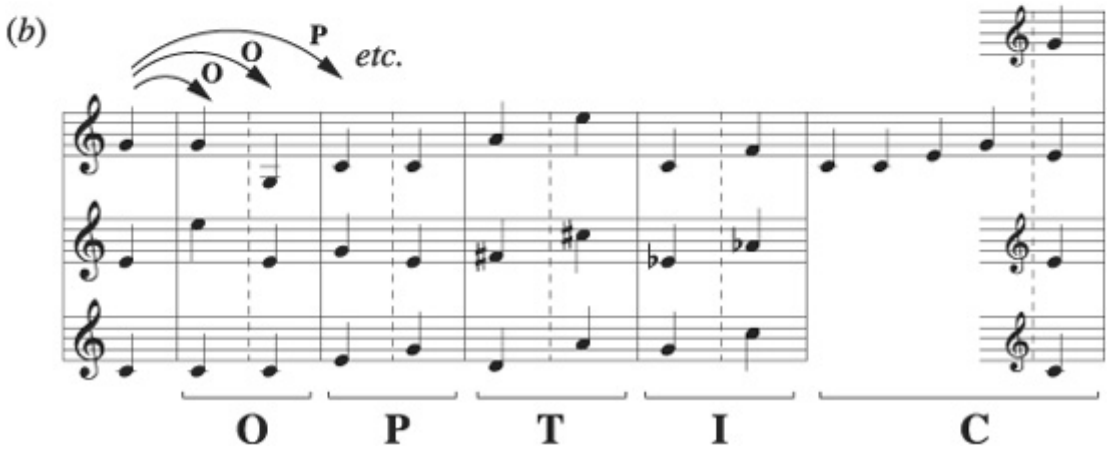
counterclockwise from G. Chords belonging to the same set class will always share the same sequence of arc lengths, although this sequence may progress either clockwise or counterclockwise around the pitch-class circle.

Figure 2.4.4 (*a*) Each of the five OPTIC operations allows you to transform a musical object in some way. Musical objects belong to the same *set class* if they can be transformed into each other (or into some third chord) by some sequence of OPTIC transformations. (*b*) These objects all belong to the same set class, and relate to the initial (C4, E4, G4) by octave shift, reordering of voices, transposition, inversion, or cardinality change.

(*a*)

Operation	Allowable Action
Octave	Move *any* note into a new octave.
Permutation	Reorder the object, changing which voice is assigned to which note.
Transposition	Transpose the object, moving *all* of its notes in the same direction by the same amount.
Inversion	Invert the object by turning it "upside down."
Cardinality change	Add a new voice duplicating one of the notes in the object.

(*b*)



A specific example will help reinforce this approach to chord classification. Figure 2.4.6 begins with the basic musical object (E4, G4, B♭4, D5)—a half-diminished chord starting on the E above middle C. We then exploit the octave symmetry to produce (E3, G4, B♭3, D4). The permutation symmetry then rearranges the voices in which the notes appear,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

transforming (E3, G4, B♭3, D4) into (E3, B♭3, D4, G4); now the order of the voices corresponds to their registral order. We then use the transposition symmetry to shift the entire object up by semitone, giving (F3, B3, D♯4, G♯4)—recognizable as the first chord in Wagner's *Tristan*. Next, we apply the I symmetry to invert the chord around the point halfway between A3 and B♭3, transforming (F3, B3, D♯4, G♯4) into (D4, G♯3, E3, B2). The octave and permutation symmetries can then be used to give us (E3, G♯3, D4, B4), which is the second chord in *Tristan*. Finally, we can use cardinality changes to create additional voices that double notes already in the chord.

Figure 2.4.5 Major and minor chords relate by inversion, and divide the pitch-class circle into arcs that are 4, 3, and 5 semitones large, proceeding either clockwise (major) or counterclockwise (minor).

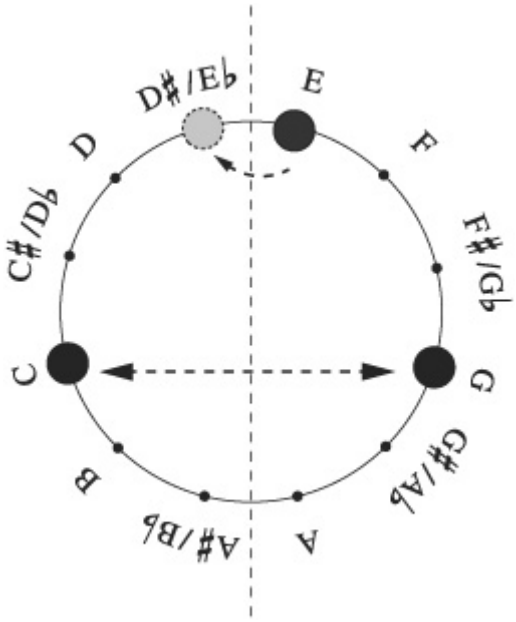
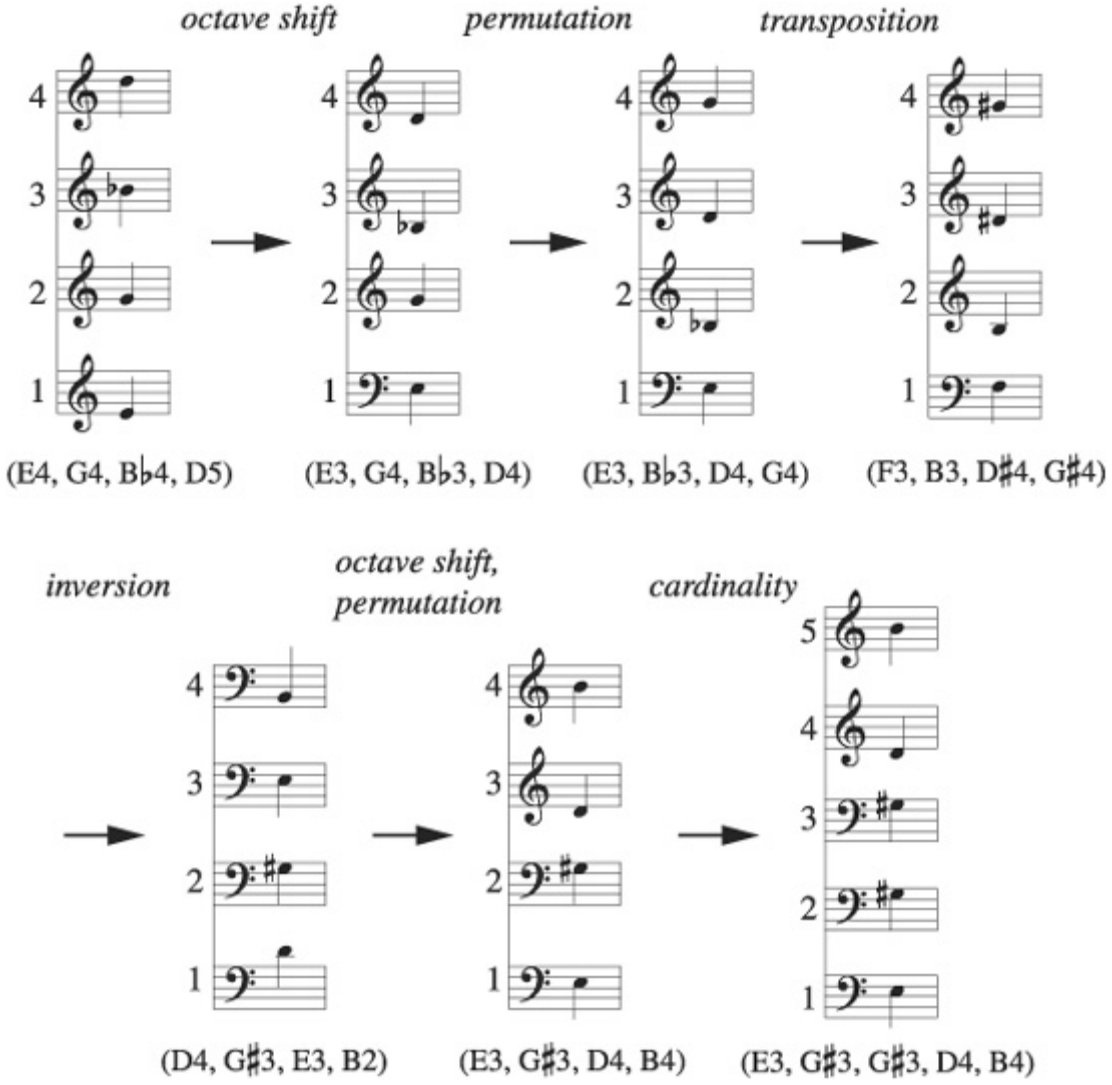


Figure 2.4.6 Using OPTIC operations to transform musical objects.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Musical classification thus proceeds by the progressive discarding of information. When we understand music as a sequence of pitches, we disregard many specific features of the acoustic signal, such as timbre, vibrato, duration, and rhythm. When we think in terms of pitch classes, we disregard the particular octave in which notes appear. To form chords, we ignore even more information, considering only the pitch-class content of a group of notes, rather than their order or multiplicity. Finally, we form chord types (or set classes) by focusing on the distances between a chord's notes rather than its pitch classes. Thus when we say that an object is a major chord, we are neglecting an enormous number of musical details, leaving behind something that is very abstract—an ordered sequence of clockwise distances around the pitch-class circle.

Although chords, chord types, and set classes are central to contemporary theory, they are not the only possibilities to consider. In certain situations it may be useful to talk about unordered sets of *pitches* rather than pitch classes ("chords" of pitches), or to distinguish the *multiset* {C, C, E, G}, which contains two copies of the pitch class C, from {C, E, G},

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

which contains only one C. We may at other times want to consider "tone rows" or *ordered* sequences of pitch classes, such as (C, E, G). (Order is particularly important when modeling melody, of course.) As shown in Figure 2.4.7, each of these terms corresponds to a unique combination of the five OPTIC symmetries. These alternatives all represent potentially useful ways of classifying musical objects, and there is no one optimal degree of abstraction: since different musical purposes require different kinds of information, we need to remain somewhat flexible about how we conceptualize music. To this end, it can be quite instructive to work out the musical significance of all 32 combinations of the OPTIC symmetries.

Figure 2.4.7 Music-theoretical terms and the symmetry operations to which they correspond. A "chord" is a group of musical objects related by octave shifts, permutations, and note-duplications, while a "multiset" is a group of objects related by octave shifts and permutation. (Thus, when we are talking about multisets, the number of times a note appears is important; when we are talking about chords it is not.) In a "tone row," as defined by Schoenberg, order is important: we are permitted only to shift octaves and introduce note duplications.

Term	Symmetry
chord	OPC
"chord type" or transpositional set class	OPTC
set class	OPTIC
multiset of pitch classes	OP
chord (of pitches)	PC
"tone row" (ordered set of pitch classes)	OC

2.5 VOICE LEADINGS AND CHORD PROGRESSIONS
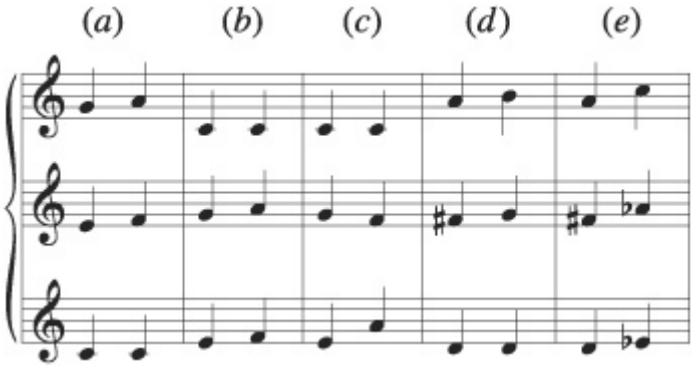
Having reviewed some basic objects of music theory, our next task is to consider *progressions*, or sequences of musical objects, as musical entities in their own right. In other words, we will shift from an object-based approach (whose subject is single chords such as {C, E, G}) to a transformational approach (whose subject is progressions such as "C4, E4,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

G4 followed by C4, F4, A4").[17] This increased level of abstraction can be a little confusing, and readers may want to brace themselves for a slight uptick in the level of theoretical difficulty.

Figure 2.5.1 (*b*) relates to (*a*) by uniform permutation. To get from (*a*) to (*b*) move all notes downward by one staff, shifting the bottom staff to the top. (*c*) relates to (*a*) by individual permutation, since it applies different permutations to each chord. Similarly, (*d*) and (*e*) relate to (*a*) by uniform and individual transposition, respectively.



We can classify progressions using the very same OPTIC symmetries described in the preceding section. The main complication is that progressions are "higher-order" constructions containing multiple individual objects, which means that each symmetry can be applied in two different ways: we can either apply the *exact same* operation to each object in a progression or we can use different versions of the same symmetry on the progression's two objects. Take the progression (C4, E4, G4)–(C4, F4, A4), whose first object is (C4, E4, G4) and whose second is (C4, F4, A4). To apply the permutation symmetry *uniformly* is to reorder each of the two objects in precisely the same way, for instance producing (E4, G4, C4)–(F4, A4, C4) (Figure 2.5.1). (Here we have moved the first note in each object to the end.) *Individual* permutations apply different reorderings to the two objects in the progression, and can produce results such as (C4, E4, G4)–(F4, C4, A4). (Here we leave the first object unchanged, while switching the first two notes in the second object.) The distinction between "individual" and "uniform" can be applied to the other symmetries as well: the progression (C4, E4, G4)–(C4, F4, A4) can be transposed uniformly to produce (D4, F♯4, A4)–(D4, G4, B4) and individually to produce (D4, F♯4, A4)–(E♭4, A♭4, C5). Since each of the five OPTIC symmetry operations can be applied uniformly, individually, or not at all, there are many more categories of progressions than
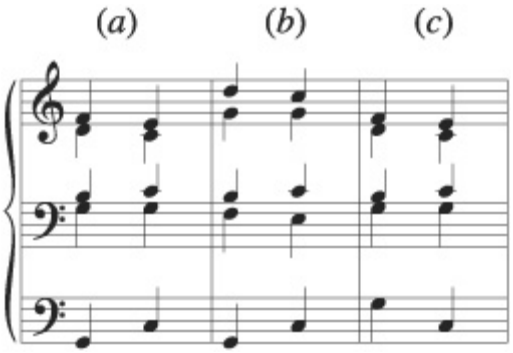
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

there are of individual objects.

Of these, two are particularly important for our purposes: *voice leadings*, which describe how individual musical "voices" move from chord to chord, and *chord progressions*, which describe successions of harmonies with no regard for musical voices. Intuitively, voice leadings can be represented by phrases such as "move the C major triad to a C♯ diminished triad by shifting the note C up by semitone," while chord progressions correspond to phrases such as "move the C major triad to the C♯ diminished triad however you want." Mathematically, it turns out that voice leadings arise from *uniform* applications of the permutation symmetry, while chord progressions arise from *individual* applications of the permutation and cardinality-change symmetries. (These individual transformations destroy the identity of musical "voices," since they change the two chords in different ways.) One way or another, voice leadings and chord progressions will occupy us for the rest of the book.

Figure 2.5.2 Three voice leadings between G^7 and C.



We'll begin with *voice leadings in pitch space*, which represent mappings from one collection of pitches to another.[18] For example, Figure 2.5.2a presents the voice leading (G2, G3, B3, D4, F4)→(C3, G3, C4, C4, E4), in which the voice sounding G2 in the first chord moves to C3 in the second chord, the voice sounding G3 in the first chord continues to sound G3 in the second, the voice sounding B3 moves up to C4, and so on. (Note that I use an arrow to represent voice leadings.) Since the overall order in which the voices are listed is not significant, one could also represent this voice leading as (F4, D4, G2, B3, G3) →(E4, C4, C3, C4, G3); what matters is simply that we describe how the notes in one chord move to those in the next.[19] Voice leadings are like the atomic constituents of musical scores, the basic building blocks of polyphonic music.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Clearly, the voice leading in Figure 2.5.2b is closely related to that in Figure 2.5.2a; all that has changed is the octave in which some voices appear. We can represent what is common to them by writing $(\text{G, G, B, D, F}) \xrightarrow{5, 0, 1, -2, -1} (\text{C, G, C, C, E})$. This indicates that one of the voices containing G, whatever octave it may be in, moves up five semitones to C; the other voice containing G is held over into the next chord; the B moves up by semitone to C; and so on. I will refer to such octave-free voice leadings as *voice leadings between pitch-class sets* or *pitch-class voice leadings*.[20] The numbers above the arrow, here (5, 0, 1, –2, –1), are paths in pitch-class space that describe how the voices move (§2. 2). The two pitch-space voice leadings in Figure 2.5.2a-b are both *instances* of the pitch-class voice leading $(\text{G, G, B, D, F}) \xrightarrow{5, 0, 1, -2, -1} (\text{C, G, C, C, E})$. Figure 2.5.2c is not an instance of this voice leading, since G moves to C by seven descending semitones rather than five ascending semitones. (The specific path matters!) For simplicity, I will omit the numbers over the arrow when the number of semitones moved by each voice lies in the range $-6 < x \leq 6$. Thus I will write (G, G, B, D, F)→(C, G, C, C, E) for the pitch-class voice leading in Figures 2.5.2a–b. This indicates that each voice moves by the *shortest possible path* to its destination, with the arbitrary convention being that tritones ascend.

Pitch-class voice leadings are abstract schemas that are central to the enterprise of Western composition. Without them, a composer would need to conceptualize the voice leadings in Figure 2.5.3 as being entirely unrelated, and this would pose overwhelming burdens on his or her memory. It is far simpler, and more practical, to understand the two voice leadings as examples of a single principle: *you can transform a C major triad into an F major triad by moving the E up by semitone and the G up by two semitones*. The concept "pitch-class voice leading" is simply a tool for formalizing such principles, and thus for modeling one important aspect of the composer's craft. Of course, in actual compositions voice leadings are always represented by specific pitches, and composers always think very carefully about register. Nevertheless, it is also true that composition requires a general sense of the various "routes" from chord to chord, and these routes are precisely what pitch-class voice leadings describe.

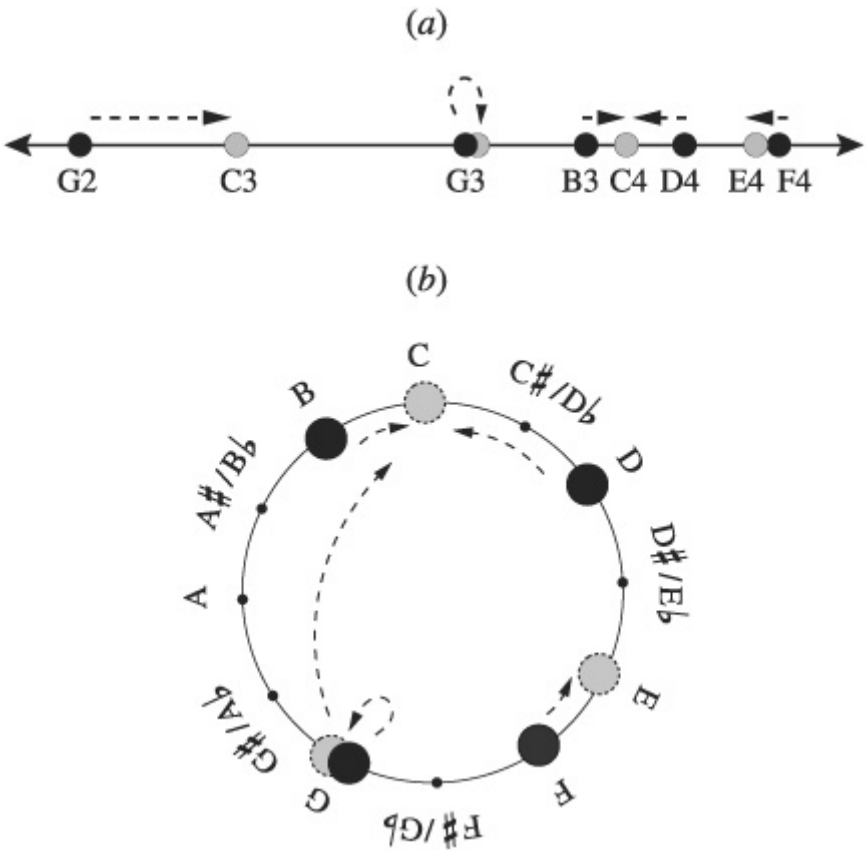
Figure 2.5.3 Two instances of the same voice-leading schema.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Figure 2.5.4 (*a*) Voice leadings in pitch space can be represented as paths on a line. (*b*) Pitch-class voice leadings can be represented as paths on a circle. These paths can move in either direction by any distance, and may complete one or more circumferences of the circle.



Geometrically, a pitch-space voice leading corresponds to a collection of paths in linear pitch space. Figure 2.5.4a represents the pitch-space voice leading in Figure 2.5.2a, with the dotted lines showing how each pitch moves to its destination. Similarly, a pitch-class voice leading can be represented as a collection of paths in circular pitch-class space. In principle these paths can be arbitrarily long, with notes taking one or more complete turns around the pitch-class circle. Indeed, any collection of paths on the pitch-class circle, between any points whatsoever, defines a voice leading in pitch-class space. In some cases, it can be useful to represent voice leadings by a continuous process (which could be shown

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

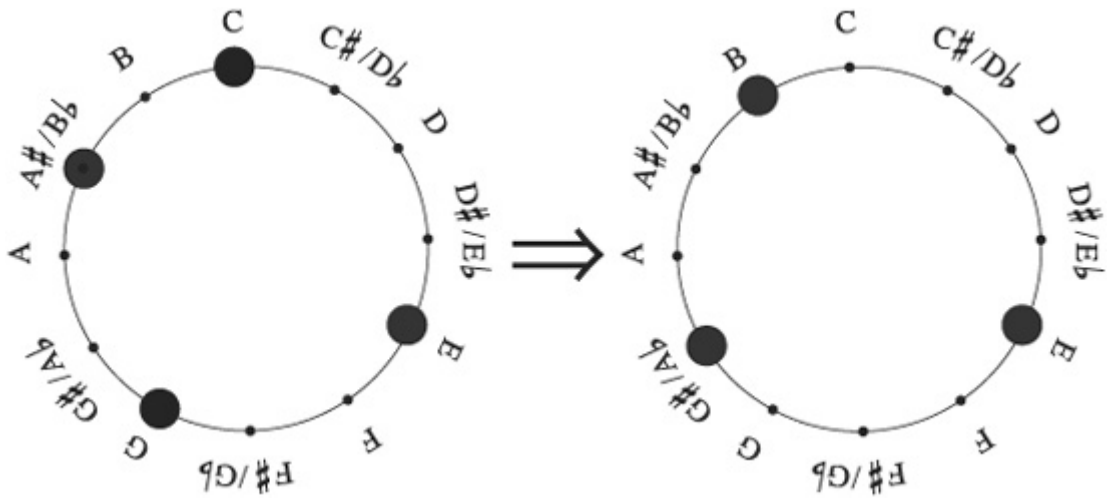
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

as a movie) in which each note glides along the appropriate path to its destination, beginning and ending at the same time.[21]

By contrast, *chord progressions* (as I define them) are simply successions of chords, with no implied mappings between their notes. Typically, the term "chord progression" is understood to refer to a sequence of unordered *pitch-class sets*. Thus {C, E, G, B♭}⇒{E, G♯, B}, or C^7⇒E, is a chord progression whose first chord is {C, E, G, B♭} and whose second chord is {E, G♯, B}, but which does *not* associate the note C with any particular note in the second chord.[22] (Note that I use the double arrow for chord progressions, reserving the single arrow for voice leadings.) However, we could also define a *pitch-space chord progression* as a pair of unordered pitch sets, as in {C4, E4, G4}⇒{C4, F4, A4}. Again, there is no implication that the note E4 moves to F4 or to any other pitch in the chord. (When we are talking about chord progressions in pitch or pitch-class space, we are focusing on the harmonies and ignoring how individual voices might happen to move.) Figure 2.5.5 shows that chord progressions can be modeled as successions of unordered points in either linear pitch space or circular pitch-class space. Geometrically, we can imagine each chord occupying its own individual frame of a movie, with the motion between them being instantaneous and giving us no reason to associate any pair of notes.

Figure 2.5.6 summarizes this discussion, showing that each of these different theoretical concepts applies the five OPTIC symmetries in different ways.

Figure 2.5.5 The chord progression {C, E, G, B♭}⇒{E, G♯, B}.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.5.6 Music-theoretical terms for progressions, along with the symmetries that generate them.

Term	Symmetry
voice leading in pitch space	uniform P
voice leading in pitch-class space	uniform OP
pitch-space chord progression	individual PC
chord progression	individual OPC
path in pitch-class space	uniform O (one-note progressions only)

2.6 COMPARING VOICE LEADINGS

We'll now start to think about how voice leadings relate to one another. The interesting wrinkle is that we'll consider not just *uniform* relationships, in which the same operation applies to both objects in a voice leading, but also their *individual* counterparts.

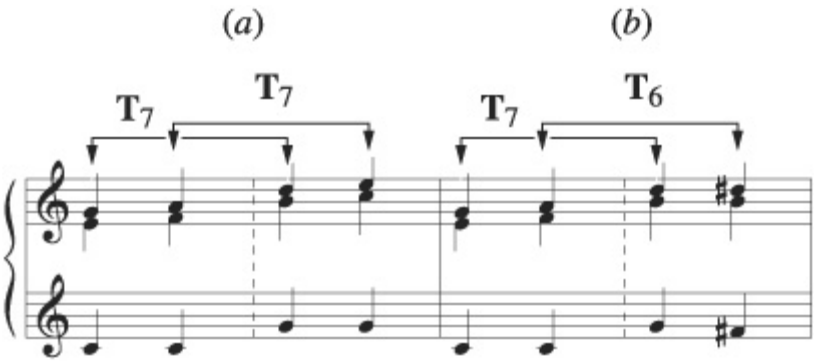
2.6.1 Individual and Uniform Transposition

The voice leadings (C, E, G)→(C, F, A) and (G, B, D)→(G, C, E) are very similar, exhibiting the same musical pattern at different transpositional levels: each holds the root of a major triad fixed, moves the third up by semitone, and the fifth up by two semitones (Figure 2.6.1). Since we transpose both chords in the same way, the two voice leadings are *uniformly transpositionally related*, or *uniformly T-related*. The voice leadings (C, E, G)→(C, F, A) and (G, B, D)→(F♯, B, D♯) are also similar, albeit slightly less so; each maps the root of the first chord to the fifth of the second, the third of the first chord to the root of the second, and the fifth of the first chord to the third of the second. Figure 2.6.1b shows that one voice leading can be transformed into the other by applying a *different* transposition to each chord: (C, E, G) is seven semitones away from (G, B, D), but (C, F, A) is *six* semitones away from (F♯, B, D♯). These voice leadings can thus be said to be *individually transpositionally related*, or *individually T-related*.

Figure 2.6.1 Uniformly and individually T-related voice leadings.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

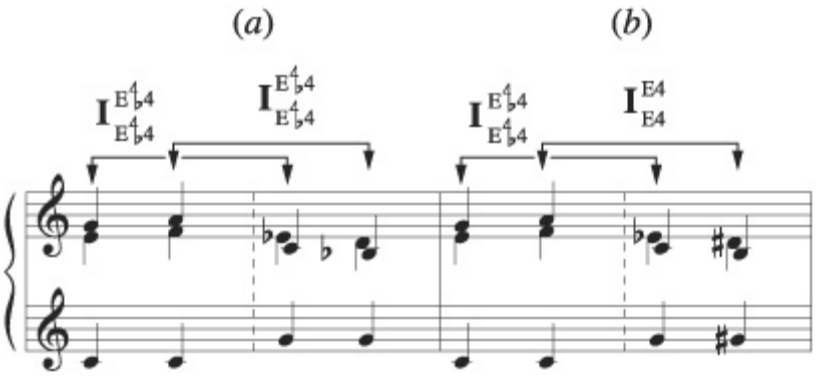
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



2.6.2 Individual and Uniform Inversion

The distinction between individual and uniform relatedness extends naturally to inversion. Figure 2.6.2a shows that (C, E, G)→(C, F, A) can be inverted to produce (G, E♭, C)→(G, D, B♭). These are *uniformly inversionally related* (or *uniformly I-related*) voice leadings. By contrast, (C, E, G)→(C, F, A) and (G, E♭, C)→(G♯, D♯, B) are *individually inversionally related*, or *individually I-related*, since it takes *two different* inversions to transform the first into the second. (Figure 2.6.2b inverts the first chord around E $^{4}_{\flat}$4, or E quarter-tone flat, while inverting the second around E4.) Note that as I am using the term, "individual I-relatedness" requires that each chord in the first voice leading be related by some inversion to the corresponding chord in the second voice leading; it is not permissible to invert only one of the two chords.

Figure 2.6.2 Uniformly and individually I-related voice leadings. Here, pitch-space inversion around E $^{4}_{\flat}$4 (E quartertone flat, halfway between E♭4 and E) sends C4 to G4, E4 to E♭4, F4 to D4, G4 to C4 and A4 to B♭3.



The significance of uniform transposition is clear, as it moves the same musical pattern to a different transpositional level. Uniform I-relatedness is somewhat more abstract: (C, E, G)→(C, F, A) sounds rather different from (C, E♭, G)→(B♭, D, G), which is uniformly I-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

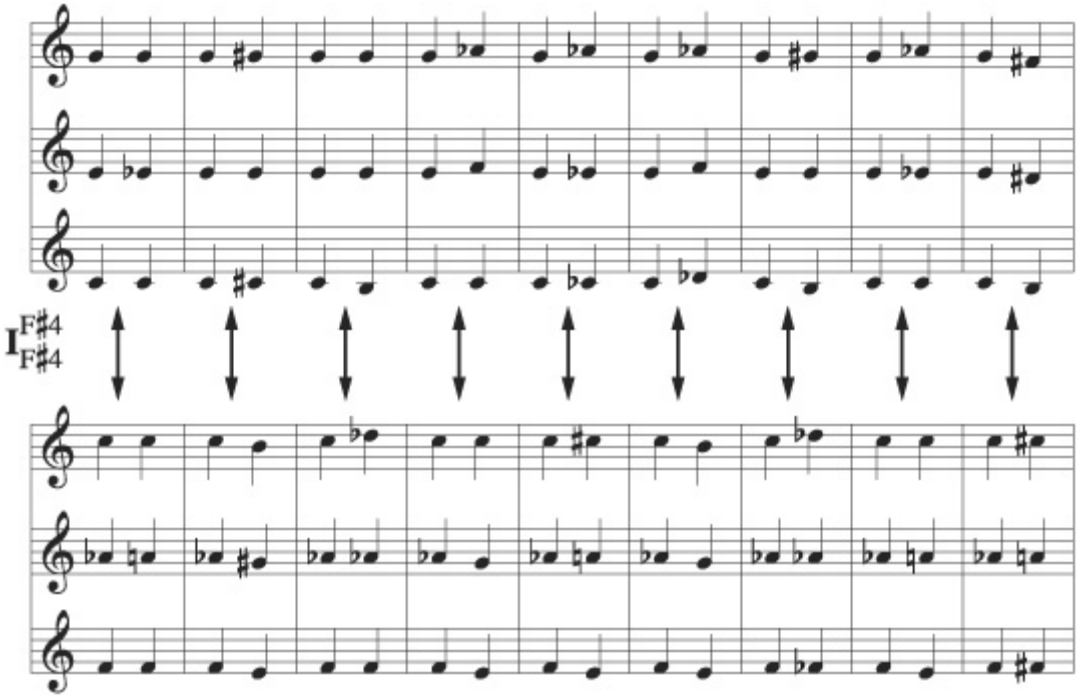
related to it. Note however that the two voice leadings are mirror images of each other: the first moves one voice by zero semitones, one voice up by one semitone, and one voice up by two semitones; the second moves one voice by zero semitones, one voice *down* by one semitone, and one voice *down* by two semitones. The distances are the same but the directions have been reversed. Some theorists have asserted that individually I-related voice leadings are perceptually or metaphysically similar.[23] From our perspective, what is more important is that the answers to theoretical questions often come in I-related pairs. For example, suppose we want to catalogue voice leadings between major and minor triads in which no voice moves by more than a semitone; since the inversion of any such "semitonal" voice leading is also semitonal, each of these voice leadings has an inversionally related partner (Figure 2.6.3). This makes it much easier to conceptualize and remember them.

The musical significance of *individual* T and I relationships is perhaps even less clear, since the voice leading $(\mathrm{C},\ \mathrm{E},\ \mathrm{G}) \xrightarrow{0,\,1,\,2} (\mathrm{C},\ \mathrm{F},\ \mathrm{A})$ sounds like a standard I–IV chord progression, while $(\mathrm{C},\ \mathrm{E},\ \mathrm{G}) \xrightarrow{-1,\,0,\,1} (\mathrm{B},\ \mathrm{E},\ \mathrm{G}\sharp)$, individually T-related to it, evokes Schubertian chromaticism. Nevertheless, there is a sense in which the two are similar: each relates structurally analogous notes (moving the root of the first chord to the fifth of the second, the third of the first chord to the root of the second, and the fifth of the first to the third of the second); and the voices in the second voice leading end up exactly one semitone lower than in the first (moving by –1, 0, and 1 semitones rather than 0, 1, and 2). Very similar points can be made about individually I-related voice leadings.[24] We will see that it is often advantageous to focus on these sorts of relationships, as they allow us to make interesting analytical observations that would otherwise be very hard to express.

Figure 2.6.3 The semitonal voice leadings between major and minor triads. Since inversion preserves the distance moved by each voice, these voice leadings can be grouped into uniformly I-related pairs.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
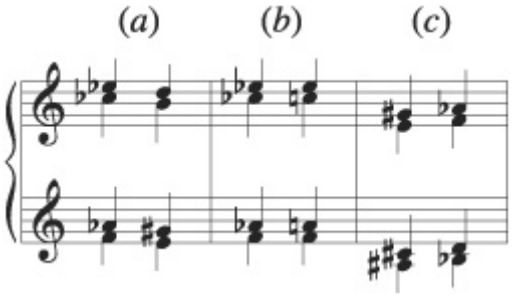


For example, Figure 2.6.4 compares the opening progressions in Wagner's *Tristan*, Brahms' Op. 76 No. 4, and Debussy's *Prelude to "The Afternoon of a Faun."*[25] I think it is likely that both Brahms and Debussy, despite their ambiguous feelings about Wagner, had *Tristan* somewhere in the backs of their minds: the opening gesture, by which a half-diminished seventh chord slides chromatically to a dominant seventh, is unmistakably Tristanesque. (Note that Brahms' opening chord even uses the same pitch classes as Wagner's.) But rather than blatantly copying *Tristan*, both Brahms and Debussy do something *similar to* yet *different from* the earlier piece. The concept of individual T-relatedness provides a precise way to describe this sense of "similarity-with-difference": what Brahms and Debussy did, essentially, was transpose the second chord in *Tristan* up by semitone. (In Debussy's case, he also transposes the entire voice leading by tritone, rearranging the voices somewhat.) In all three cases, the result is a two-semitone voice leading between half-diminished and dominant seventh chords, an unfamiliar chromatic motion linking familiar chords. Furthermore, all three voice leadings move root to root, third to third, fifth to fifth, and seventh to seventh. Yet at the same time, the connection to *Tristan* has been disguised: what has been borrowed is not the progression itself, but rather something more abstract—a general voice-leading schema rather than a particular way of resolving the half-diminished chord. (Thus we see how theoretically savvy composers can steal without getting caught!) Whether this is a matter of defusing, one-upping, or "correcting" Wagner is an interesting critical question.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.6.4 The opening progressions in Wagner's *Tristan* (*a*), Brahms' Intermezzo Op. 76 No. 4 (*b*), and Debussy's *Prelude to "The Afternoon of a Faun."*

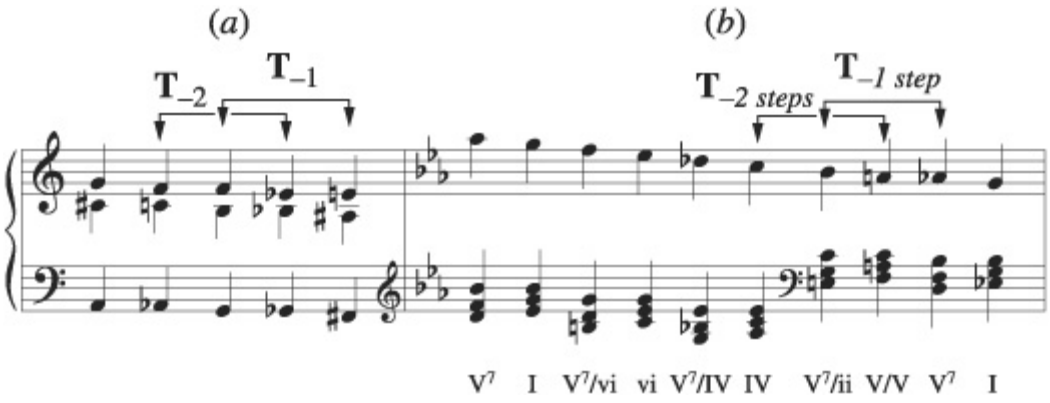


In classical music, individually T-related voice leadings often appear in sequential contexts. Figure 2.6.5a presents a nice example from the opening of Mozart's C minor Fantasy, K. 475. The music invokes a relatively standard chromatic sequence in which iv^6–V^7 chord progressions move downward by major second: beginning with A^7, we have f^6–G^7 followed by what we expect to be $e\flat^6$–F^7. But Mozart transposes the last chord up by semitone so that the final pair is individually T-related to the pair preceding it, with the sequence ending on $F\sharp^7$ rather than the expected F^7. This maneuver, which transforms a minor-key iv^6–V^7 progression into a major-key iii^6–V^7 progression, is reasonably common in tonal music. Figure 2.6.5b cites an analogous progression from the recapitulation of the first movement of Beethoven's E♭ major Piano Sonata, Op. 31 No. 3.[26] Here the sequence moves downward by diatonic thirds, with each chord preceded by its applied dominant. If the passage were strictly sequential, it would cycle through all seven diatonic triads before returning to the tonic E♭; however, Beethoven arranges an earlier return by inserting a descending diatonic *second* into the sequence of thirds. Once again, the structural similarity of the voice leadings masks the deviation from strict sequential procedure: we hear successive voice leadings as being similar, even though they are not related by exact transposition. Again, the concept of "individual T-relatedness" provides a precise tool for describing this sort of similarity.

Figure 2.6.5 Individually T-related voice leadings in sequential passages from Mozart (*a*) and Beethoven (*b*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



2.7 VOICE-LEADING SIZE

Polyphonic music uses independent melodies to articulate efficient voice leadings between meaningful harmonies (Figure 2.7.1). To be sure, it also involves following many other rules that are particular to specific styles and time periods: in some genres, for example, parallel perfect fifths are forbidden, while in others they are allowed. (Similarly, in some genres the bass is more likely to move by leap, while in others the bass is essentially similar to the other voices.) Later chapters will deal with these style-specific nuances. For now, I want to focus on the basic conditions under which polyphony itself is possible.

Clearly, to write this sort of music, composers need to be able to compare the overall efficiency, or "size," of different voice leadings. But how is this done? In some situations, comparisons seem unproblematic: for instance, a voice leading that moves just one note by just one semitone seems obviously smaller than a voice leading that moves three notes by six semitones each. But other cases are considerably more difficult. Is (C, E, G)→(D♭, F, A♭) larger or smaller than (C, E, G)→(C, E, A)? The first voice leading moves its voices by a greater total distance (three semitones vs. two semitones), while in the second, the largest distance moved is greater (two semitones vs. one semitone). Which is more important? Is the smaller voice leading the one that minimizes the total amount of motion or the largest distance moved by any voice?

Although pedagogues have long enjoined their students to use small voice leadings, they have never bothered to provide specific answers to these sorts of questions. (Students, it seems, are supposed to intuit which of the available voice leadings is the smallest.) More recently, theorists have proposed a variety of very precise methods of measuring voice-leading size. Unfortunately, musical practice does not allow us to adjudicate between these

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

different proposals, leaving us with many plausible but incompatible solutions to the same problem. Rather than adopting one of these, my strategy here is to try to remain as neutral as possible about the issue. (Precision is a virtue only when it is not arbitrarily imposed!) Happily, it turns out that "reasonable" measures of voice-leading size agree about a number of important cases.

Figure 2.7.1 The opening phrase of the Bach chorale "O Herzensangst." This music can be represented as the sequence of voice leadings (E♭, G, E♭, B♭)→(E♭, B♭, E♭, G)$\xrightarrow{-7,-2,0,5}$(A♭, A♭, E♭, C)→(A♭, F, D, B♭). I have ignored the starred ("nonharmonic") note at the end of the first measure.
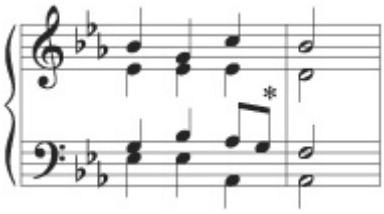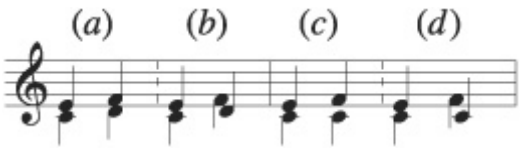


Figure 2.7.2 The principle of avoiding crossings tells us that (*a*) should be no larger than (*b*) and (*c*) should be no larger than (*d*).
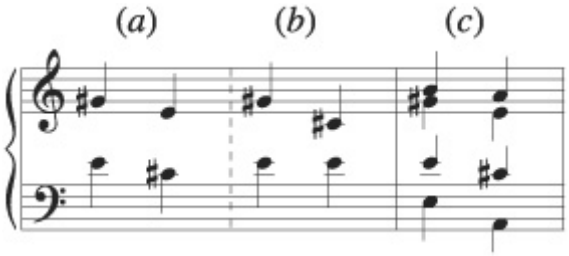


I consider a measure of voice-leading size to be "reasonable" if it satisfies two basic requirements: first, it should depend only on *how far* the individual voices move, with larger motion leading to larger voice leadings; and second, it should judge voice leadings with *voice crossings* to be larger (or at the very least, no smaller) than their natural uncrossed alternatives. This is illustrated by Figure 2.7.2. The first requirement is basic to the very notion of voice-leading size; the second mandates a kind of musical "division of labor," according to which it is preferable to have more voices moving by small amounts, rather than fewer voices moving by larger amounts. (The connection between avoiding voice crossings and the division of musical labor is not obvious, although it can be proved mathematically.[27]) Thus Figure 2.7.3a, in which two voices move by three and four semitones, should be at least as small as Figure 2.7.3b, in which one voice moves by seven

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

semitones. Both principles are consistent with Western musical practice, and to my knowledge no composer, theorist, or pedagogue has ever proposed a method of measuring voice leading that violates either principle.[28]

Figure 2.7.3 Generally speaking, Western composers prefer smaller amounts of motion in more voices (*a*) to larger amounts of motion in fewer voices (*b*). The pattern in (*a*) is common in the inner voices in classical music; (*c*) provides an example from the Bach chorale "Nun lob' mein' Seel' den Herren."



Appendix A summarizes the technical issues involved in measuring voice leading. For our purposes, the important point is that the two constraints are powerful enough to imply a number of theoretical results: for instance, they imply that the more evenly a chord divides the octave, the smaller the voice leadings to its various transpositions; they also imply that E and A♭ major triads can be linked to the C major triad by smaller voice leadings than any other major triads. Indeed, the constraints are so powerful that, in many contexts, it is unnecessary to specify a particular method of measuring voice leading, because all reasonable metrics will agree. For most of this book, we will therefore be able to speak of "large" and "small" voice leadings without getting into the details of how to measure voice leading. Readers should feel comfortable trusting their musical intuitions, interpreting the term "efficient voice leading" as a synonym for "voice leadings in which all voices move by short distances."

2.8 NEAR IDENTITY

So far we have grouped chords into categories based on transpositional and inversional equivalence. But this insistence on *exact* equivalence is actually quite restrictive: in many contexts, it is useful to describe two chords as being similar even though they are not exactly the same.[29] This is particularly important when investigating matters of intonation

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

and tuning: for instance, music theorists might speak about the "acoustically pure perfect fifth {C, G}," the "equal-tempered fifth {C, G}," the "quarter-comma meantone perfect fifth {C, G}" and so on. The language suggests that these different tunings are variations on the same object, which may come in a variety of closely related forms. Yet from our perspective the different tunings of {C, G} belong to different chord types, since no transposition or inversion relates them. Thus there is a mismatch between ordinary musical thinking and the extremely precise terminology we have been developing.

The problem is not limited to matters of tuning and intonation. The dominant seventh chord {C, E, G, B♭} is in the grand scheme of things rather similar to the minor seventh {C, E♭, G, B♭}, which differs only in that it has an E♭ rather than E♮. Although they do not belong to the same set class, the two chords are clearly more similar to each other than either is to the dissonant cluster {F♯, G, A♭, A}. Consequently, {C, E, G, B♭}⇒{C, E♭, G, B♭} feels like a progression between two more-or-less similar chords, while {C, E, G, B♭}⇒{F♯, G, A♭, A} is quite jarring, connecting very different harmonies. Our discussion, rather than trying to model to this fact, has instead treated harmonic identity as an all-or-nothing affair.

It therefore seems that exact identity stands at one end of a *continuum* of musical relatedness. The equal-tempered chord {C, E, G, B♭} is exactly identical to itself, and is very closely related to the just-intonation chord {C, E, G, B♭}. Both of these chords are somewhat related to the minor-seventh chord {C, E♭, G, B♭}, and are not very similar at all to the chromatic cluster {F♯, G, A♭, A}. Notice that these "degrees of relatedness" seem to mirror facts about voice leading: there is a very small voice leading from the equal-tempered {C, E, G, B♭} to its just-intonation counterpart, and a reasonably small voice leading from {C, E, G, B♭} to {C, E♭, G, B♭}, but it takes a relatively large voice leading to get from {C, E, G, B♭} to {F♯, G, A♭, A}. This suggests that we may be able to use voice leading to model at least some of our intuitions of musical similarity.[30]

Metaphors of distance are useful here. We can think of the acoustically pure perfect fifth {0, 7.01} as being *close* to the equal-tempered fifth {0, 7}, as if the two chords were nearby in some abstract "space of all chords." Conversely, the chord {3, 4} is much farther from {0, 7} than {0, 7.01} is, since it takes a larger voice leading to connect {3, 4} to {0, 7}. Chapter 3 will show that this talk of distance is not just a metaphor: it is possible to

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

describe geometrical spaces containing all chords with a particular number of notes, and in which the distance between any two chords corresponds to the size of the minimal voice leading between them. Musical "similarity" can therefore be represented by distance in these spaces. In anticipation of this discussion, I will say that chords are "near" or "close to" each other when they can be linked by efficient voice leading.

We can extend these ideas to chord types as well. Two chord types can be said to be similar if there is a relatively small voice leading between their transpositions. Thus the diminished triad is closer to the minor triad than to the chromatic cluster, since there is a very small voice leading taking C diminished to C minor, but no similarly small voice leading connecting any diminished triad to any chromatic cluster.[31] Another way to make the point is to say that any particular diminished triad is *nearly transpositionally related* to any minor triad: for example, C diminished is nearly transpositionally related to F minor, since {C, E♭, G♭} is *near* {C, E♭, G} which is in turn transpositionally related to {F, A♭, C}. In the same spirit, we could say that {C, D♭, G} is *nearly inversionally related* to {E♭, G, A}, meaning that {C, D♭, G} is near {C, D, F♯}, which is inversionally related to {E♭, G, A}. What results is a more flexible conception of harmonic similarity: rather than considering two chords to be similar only when they are *exactly* related by transposition or inversion, we can consider them similar when they are nearly (or *approximately*) related by these same transformations. This is useful insofar as actual composers are often more concerned with approximate than with exact musical relationships.

2.9 HARMONY AND COUNTERPOINT REVISITED
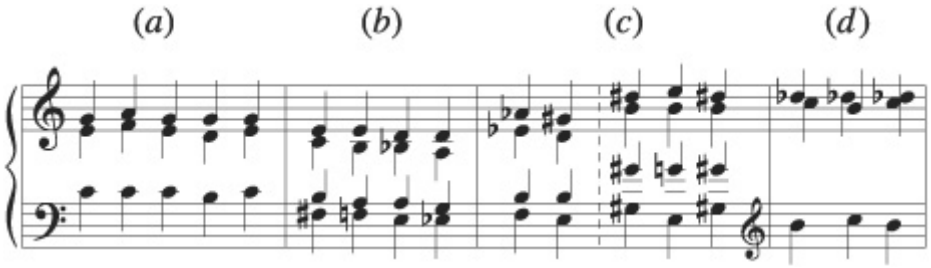
In Chapter 1, we observed that Western music has a kind of two-dimensional coherence: vertical or harmonic principles dictate that chords should be audibly similar, while horizontal or contrapuntal principles dictate that chords should be connected by efficient voice leading. Figure 2.9.1 shows these constraints operating in a wide range of different Western styles, including classical music, contemporary jazz, late-nineteenth-century chromaticism, and even modern atonality. To be sure, composers often deviate from these norms for expressive reasons—just as, when driving, one sometimes takes a scenic detour

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

rather than following the most direct route. And to be sure, different styles always involve additional constraints over and above these very basic norms. But the norms themselves are relatively robust, shared by a wide range of styles spanning several centuries.

Figure 2.9.1 Harmonic consistency and efficient voice leading in a range of styles. (*a*) A common upper-voice pattern for the classical I–IV–I–V–I chord progression. (*b*) A common jazz-piano "left-hand voicing" for the descending-fifths progression D^7–G^7–C^7–F^7. The voicings add ninths and thirteenths and omit roots, fifths, and elevenths, as is common in jazz. (*c*) Two celebrated examples of Wagnerian chromaticism; the first is a simplification of the opening of *Tristan* and the second is the opening of *The Ring's* "Tarnhelm" motif. (*d*) Chromatic clusters of the sort often found in late twentieth-century music, particularly in the music of Ligeti and Lutosławski.



So under what circumstances is it possible to find an efficient voice leading between nearly transpositionally or inversionally related chords? In other words, how is it possible to write music that is both harmonically and melodically coherent?

Chapter 1 noted that this is possible when a chord divides the octave very evenly or very unevenly. Here I want to replace the contrast "even/uneven" with the more powerful concept of *near symmetry*. It turns out that one can combine harmonic consistency and efficient voice leading only by using *nearly symmetrical* chords, which are typically near chords that are *completely symmetrical* under transposition, inversion, or reordering. Nearly even chords (such as the major triad) are near the symmetrical chords that divide the octave perfectly evenly (such as the augmented triad). Clustered chords (such as {B, C, C♯}) are near the symmetrical chords that contain many copies of one specific pitch class (such as the "triple unison" {C, C, C}). But there are other symmetrical chords as well: transpositionally symmetrical chords such as {C, C♯, F♯, G}, which do not divide the octave particularly evenly, and inversionally symmetrical chords such as {C, D, E}, which are neither particularly even nor particularly uneven. Thus an approach based on

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

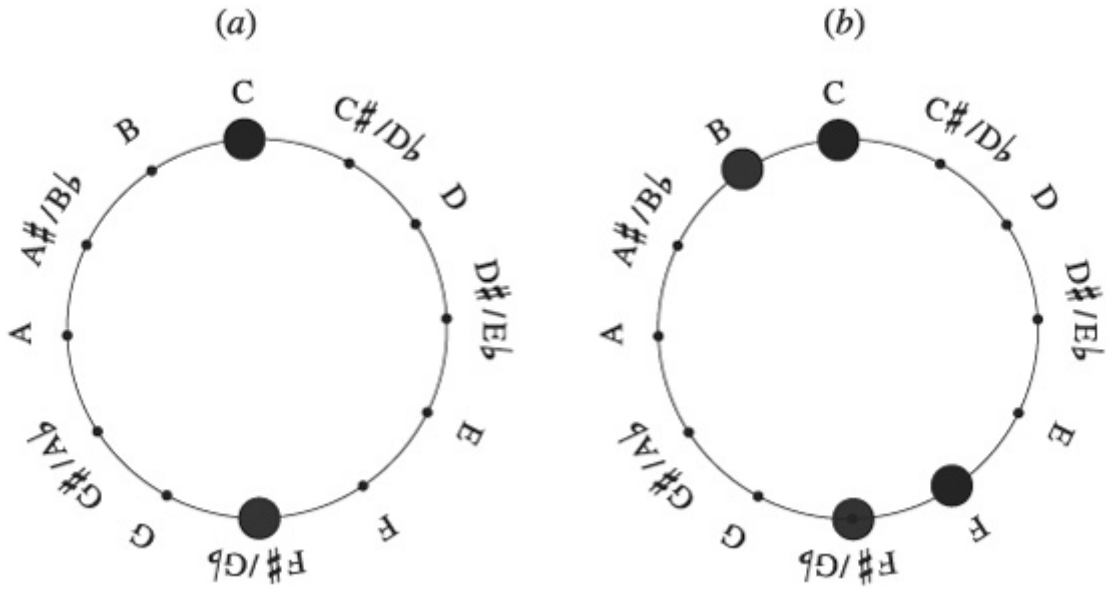
symmetry leads to a much deeper understanding of the fundamental connection between harmony and counterpoint—opening our eyes to new compositional possibilities and revealing regularities latent in the music of the past.

We will begin by considering efficient voice leading between chords *exactly* related by transposition or inversion; then, at the end of the section, we will broaden the discussion to include near transposition or near inversion.

2.9.1 Transposition

A chord is transpositionally symmetrical if it is unchanged by some transposition. Thus, when we transpose an augmented triad by a major third, we produce the same pitch classes we started with. Figure 2.9.2 shows that there are two kinds of transpositionally symmetrical chords, those that divide the pitch-class circle evenly and those that can be decomposed into equal-size subsets that do so.[32] For example, {B, C, F, F♯} does not divide the circle into four even parts, but it can be decomposed into a pair of tritones that each divide the circle in half.

Figure 2.9.2 Transpositionally symmetrical chords either divide the octave evenly (*a*), or can be decomposed into equal-size subsets that themselves do so (*b*). Here, the transpositionally symmetrical chord {C, F, F♯, B} can be decomposed into tritones {C, F♯} and {B, F}.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

If a chord is near one of these symmetrical chords, then it can be linked by efficient voice leading to some of its transpositions. Consider {C, E, G}, which is near {C, E, G♯} since the voice leading (C, E, G)→(C, E, G♯) is small. It follows that the E major chord must also be near the C augmented triad. This is because we can transpose the small voice leading (C, E, G)→(C, E, G♯) uniformly by ascending major third, obtaining (E, G♯, B)→(E, G♯, C). This is an efficient voice leading from E major to C augmented. Figure 2.9.3 shows that we can retrograde this second voice leading and attach it to the earlier one, producing (C, E, G)→(C, E, G♯)→(B, E, G♯). Here we have the conjunction of two small voice leadings. Removing the middle chord will give a small voice leading from C major to E major: (C, E, G)→(B, E, G♯). In other words, if C major is close to an augmented triad, then its major third transposition must also be close to that same augmented triad, and since C and E major are both close to the same chord, they must also be close to each other.

Of course, there is nothing particularly special about major or augmented chords. Take any chord *A* close to a symmetrical chord *S* that is unchanged by transposition $\mathbf{T}_x$. Then $\mathbf{T}_x$ (*A*) will also be close to $\mathbf{T}_x$ (*S*), which is the same as *S;* and since *A* and $\mathbf{T}_x$ (*A*) are both near *S*, they are also near each other. Speaking somewhat metaphorically, the symmetrical chord passes on its symmetry to nearby chords in the form of efficient voice leading. These nearby chords are not themselves symmetrical, but they are *nearly* so, which is to say that they can be linked to their *x*-semitone transpositions by efficient voice leading. It follows that transpositional symmetry is a limiting case of efficient voice leading: as chord *A* moves closer and closer toward a transpositionally symmetrical chord, the size of the minimal voice leading between *A* and $\mathbf{T}_x$ (*A*) decreases, reaching zero when the chord is exactly symmetrical.

Figure 2.9.3 Any chord that is near an augmented triad can be connected to its major-third transposition by efficient voice leading.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

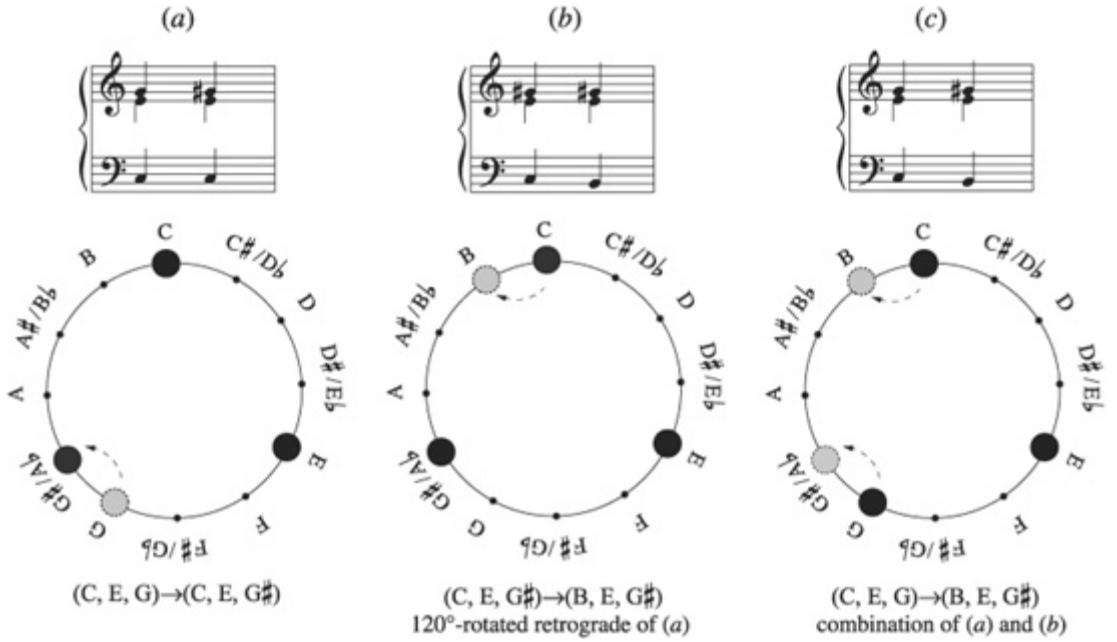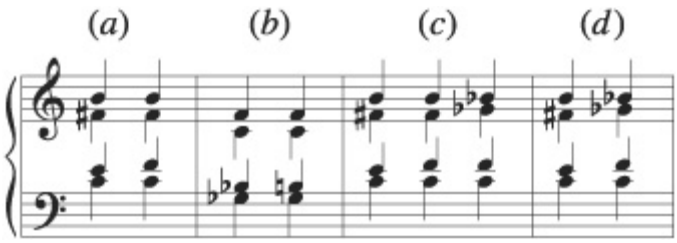


Figure 2.9.4 Since both {C, E, F♯, B} and its tritone transposition are close to {B, C, F, F♯} (*a*–*b*), they are close to each other. In (*c*), we retrograde the voice leading in (*b*) and attach it to (*a*). Removing the middle chord gives us the voice leading in (*d*).



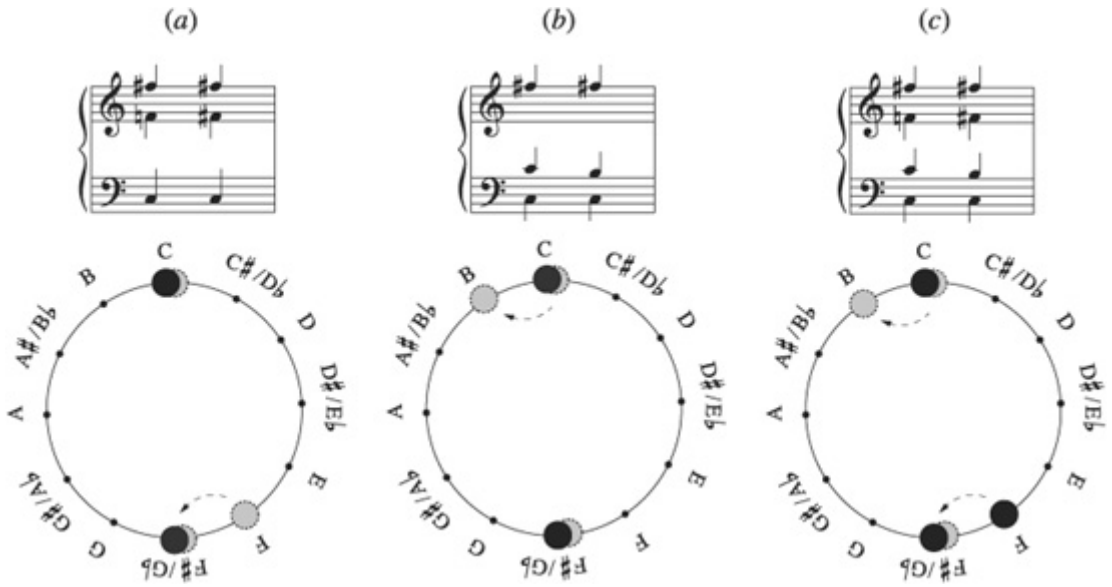
Let's consider a second example. The chord {B, C, E, F♯} appears commonly in jazz, but does not divide the octave particularly evenly. However, it is near the trans-positionally symmetric {B, C, F, F♯}, which contains two separate tritones. Beginning with the small voice leading (B, C, E, F♯)→(B, C, F, F♯), we transpose by tritone to produce (F, G♭, B♭, C) →(F, G♭, B, C) (Figure 2.9.4). Retrograding this and gluing it to the original voice leading, we obtain (B, C, E, F♯)→(B, C, F, F♯)→(B♭, C, F, G♭). Removing the middle chord gives (B, C, E, F♯)→(B♭, C, F, G♭), which efficiently links tritone-related chords. Chapter 10 will show that this sort of voice leading plays a role in jazz "tritone substitutions." For now, however, the focus should be on the general form of the argument: since chord *A* is close to the tritone-symmetrical chord *S*, then so is the tritone transposition of *A;* and since *A* and its tritone transposition are close to the same chord, they must also be close to each other.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

There is one complication that should be mentioned here. Suppose a three-note chord, such as {C, F, F♯}, is near a transpositionally symmetrical two-note chord, such as {C, F♯} (Figure 2.9.5). Proceeding as we did earlier, we begin with the three-voice voice leading (C, F, F♯)→(C, F♯, F♯); retrograding and transposing by tritone we obtain (F♯, C, C)→(F♯, B, C). Note that we cannot simply glue these two voice leadings together, since (C, F♯, F♯) is not the same as (F♯, C, C). However, if we double a voice in each voice leading, we can attach them. What results is a *four-voice* voice leading between {C, F, F♯} and its tritone transposition, (C, C, F, F♯)→(C, C, F♯, F♯)→(B, C, F♯, F♯) or (C, C, F, F♯)→(B, C♯, F♯, F♯) upon removing the middle chord. The fourth voice is the price the three-note chord has to pay for exploiting the symmetry of a two-note chord. In general, extra voices will *always* be necessary whenever a larger chord exploits the symmetries of a smaller chord. In Chapter 7 we will see that there are common tonal routines, such as the voice leading (C, C, E, G)→(A, C, F, F), which exploit this very procedure (§7.2). It follows that the familiar major triad is special not just because it divides the octave into three nearly even parts, but also because it is reasonably close to the tritone.

Figure 2.9.5 A larger chord can take advantage of the symmetries of a smaller chord, but it requires additional voices.



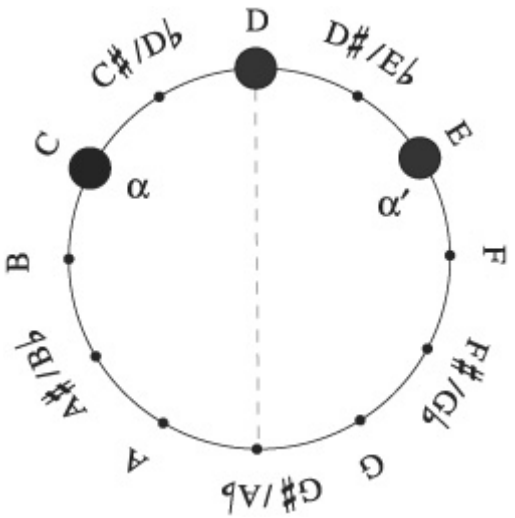
2.9.2 Inversion

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

A chord will be unchanged by inversion if its notes are arranged symmetrically around an "axis of symmetry" crossing the pitch-class circle at two antipodal points, with tones placed freely on either of these points. For example, Figure 2.9.6 shows that the axis of the inversionally symmetrical {C, D, E} contains the notes D and A♭; since C and E are arranged symmetrically around this axis, inversion around D leaves the chord unchanged. The nonsymmetrical chord {C, D♯, E} is near {C, D, E}, since the voice leading (C, D♯, E)→(C, D, E) is small. Inverting this entire voice leading around the fixed point D produces (C, D♭, E)→(C, D, E). Reversing this voice leading and gluing it to the earlier one produces (C, D♯, E)→(C, D, E)→(C, D♭, E). Removing the middle chord gives us a small voice leading between inversionally related chords: (C, D♯, E)→(C, D♭, E) (Figure 2.9.7). As in the transpositional case, the inversionally symmetrical chord passes its symmetry on to nearby chords: if chord *A* is near a chord *S* that is invariant under some inversion I_y^x, then I_y^x (*A*) will also be near *S*, and hence *A* and I_y^x (*A*) can be linked by efficient voice leading.

Figure 2.9.6 To form an inversionally symmetrical chord, choose two antipodal points to act as an axis of symmetry—here, {D, A♭}. One can place notes on either axis point (or both), as they will be unaffected by the inversion. For any points not lying on the axis, such as a, one must also add its inversional partner, symmetrically placed on the other side of the axis. Here, a is 60° counterclockwise from the upper axis point, while α' is 60° clockwise from that point. The antipodal points can also lie *between* equal-tempered pitch classes.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.9.7 Any chord that is near an inversionally symmetrical chord can be connected to its inversion by efficient voice leading.

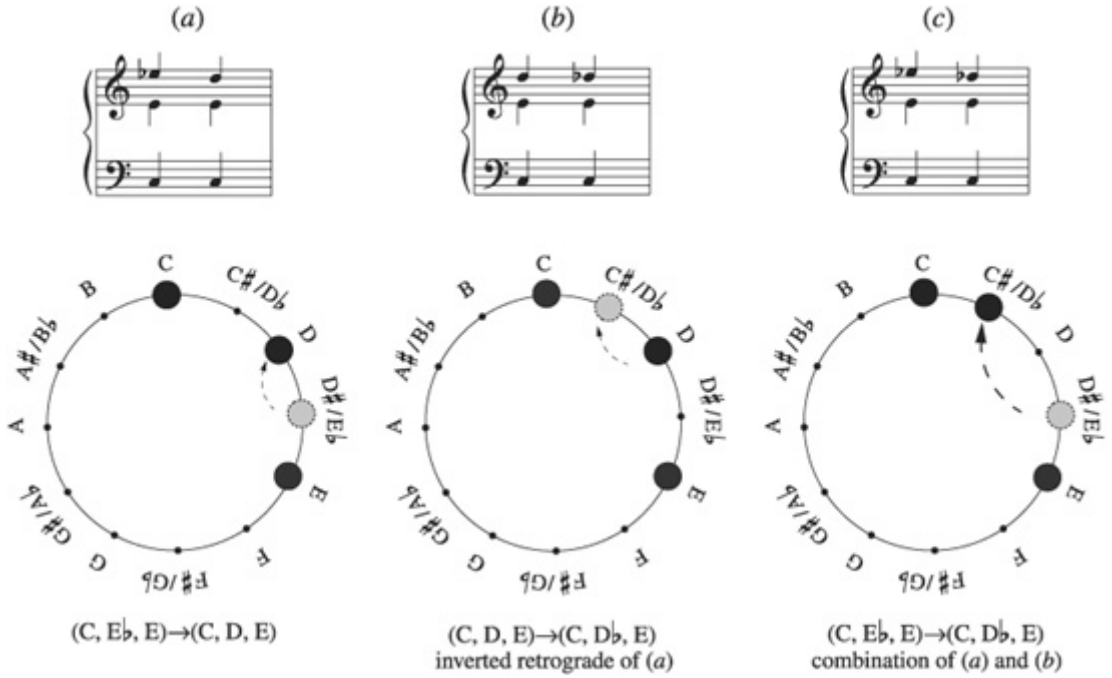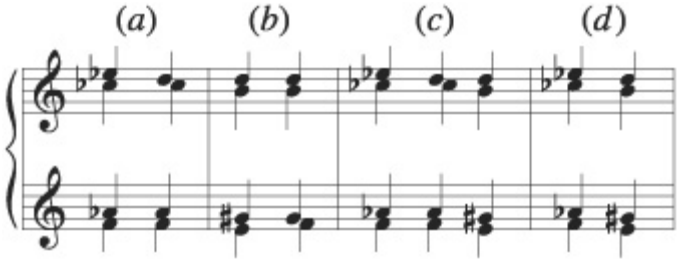


Figure 2.9.8 provides a real-world example. We begin with the efficient voice leading (F, A♭, C♭, E♭)→(F, A♭, C♭, D), which uses a single-semitone motion to transform the F half-diminished chord into a fully diminished seventh. The next voice leading inverts the first uniformly around A4/B♭4, producing an efficient voice leading from the E dominant seventh to the same diminished seventh chord. In Figure 2.9.8c we reverse this and glue it to the first voice leading. Removing the symmetrical chord gives us an efficient voice leading from F half-diminished to E^7, strongly reminiscent of the opening of *Tristan*. We conclude that this celebrated nineteenth-century progression exploits the near I-symmetry of the half-diminished and dominant seventh chords.

Figure 2.9.8 The F half-diminished chord is close to a diminished seventh chord (*a*). Inverting this voice leading uniformly around A4/B♭4, we get an efficient voice leading from E^7 to the same diminished chord. In (*c*), we retrograde the voice leading in (*b*) and attach it to (*a*). Removing the middle chord (*d*) gives us an efficient voice leading from the F half-diminished chord to E^7, a voice leading that plays a central role in Wagner's *Tristan*.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



2.9.3 Permutation

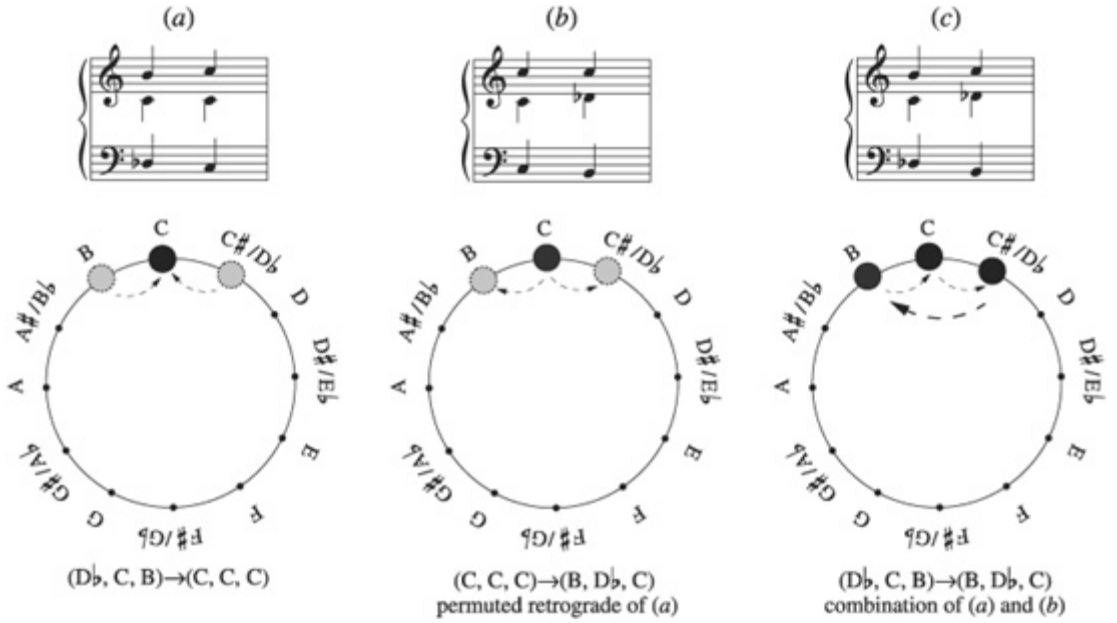
The third symmetry is a little bit different from the others. Suppose we model chords as *multisets* that may contain multiple copies of a single note. From this point of view, the chord {C, C, C} is different from {C, C}, since they have a different number of copies of the note C. Let's also imagine that (contrary to fact) we could somehow keep track of these different copies, which we can represent with subscripts like $\{C_a, C_b, C_c\}$. Now consider $(C_a, C_b, C_c) \rightarrow (C_c, C_a, C_b)$, a voice leading that permutes the notes in the chord {C, C, C}. Since none of the voices in this voice leading actually moves, we can say that {C, C, C} is *permutationally symmetrical*. "Permutation symmetry" is just a fancy synonym for "has multiple copies of one or more of its notes."

Permutationally symmetrical chords are themselves rather boring, but the *nearby* chords are quite interesting. The chromatic cluster {B, C, D♭} is near {C, C, C} since (B, C, D♭)→(C, C, C) is small. Retrograding and reordering voices, we obtain (C, C, C)→(C, D♭, B). Gluing the two voice leadings together gives us (B, C, D♭)→(C, C, C)→(C, D♭, B); upon removing the middle chord, we obtain (B, C, D♭)→(C, D♭, B), a small voice leading from {B, C, D♭} to itself. (Figure 2.9.9 attempts, somewhat lamely, to illustrate.) Once again, the symmetrical chord passes its symmetry on to nearby chords: given a small voice leading $A \rightarrow S$, from chord A to the permutationally symmetrical chord S, we can permute both chords uniformly to obtain the small voice leading $\boldsymbol{P}(A) \rightarrow \boldsymbol{P}(S)$, or $\boldsymbol{P}(A) \rightarrow S$, since S is permutationally symmetrical. Retrograding and gluing together gives us $A \rightarrow S \rightarrow \boldsymbol{P}(A)$; upon removing the middle chord, we have $A \rightarrow \boldsymbol{P}(A)$, an efficient voice leading between A and itself. The general procedure here is precisely the same as in the cases of transposition and inversion; the only difference is the nature of the symmetry.

Figure 2.9.9 Any chord that is near a chord with pitch-class duplications can be connected to its untransposed form by efficient voice leading.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



It may seem surprising to place permutation alongside transposition and inversion, since we are accustomed to thinking of chords as inherently unordered objects. But the discussion in §2.4 should ameliorate any feelings of unease: we form chords by abstracting away from the order of the notes in a musical object, just as we form set classes by abstracting away from the distinction between transpositionally and inversionally related objects. And just as transpositionally symmetrical chords are special (because there are fewer transpositions to abstract away from) so too are permutationally symmetrical chords (because there are fewer orderings to abstract away from). Voice leading reveals this "specialness" by providing a musical mechanism for rearranging the notes in an unordered chord. As a result, it should not be too surprising that there is a close analogy between permutational symmetry and the more familiar symmetries considered previously. Transpositional near-symmetry guarantees efficient voice leading between transpositionally related chords; inversional near-symmetry guarantees efficient voice leading between inversionally related chords; and permutational near-symmetry guarantees efficient voice leading between a chord and itself. Here we see the advantages of a more flexible approach to chord classification, in which we embrace a wide variety of musical transformations. From our point of view, permutation is just another kind of symmetry, with its own distinctive collection of musical consequences.

The arguments in this section show that *if* a chord is near a symmetrical chord then it will inherit this symmetry in the form of efficient voice leadings. One might also wonder whether the converse is true: is it the case that *if* there is an efficient voice leading from a chord *A* to one of its transpositions or inversions, then chord *A* is nearly symmetrical

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

under transposition, inversion, or permutation? That is, is it the case that our three symmetries are *the only symmetries there are*—the only symmetries we need to understand if we want to know how to combine efficient voice leading with harmonic consistency? Basically, the answer is "yes," at least insofar as we understand the term "harmonically consistent" to mean "using chords that are approximately related by transposition or inversion."[33] It follows that every instance of efficient voice leading between structurally similar chords—in any scale or tuning system—can be understood in terms of the three basic symmetries we have just discussed. This is precisely what will allow us, in Part II, to interpret a large number of Western musical practices as variations on the same basic techniques.

It is very important to realize that the symmetrical chords relevant to voice leading will *not* necessarily lie on the ordinary piano keyboard. The familiar pentatonic scale, {C, D, F, G, B♭} or {0, 2, 5, 7, 10}, is quite close to the chord that divides the octave into five equal pieces: {0, 2.4, 4.8, 7.2, 9.6}. Consequently, the pentatonic scale can be efficiently linked to its transpositions, as in (C, D, F, G, B♭)→(C, D, F, G, A). While both of these chords belong to the equal-tempered universe, and can be played on an ordinary piano, the perfectly even five-note chord does not. Yet this missing chord nevertheless exerts its influence on twelve-tone equal-tempered music; indeed, twelve-tone equal-tempered five-note chords can be linked to their transpositions by efficient voice leading only if they are close to this phantasmic and elusive chord. This is just one of many cases where a deep understanding of the discrete musical universe requires us to think in continuous terms. Even though we are ultimately concerned with equal-tempered music, we find that we can understand this music best when we consider the continuous space in which equal temperaments are embedded.[34]

Finally, note that nothing really changes when we adopt the more flexible notion of harmonic relatedness discussed in §2.8. For suppose we would like to find an efficient voice leading between two chords A and B that are *nearly*—rather than precisely—transpositionally or inversionally related. Since A is nearly transposition-ally or inversionally related to B, there is a small voice leading $\mathcal{F}(A)\rightarrow B$, where $\mathcal{F}$ is some transposition or inversion. And if the voice leading $A\rightarrow B$ is small, then the voice leading $A\rightarrow\mathcal{F}(A)$ must also be small, as both A and $\mathcal{F}(A)$ will be close to B. In the cases of musical interest, $\mathcal{F}$ will be some nonzero transposition or inversion, which means that A is itself

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

nearly symmetrical.[35] (The same argument shows that *B* must also be nearly symmetrical.) Thus, expanding our notion of "harmonic consistency" does not expand the realm of musical possibilities: a chord *A* can be linked by efficient voice leading to one of its *near*-transpositions or *near*-inversions only if *A* is itself nearly symmetrical under transposition, inversion, or permutation.

2.10 ACOUSTIC CONSONANCE AND NEAR EVENNESS

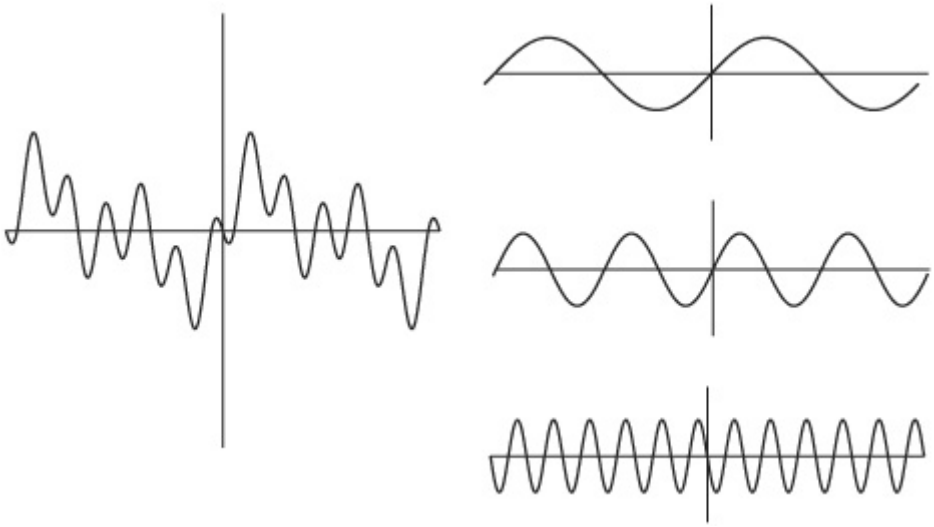
We now return to acoustic consonance, the second of our five basic components of tonality. Recall that Western listeners tend to agree that certain chords sound stable and restful, or consonant, while others sound unstable and harsh, or dissonant. Many, though not all, Western styles exploit this difference: consonant sonorities tend to appear as musical destinations or at points of rest, while dissonant sonorities tend to be more active and unstable. To be sure, the distinction between consonance and dissonance is extremely complex, and it maps only indirectly onto the concept of musical stability. (There are situations when an acoustically consonant sonority behaves as if it were musically unstable.) But to a first approximation, it is reasonable to say that in many musical styles, dissonant sonorities tend to resolve to consonant sonorities.

Acoustic consonance is somewhat imperfectly understood, and there are a number of theories about what produces it.[36] Nevertheless, there is broad agreement on some basic principles. The eighteenth-century mathematician Jean-Baptiste Fourier showed that any periodic mathematical function can be represented as the sum of sine and cosine waves whose frequencies are integer multiples of a single frequency *f* (Figure 2.10.1). In the case of a musical sound, these sine waves are called *partials*. The nineteenth-century German psychologist Hermann von Helmholtz proposed that two pitches are consonant if their partials are aligned, or else are sufficiently far apart so as not to interfere. Dissonance is caused when partials are close together, but not perfectly aligned; this creates a sense of "roughness" or harshness. (Twentieth-century auditory physiologists have shown that the ear's basilar membrane decomposes incoming sound waves into sine waves, more or less as Fourier described.) Although there is much disagreement about just how complete this story is, most contemporary psychologists believe that it is at least partly right.[37]

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 2.10.1 The periodic sound on the left has frequency *f*. (Two repetitions of the waveform are shown.) This sound can be analyzed as the sum of the sine waves on the right, with periods *f*, 2*f*, and 6*f*.



Western instruments typically produce periodic sounds that, when analyzed as Fourier described, have relatively strong lower partials (*f*, 2*f*, 3*f*, 4*f*, etc.). The partials of several such "harmonic" sounds will match when their fundamental frequencies are related by simple whole-number ratios.[38] Consequently, as Pythagoras discovered roughly 2700 years ago, the most consonant intervals have fundamental frequencies in simple ratios: 2:1 (octave), 3:2 (perfect fifth), 4:3 (perfect fourth), 5:4 (major third), and 6:5 (minor third). A larger chord will be consonant when it contains a preponderance of consonant intervals, and this in turn requires that its notes be relatively evenly distributed in pitch-class space.

For small chords there is a particularly elegant connection between near evenness and acoustic consonance. The most consonant three-note chord, the major triad, contains the first three distinct pitch classes in the harmonic series, with fundamental frequencies in a 3:4:5 ratio. Figure 2.10.2 shows that the triad divides the frequency-space octave into three exactly equal pieces. When we take the logarithm of these frequencies, passing from frequency space into pitch space (§2.1), we move from an equal division of the (frequency-space) octave into a nearly even division of the (log-frequency, pitch-space) octave. (A similar point holds for the perfect fifth, which contains the first two distinct pitch classes in the harmonic series and evenly divides the frequency-space octave.) Thus, for small

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

chords, maximal consonance directly implies near evenness. And though the situation is somewhat more complicated in the case of larger chords, it remains true that highly consonant chords always divide the octave relatively evenly (Figure 2.10.3).[39]

Figure 2.10.2 The frequencies {330, 440, 550} divide a frequency-space octave (between 330 and 660) perfectly evenly. In pitch space, the chord divides the octave *nearly* evenly, into pieces that are approximately 5, 4, and 3 semitones large. Note that the acoustically pure C♯ (72.9) is not quite the same as the equal-tempered C♯ (73).

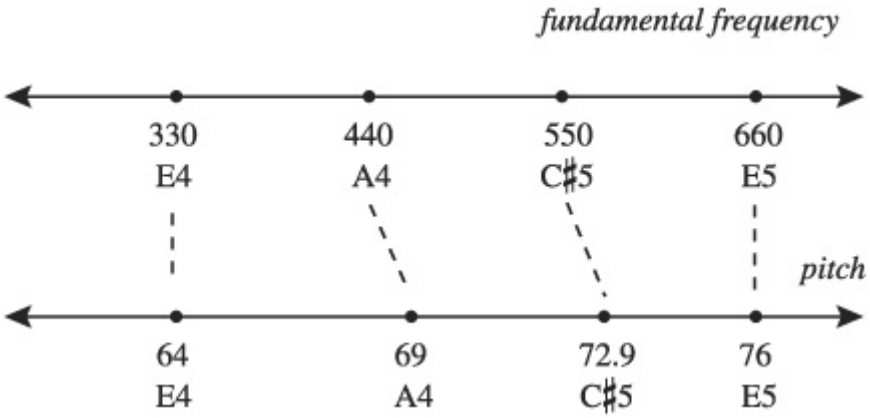


Figure 2.10.3 Highly consonant chords divide the octave nearly evenly.

Notes	Name
{C, G}	Perfect fifth
{C, E, G}	Major triad
{C, E♭, G}	Minor triad
{C, E, G, A}	Major triad "add six"
{C, E, G, B♭}	Dominant seventh chord
{C, E, G, B}	Major seventh chord
{C, D, E, G, A}	Pentatonic scale
{C, E, G, B♭, D}	Dominant ninth chord
{C, D, E, F, G, A}	Diatonic hexachord
{C, E, G, B♭, D, F♯}	Dominant "sharp eleven"
{C, D, E, F, G, A, B}	Diatonic scale
{C, D, E, F♯, G, A, B♭}	Melodic minor ascending

In other words—and this being one of the most important ideas in the entire book, it is worth a little interjective buildup—the basic sonorities of Western tonal music are optimal for two distinct reasons: considered as individual sonic objects, they are acoustically consonant and hence sound pleasing in their own right; but since they divide the pitch-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

class circle nearly evenly, they can also be connected to their transpositions by efficient voice leading. Any composer who cares about harmonic consistency would therefore have reason to choose these chords, even if he or she did not care a whit about acoustic consonance. Or to put the point in the form of a parable: suppose God asked you, at the dawn of time, to choose the chords that humanity would use in its music. There are two different choices you might make. You might say, "Well, God, I'm a somewhat cerebral type and I'd like to combine efficient voice leading and harmonic consistency, since this will allow me to praise You with glorious polyphony." And God would hand you a suitcase containing nearly even chords, including the perfect fifth, the major triad, and dominant seventh chord.[40] On the other hand, you might say, "You know, God, I'm kind of a hedonist, and at the end of a hard day of hunting and gathering I'd really like to hear chords that *sound good*—chords whose intrinsic consonance will put a smile on my face." And in this case, God would hand you a suitcase containing . . . the perfect fifth, the major triad, the dominant seventh chord. In other words, he would hand you *the very same chords* , no matter which choice you made.

It follows that the development of Western counterpoint is something of an amazing accident. No doubt musicians originally chose consonant chords because they "sounded good" in some primary sense—they were acoustically consonant and hence felt restful and harmonious. But during the early history of Western music it gradually became apparent that these chords were special in another way as well: they could be used to write *contrapuntal* music that was also *harmonically consistent*. This realization led to the marvelous efflorescence of Western contrapuntal technique, as composers over the centuries explored an ever-increasing range of voice-leading possibilities between an ever-increasing range of consonant chords. (This efflorescence, and its numerous variations and ramifications, is the subject of Part II.) Ultimately, this was possible because of the nonobvious connection between efficient voice leading, harmonic consistency, and acoustic consonance—a connection that we now understand as a simple consequence of the hidden symmetries of circular pitch-class space.[41]

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

CHAPTER 3

A Geometry of Chords

We now turn to geometrical models in which chords are represented as points in higher dimensional spaces. As we will see, these "chord spaces" are a good deal more interesting than the plain-vanilla space of ordinary Euclidean experience, containing twists, mirrors, Möbius strips, and their higher dimensional analogues. Besides being extraordinarily beautiful, these geometrical models clearly illustrate musical principles that can otherwise be quite difficult to grasp.[1] In particular, we can use them to construct "voice-leading graphs" that depict the basic contrapuntal relations between the familiar chords and scales of Western music, graphs that will serve as the principal analytical tools of the second half of this book.

3.1 ORDERED PITCH SPACE

Chapter 2 introduced the *basic musical object*, or ordered sequence of pitches. Figure 3.1.1 shows that there are two ways to represent these objects geometrically. In Figure 3.1.1a we represent ordered pairs of notes using a gray circle for the first note and a black circle for the second. (We need different colors because the pairs are ordered, and we need to differentiate the first note from the second.) In Figure 3.1.1b, the horizontal axis represents the first pitch while the vertical axis indicates the second. Each model therefore has two *degrees of freedom:* in the first, we can move either circle independently of the other; in the second, we can move along one axis while holding fixed our position on the other. It is more or less a matter of convenience whether we wish to represent two-note objects using two points in a one-dimensional space, as we did in Chapter 2, or one point in a two-dimensional space. However, the models are different in at least one important respect. It is easy to see how to ignore octave and order in the first, linear model: we simply represent all notes using same color (thereby ignoring order) and wrap the line into a circle (thereby ignoring octaves and reproducing pitch-class space). But it is not at all clear how we should modify the two-dimensional representation in those situations where we wish to ignore these musical parameters.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.1.1 Two ways to represent the ordered pair (C4, E4). The first uses two points in a one-dimensional space—here, the gray circle represents the first note, and the black ball the second. The second uses a single point in a two-dimensional space.

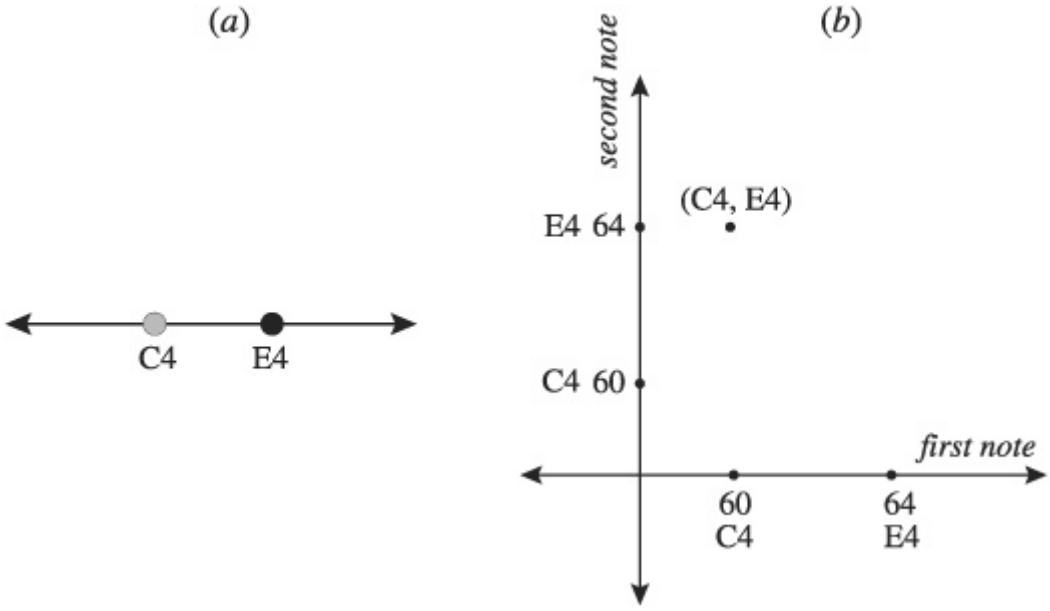
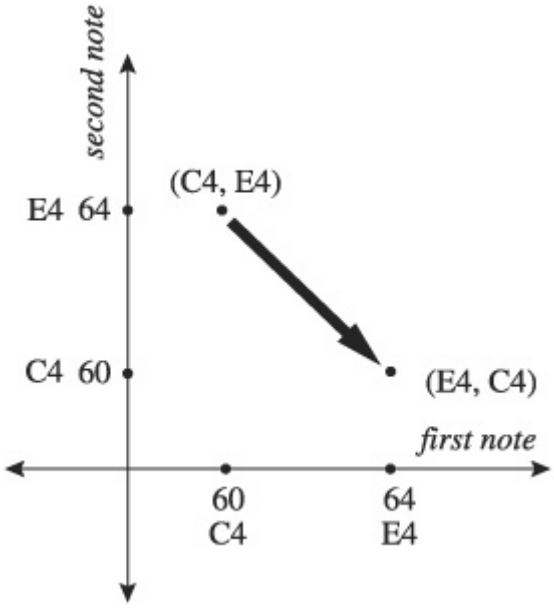


Figure 3.1.2 A voice leading can be represented as a line segment in the plane. Here, the voice leading (C4, E4)→(E4, C4), in which the first and second voices trade notes.



In the one-dimensional space, progressions between objects are represented by collections of paths (cf. Figure 2.5.4). In the two-dimensional space, progressions can be represented by line segments such as that in Figure 3.1.2, which depicts a situation in which the voices trade notes. Figure 3.1.3 uses line segments to model a two-voice passage from Josquin's *Missa l'homme armé*. With a little practice, it becomes easy to translate two-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

voice music into a series of line segments in the space. Any two-voice passage of music, in any style whatsoever, can be depicted in this way.

Horizontal and vertical line segments represent motion in a single voice (Figure 3.1.4). Parallel motion, in which the two voices move in the same direction by the same amount, is represented by lines parallel to the 45° NE/SW diagonal, while perfect *contrary* motion—in which the voices move the same distance in opposite directions—is represented by lines parallel to the 45° NW/SE diagonal. It turns out to be somewhat more convenient to rotate the space clockwise by 45°, so that parallel motion is horizontal and perfect contrary motion is vertical. Chords on the same horizontal line now relate by transposition; chords on the same vertical line sum to the same value when pitches are represented numerically. (This follows from the fact that perfect contrary motion subtracts from one voice what it adds to the other.) Oblique motion now moves along the 45° diagonals: motion in the first voice occurs along the NW/SE diagonal, while motion in the second occurs along the NE /SW diagonal. This rotation does not change the space in any way; we are simply changing our perspective on it.

Figure 3.1.3 A passage from Josquin represented in two-dimensional space.

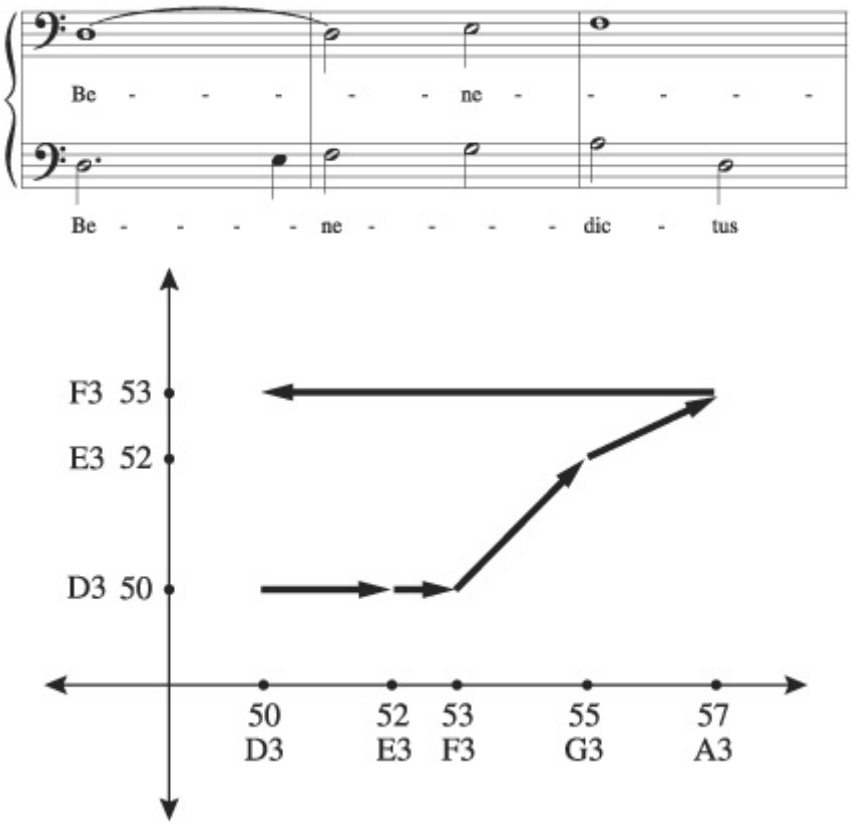


Figure 3.1.4 Rotating two-note ordered pitch space.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

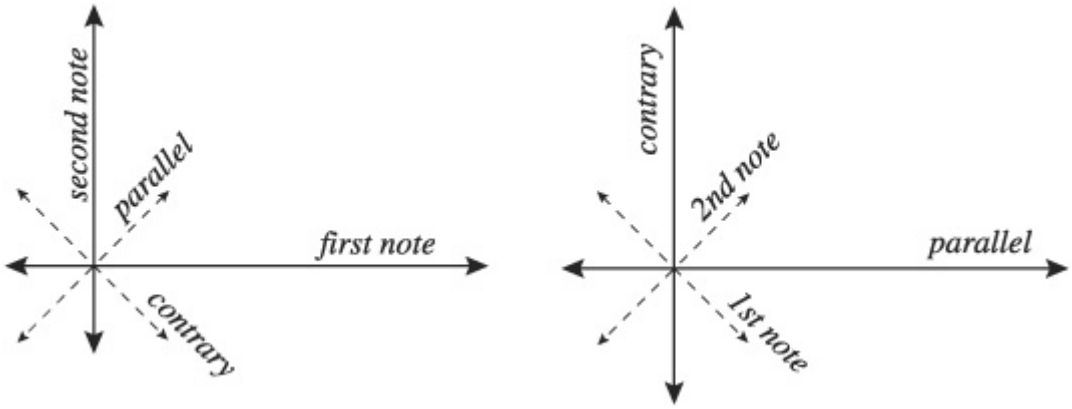


Figure 3.1.5 depicts an extended portion of two-dimensional ordered pitch space, with the axes rotated as just described. The most striking feature of this graph is its *periodicity:* like a piece of wallpaper, it consists of a single pattern (or "tile") that is repeated to cover the larger plane. The figure contains four complete tiles. The points in the lower left quadrant are related by octave transposition to the corresponding points in the upper right: the first element in each pair is the same, while the second element in the lower left pair is one octave below the second element in the upper right pair. Moving from the lower left quadrant to the corresponding point in the upper right will therefore transpose the second note up by an octave. Similarly, moving from the upper left to the corresponding point in the lower right shifts the *first* element up by an octave. Since any quadrant can be connected by a series of diagonal motions to one of the quadrants in Figure 3.1.5, and since diagonal motion between quadrants always corresponds to octave transposition in one voice, the rest of the infinite space can be generated from this figure.

Figure 3.1.5 A portion of infinite, two-dimensional ordered pitch space.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



What about the lower left and upper left quadrants? Here, the relationship is somewhat harder to grasp. Imagine that there were a hinge connecting them, so that the bottom-left quadrant could be lifted out of the paper and flipped onto the upper left. This transformation maps each pair in the lower quadrant onto a pair with the same pitch content, but in the reverse order. Geometrically, the transformation is a *reflection:* a pair in the lower left quadrant gets sent to the spot where its reflection would appear if the common border were a mirror. When we move a dyad from the lower left quadrant to its reflected image in the upper left, we therefore switch the order of its notes. The two rightmost quadrants are related in exactly the same way.

Figure 3.1.6 depicts symbolically the relationship between the four tiles, using a right-side-up human face to represent the lower left tile. The upper left quadrant is upside down relative to the lower left: if they were connected by a hinge along their common border, then either could be flipped over so as to coincide with the other. Similarly, the lower right tile is upside-down relative to the upper right. The lower right tile is also upside-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

down relative to the lower *left* tile, but in this case they cannot be related by reflection along their common border—that process would exchange left and right, but not up and down.

Figure 3.1.6 Ordered pitch space is like a piece of wallpaper.



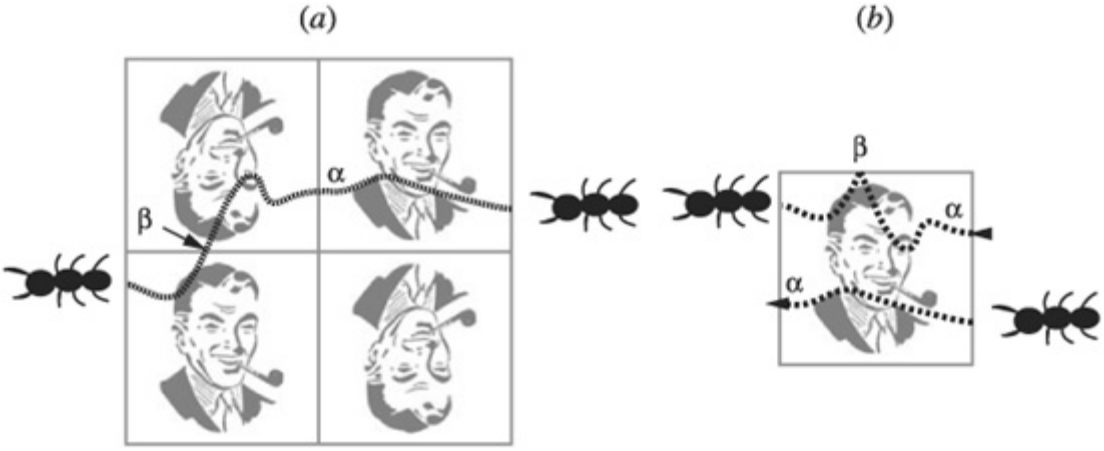
3.2 THE PARABLE OF THE ANT

Imagine now that an ant is walking along the wallpaper in Figure 3.1.6. Suppose that you and I are gambling types and decide to bet on whether the ant will touch a pipe in the next 30 seconds. For the purpose of settling the bet, it does not matter which tile the ant is on; what matters is whether it touches *any* pipe in *any* tile. We could therefore represent the ant's trajectory on a single tile, as in Figure 3.2.1. (Perhaps we use a single tile because we want to record our game for posterity while being environmentally conscious and using the minimum amount of paper.) That is, we arbitrarily select one tile from Figure 3.1.6 and take the ant's position to be the point on this tile corresponding to its position on the tile it actually occupies.

Figure 3.2.1 An ant's path can be represented on a single tile.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Although the underlying idea is simple, the structure of the resulting single-tile space is complex. For example, at the point marked a, the ant disappears off the lower left edge, only to reappear on the upper right. This is reminiscent of early video games such as *Asteroids* or *Pac-Man*, in which objects could move off one side of the screen to reappear on the other. Unlike those games, however, the ant leaves the *lower* half of the figure only to reappear on the *upper* half and vice versa, as if the left edge were attached to the right in a twisted fashion. The mathematical name for such a space is a *Möbius strip*.

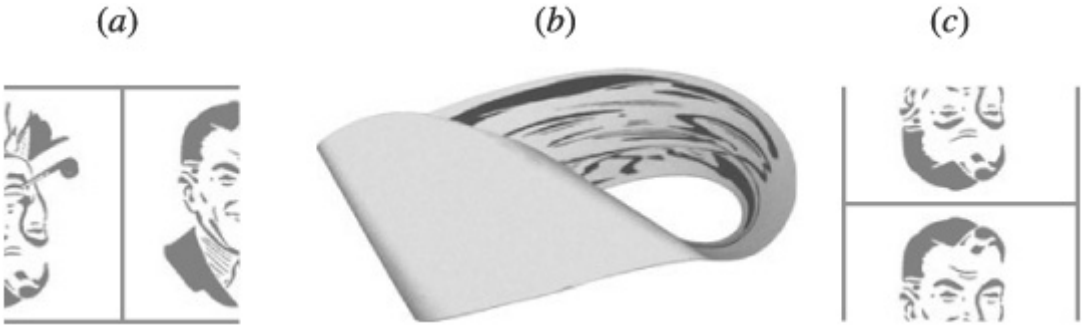
Now consider point β in the ant's trajectory. In the single-tile representation the ant appears to "bounce off" the figure's upper edge, as if it were a mirror, or the bumper of a pool table. But we can see from Figure 3.2.1a that the ant's actual trajectory is straight. Nothing intrinsic to the ant's motion produces the change in direction—rather, it is the structure of the wallpaper on which it walks. The ant, being unaware of the wallpaper pattern, would have no knowledge that anything unusual was occurring.

Note that the placement of the left and right boundaries is essentially arbitrary. For the purpose of settling the bet, one could equally depict the ant's trajectory on Figure 3.2.2a, which contains two half faces, one upside down and the other right side up. Though bizarre from an artistic point of view, it is perfectly adequate for gambling, since all parts of the figure are represented. If we were to stretch the figure horizontally, forming a rectangle rather than a square, we could even represent the tile in three dimensions *without any left or right boundaries whatsoever*—as in Figure 3.2.2b.[2] By contrast, the location of the upper and lower boundaries is *not* arbitrary. Figure 3.2.2c shows that we cannot settle our bet using the space consisting of two halves of two vertically adjacent tiles, since that space contains two copies of the top half of the figure but none of the lower half.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.2.2 (*a*) The choice of left and right boundaries is arbitrary. (*b*) We could even represent the wallpaper without any left and right boundaries at all, if we stretched it horizontally and used the third dimension to attach the two edges. (*c*) The choice of upper and lower boundaries is not arbitrary, since this figure has no pipe.



All this talk about ants, wallpaper, and gambling may seem like a distraction from the serious goal of understanding music. But in fact, the Parable of the Ant has introduced the fundamental concepts needed in the rest of this chapter. The key idea is that we can form a similar structure by "folding up" the two-dimensional *musical* space in Figure 3.1.5. In this case, ignoring what tile we are on corresponds to ignoring the *order* and *octave* of a pair of notes. In other words, the resulting space represents two-note chords as musicians are accustomed to thinking of them. Thus the Parable of the Ant, rather than being a frivolous digression, has actually marked the beginning of our investigation into a remarkable convergence between music theory and contemporary geometry.

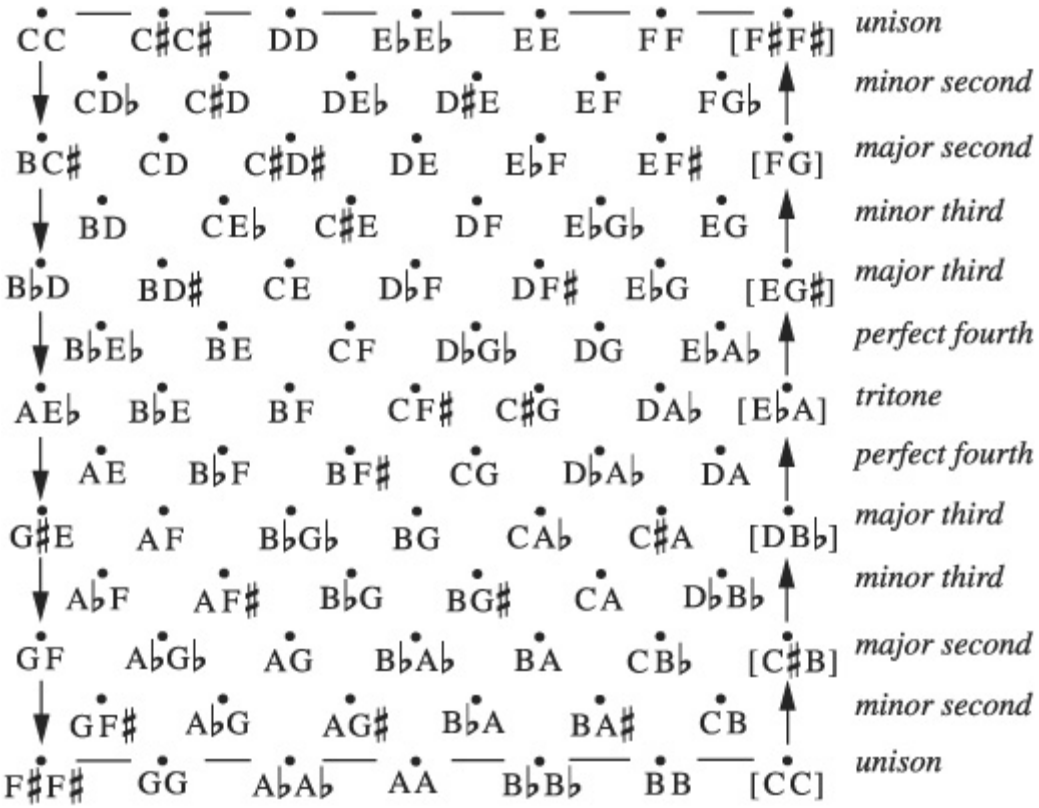
3.3 TWO-NOTE CHORD SPACE

Our goal is to "fold" the infinite space of Figure 3.1.5 so as to glue together all the different points representing the same chord: (C4, E4), (E4, C4), (E5, C2), and so on. The result will be a new geometrical space, analogous to our single tile of wallpaper, in which points represent two-note *chords*—or unordered pairs of pitch classes—rather than ordered pairs of pitches.

Figure 3.3.1 Two-note chord space. The left edge is "glued" to the right, with a twist.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Happily, the detailed work has already been done. The “wallpaper space” described in the preceding section was modeled on Figure 3.1.5, and we can immediately adapt our earlier results.[3] Since each of the quadrants in Figure 3.1.5 contains precisely one point for every unordered set of pitch classes, any of them can be used to represent two-voice music. The *two-note chord space* shown in Figure 3.3.1 consists in a single quadrant of Figure 3.1.5 , with octave designations removed to reflect the fact that we have discarded octave information. Although the points are labeled using ordered pairs, this ordering is not significant: the point CD♭ represents the ordering (D♭, C) just as much as it represents the ordering (C, D♭). Accordingly, I will sometimes refer to this point as {C, D♭} or {D♭, C}.

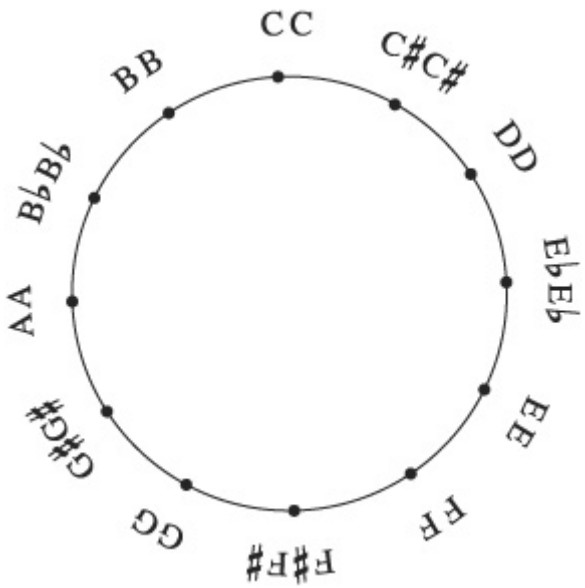
In §3.1, we rotated the coordinate axes so that parallel musical motion is represented by horizontal geometrical motion. This is again true in our new, “folded” space: horizontal motion represents parallel motion in both voices, while vertical motion represents contrary motion in which the two voices move in opposite directions by the same distance. (Oblique motion, in which one voice stays fixed, is again represented by the 45° diagonals.) It follows that chords on the same horizontal line of Figure 3.3.1 are related by transposition: the top edge of the figure, for example, contains unisons—two-note chords in which both voices sound the same pitch class.[4] Unisons can also be found on the bottom edge of the figure, with two of them—{C, C} and {F♯, F♯}—appearing on both edges. (Chords on the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

right side of the figure are enclosed in brackets, indicating that they also appear on the left side.) These duplications are different representations of the same chord, artifacts of the doomed attempt to depict a Möbius strip in two Euclidean dimensions. *Intrinsically*, the Möbius strip has only one edge: what appear as distinct line segments containing unisons are actually two halves of an abstract "circle," since horizontal motion along the edge eventually returns to its starting point (§2.2). Figure 3.3.2 shows that this abstract circle is very similar to pitch-class space, with the only difference being that its points represent not single pitch classes but musical states in which two voices articulate the same pitch class.

Figure 3.3.2 The boundary of two-note chord space is an (abstract) circle, since you can move in one direction to return to your starting point. The circle is very similar to pitch-class space.



The other intervals are also found on horizontal line segments, one in the top half of the figure and one in the bottom. The horizontal line segment just below the top edge, and just above the bottom edge, contains minor seconds. (Again, these apparently different line segments are attached at their endpoints, forming two halves of an abstract circle.) Major seconds are just below the minor seconds on the top half and just above the minor seconds on the bottom. As the interval between the two notes gets larger, dividing the octave more evenly, the lines representing them move toward the center. Consequently, the horizontal line at the midpoint of the figure contains tritones, which divide the pitch-class circle into two halves.[5]

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Chords on the same vertical line can be linked by exact contrary motion: for instance, the voice leading (C, F♯)→(D♭, F) moves C up by semitone and F♯ down by semitone, and joins two chords on the same vertical line. It follows that every dyad lies on the same vertical line as its tritone transposition.[6] Since exact contrary motion preserves the sum of a chord's pitch classes, chords on the same vertical line always sum to the same value. Thus {C, F♯}, {G, B}, and {A♭, B♭}, which lie on the vertical line at the center of Figure 3.3.1, all sum to six using pitch-class arithmetic.[7] (*Note:* this fact will be important later on, so make sure you understand it.) The pitch classes on the left edge of Figure 3.3.1 sum to zero. The sum of the chords in the vertical cross section increases as one moves rightward until we reach the right edge, whose chords again sum to zero (equivalent to twelve in pitch-class arithmetic). Finally, motion along the 45° diagonals alters just one note, so that (C♯, E) lies 45° northeast of (C, E). (Note, however, that diagonal motion can no longer be thought of as moving the "first note" rather than the "second note" because we are now neglecting order.[8]) It is worth emphasizing that two-note chord space is *continuous*, containing every conceivable chord in every conceivable tuning system. A chord like (C𝄲, E) (C quarter-tone sharp, E), though not labeled on the example, lies 45° northeast of (C, E), halfway to (C♯, E).

3.4 CHORD PROGRESSIONS AND VOICE LEADINGS IN TWO-NOTE CHORD SPACE

Our Möbius strip can be used to represent any chord progression and any voice leading between two-note chords. To represent a progression like {C, E}⇒{F, A} simply identify the initial point, {C, E}, and the final point, {F, A}. It is not necessary to choose any particular path between them, because a chord progression does not specify exactly *how* the individual notes move (§2.5). It is as if the music magically teleports, disappearing at {C, E} and instantaneously reappearing at {F, A}, without occupying any of the places in between.

A voice leading, by contrast, is represented by a particular *path* between chords, with the *size* of the voice leading corresponding to the *length* of the path. The simplest way to find this path is to imagine each voice making a continuous glissando, beginning and ending at the same time. For example, (C, E)→(E♭, G) moves each voice up by three

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

semitones. If each voice made a continuous glissando, the music would pass through a series of major thirds: (D♭, F), (D, F♯), and countless others not accessible with an ordinary piano. These all lie on the horizontal line segment shown on Figure 3.4.1. Similarly, the voice leading (B, D)→(A♭, F) moves each voice by three semitones in contrary motion. This voice leading passes through the chords {B♭, E♭}, {A, E}, and a variety of other non-equal-tempered chords. It is therefore represented by the vertical line segment on the figure.

A little experimentation will show that the boundaries of the figure behave exactly like the boundaries of our earlier wallpaper space. To see this, imagine making a smooth glissando from (E♭4, G4) to (F4, A4). Figure 3.4.2 indicates that the glissando is initially represented by rightward motion on the top half of the strip, from {E♭, G} to {E, G♯}. However, the point {E, G♯} is represented *both* on the top half of the right edge and on the bottom half of the left edge. As we move from (E4, G♯4) to (F4, A4), the glissando now appears on the bottom half of the figure, moving from the lower-left {E, G♯} to {F, A}. Thus, as in the Parable of the Ant, the music disappears off the upper right edge only to reappear on the lower left. Alternatively, consider the smooth glissando from (C4, D4) to (E4, D4)—in which the first voice moves from C4 to E4 while the second voice remains fixed. The resulting path, illustrated in Figure 3.4.2, is represented by a line segment that begins at {C, D} and moves diagonally northeast to {D, D}. (Recall that motion in just a single voice always lies along a 45° diagonal.) However, the second half of the path is represented by a line segment moving diagonally *southeast* from {D, D} to {E, D}. As in the Parable of the Ant, what causes the apparent "change in direction" is the structure of the underlying space: musically, the first voice moves smoothly upward from C4 to E4, and does not do anything strange at D4.

Figure 3.4.1 Voice leadings are represented by line segments. Parallel motion is horizontal, while perfect contrary motion is vertical. The voice leadings (C, E)→(E♭, G) and (B, D) → (A♭, F) are shown.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

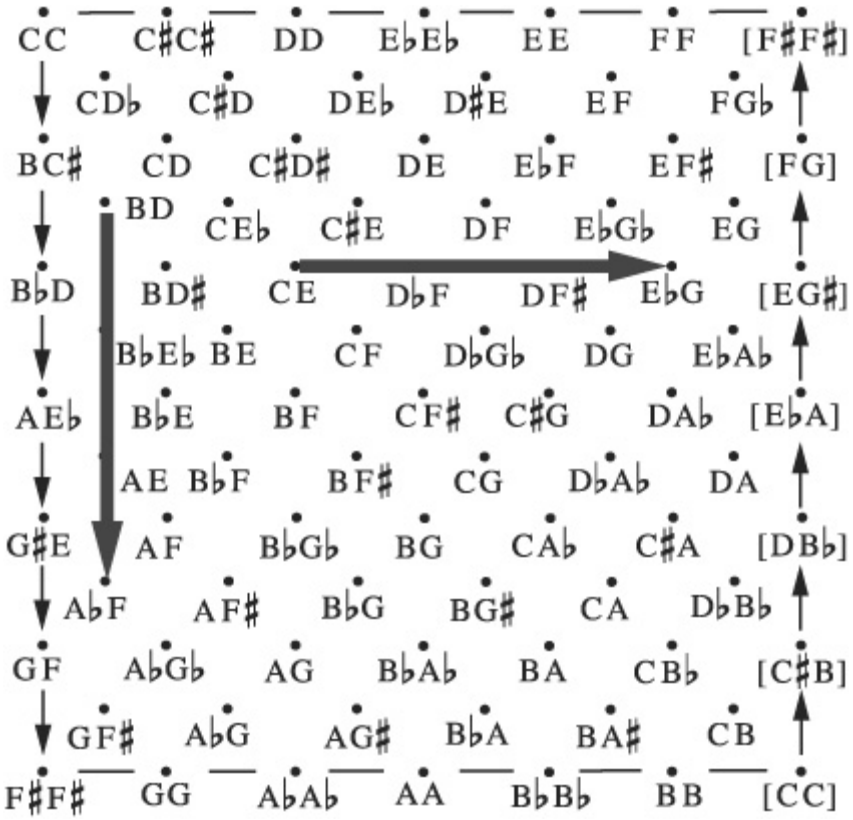
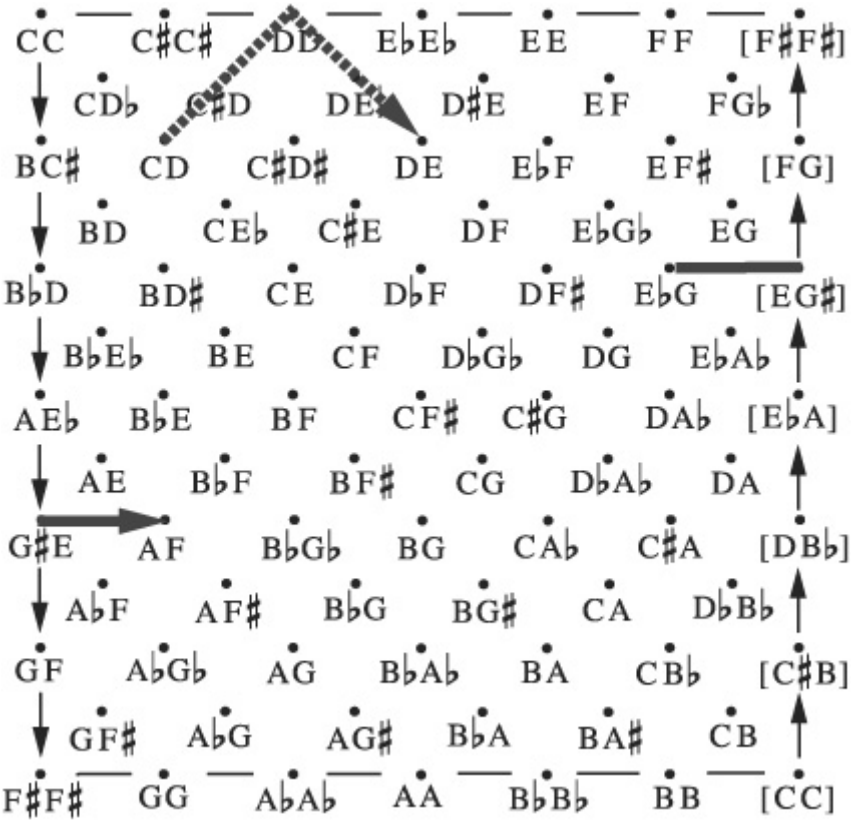


Figure 3.4.2 The horizontal boundaries act like mirrors, whereas the vertical boundaries are glued together with a "twist." Voice leadings thus disappear off the left edge to reappear on the right, and vice versa. Here, the voice leadings (E♭, G) → (F, A) and (C, D) → (E, D) are shown.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Voice leadings are therefore represented by what might be called "generalized line segments" that can bounce off a mirror boundary, or disappear off one side of the figure only to reappear on the other. There is, in fact, a one-to-one correspondence between these "generalized line segments" and voice leadings: for every generalized line segment there is a voice leading in pitch-class space, and for every voice leading in pitch-class space there is a generalized line segment. In principle, there can be infinitely many generalized line segments linking any two chords, each represented by a different pattern of reflections off the horizontal boundaries and "disappearances" off the left and right edges.

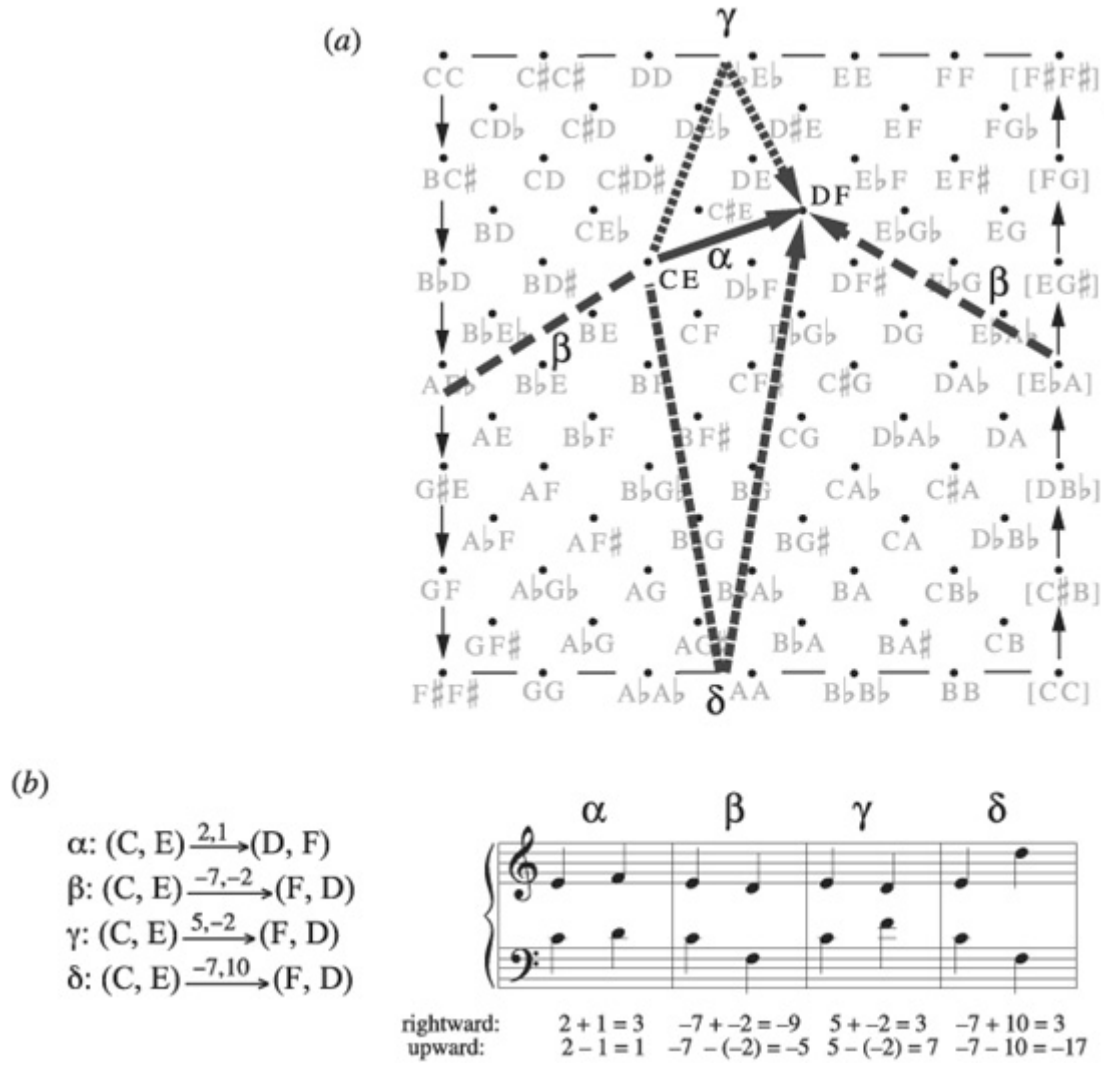
The total amount of horizontal motion in a voice leading can be calculated algebraically by adding the pitch-class paths in the two voices. A voice leading like $(\mathrm{C}, \mathrm{E}) \xrightarrow{2,\ 1} (\mathrm{D}, \mathrm{F})$ moves its two voices by a total of $2 + 1 = 3$ semitones, and is therefore represented by a generalized line segment whose horizontal component moves 3 units to the right.[9] $(\mathrm{C},\ \mathrm{E}) \xrightarrow{-7,\ -2} (\mathrm{F},\ \mathrm{D})$ moves its voices by $-7 + -2 = -9$ semitones, and therefore moves 9 units to the left (Figure 3.4.3). Similarly, if one writes the voice leading so that the first two notes are in the same order as they are on Figure 3.3.1, then the total amount of *vertical* motion can be determined by subtracting the second path from the first. For example, $(\mathrm{C}, \mathrm{E}) \xrightarrow{2,\ 1} (\mathrm{D}, \mathrm{F})$ moves by a total of $2 - 1 = 1$ unit upward, while $(\mathrm{B},\ \mathrm{D}) \xrightarrow{-3,\ 3} (\mathrm{A}\flat,\ \mathrm{F})$ moves its voices by $-3 - 3 = -6$ units upward, or 6 units downward.[10] (The first of these voice leadings is shown in Figure 3.4.3, the second in Figure 3.4.1.) However, contact with *any* of the four "edges" of the strip exchanges "downward" and "upward": $(\mathrm{C},\ \mathrm{E}) \xrightarrow{5,\ -2} (\mathrm{F},\ \mathrm{D})$ moves its voices by $5 - (-2) = 7$ upward units, but it reaches the upper mirror boundary after just four of these; the remaining three steps are therefore taken in a descending direction (Figure 3.4.3). Similarly, the voice leading $(\mathrm{C},\ \mathrm{E}) \xrightarrow{-7,\ -2} (\mathrm{F},\ \mathrm{D})$ moves its voices by $-7 - (-2) = -5$ units, but when the path disappears off the left edge to reappear on the right, the direction turns upward. These algebraic principles provide a more systematic alternative to the method of imagining each voice making a continuous glissando. For an even simpler method readers can download a free computer program that plots voice leadings on the Möbius strip automatically.[11]

Figure 3.4.3 Four voice leadings between {C, E} and {D, F}, as represented on the Möbius strip. The amount of rightward motion can be calculated by adding the paths in the two

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

voices; the amount of upward motion can be calculated by subtracting the second path from the first.



3.5 GEOMETRY IN ANALYSIS

Figure 3.5.1 uses the Möbius strip to graph the eighth phrase of the *Allelujia Justus et Palma* , one of the earliest examples of Western counterpoint. The music begins by tracing out a pair of triangles, each of which represents a four-chord unit that returns to its starting point. (The triangles share a side because the last two chords of the first unit are also the first two chords of the second.) Having completed the second triangle, the music moves back to the initial {G, D} dyad, disappearing off the left edge and reappearing on the right. The final voice leading retraces the (G, D)→(E, E) motion that begins the phrase.

All of this, I believe, is considerably easier to see in Figure 3.5.1b than in traditional musical notation. The geometrical patterns virtually jump off the page, without any effort

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

or concentration on the analyst's part, whereas the corresponding musical patterns are much harder to identify. This is largely because our visual system is optimized for perceiving geometrical shapes such as triangles, but not for perceiving musical structures as expressed in standard musical notation. Translating the music into geometry thus allows us to bring our formidable visual pattern-matching skills to bear on musical analysis.

Figure 3.5.1 Plotting a phrase from the *Allelujia Justus et Palma* (*a*) on the Möbius strip (*b*) reveals interesting musical structure (*c*).

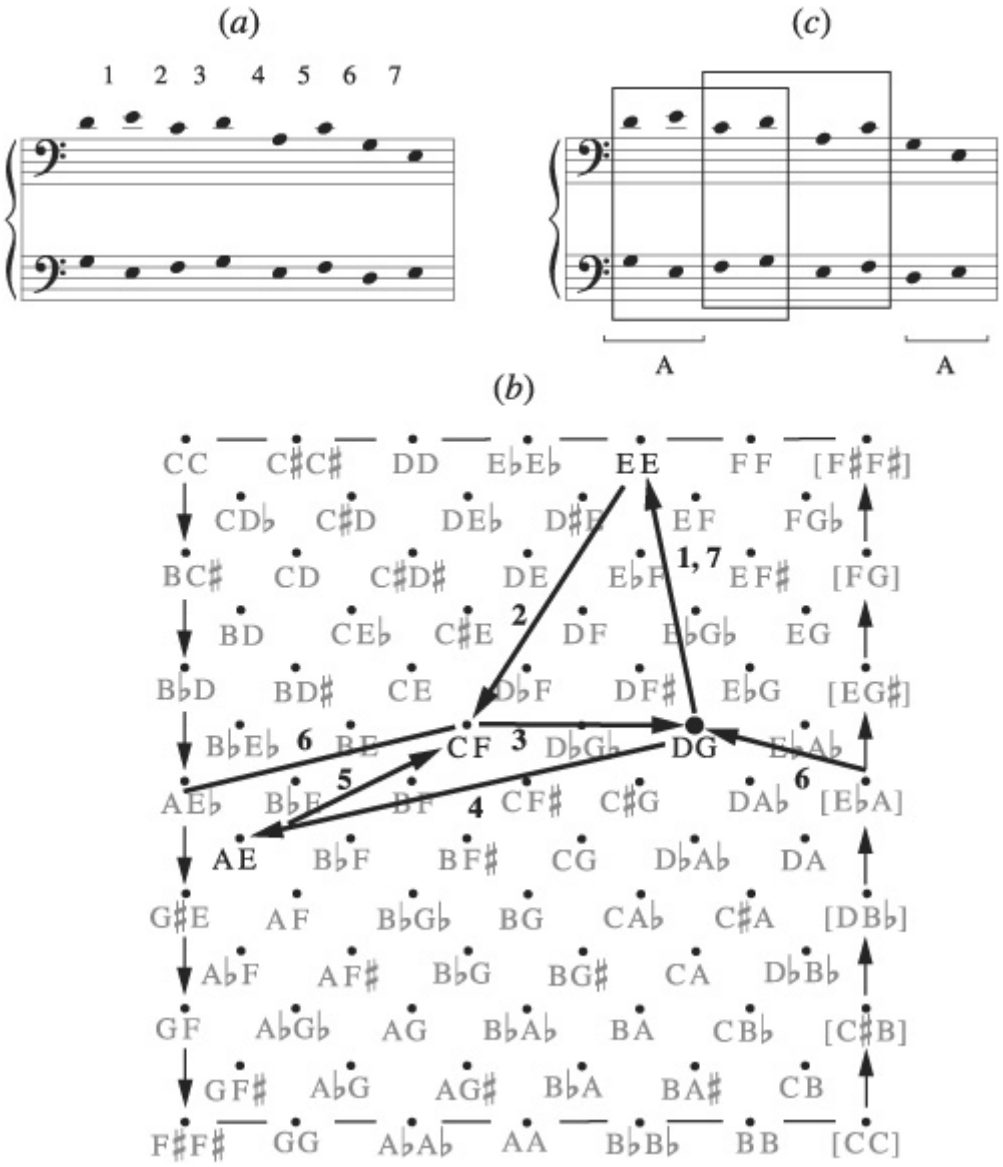


Figure 3.5.2 provides a more sophisticated example.[12] We begin with four voice leadings from the opening of Brahms' Intermezzo, Op. 116 No. 5. Figure 3.5.2b graphs the voice leadings in two-note chord space, representing each measure as a pair of line segments forming an open angle. We can imagine sliding the pair {X1, X2} so that X1

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
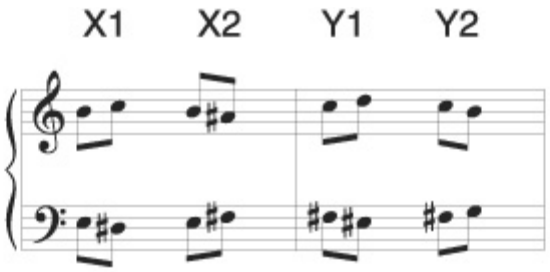
nearly coincides with Y1, and X2 nearly coincides with Y2. This represents the most obvious analysis of the passage, according to which voice leading Y1 is a slight variation of X1, and Y2 is a slight variation of X2. (That is, X1 moves its two voices by semitonal contrary motion, whereas Y1 moves by slightly skewed contrary motion; X2 moves in a skewed fashion, while Y2 moves in pure contrary motion.) Geometrically, however, it is clear that (Y1, Y2) is also the *mirror image* of (X1, X2). Hence we can move the pair (X1, X2) off the left edge so that it exactly coincides with (Y1, Y2), as in Figure 3.5.2c. (Remember that an upward-pointing arrow becomes a downward-pointing arrow when it moves off one side of the figure to reappear along the other.) On this interpretation, Y2 is exactly equivalent to X1, and Y1 is exactly equivalent to X2. Figure 3.5.2d represents this musically, heightening the comparison by switching hands and reordering dyads. Now both pairs begin with perfect contrary motion and move to less perfectly balanced motion, with melodies in each staff being transpositionally related.

Figure 3.5.2 (*a*) Two passages from Brahms' Op. 116 No. 5, as plotted on the Möbius strip (*b*). It is natural to view Y1 and Y2 as variants of X1 and X2, respectively. However, it is clear from the graph that Y2 and Y1 are also mirror images of X1 and X2. This means we can move the pair {X1, X2} off the left edge of the figure so that it coincides with {Y1, Y2} (*c*). On this reading, Y2 is related to X1, and Y1 is related to X2 (*d*). Here, "P" and "N" stand for "perfect" and "near perfect" contrary motion.

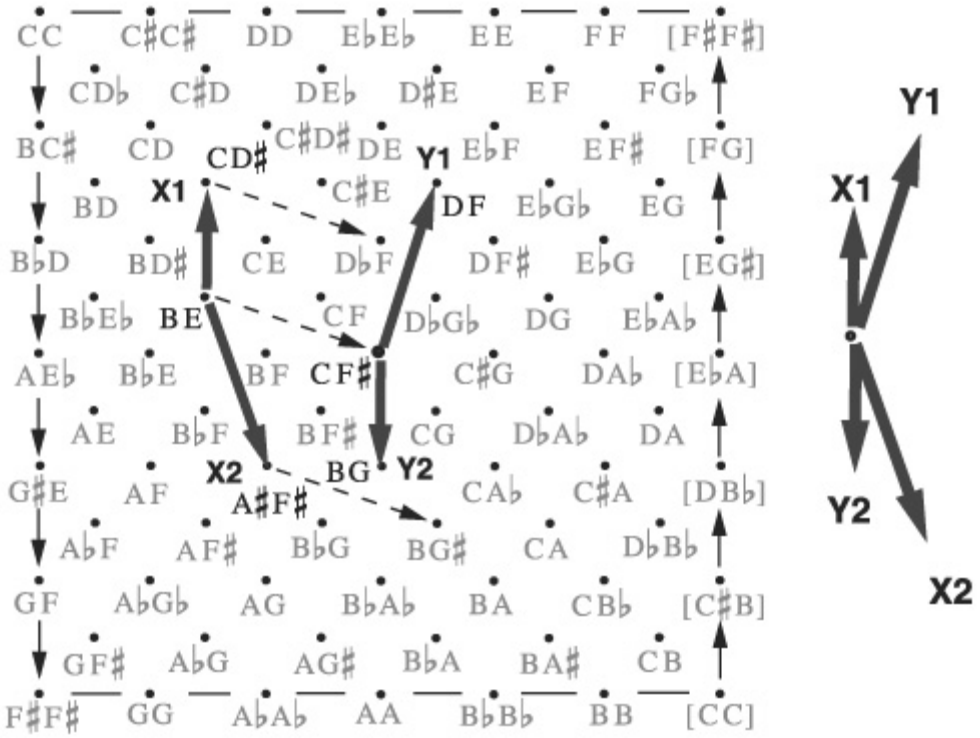
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

(a)

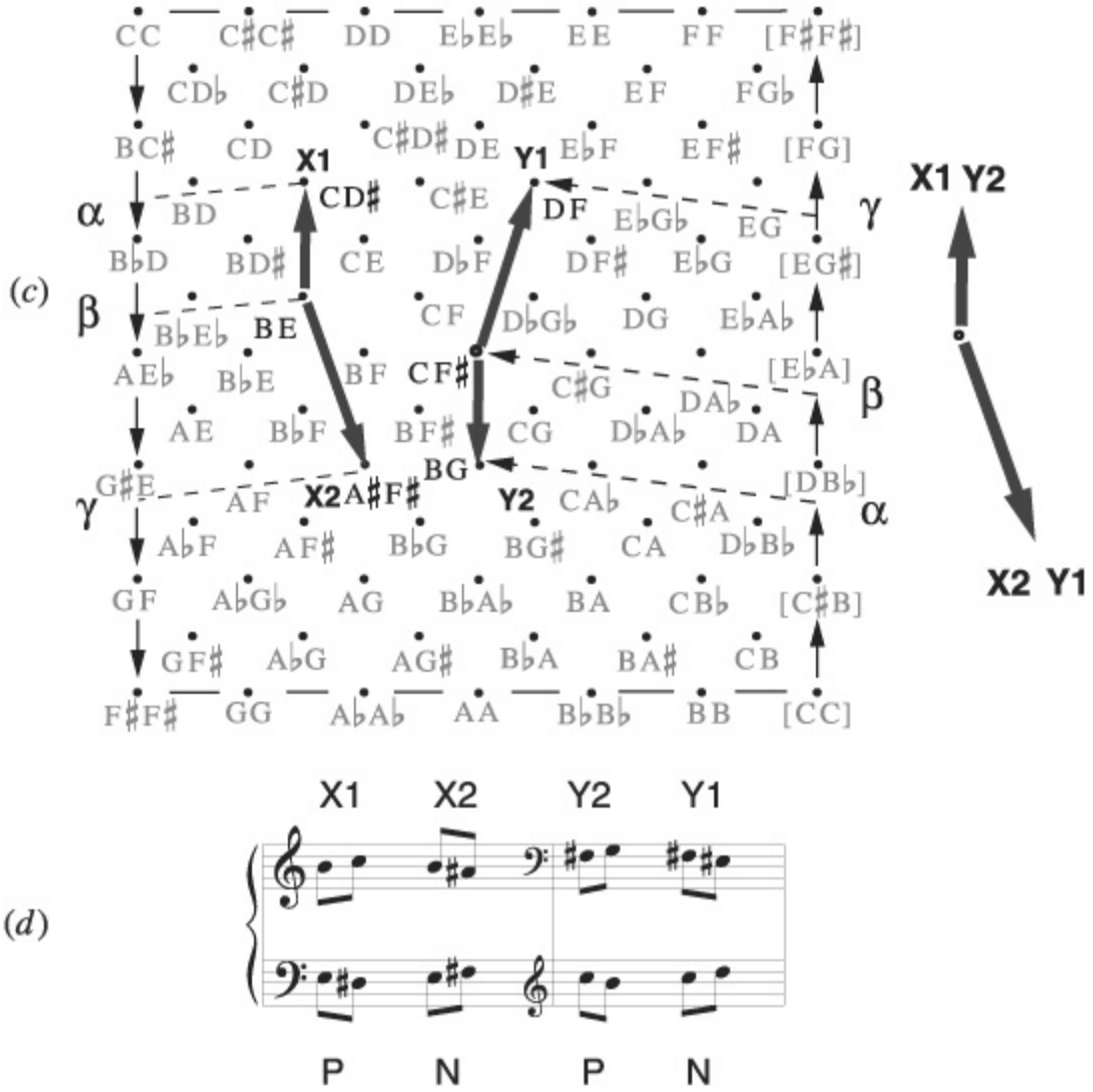


(b)



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Playing through Brahms' piece, I have always been dimly aware of these relationships—you feel in your fingers the difference between the precisely contrary motion and the less balanced motion, and you feel the purely semitonal motion in the right hand of measures 1–2 moving to the left hand of mm. 3–4. However, it is quite difficult to see the relationship in the musical notation, and rather hard to describe it using traditional terminology. By contrast, Figure 3.5.2b makes it obvious that the passages are mirror images—a fact that Brahms, with all his love for invertible counterpoint and other forms of compositional trickery, would no doubt have enjoyed.
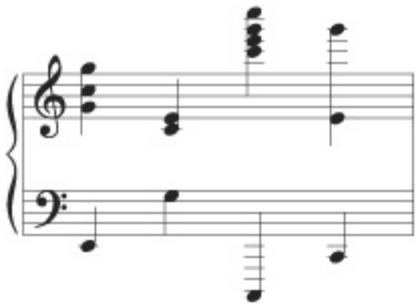
In these sorts of cases, geometry can help to sensitize us to relationships that might not be immediately apparent in the musical score. Ultimately, this is because conventional musical notation evolved to satisfy the needs of the performer rather than the musical thinker: it is designed to facilitate the translation of musical symbols into physical action, rather than to foment conceptual clarity. This is precisely why one and the same chord can

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

be notated in such a bewildering variety of different ways (Figure 3.5.3). Learning the art of musical analysis is largely a matter of learning to overlook the redundancies and inefficiencies of ordinary musical notation. Our geometrical space simplifies this process, stripping away musical details and allowing us to gaze directly upon the harmonic and contrapuntal relationships that underlie much of Western contrapuntal practice.

Figure 3.5.3 There are many ways to notate a C major chord.



3.6 HARMONIC CONSISTENCY AND EFFICIENT VOICE LEADING

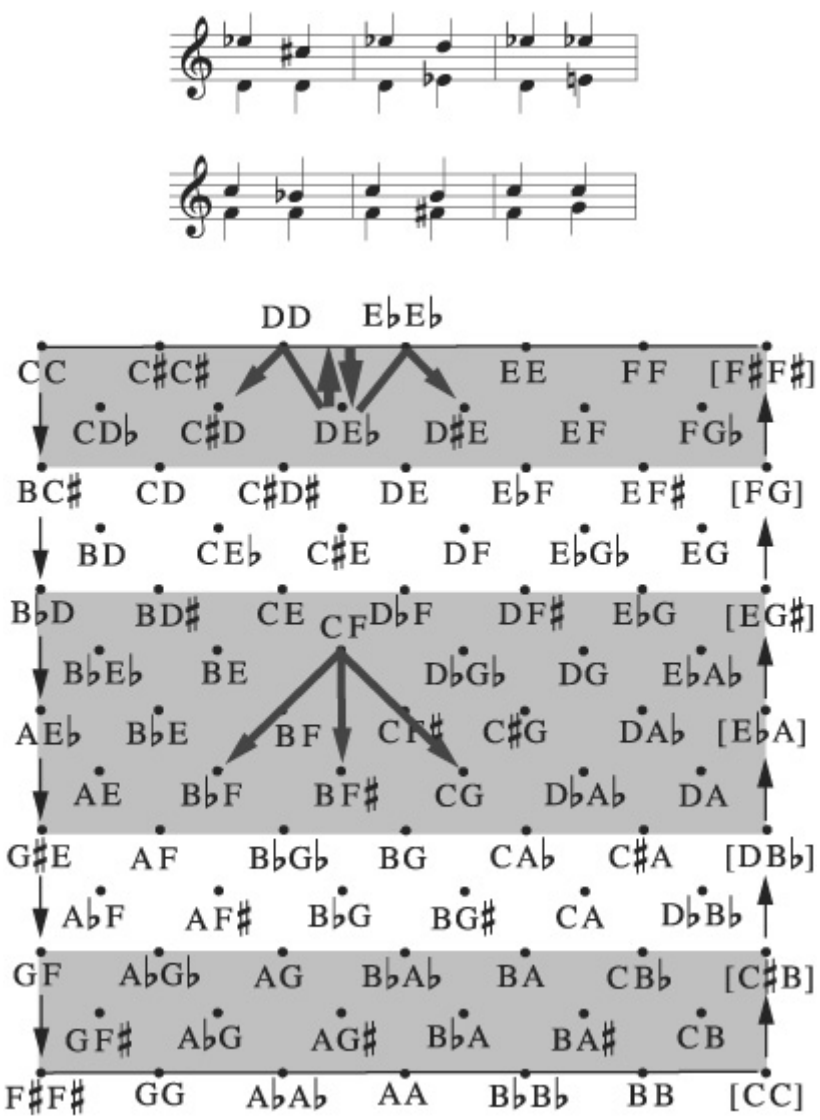
Let's pursue this further by returning to a more abstract question. Suppose you are a composer who wants to write two-part note-against-note counterpoint, in which transpositionally related dyads are linked by "stepwise" voice leading (or voice leading in which no voice moves by more than two semitones). Under what circumstances is this possible? In other words, how can we combine harmonic consistency and efficient voice leading with just two voices?

The question can be answered simply by inspecting two-note chord space. Transpositionally related dyads can be found on two horizontal line segments, one in the lower half of the strip and one in the upper half, equidistant from the central line of tritones; we are therefore looking for short line segments connecting points on these line segments. One solution is immediately clear, since we can move any chord horizontally to its transpositions. Unfortunately, these voice leadings involve parallel motion in the two voices, and are therefore somewhat unsatisfactory. If the goal is to write contrapuntal music that suggests *independent* melodies, then it will not do to move the voices in the same direction all the time; this would sound like a single melody being doubled at two pitch levels, rather than two distinct melodic voices.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.6.1 Even dyads and uneven dyads can both be connected to their transpositions by stepwise voice leading.



A second solution involves nearly even chords, which occupy horizontal line segments near the center of the space. Thus tritone-related perfect fifths can be linked by semitonal voice leading, represented geometrically by a vertical arrow (Figure 3.6.1, second line of music). In addition, they can be linked to their fifth transpositions by oblique voice leading in which one voice moves by two semitones. Major thirds, since they lie farther away from the center of the figure, can be linked only to their tritone transpositions by stepwise voice leading. Analogous voice leadings are available for the non-twelve-tone-equal-tempered intervals larger than a major third: in each case, the voice leading between tritone-related chords is the shortest, and is represented by a vertical line, with the other acceptable transpositions represented by slightly diagonal lines.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

There is also a third and even less obvious possibility: chords that divide the octave very *unevenly*—such as minor and major seconds—can be linked to themselves by short line segments reflecting off the nearby mirror boundary. For instance, the voice leading $(\mathrm{D}, \mathrm{E}\flat) \rightarrow (\mathrm{E}\flat, \mathrm{D})$ connects a minor second to itself by stepwise contrary motion, and is represented by a vertical arrow. (Analogous voice leadings again exist for any of the non-twelve-tone-equal-tempered intervals smaller than the major second.) Given any such voice leading, we can always transpose the second chord by a small amount without much increasing the voice leading's size. Consequently, small dyads can also be connected by stepwise voice leading to their one-semitone (or smaller) transpositions. For these chords the short vertical arrows connect a chord to itself, while for nearly even dyads the vertical arrows connect a chord to its tritone transposition (Figure 3.6.1). Geometrically, the clustered chords reach their nearby transpositions by bouncing off the mirror, while nearly even chords cross the center of the Möbius strip, thereby taking advantage of its "twist."[13] Thus we see that our earlier conclusions about symmetry are manifest in the structure of the two-dimesional space.

3.7 PURE PARALLEL AND PURE CONTRARY MOTION

Are we completely certain that these three possibilities are the *only* ones? Some readers might think this is obvious, while others may (justifiably) wonder whether there might be additional features of Möbius-strip geometry that we have overlooked. How do we know we have accounted for all the relevant musical alternatives?

Let me address this worry by noting that any two-voice voice leading can be decomposed into two components, one involving pure contrary motion and the other involving pure parallel motion (Figure 3.7.1). The "pure contrary" component moves the notes by the same amount in opposite directions, while the "pure parallel" component moves the notes by the same amount in the same direction. Geometrically, the "contrary" component will be confined to a vertical slice of the Möbius strip, such as that in Figure 3.7.1b, while the transpositional component will move horizontally. This decomposition into parallel and contrary is nothing other than high school vector analysis, which dissects

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

arbitrary vectors into their *x* and *y* components.[14] Musically, the goal of the decomposition is to focus on the *relative* motion among the voices, ignoring the parallel motion they all share.

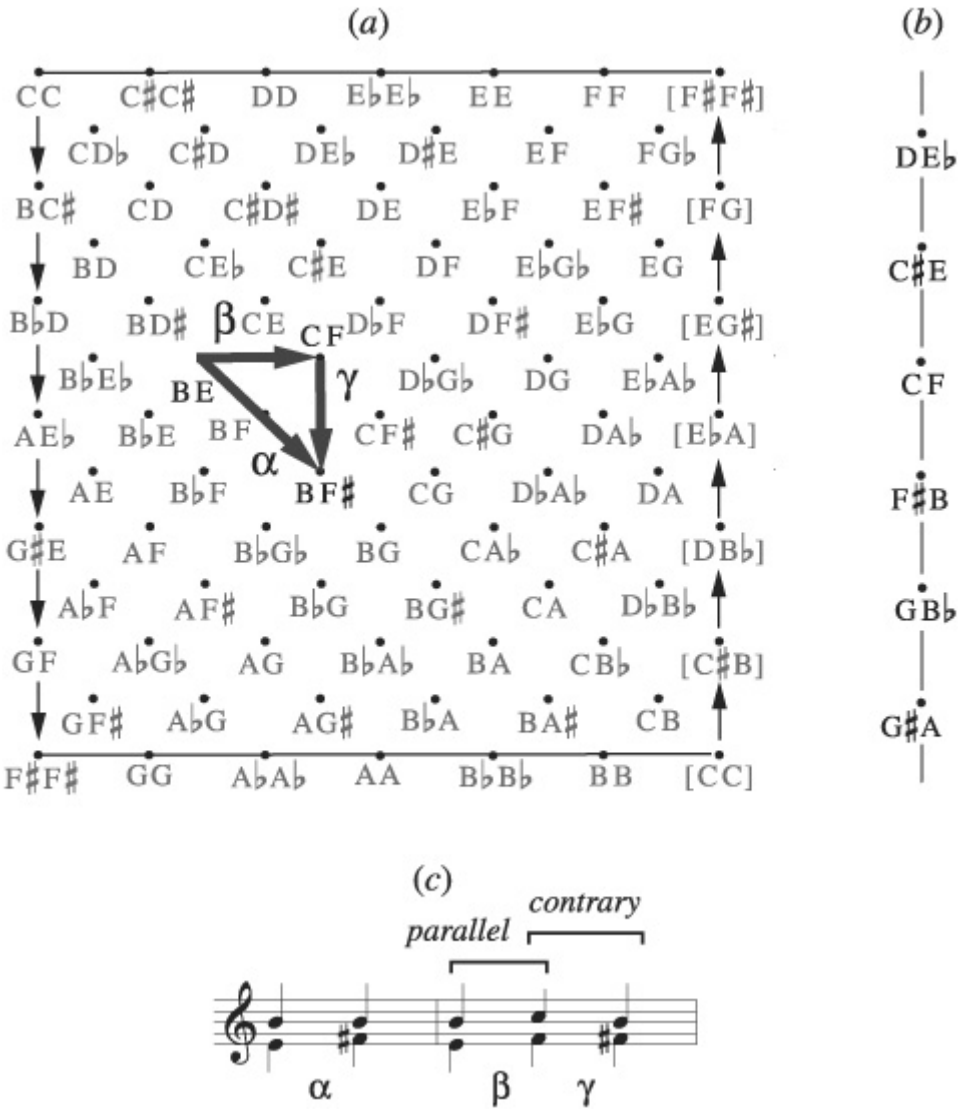
Now consider any efficient voice leading between transpositionally related dyads. The purely contrary component of this voice leading will connect transpositionally related dyads lying in the same vertical slice of the Möbius strip. It is visually obvious that the contrary component must either bounce off the mirror or cross the midpoint of the line. In the former case, it will be smaller when the chord is uneven; in the latter case, it will be smaller when the chord is even. Simple examination of the cross section shows that there are no other possibilities: if a dyad is neither particularly even nor particularly uneven, then it simply cannot participate in purely contrary voice leadings of the sort we are interested in.

We will periodically find it useful to abstract away from a voice leading's parallel component in this way. Geometrically, this involves restricting our attention to a cross section of chord space—the vertical line in Figure 3.7.1b rather than the entire Möbius strip. However, although every cross section of the space contains precisely the same chord types, not all of these appear in twelve-tone equal-tempered forms. Consider the chord {C𝄲, E𝄲} (C quarter-tone sharp, E quarter-tone sharp), which lies halfway between {C♯, E} and {C, F}.[15] Although this chord cannot be played on an ordinary piano, it is a major third, transpositionally related to familiar chords like (C, E). (Geometrically, {C𝄲, E𝄲} lies on the same horizontal line as {C, E} and {D♭, F}, halfway between them.) Clearly, no vertical cross section of the Möbius strip will contain twelve-tone equal-tempered major and minor thirds, even though *every* such cross section contains some representative of every chord type.

Figure 3.7.1 Any voice leading can be decomposed into parallel and contrary components. Here, the voice leading (E, B)→(F♯, B) (α) combines the parallel (E, B)→(F, C) (β), which moves both voices up by semitone, with the contrary (F, C)→(F♯, B) (γ), which moves the voices semitonally in opposite directions. Pure parallel motion is represented geometrically by horizontal lines, while pure contrary motion is vertical. This means that the pure contrary component of any voice leading will remain within a cross section of the space (*b*), containing dyads whose pitch classes sum to the same value.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



When restricting our attention to a single cross section, it is therefore convenient to relabel its points so that all twelve-tone equal-tempered intervals are on equal footing. Figure 3.7.2a identifies points according to the interval they represent: 01 for minor second, 02 for major second, and so on, with the symbols "T" and "E" referring to ten and eleven, respectively. These new labels can be understood as referring to chord *types*, rather than particular chords: the label 04, for instance, refers to any major third, rather than the particular pitches C and E.[16] Note that the cross section is redundant: major thirds can be represented both by 04 or 08, just as perfect fifths are represented both by 05 and 07. (Remember that our chords are unordered.) This means that the contrary component of any voice leading can be represented in this abstract space in two different ways (Figure 3.7.3). Removing this redundancy produces a true "set class" space in which each point corresponds to a chord type; however, this involves some unpleasant mathematical complexities, so we will need to learn to live with the redundancy.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

It turns out that line segments in the cross section represent collections of voice leadings, all "individually T-related" to one another (§2.6). This is because the operation of "individual transposition" alters the horizontal component of a voice leading while leaving the vertical component unchanged. Figure 3.7.4 shows that the voice leading (E, B)→(F, B♭) is represented by a purely vertical line segment, with no horizontal component whatsoever. The voice leading (G, D)→(F♯, B) is related to it by individual transposition: we transpose the first chord up by minor third, which corresponds to sliding it rightward by three places, while we transpose the second chord up by one semitone, which amounts to sliding it rightward by one place. This introduces a purely horizontal component into the voice leading while simply shifting the contrary component rightward. Consequently both (G, D)→(F♯, B) and (E, B)→ (F, B♭) are represented in the cross section by the line segment (0, 7)→(0, 5). Thus we see that limiting our attention to the cross section of the Möbius strip is equivalent to focusing on the qualities shared by individually T-related voice leadings. This is just one of many cases where we find unexpected connections between purely musical ideas and relatively straightforward geometrical operations.

Figure 3.7.2 (*a*) It is useful to label the cross sections such that every equal-tempered interval is represented. (*b*) To translate these abstract labels into labels for particular chords in a cross section, subtract a constant from each number in the label so that its elements sum to the same value. Here, for example we recover the sum-0 cross section from the more abstract labels.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

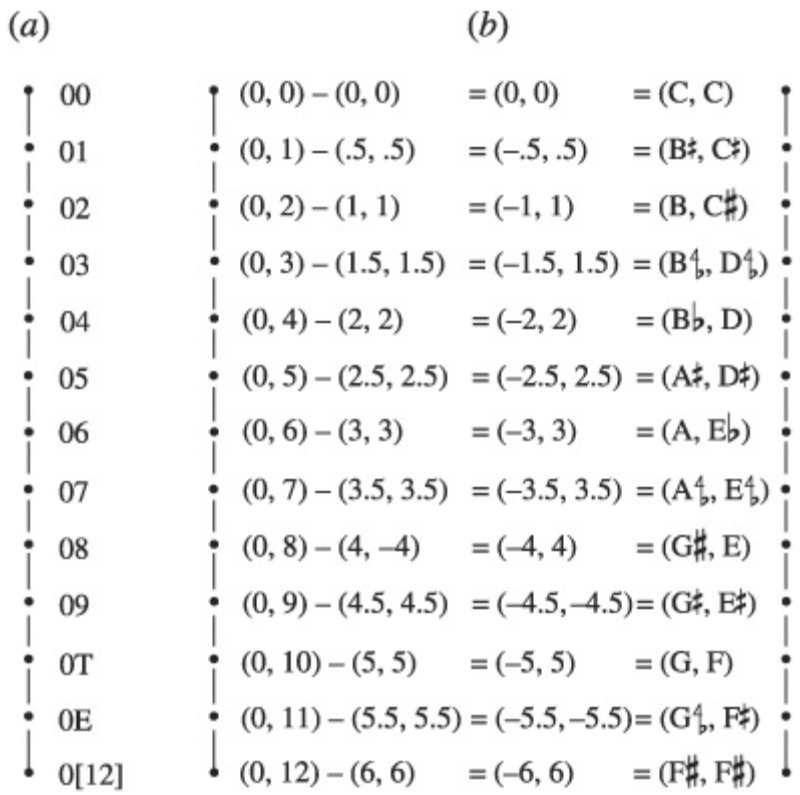


Figure 3.7.3 There are always two equally good ways to move the contrary component of a voice leading into a particular cross section. Either of the arrows in (*b*) can represent the voice leadings in (*c*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

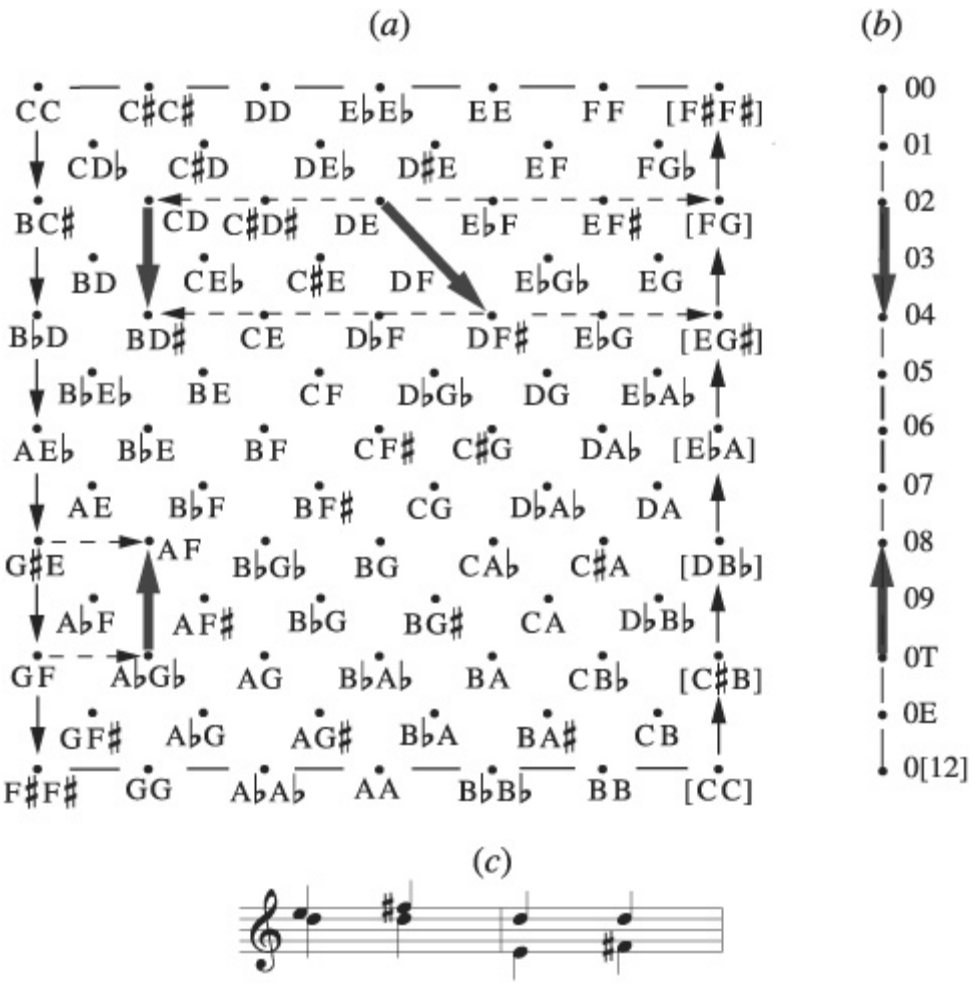
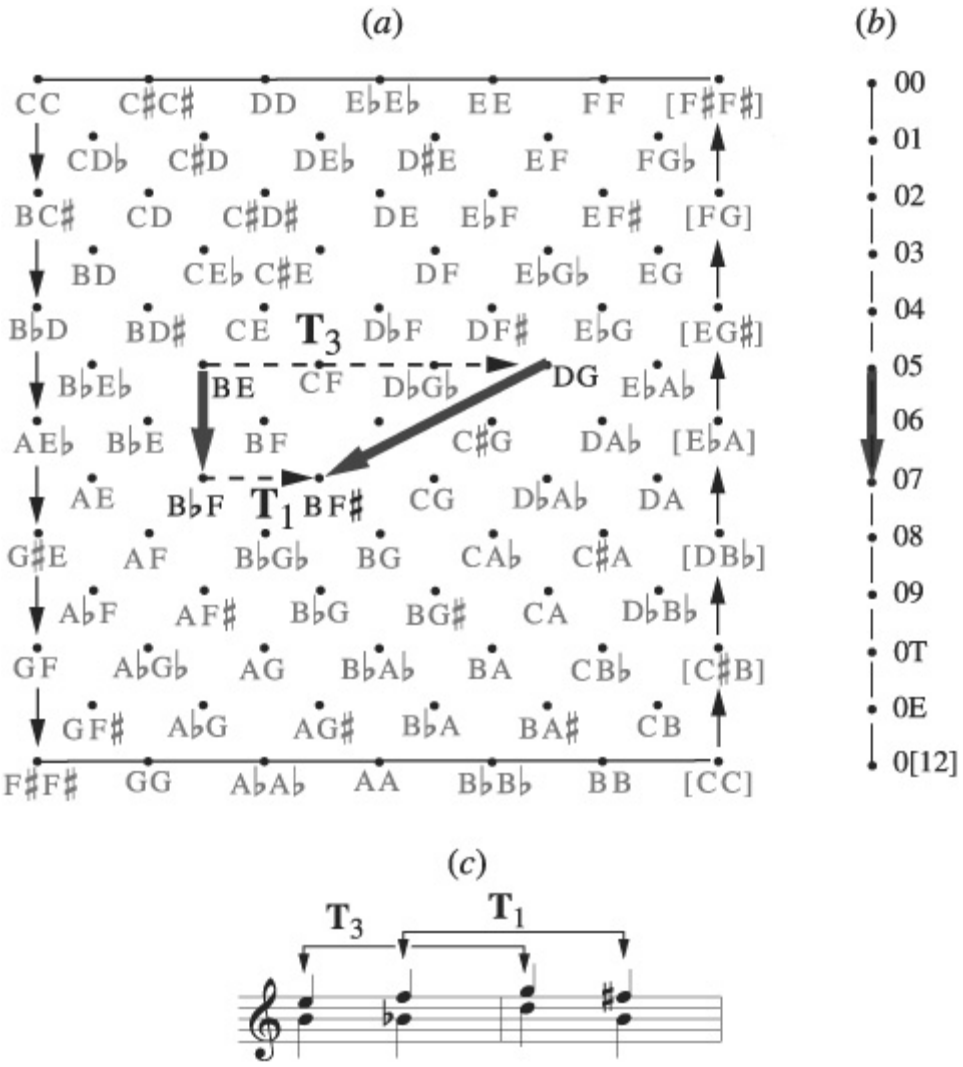


Figure 3.7.4 Individually T-related voice leadings can be moved into the same cross section so that their purely contrary components coincide. Here, the voice leadings in (*c*) share the same pure contrary component (*b*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



3.8 THREE-DIMENSIONAL CHORD SPACE

Having considered two-note chords, the next task is to describe the analogous spaces containing three-note chords, four-note chords, and so on. These higher-dimensional spaces are fundamentally similar to our two-dimensional Möbius strip, and it is usually possible to reason about them by extrapolating from the simpler space we have just explored. Readers might therefore want to make sure they are comfortable with the Möbius strip before proceeding onward.

Suppose you wanted to model music consisting of *three* distinct voices—say trumpet, saxophone, and trombone. Figure 3.8.1 shows that it takes three dimensions to record the possible musical states of the trio, with the *x*-axis representing the trumpet, the *y*-axis the saxophone, and the *z*-axis the trombone. If you were to examine a large volume of this three-note ordered pitch space, you would find that it again has a repeating, periodic structure—being divisible into three-dimensional "tiles" containing one representative of

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

every three-note chord, exactly like Figure 3.1.5. Once again, focusing on a tile of the space is the geometrical analogue to ignoring octave and order information. But since the "tile" has three dimensions, our musical space is now a three-dimensional piece of wallpaper—perfect for covering the walls of your four-dimensional bedroom.[17]

Figure 3.8.2 depicts one of these tiles. The space is a triangular prism. Augmented triads, dividing the octave into three equal parts, lie on the vertical line at its center, and are represented as dark cubes. (For clarity, only twelve-tone equal-tempered triads are shown.) Chords that divide the octave nearly evenly are found near the center, with the equal-tempered major and minor triads depicted as dark and light spheres, respectively.[18] More and more uneven chords are found farther and farther from the center. At the boundary are chords with multiple copies of some note: chords with two copies of some pitch class are found on the sides of the prism, while triple unisons are found on its edges. (Triple unisons are maximally uneven, containing three copies of a single pitch class.) And although the illustration depicts only a few familiar equal-tempered chords, the space itself is again continuous: *any* possible unordered set of pitch classes corresponds to some point in the prism, including microtonal chords such as {0.17, 2.5, 8.999}.

Figure 3.8.1 It takes three dimensions to model three instruments.

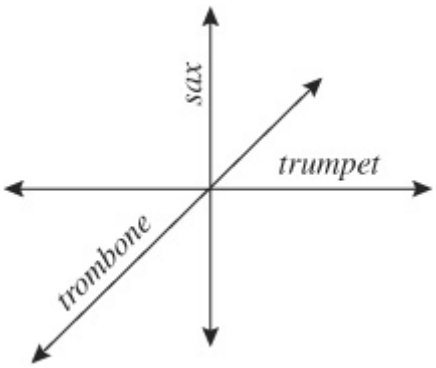
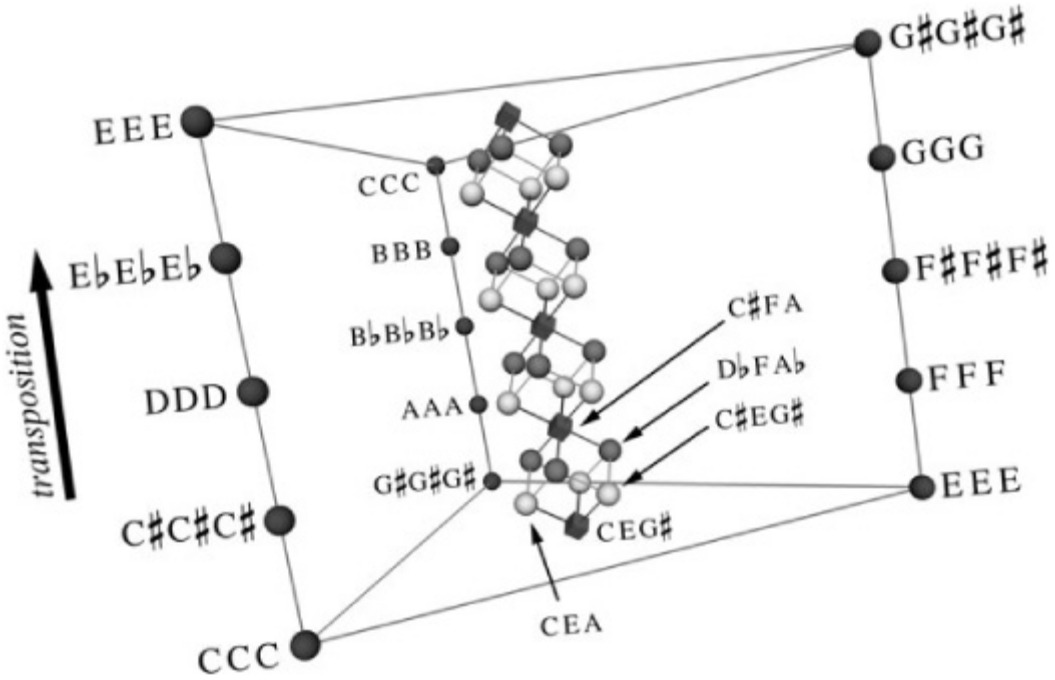


Figure 3.8.2 A single "tile" of three-note chord space is a triangular prism. Minor triads are light spheres, major triads are dark spheres, and augmented triads are cubes. The dark spheres on the edges of the prism are triple unisons. The lines in the center of the space connect chords that can be linked by voice leading in which only a single voice moves, and it moves by only a single semitone.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



As before, voice leadings are "generalized line segments" that stretch from one point to another. Ascending parallel motion in all three voices corresponds to ascending vertical motion on the prism.[19] Here, however, line segments disappear off the top face and reappear on the bottom, rotated by one third of a turn. Thus as the three voices ascend in parallel from {C, C, C} to {E, E, E}, the associated line segment in Figure 3.8.2 ascends along the left edge of the figure. When the line segment reaches {E, E, E}, it reappears on the bottom face of the figure, on the right corner. It then climbs up the front right edge to {G♯, G♯, G♯}. When it reaches the top of this edge, it moves up the rear edge of the figure and begins to climb again until reaching {C, C, C}. This means that we should imagine the triangular faces to be "glued together" with a 120° twist. Since the line containing triple unisons returns eventually to its starting point, it can be considered an (abstract) circle. The same is true for the three line segments containing all the transpositions of any other chord.

Once again, three-dimensional chord space can help clarify musical relationships that would otherwise be very hard to describe. Figure 3.8.3a presents four chromatic sequences from the first movement of Brahms' C minor Piano Quartet, each of which uses descending semitonal voice leading to connect major and minor triads. In the first two sequences, Brahms lowers the major triad's root to form a minor triad (as indicated by "r^{-1}") while in the last two sequences, he lowers the major triad's third ("t^{-1}"). Note that the sequences repeat at three different transpositional levels, as can be seen by comparing successive

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

major or minor triads: the first and last sequences descend by semitone, the second ascends by minor third, and the third ascends by perfect fifth. At first glance, there does not seem to be any clear structure here; the impression is of a kind of intuitive play, as if Brahms had arbitrarily chosen four unrelated sequences from among countless essentially similar possibilities.

When we consider the music geometrically, however, we see that Brahms is actually moving quite systematically along the lattice at the center of three-note chord space (Figure 3.8.3b). Starting from any major triad, there are only two semitonal descents that will produce a minor triad (labeled "*a*" and "*b*" on the lattice); the two available paths correspond to the possibility of lowering the major triad's root or third. From here, one must move the root downward by semitone to form an augmented triad. (Brahms typically skips this triad, though it does appear in the second sequence.) There are then three geometrical paths to choose from (labeled "1," "2," and "3" on the lattice), corresponding to the three major triads that can be reached by lowering a note of the augmented triad, and producing three different root relationships to the starting chord. (The possible relationships are descending semitone, ascending minor third, and ascending perfect fifth, precisely as in Brahms' piece.) Clearly, there are six sequences that can be formed in this way, since there are two possibilities for the initial move and three possibilities for the second (Figure 3.8.3c). What is interesting is that Brahms uses four of these six options, composing a group of sequences that are *contrapuntally* similar (by virtue of using descending stepwise voice leading) while *harmonically* distinct (by virtue of using a variety of different root progressions). The result is a sort of developing or continuous variation, in which the same basic procedure (descending semitonal voice leading among major and minor triads) produces subtly different results. And even though this process of developing variation might at some level be intuitive or improvi-sational, it nevertheless exhibits very clear structure. By showing us the field in which Brahms' intuition necessarily operates, geometry helps us understand this structure, while also leading us to expect that we will find these same sequences in other pieces as well. Chapter 8 will return to this thought, using the very same lattice to interpret chromatic procedures in music as diverse as that of Schubert and Jimi Hendrix.

Once again, it is sometimes useful to consider just the horizontal cross section of our three-note chord space—which, being two-dimensional, is much easier to draw. As before,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

these horizontal cross sections contain chords whose pitch classes sum to the same value (Figure 3.8.4).[20] If a chord appears on the triangular cross section, then so do its transpositions by ascending and descending major third. This is because transposing a three-note chord by four semitones does not change the sum of its pitch classes: $4 + 4 + 4 = 0$ in pitch-class arithmetic. The three transpositions of each chord are arranged symmetrically around the triangle, so that 120° rotation transposes each by major third. Chords in the same triangular cross section can be linked by "pure contrary" voice leadings in which the amount of ascending motion exactly balances the amount of descending motion. For example, Figure 3.8.5 shows that {E, A, B} and {F, G♭, D♭} can be linked by the voice leading (E, A, B)→(F, G♭, D♭), which is represented by a line segment lying entirely within the triangle. Here, one voice ascends by semitone, one voice ascends by two semitones, and one voice descends by three semitones, so that the total amount of ascending motion is $1 + 2 - 3 = 0$.

Figure 3.8.3 (*a*) Descending sequences from the first movement of Brahms' C minor Piano Quartet, Op. 60. (*b*) These result from moving downward along the equal-tempered lattice at the center of chord space. (Major chords are dark spheres, minor chords light spheres.) (*c*) There are six basic sequences that can be formed in this way, depending on whether one lowers the root or the third of the initial major triad, and whether the sequence descends by semitone (D1), ascends by seven semitones (A7), or ascends by three semitones (A3).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

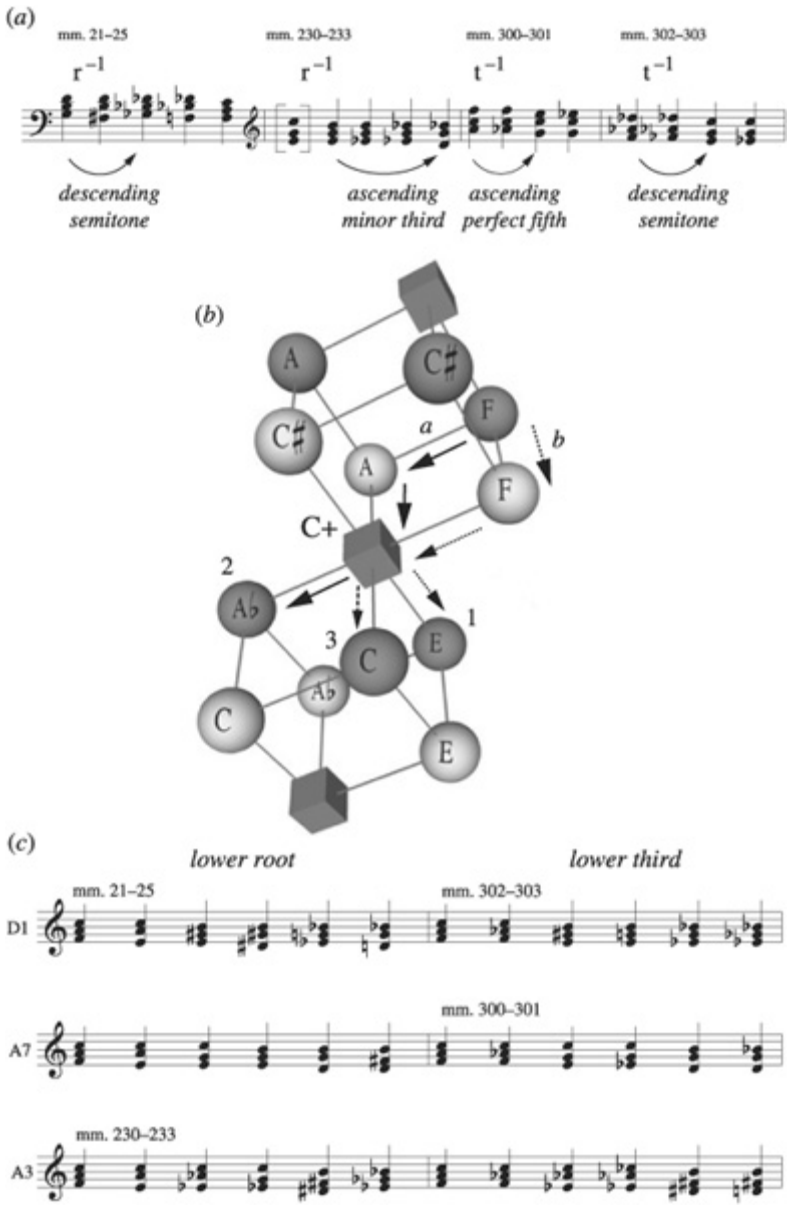


Figure 3.8.4 (*a*) A horizontal slice of three-note chord space is a triangle containing chords summing to the same value; here the chords sum to 0. (Only equal-tempered chords are shown.) (*b*) These triangular cross sections are analogous to the vertical "slices" of two-note chord space, which also contain chords summing to the same value.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

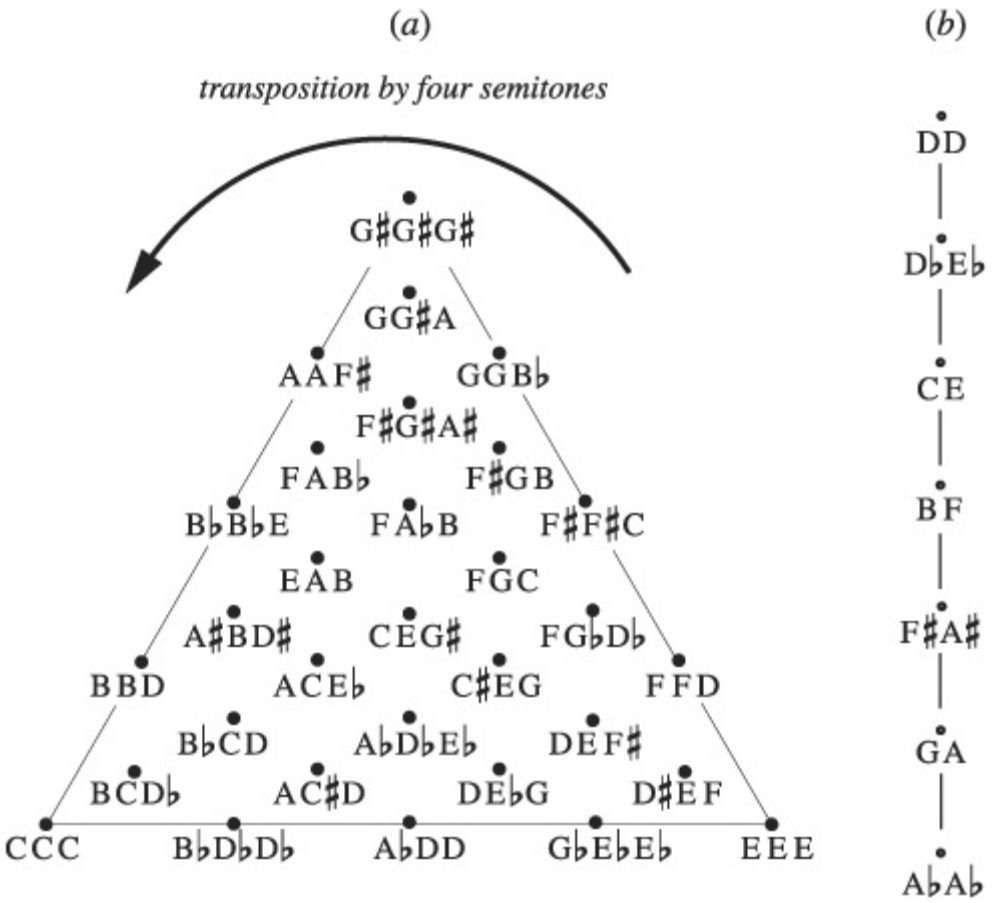
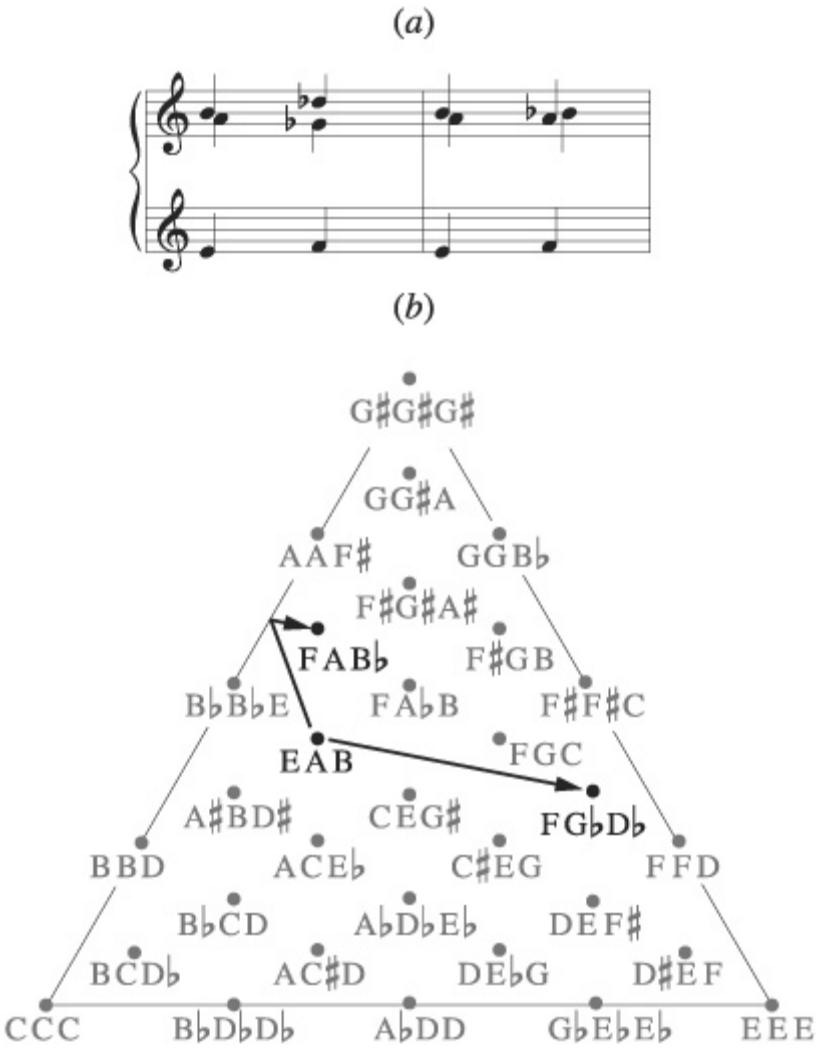


Figure 3.8.5 (*a*) Two voice leadings in which the total amount of ascending motion perfectly balances the total amount of descending motion. (*b*) These voice leadings are contained in a cross section of three-note chord space. The second voice leading, which has voice crossings, bounces off the triangle's mirror boundary; the first does not.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

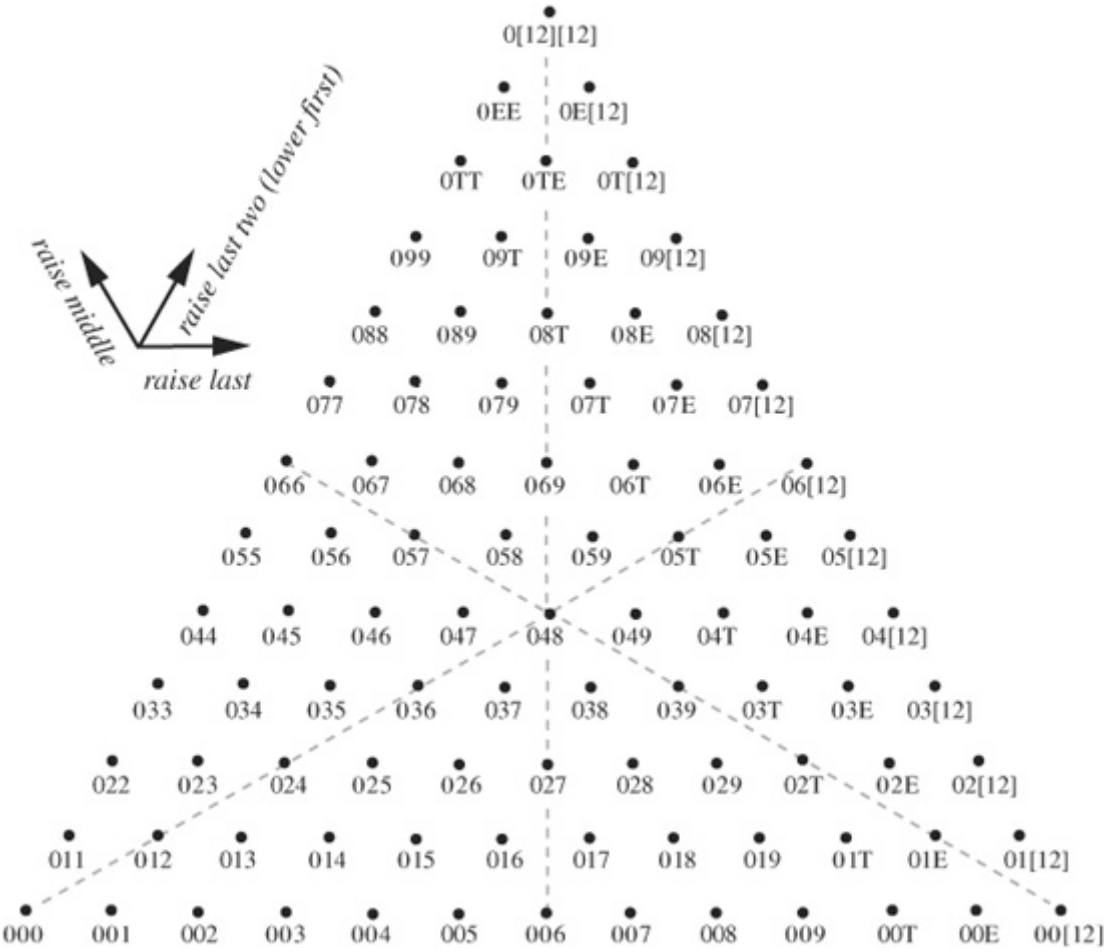


Since Figure 3.8.4 shows only equal-tempered chords summing to zero, certain chord types do not appear. (In particular, there are no major triads.) Again, though, every *chord type* appears in every cross section: if we were to transpose {C, E, G} up by one third of a semitone, we would obtain a major triad whose pitch classes sum to zero.[21] (This chord would appear on Figure 3.8.4a precisely in the same place that the C major triad appears on its cross section.) As in §3.7, we can therefore construct an abstract representation of the cross section in which all equal-tempered chord types appear (Figure 3.8.6). The arrows to the left of the triangle show how to interpret motion in the space: horizontal motion raises or lowers the last note in the chord; motion along the up-and-left diagonal raises or lowers the middle note in the chord; and motion along the other diagonal raises and lowers the last *two* notes in the chord. (Of course, raising two notes is equivalent to lowering the other when we factor out pure parallel motion, which is what we are doing when we consider only the cross section.[22]) Figure 3.8.7 explains how to plot the pure contrary component of a voice leading in this space.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.8.6 An abstract representation of the cross sections of three-dimensional chord space. In this abstract representation, the labels do not sum to the same value. (Compare Figure 3.7.2.)



Finally, observe that the structure of the space again demonstrates the relation between evenness and efficient voice leading. Suppose you want to find an efficient (three-voice) voice leading between transpositionally related three-note chords. As in §3.7, we decompose our voice leading into pure parallel and pure contrary components. Since the original voice leading is small, then both the pure parallel and pure contrary components will also be small. It is clear from Figure 3.8.8 that there are just two basic possibilities: either the chord is near the center and the pure contrary component links it to its four- or eight-semitone transposition, or it is near the edge and the line segment links it to itself.[23] The situation is precisely analogous to the two-dimensional case, except that here the contrary component connects *major-third*-related (rather than tritone-related) chords. This fact has important musical consequences, as we will shortly see.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.8.7 To plot the purely contrary component of a voice leading in the triangular cross section, reorder it (uniformly) so that the first chord is in ascending order spanning less than an octave, then transpose the chords (individually) so that they both start on C, with the first voice moving by zero semitones. One can then use the arrows to the left of the figure to determine the path representing the voice leading. For example, to represent (C, E, G)→(C, F, A) we need to raise the middle voice by one semitone, represented by one-step diagonal NW motion; we then raise the third voice by two semitones, which requires moving two units to the right. Combining these vectors gives us the path representing the voice leading's pure contrary component (*center of figure*). Contact with a boundary exchanges the position of two voices: for example, (C, F, F♯)→(C, G, F♯) can be represented by two diagonal steps to the NW from 056 (*left side of figure*). However, the first of these steps brings us to the boundary, which exchanges the position of the second and third voices. Thus, instead of moving the middle voice up by semitone, we now have to move the *third* voice up by semitone; this is represented by a one-step rightward motion. Geometrically, it looks as if our arrow has reflected off the mirror boundary. Note that we cannot represent this voice leading with a direct line from 056 to 067, since the voice crossing requires that the line reflect off the mirror boundary: thus where the two arrows 047→057 and 057→059 can be combined into a single arrow, the two arrows 056→066 and 066→067 cannot.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

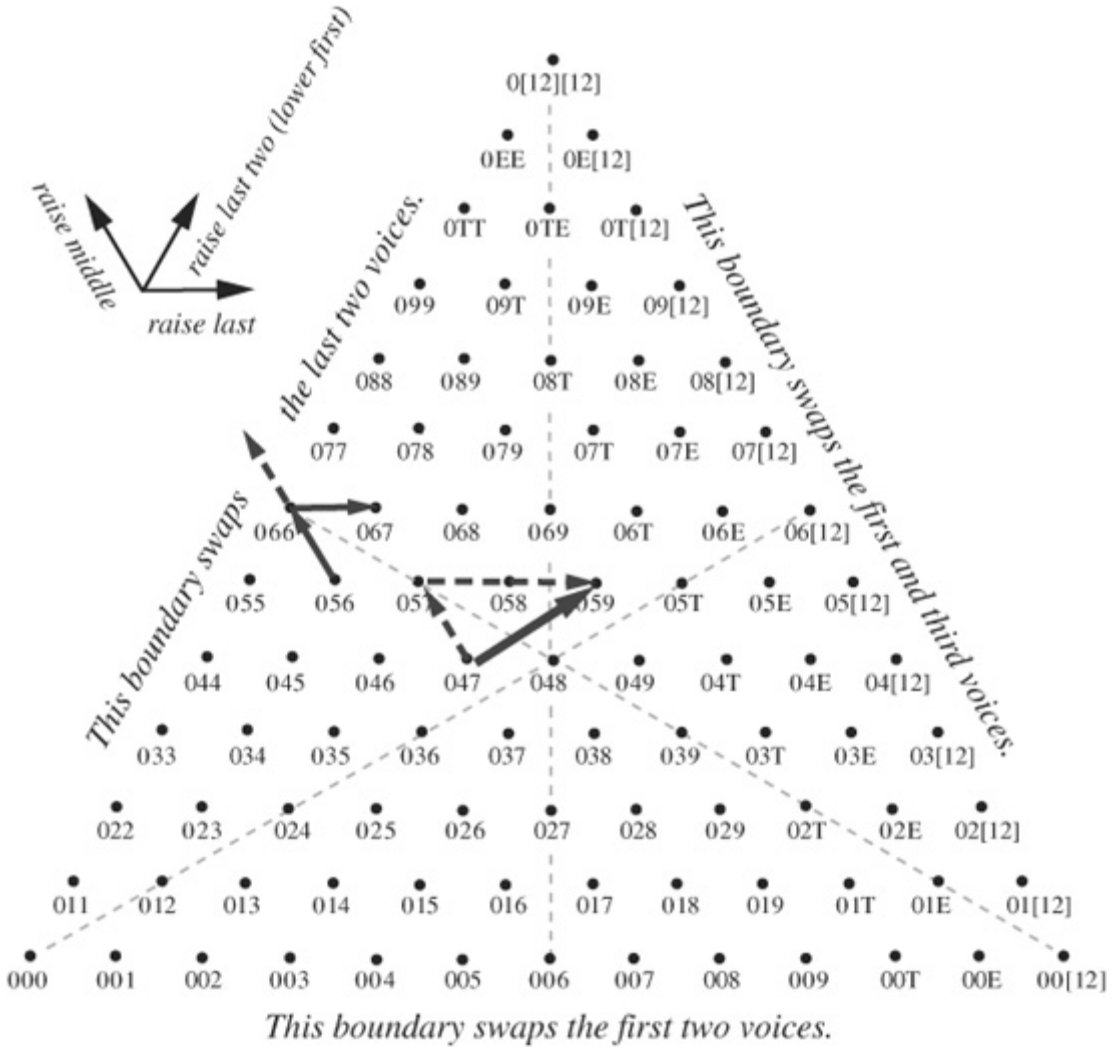
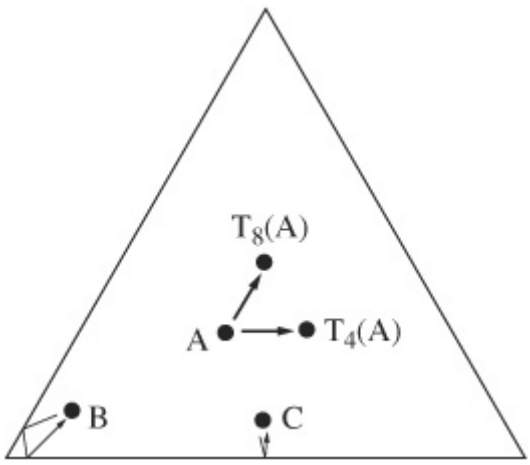


Figure 3.8.8 Efficient pure contrary voice leading between transpositionally related three-note chords. Chord A divides the octave nearly evenly; it and its transpositions are near the center of the space. Chords B and C have notes that are clustered together, and are near the boundary.



3.9 HIGHER DIMENSIONAL CHORD SPACES

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Attempting to visualize higher dimensional spaces is an exercise in diminishing returns: while one might barely manage four dimensions, it becomes increasingly difficult to picture five, six, or more. Nevertheless, it will be useful in what follows to have a rough understanding of the geometry of the higher dimensional chord spaces. Accordingly, I will ask the reader's indulgence as I try to describe the spaces representing four-note, five-note, and even larger chords.

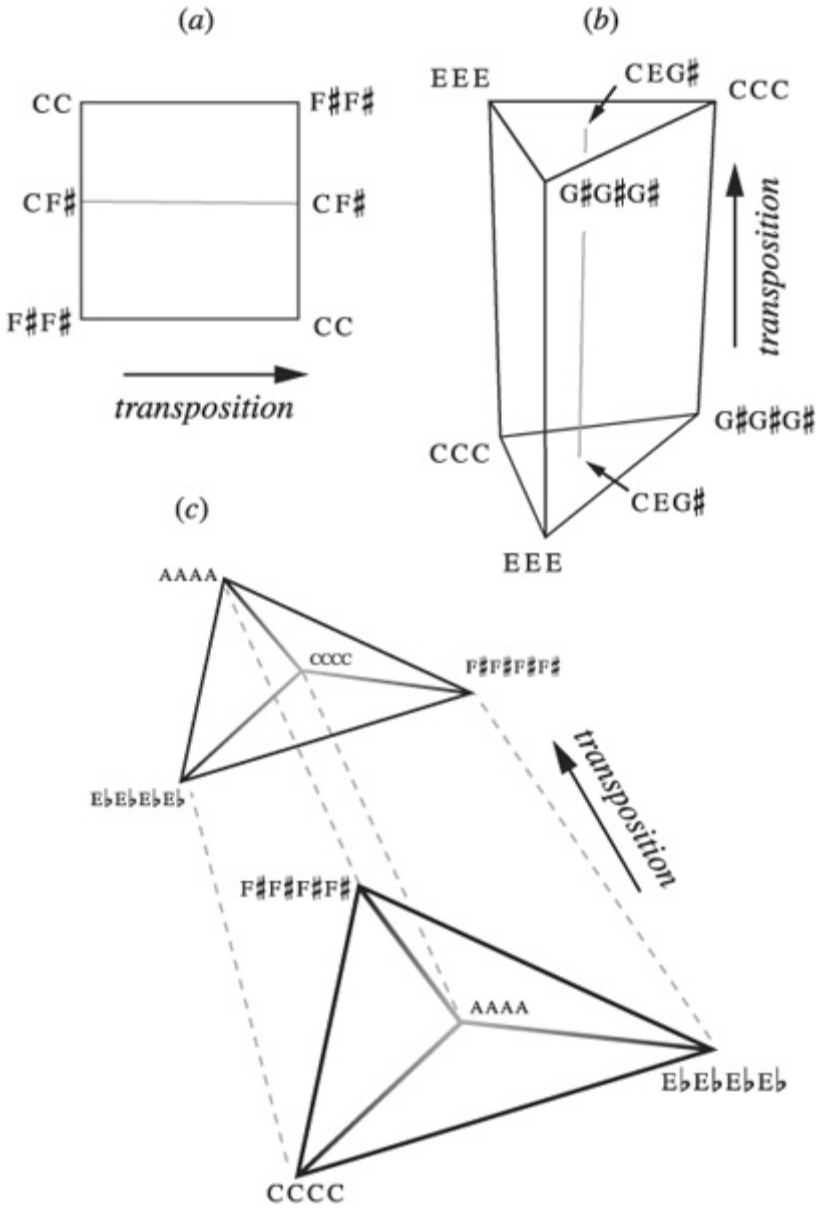
We have seen that it takes a separate dimension to model each individual musical voice. Thus, it takes four dimensions to represent the state of a four-instrument ensemble, five dimensions to represent the state of a five-instrument ensemble, and so on. When we abstract away from order and octave, we find that each of these spaces is periodic, a higher dimensional piece of wallpaper consisting of many equivalent copies of the same pattern or tile.

Figure 3.9.1 displays the tiles in the two-, three-, and four-dimensional cases. The two-dimensional tile is our familiar Möbius strip, a rectangle whose left edge is glued to its right with a half twist. For the sake of generality, however, it is useful to forget the term "rectangle," and to imagine the space as a *two-dimensional prism*—the shape that results when we drag the (one-dimensional) left edge horizontally rightward to form a two-dimensional figure. The resulting structure has two "faces" (here, the left and right edges), which are one-dimensional line segments. Chords on the same vertical line sum to the same value; hence every chord is on the same vertical line segment as its tritone transposition. As we drag the left face horizontally rightward, we transpose each chord upward. We can obtain new chords in this way until we have transposed by six semitones, at which point the left and right faces contain the same chords. These two faces need to be glued together with a "twist" that gives the space a circular structure: because of this twist, we can eventually return to our starting point by moving perpendicular to the face (§3.3).

Figure 3.9.1 Chord space in two, three, and four dimensions.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



The three-dimensional tile is also a prism, the shape that results from dragging the bottom face, a triangle, vertically upward. The resulting structure again has two faces, which are now two-dimensional triangles rather than one-dimensional line segments. Each triangular face contains three-note chords summing to the same value, so that every chord is on the same face as its major-third transposition. Dragging the bottom triangle upward transposes each chord upward; we drag it until the chords have been transposed by four semitones, at which point the top face contains the same chords as the bottom. These two faces must then be glued together with a "twist" that matches the appropriate chords—here, a 120° rotation. This "gluing" produces a triangular doughnut and introduces a circular structure in to the space: once again, if we move perpendicular to the triangle, we eventually return to where we started.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.9.1c illustrates the four-dimensional shape that tiles the space of four-note chords. Readers will not be surprised to find that it, too, is a prism. Here the faces are *tetrahedra*, the three-dimensional analogues of triangles. Each tetrahedral cross section contains pitch classes that sum to the same value. Consequently, if a chord is on the cross section, then so is its minor third transposition.[24] The four vertices of the tetrahedron contain quadruple unisons related by *minor* third—here, {C, C, C, C}, {E♭, E♭, E♭, E♭}, {F♯, F♯, F♯, F♯} and {A, A, A, A}—while the chord at the center divides the octave perfectly evenly and contains the pitch classes {C, E♭, F♯, A}. To form the prism, drag the tetrahedron into the *fourth* dimension, represented by dotted lines on Figure 3.9.1c, thereby transposing every chord upward. We can drag it until its chords have been transposed by minor third, at which point the transposed tetrahedron contains the same chords as the initial one. These again need to be attached by a "twist" that connects the appropriate chords and transforms the prism into a kind of four-dimensional doughnut.

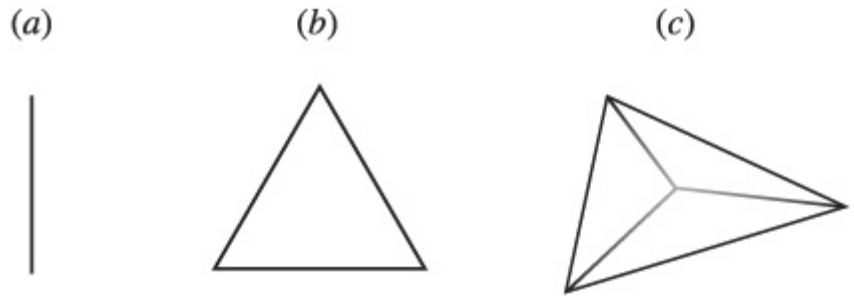
The general pattern should now be clear. In each dimension, chord space is an n-dimensional prism, formed by dragging a "generalized triangle" through an additional dimension (Figure 3.9.2).[25] The generalized triangle is the *face* of the prism, containing chords summing to the same value. Consequently, each n-element chord is on the same face as its transposition by $12/n$ semitones. (This is because transposing n notes by $12/n$ adds $12n/n = 12$ to their sum, and in pitch-class arithmetic we discard multiples of 12.) Dragging the face into the additional dimension transposes each chord upward; it can be dragged until the chords have been transposed by $12/n$ semitones, at which point it contains the same chords as those on the initial face. The two duplicate faces are glued together with a "twist" that matches the appropriate chords, introducing a circular structure and creating the higher dimensional analogue of a twisted doughnut.

Nonmathematical readers should not wrack their brains trying to picture these spaces in too much detail. Instead, focus on the important structural features they all share:

Figure 3.9.2 A simplex (or "generalized triangle") is bounded by n points, all connected by line segments. One-, two-, and three-dimensional simplexes are line segments, triangles, and tetrahedra, respectively.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



1. In each space, points correspond to chords and line segments correspond to voice leadings, with the length of the line segment representing the size of the voice leading. Consequently, nearby chords can be linked by efficient voice leading, while distant chords cannot.
2. A chord's "evenness" determines its distance from the center. Chords that divide the octave precisely evenly are found at the very center of the space, while nearly even chords are nearby. Chords whose notes are clustered together are found near the boundary of the space, as far from the center as possible.
3. Chord space is made up of "layers," or cross sections, each containing chords summing to the same value. There are n instances of every n-note chord type in each layer. These can be obtained by repeatedly transposing by $12/n$ semitones. The transpositions are arranged symmetrically around the center of the cross section, with their distance from the center determined by the chord's evenness. The geometrical *symmetries* of the cross section relate the different forms of each chord type.
4. Each cross section of the space is a generalized triangle. The boundaries of the generalized triangle contain chords like {C, C, E, G}, which have multiple copies of some pitch class.[26]
5. Nearly even chords can be linked to their transpositions by very efficient voice leading. An n-note chord is particularly close to its transposition by $12/n$ semitones, since they are near the center of the same horizontal cross section.
6. Voice leading that remains within the layers represents pure contrary motion in which the amount of ascending motion perfectly balances the amount of descending motion. Moving perpendicular to these layers corresponds to pure parallel motion,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

in which every note moves in the same direction by the same amount. Any voice leading can be decomposed into parallel and contrary components, as in high school vector analysis (§3.7).

7. The top and bottom "layers" of the prism are glued together in a twisted fashion, endowing the space with a circular, doughnut-like structure. For this reason, a voice leading in which every voice ascends by octave will be represented by a line segment that passes through each cross section *n* different times, moving through each of its transpositions before returning to its starting point. The trajectory it follows will be a circle in the abstract sense.
8. The remaining boundaries of the figure—the ones that are not glued together—act like mirrors. Voice leadings bounce off these mirrors, like balls off of the bumpers of a pool table. "Clustered" chords near the edges of the space can be linked to their untransposed forms by efficient voice leading, bouncing off nearby mirrors to return to their starting point.

Appendix B provides a more detailed mathematical description of the spaces, for readers who are so inclined.
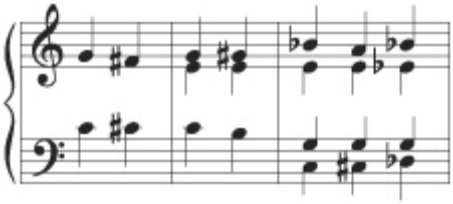
3.10 TRIADS ARE FROM MARS; SEVENTH CHORDS ARE FROM VENUS

Musically, the most important consequence of all this geometry is the following: since a nearly even *n*-note chord occupies the same cross section of the space as its transposition by 12/*n* semitones, such chords can be linked by particularly efficient voice leading.[27] Thus, nearly even two-note chords are very close to their six-semitone transpositions, nearly even three-note chords are very close their four- and eight- semitone transpositions, nearly even four-note chords are very close to their three-, six-, and nine-semitone transpositions, and so on. This is illustrated in Figures 3.10.1 and 3.10.2.

Figure 3.10.1 Two-note chords are particularly close to their tritone transpositions; three-note chords are particularly close to their major-third transpositions; and four-note chords are particularly close to their minor-third and tritone transpositions.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Suppose, then, that a composer is interested in exploiting efficient voice leadings between two major triads or two dominant seventh chords. Theoretically, we should expect an abundance of major-third relations between the triads and minor-third relations between the dominant sevenths. And when we look at actual music, we see that this is indeed the case. Direct juxtapositions of minor-third-related dominant seventh chords already occur in baroque and classical music, while direct juxtapositions of major-third-related seventh chords are very rare; similarly, major-third-related triads sometimes occur across phrase boundaries, as when V/vi moves directly to I, while there is no analogous convention associating minor-third-related triads.[28] Figure 3.10.3 indicates that this asymmetry also characterizes the more chromatic music of the nineteenth century. In Schubert, major-third-related triads occur about 1.5 times more often than minor-third-related triads, while minor-third-related dominant seventh chords occur about 14 times more often than major-third-related dominant seventh chords.[29] (In Chopin, major-third-related major triads occur about 1.6 times more often than minor-third-related triads, and minor-third-related dominant-seventh chords occur about 2.2 times more often than major-third-related dominant sevenths.) This suggests that composers' harmonic choices are indeed guided by the voice-leading relationships we have been exploring. To be sure, composers are not guided *exclusively* by voice-leading considerations, and individual artistic preference no doubt plays a significant role. Nevertheless, there is clear evidence that Western musicians are sensitive to the relationships that are modeled by our geometrical spaces.

Figure 3.10.2 A geometrical representation of the closeness between (*a*) tritone-related perfect fifths and (*b*) major-third-related triads. These figures show just a portion of the relevant geometrical spaces, and connect chords by lines if they can be linked by singlesemitone voice leading.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

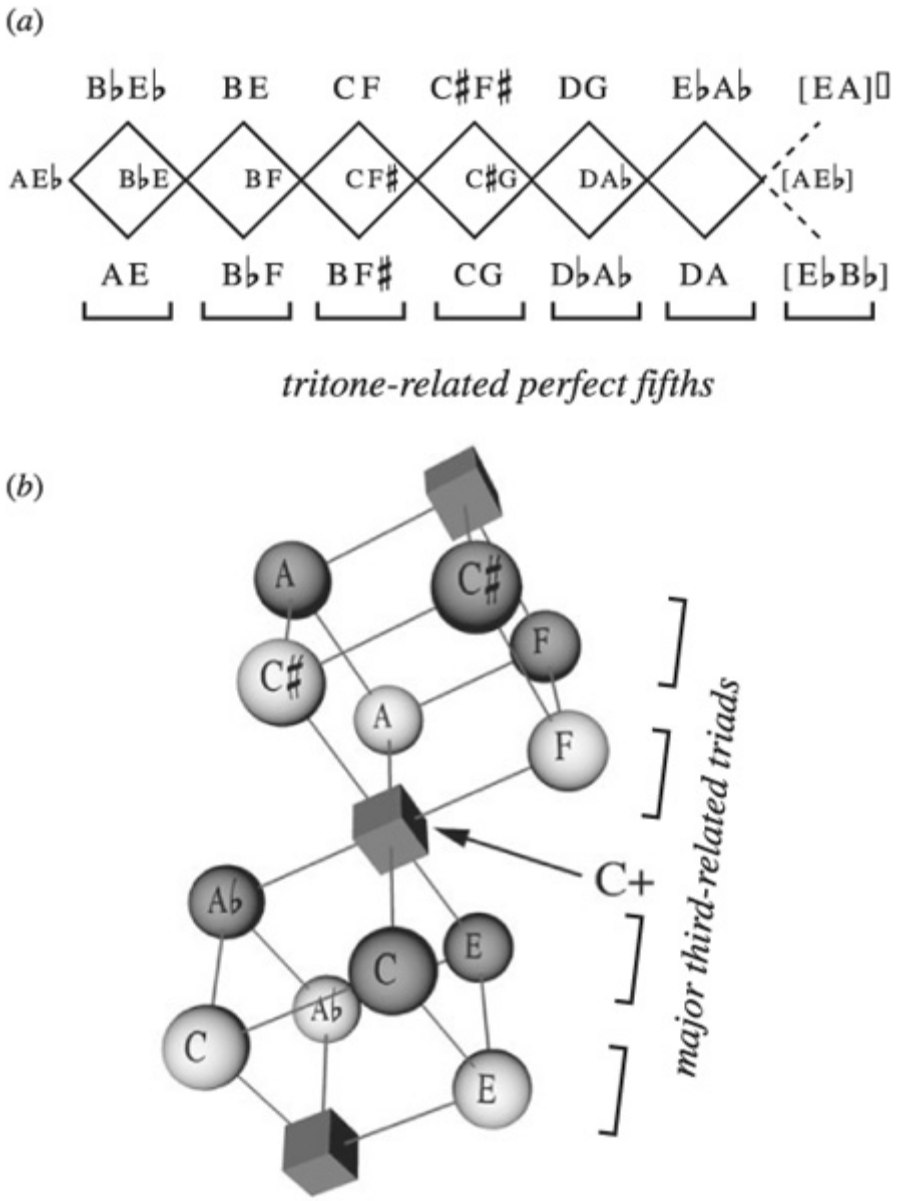


Figure 3.10.4 turns to a very different sort of music, analyzing the basic three-chord progression of Nirvana's "Heart-Shaped Box." The first two chords, A and F, are pure triads related by major third.[30] The final D^7 chord has four notes, retaining the fifth of the F major triad as the seventh of D. (Note that every chord contains the note A, which appears as root, third, and fifth in turn.) We can understand this last voice leading as an incomplete manifestation of the efficient voice leading between F^7 and D^7 chords. This is not to say that Kurt Cobain necessarily conceived it as such; rather, it is to say that the voice leading between the last two chords is contained within a very familiar seventh-chord schema and exploits the same musical facts that make the more familiar schema possible. From our point of view, the interesting fact is that Cobain's switch from triads to seventh chords goes hand in hand with the switch from major-third motion to minor-third motion. There are countless other examples of this phenomenon: for instance, the Beatles'

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

"Glass Onion" juxtaposes major-third-related triads in the verse and minor-third-related dominant sevenths in the chorus. Figure 3.10.5 presents a similar progression from Schumann, which alternates between the "major-third system" of closely related triads (represented geometrically by the cubic lattice of Figure 3.10.2b) and the "minor-third system" of closely related seventh chords (to be discussed shortly).[31] As we will see, a large amount of chromatic music can be understood in this way (Figure 3.10.6).

Figure 3.10.3 Nondiatonic chord progressions in a large number of pieces by Schubert and Chopin. The leftmost column indicates root motion: thus "+1" indicates a progression where the root moves upward by semitone, while "-3" refers to one where the root moves downward by three semitones. The next two pairs of columns distinguish progressions between two major triads and two dominant seventh chords. The boldface numbers show that major triads are more likely to be connected by major third than by minor third, while the reverse is true for dominant sevenths.

Root Motion	Schubert		Chopin	
	Maj. triads	Dom. 7th	Maj. triads	Dom. 7th
+1	22%	5%	32%	1%
–1	10%	1%	22%	4%
+2	-	9%	-	4%
–2	-	10%	-	6%
+3	10%	**17%**	8%	**5%**
–3	16%	**25%**	7%	**15%**
+4	**18%**	0%	**13%**	1%
–4	**21%**	3%	**12%**	8%
+5	-	26%	-	52%
–5	-	2%	-	5%
6	3%	1%	6%	1%
#	**560**	**163**	**630**	**239**

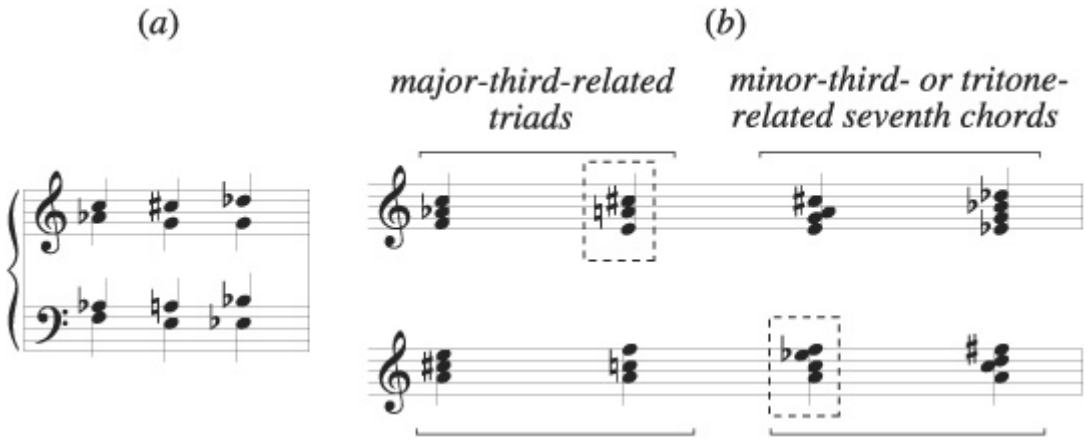
Figure 3.10.4 Nirvana's "Heart-Shaped Box" switches from major-third root motion to minor-third root motion when it switches from triads to a seventh chord. The upper three voices in the last voice leading of (*a*) are contained within the common four-voice voice leading in (*b*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
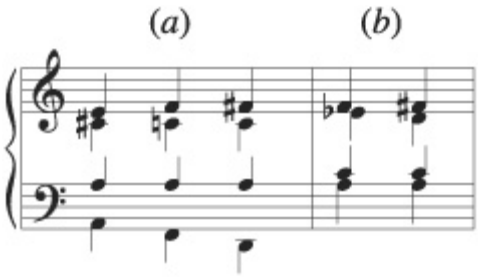


Figure 3.10.5 (*a*) A voice leading from the "Chopin" movement in Schumann's *Carnaval*. Here, an F minor triad moves by major third to an A dominant seventh. (The bass, tenor, and soprano give us a pair of major-third-related triads, with the alto moving semitonally from the doubled third to the seventh.) The next progression moves the A^7 chord to an E♭^{7}. Once again the shift from triads to seventh chords accompanies a shift in root motion. (*b*) A comparison of Schumann's "Chopin" and Nirvana's "Heart-Shaped Box." In Schumann's progression, the switch to seventh chords occurs at the second chord in the passage: doubling the third of the F minor triad, he uses the semitonal motion A♭→G to create an A^7 chord. In the Nirvana progression, the switch does not occur until the final chord, where the C of F major is held over to become the seventh of D^7.



Of course, it is always possible to add a small purely parallel component without drastically increasing the size of a voice leading. So, for example, starting with the pure contrary voice leading (C, G)→(C♯, F♯), one can transpose the second chord up by semitone to produce (C, G)→(D, G), a relatively efficient voice leading between fifth-related perfect fifths. Applying the same procedure to (E, G♯, B)→(E♭, A♭, C) yields (E, G♯, B)→(E, A, C♯) (Figure 3.10.7). Figure 3.10.8 identifies, for chords with two to five notes, the transpositions that can be obtained by combining pure contrary motion with a small parallel component. The perfect fourth (or equivalently, the fifth) is the only interval

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

appearing in every row of the table, which means that nearly even chords of any size can be linked to their perfect-fourth transpositions by relatively efficient voice leading. (Here again we have an example of a familiar musical object being optimal for multiple reasons: the perfect fourth is both extremely consonant while also being a uniquely useful interval of transposition, allowing for efficient voice leading in a broad range of circumstances.[32]) This fact is of obvious relevance to tonal music, which exploits fourth-progressions between triads, seventh chords, ninth chords, and diatonic scales.

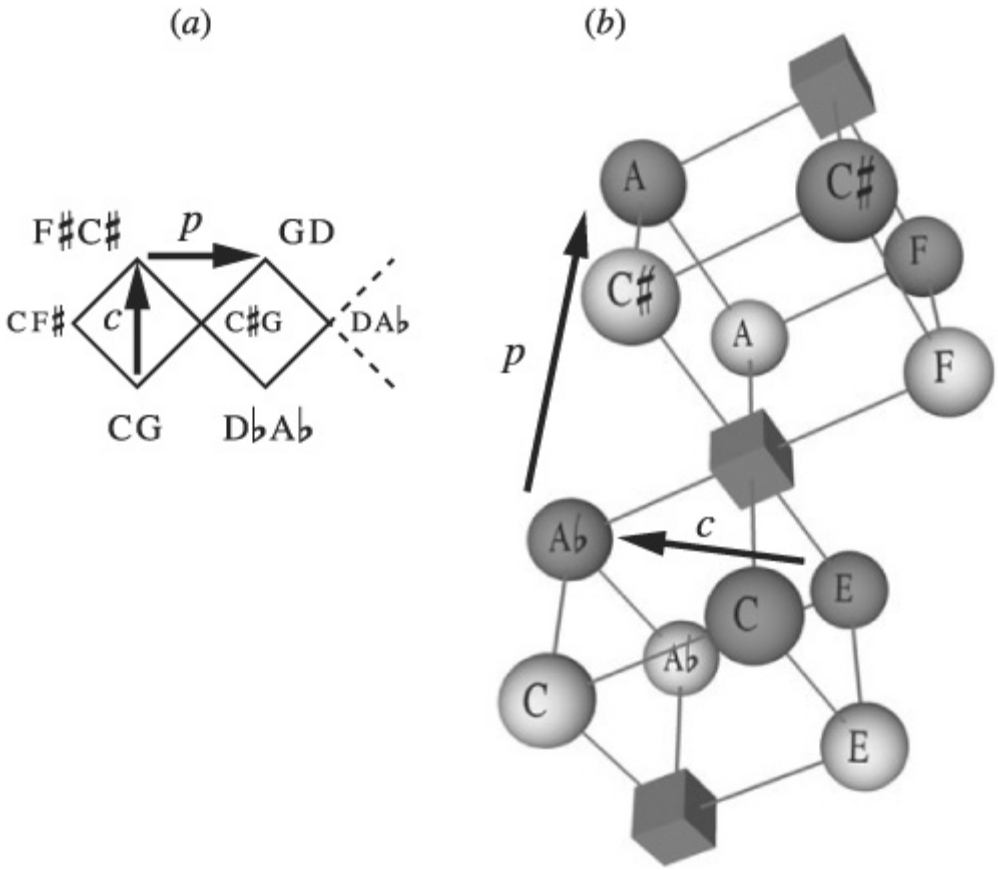
Figure 3.10.6 Triads and seventh chords in mm. 18–20 of the first movement of Schubert's B♭ major Piano Sonata (*a*), mm. 14–15 of the third-movement trio (*b*), the "Tarnhelm" motive from mm. 37–39 of Scene III of Wagner's *Das Rheingold* (*c*), mm. 11–12 of Mozart's C minor Fantasy, K. 475 (*d*), the eighteenth-century "omnibus" progression (*e*), and mm. 44–45 of Chopin's *Nocturne*, Op. 27 No. 2 (*f*).
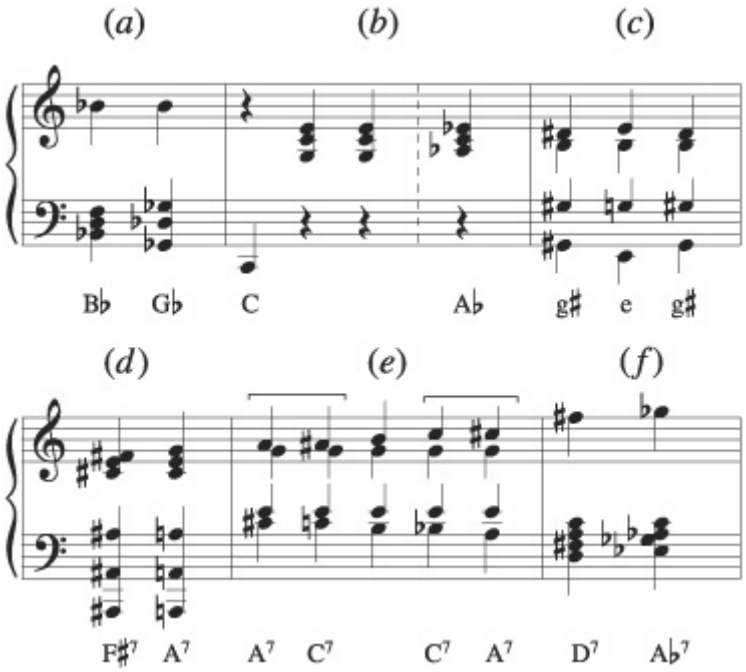


Figure 3.10.7 Adding a small parallel component (*p*) to a small purely contrary voice leading (*c*) still produces an efficient voice leading.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



One feature of Figure 3.10.8 deserves further comment. A five-note chord can be linked by pure contrary motion to its 2.4, 4.8, 7.2, and 9.6 semitone transpositions. Since none of these transpositions lie in twelve-tone equal temperament, any voice leading between familiar transpositionally related five-note chords must have at least some parallel component. If the purely contrary voice leading involves transposition by 2.4 semitones, then the purely parallel component must be at least 0.4 semitones large. However, if the purely contrary voice leading involves transposition by 4.8 semitones, then it is necessary to add only 0.2 semitones of pure parallel motion. It follows that from a twelve-tone equal-tempered perspective, very even five-note chords will seem to be closest to their five-semitone transpositions. (And indeed, the pentatonic scale can be linked to its five-semitone transposition by single-semitone voice leading.) In continuous pitch-class space, we would see that the pentatonic scale is *also* quite close to its 2.4-semitone transposition, but this symmetry disappears when we view music through a twelve-tone equal-tempered grid. Here again we see that it is sometimes useful to adopt the continuous perspective, even when we are primarily interested in equal-tempered music.

Figure 3.10.8 The transpositions that permit pure contrary motion for chords of various cardinalities. Nearby equal-tempered transpositions are also shown.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Cardinality	Pure Contrary Motion	Nearby Equal-Tempered Transpositions
2-note chords	6 semitones	perfect fourth
3-note chords	±4 semitones	minor third, perfect fourth
4-note chords	±3, 6 semitones	major second, major third, perfect fourth
5-note chords	±2.4, ±4.8 semitones	perfect fourth

In later chapters, we will find that these sorts of voice-leading relationships underwrite three major compositional practices. First, as discussed above, composers often use maximally efficient voice leading to move from one chord to another, creating an abundance of major-third relations among triads and an abundance of minor-third relationships among seventh chords. Second, composers often use maximally efficient voice leading to *replace* one chord with another, as when a jazz musician replaces a dominant seventh chord with its tritone transposition (e.g. B^7 replacing F^7). Here, rather than connecting two actually existing chords, efficient voice leading is used to substitute a surprising chord for one that we would otherwise expect; the jazz "tritone substitution" is the most familiar example of this practice, though Chapter 7 will uncover a similar procedure ("third substitution") in classical harmony. Finally, composers often combine efficient voice leading with small amounts of *descending* parallel motion, producing harmonic sequences where all voices slowly sink downward. In the triadic case, this leads to sequences such as those in Figure 3.8.3, with roots moving by descending minor second, ascending minor third, or ascending fifth.[33] In Chapter 8 we will study their seventh-chord analogues in the music of Chopin and Wagner, and in Chapter 9 we will observe Shostakovich and Reich applying the same procedure to seven-note scales. The value of geometry lies in the way it provides a unified perspective on all of these seemingly disparate musical activities, helping us to see that they involve different ways of exploiting the fundamental geometry of musical space.

3.11 VOICE-LEADING LATTICES

Obviously, the problem with higher dimensional chord spaces is that we can't visualize them; since each individual musical voice requires its own dimension, we quickly lose the ability to present musical information in a clear and intuitive manner. One solution is to

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

depict only the portion of the space directly relevant to our musical needs. In this book we will typically be interested in voice-leading relationships among nearly even chords belonging to familiar scales. Happily—and somewhat surprisingly—it turns out that there are simple three-dimensional lattices representing all of the relevant musical relationships.[34]

There are two types of graph most directly relevant to our needs. The first depicts chords whose size (number of notes) exactly divides the size of the scale containing them. These can be used to represent nearly even two-, three-, four-, or six-note chords in twelve-note chromatic space. (This is because two, three, four, and six all divide twelve.) These lattices are circular "necklaces" of cubes of various dimensions, each linked to its neighbors by shared vertices. The second kind of structure arises when the number of notes in the chord is *relatively prime* to the number of notes in the scale—which is to say that no positive integer other than one divides both numbers. Here the relevant voice-leading facts are represented by a necklace of cubes sharing a common *face* with their neighbors. This type of lattice can be used to represent five- or seven-note chords in twelve-tone chromatic space, and chords of any size in any seven-note scale.[35]

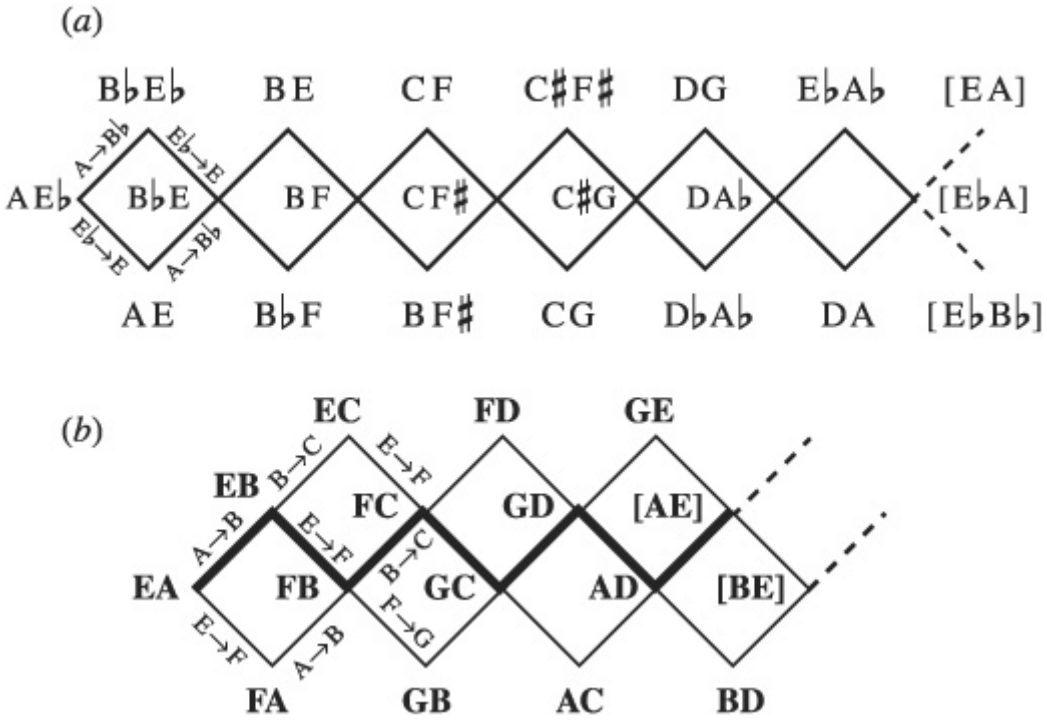
Figure 3.11.1 presents a pair of examples. The first represents single-semitone voice leadings among perfect fifths and tritones. (Consult Figure 3.3.1 to see how this figure sits in continuous space.) Each square contains two perfect fifths and two tritones, with tritones being common to adjacent squares. (The square on the right edge shares a vertex with that on the left, forming a "circle of squares.") As shown in the figure, there are two possible paths from one shared vertex to the next, representing the two ways of sequentially raising or lowering the tritone's notes.[36] Contrast this with Figure 3.11.1b, which depicts single-step voice leadings among thirds and fourths in the C diatonic scale. (In making this graph, I have represented diatonic scale steps, such as C-D or E-F, as having equal size; Chapter 4 explores this idea in detail.) Where the squares in the earlier figure are linked by a common vertex, those in Figure 3.11.1b share a common *edge*. Here, each square contains three diatonic fourths and one diatonic third: fourths zigzag through the center of the figure like the stripe on Charlie Brown's sweater, while thirds lie on the external vertices. Once again, the voice leadings within each square represent different

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

ways of ordering the same pair of single-step motions: for example, to move from {E, A} to {F, B}, we could either raise E to F and then A to B; or we could raise A to B and then E to F. The first path goes by way of {F, A} while the second moves through {E, B}.

Figure 3.11.1 The two fundamental kinds of discrete lattices. The first is a circle of squares linked by shared vertices, representing single-semitone voice leading among equal-tempered tritones and perfect fifths. The second is a circle of squares linked by common edges, representing single-step voice leading among *diatonic* fourths and thirds.



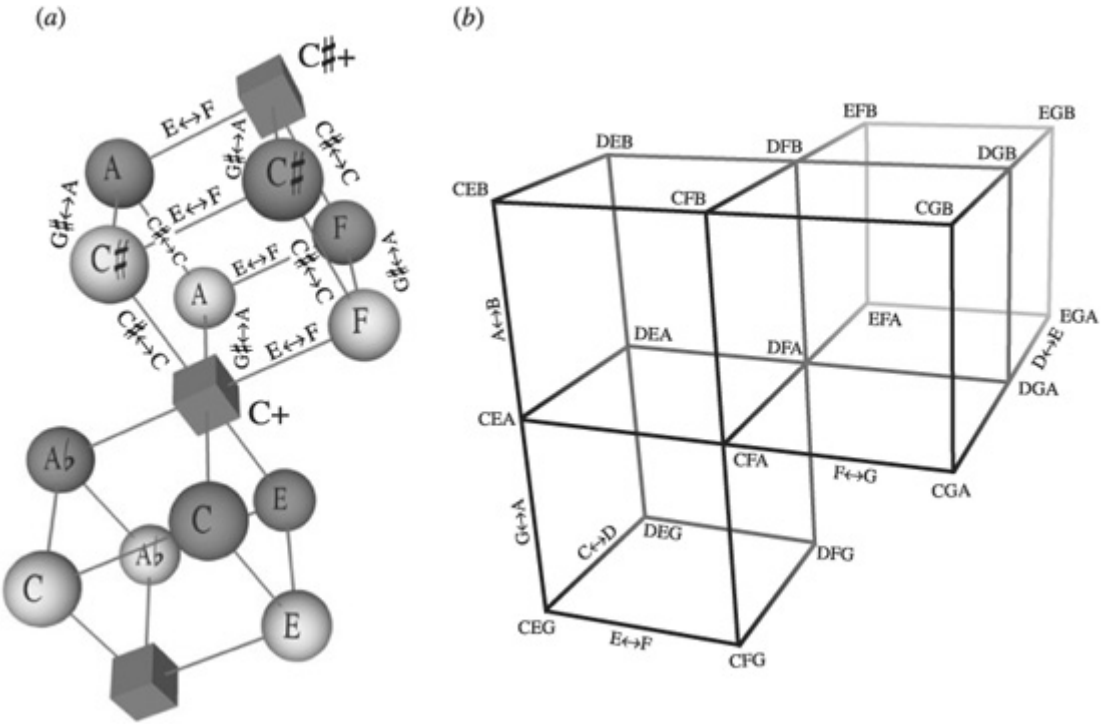
One finds closely analogous structures in higher dimensions. For example, suppose we want to graph voice-leading relationships among nearly even three-note chords in the chromatic scale. Since three evenly divides twelve, the lattice contains three-dimensional cubes linked by a common vertex (Figure 3.11.2a). As we have learned, the shared vertex is an augmented triad, with the edges of the cubes representing different ways that the augmented triad's notes can be raised or lowered by semitone: for instance, we can move from C♯ augmented to C augmented by first lowering F, then C♯, then A; alternatively, we could first lower A, then C♯, then F. Each of these sequences traces out a different series of edges on the top cube of Figure 3.11.2a. (Note that the chords on a single cube draw their notes from two semitonally adjacent augmented triads and hence belong to the same hexatonic scale.[37]) By contrast, nearly even three-note *diatonic* triads are represented by a lattice of the second kind, since three and seven have no common divisor. The resulting

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

graph, shown in Figure 3.11.2b, consists of a series of stacked cubes, each sharing a *face* with its neighbors. This graph is analogous to the linked squares of Figure 3.11.1b, though its structure is somewhat more complex. We will return to it momentarily.

Figure 3.11.2 (*a*) The cubic lattice at the center of three-note chromatic space. The shared vertex is the augmented triad; the edges represent different ways of raising or lowering the augmented triad's notes by semitone. (*b*) The cubic lattice at the center of three-note diatonic space. Here, the cubes share a face with their neighbors.



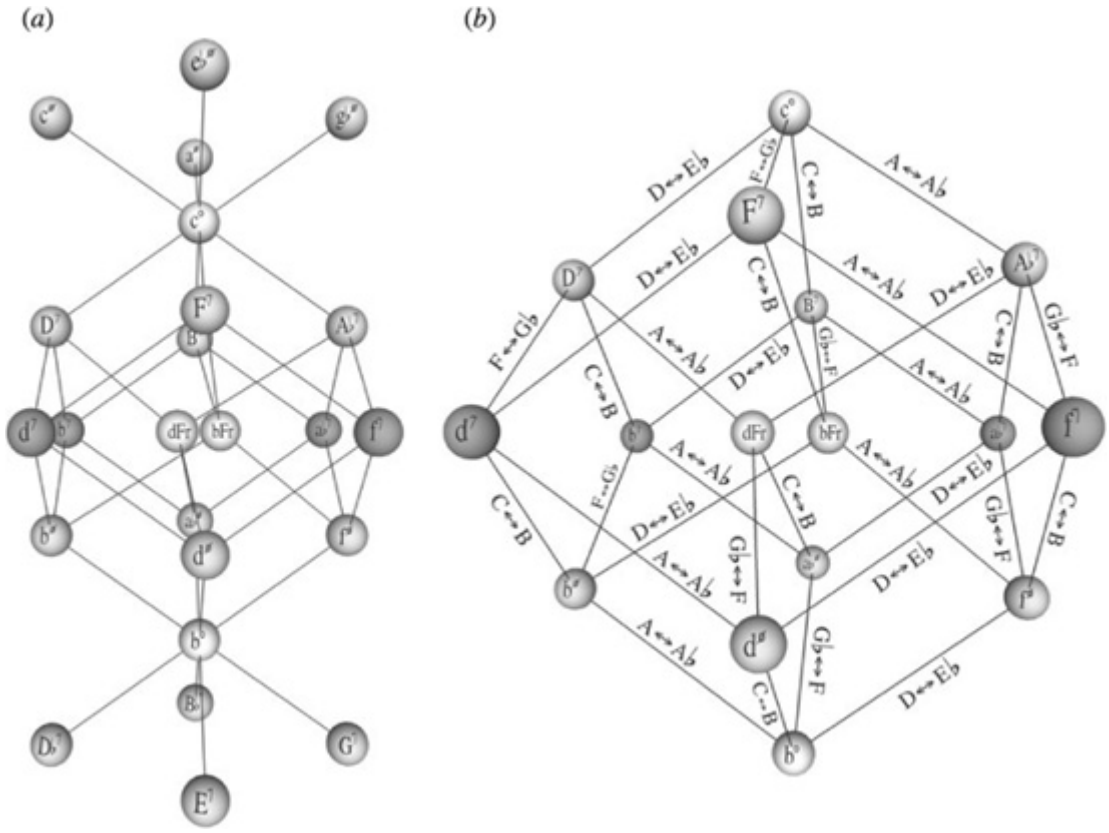
The first type of lattice is relatively easy to understand. If the number of notes in the chord exactly divides the number of notes in the scale, then the scale contains chords that are *perfectly* even.[38] These sit at the very center of (continuous) chord space and form the vertices shared by the lattice's adjacent cubes. The remaining edges on the lattice identify the various ways in which one can successively raise or lower each note of this perfectly even chord.[39] Thus the edges in Figure 3.11.1a show how one can successively raise or lower the notes of the tritone by semitone, while those in Figure 3.11.2a show how one can raise or lower the three notes of the augmented triad. Figure 3.11.3 illustrates the four-dimensional case. Here we have a series of four-dimensional cubes—also known as "tesseracts"—linked to their neighbors by shared vertices. (Figure 3.11.3 projects this four-dimensional structure into what looks like three Euclidean dimensions, much as we can

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

draw the outlines of a three-dimensional cube using a two-dimensional piece of paper.[40]) As expected, the shared vertices are diminished seventh chords, with the edges recording the various ways of sequentially raising or lowering their notes. Chords on a single "tesseract" draw their notes from two semitonally adjacent diminished seventh chords, and hence belong to the same octatonic scale. Thus chords on the hypercube containing F^7, D^7, B^7, and $A\flat^7$ all belong to the octatonic scale D-E♭-F-G♭-A♭-A-B-C. Readers may be relieved to learn that this is the most complicated lattice we will need to use in this book. Although an analogous (six-dimensional!) structure represents nearly even six-note chromatic chords, its chords can typically be interpreted as incomplete forms of familiar seven-note scales.

Figure 3.11.3 (*a*) The lattice at the center of four-note chromatic space is a circle of four-dimensional cubes linked by a shared vertex. The shared vertex is the diminished seventh chord. The labels "bFr" and "dFr" designate "French sixth" chords on D and B; these points appear to lie close to each other on the interior of the cube, but this is an artifact of the three-dimensional representation. (*b*) The edges of the four-dimensional cube represent the different ways of raising or lowering the notes of the diminished seventh chord.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

The second type of lattice appears when the size of the chord and the size of the scale have no common divisor. In this case, the scale will not contain any *perfectly* even chords; instead, it will contain *nearly* even chords that form a *near interval cycle*—that is, a circular sequence of notes all but one of which are linked by the same interval, with the unusual interval being either one scale step larger or smaller than the rest.[41] Three examples are shown in Figure 3.11.4: two- and three-note chords in diatonic space, and seven-note chords chromatic space. In each case, it is possible to shift the position of the unusual interval by moving a single note by just one scale step. Consequently, these chords can be linked by a sequence of single-step voice leadings somewhat analogous to the familiar circle of fifths (Figure 3.11.5).[42] This "generalized circle of fifths" will lie *as close as possible* to the center of chord space (at least for chords belonging to that particular scale), but it will not be exactly *at* the center (Figure 3.11.6).

Figure 3.11.4 When the size of the chord and the size of the scale are relatively prime, the maximally even chord is a "near interval cycle"—a circle of notes all but one of which are linked by the same interval, with the unusual interval being just one step different from the others. Because of this, the chords can be linked by a chain of single-step voice leadings. (*a*) Diatonic fifths linked by single-step voice leading. (*b*) Third-related diatonic triads linked by single step voice leading. (*c*) Diatonic scales linked by single-semitone voice leading.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

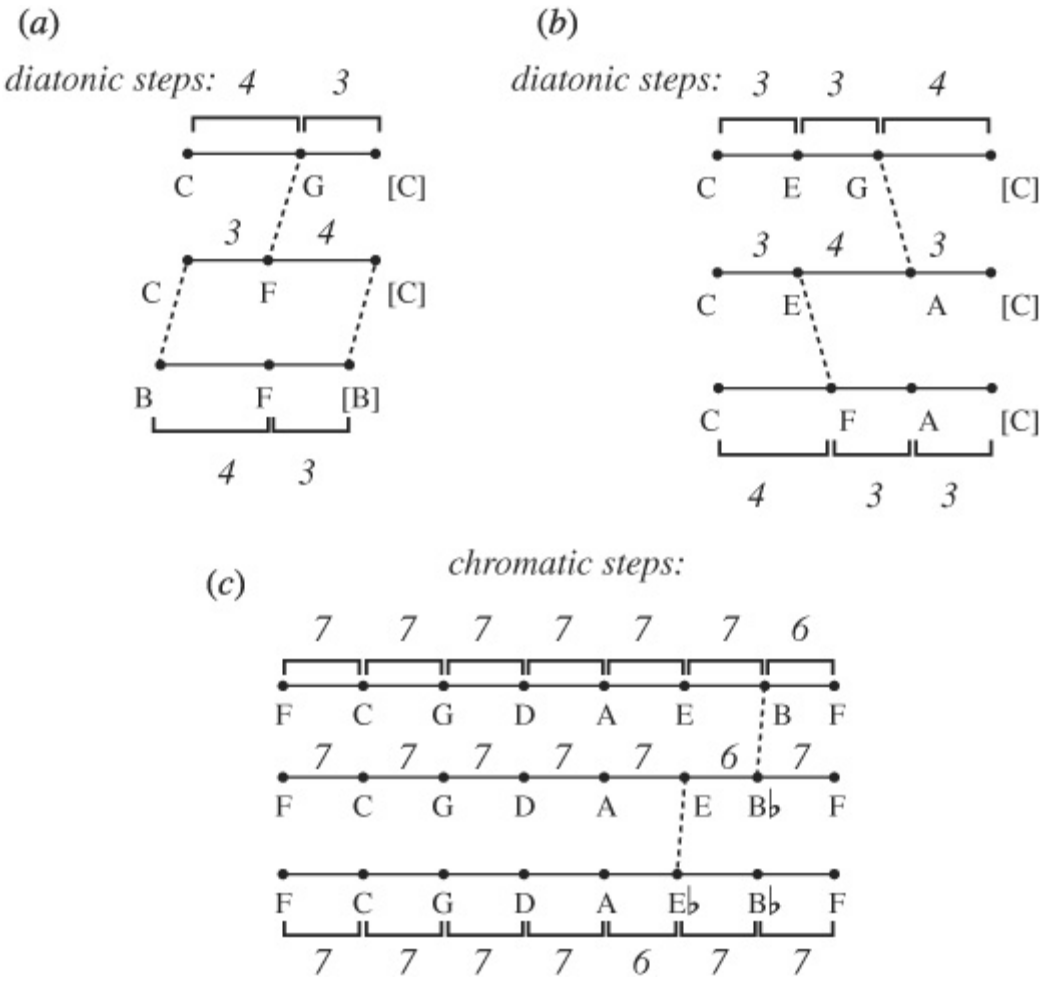
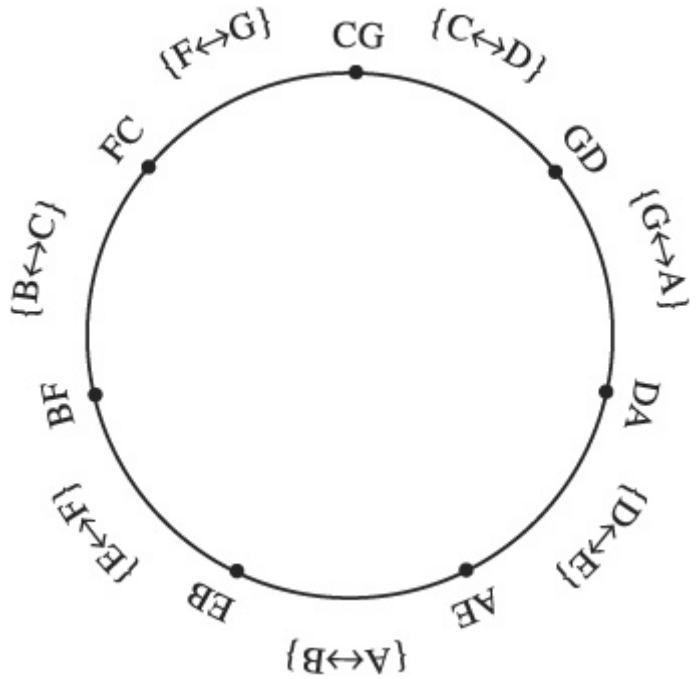


Figure 3.11.5 Single-step voice leading links near interval cycles into a circle, structurally analogous to the familiar circle of fifths. Here, the circle of two-note diatonic fifths, each linked to its neighbors by single-step motion.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 3.11.6 A "generalized circle of fifths" is near, but not exactly at, the center of chord space. In (*a*), the circle of diatonic fifths zigzags through the center of the Möbius strip. In (*b*), the circle of diatonic triads zigzags three-dimensionally through the center of three-note chord space.

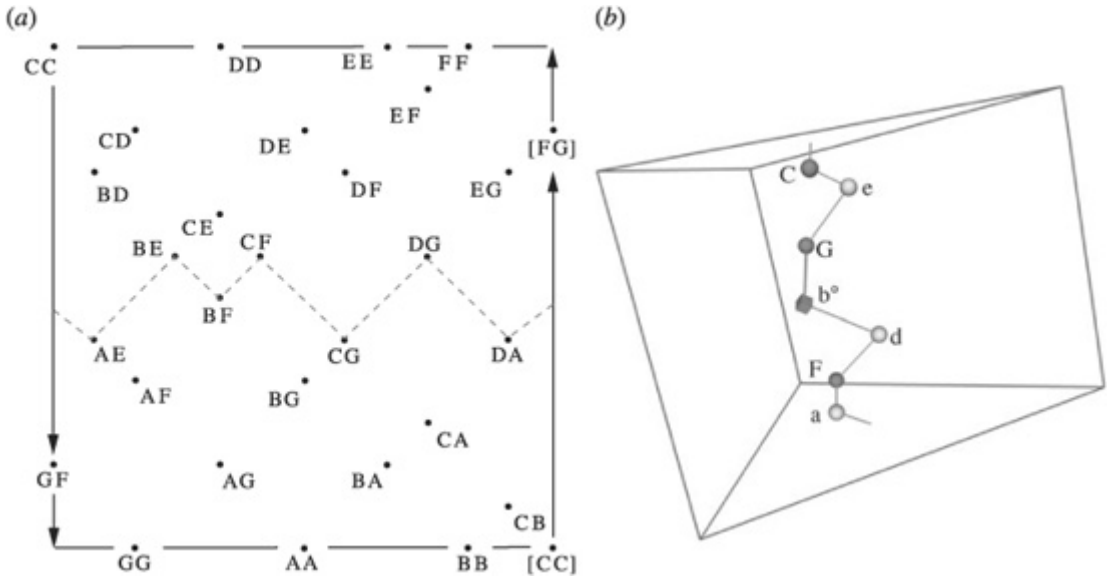
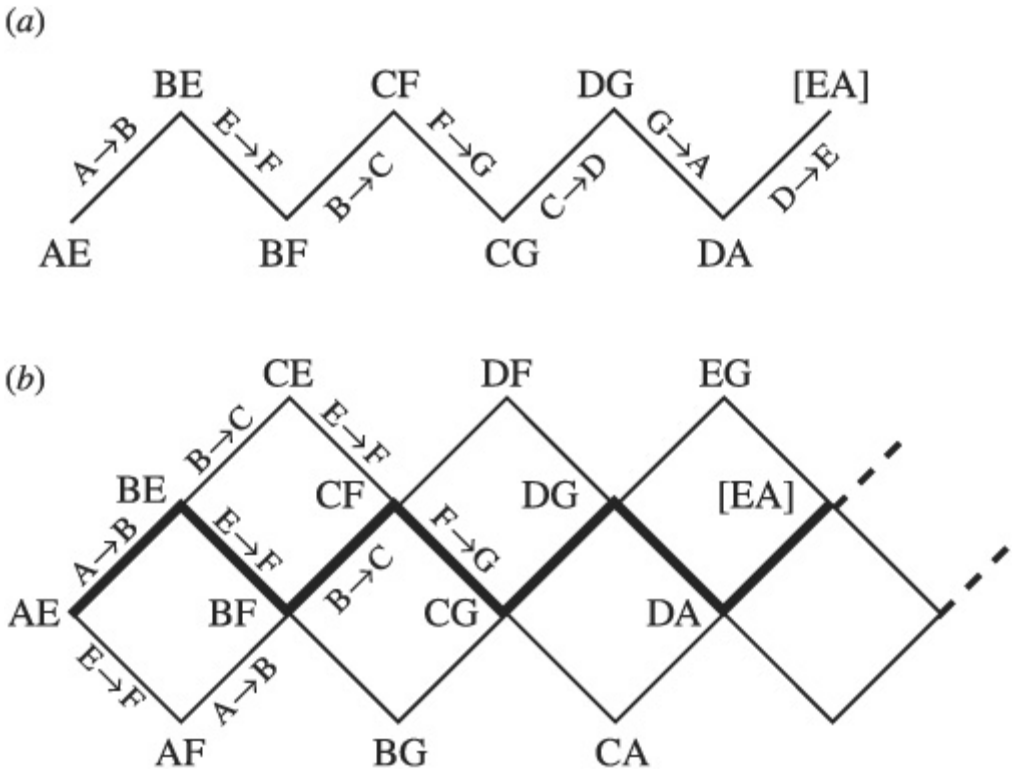


Figure 3.11.7 arranges our circle of diatonic fifths in a regular zigzag pattern.[43] These diatonic fifths are the most nearly even two-note chords in diatonic space. To add the second-most even chords, we can *rearrange the order of the voice leadings on the central zigzag* . For example, we move from {A, E} to {B, F} along the zigzag by way of the sequence (A, E)→(B, E)→(B, F), first raising A to B, and then E to F. But suppose we were to reorder these two single-step shifts: raising the E in {A, E} produces the diatonic third {A, F}, while raising the A in {A, F} brings us to {B, F}. Geometrically, these reordered voice leadings complete the square on the left side of Figure 3.11.7b. By continuing this process we can generate all the single-step voice leadings among diatonic thirds and fourths, thus reconstructing the lattice in Figure 3.11.1b.

Figure 3.11.7 (*a*) The two-note diatonic "circle of fifths," represented as a regular zigzag. (*b*) To include diatonic thirds, simply reverse the order of every pair of voice leadings, for instance by letting E→F operate on the fifth {A, E}. This converts the zigzag into a series of squares each sharing a common edge.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Remarkably, this same process of "rearranging" is sufficient to construct all the voice-leading lattices of the second type. By way of illustration, let's generate a voice-leading graph representing nearly even seven-note chords in chromatic space. The diatonic circle of fifths is shown in Figure 3.11.8a: here we move from C diatonic to G diatonic by the semitonal shift F→F♯, from G diatonic to D diatonic by the shift C→C♯, and so on. This circle lies as close as we can get to the center of seven-note chord space, given twelve-tone equal temperament. Figure 3.11.8b adds the second-most even seven-note chord by rearranging adjacent voice leadings as described in the previous paragraph: C→C♯ takes C diatonic to the G "acoustic scale" (or D melodic minor ascending) while F→F♯ takes G acoustic to D diatonic. (Note that the resulting graph is two-dimensional, even though seven-note chords live in seven dimensions!) We can even create a three-dimensional version of the graph by scrambling the order of *three* successive voice leadings on the diatonic circle of fifths. Here we begin with a three-dimensional zigzag that moves alternately up, right, and into the paper. We then apply all six reorderings of three adjacent voice leadings to a generate a cube: for instance, the scales on the lower left cube in Figure 3.11.9 result from applying to C diatonic the six different orderings of F→F♯, C→C♯, and G→G♯. (Note that F→F♯ is always represented by a vertical line, while C→C♯ is always horizontal, and G→G♯ always moves into the paper.) The resulting three-dimensional lattice, which describes voice-leading relationships among the four most even

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

twelve-tone equal-tempered chords in seven-dimensional space, is structurally analogous to the graph of three-note diatonic triads shown in Figure 3.11.2b. Here the circle of fifths zigzags three-dimensionally through the center of the space, much as the circle of thirds zigzagged through the two-dimensional Figure 3.11.8.[44]

Figure 3.11.8 (*a*) The familiar circle of fifths connects fifth-related diatonic scales by singlesemitone voice leading. (*b*) To include the second-most even five-note chords scramble the order of adjacent voice leadings, for instance by letting C→C♯ operate on C diatonic before F→F♯ does.

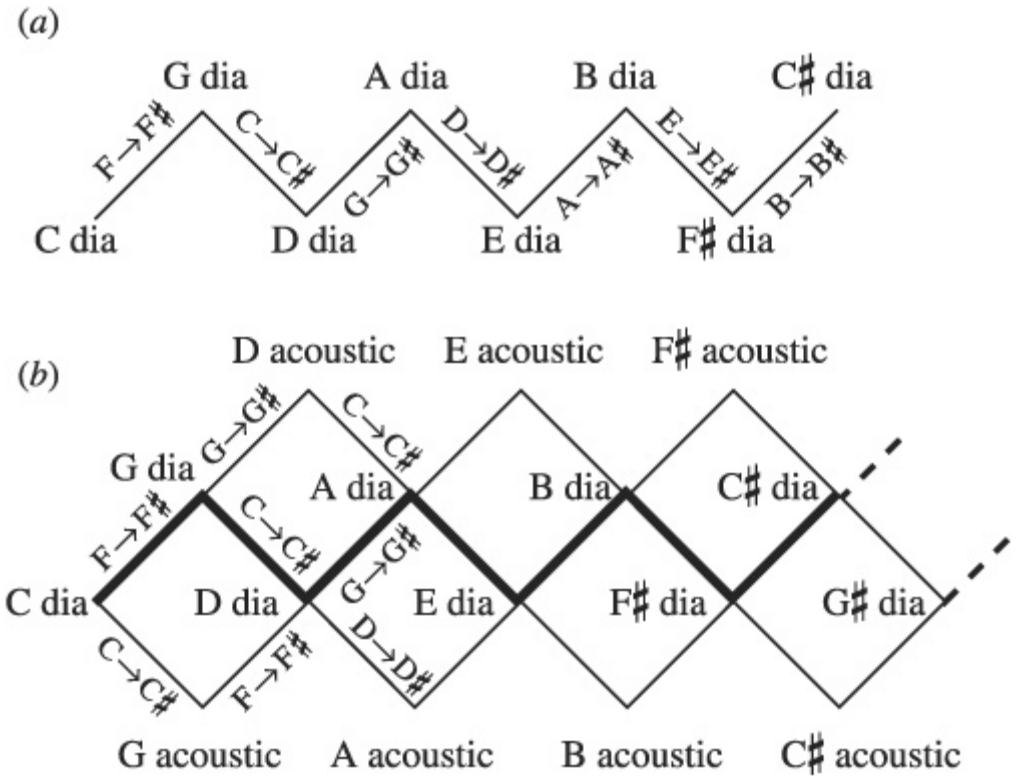
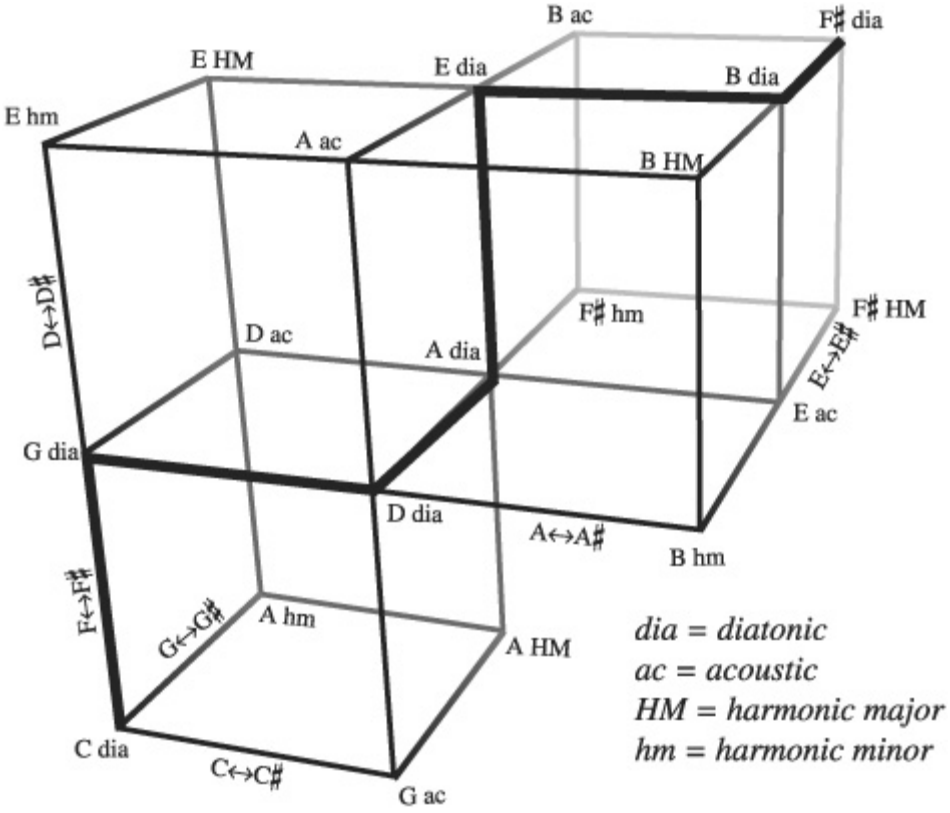


Figure 3.11.9 We can extend the scale lattice to three dimensions. The result depicts single-semitone voice leading among the four most even seven-note chromatic chords. We will discuss these scales in the next chapter.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Again, the key point is that the lattices in our second category—representing objects as diverse as diatonic seventh chords, octatonic triads, and familiar seven-note scales—are all structurally similar. At their center is a generalized circle of fifths that zigzags through two, three, or more dimensions. These chords are *near interval cycles* linked by a chain of single-semitone voice leadings. The rest of the lattice is generated by rearranging its voice leadings: for a two-dimensional graph, begin with a two-dimensional zigzag and switch the order of each pair of adjacent voice leadings, producing a sequence of squares sharing a common edge (as in Figures 3.11.7 and 3.11.8b); for a three-dimensional graph, begin with a three-dimensional zigzag (up, right, in) and scramble the order of every three consecutive voice leadings, producing a series of stacked cubes sharing a common face (as in Figure 3.11.9). This process of "scrambling" adds successively less even chords to the lattice, extending it farther and farther from the center of chord space. (Thus, where the dimension of the first kind of graph is controlled by the size of the chords we want to represent, the dimension of this second kind of graph is controlled by the number of distinct chord types we are interested in.) We will return to these structures throughout the book, using the "scale lattice" to represent voice leadings among familiar seven-note scales and using the three-note diatonic lattice to represent familiar classical voice-leading patterns.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Music theorists tend to take it for granted that we can use discrete lattices to represent voice-leading relationships, in large part because these discrete structures preceded the more comprehensive geometrical spaces we have discussed in this chapter. But on reflection, it is actually quite remarkable that (say) the one-dimensional circle of fifths can faithfully represent voice-leading relationships among seven-note diatonic scales. After all, diatonic scales are seven-note objects inhabiting a *seven-dimensional space*, and we have no reason to expect that six of these dimensions would turn out to be irrelevant. And yet it turns out that for most practical purposes we can make do by representing diatonic voice leadings on the familiar circle of fifths. This suggests that the circle of fifths is quite a special structure which lies within seven-note chord space in a very particular way. Other lattices are *not* special in this way and do *not* accurately model voice-leading distances, even though they might appear to do so. In general, it is extremely difficult to distinguish "faithful" lattices (such as the circle of fifths) from unfaithful lattices (such as the familiar *Tonnetz*). To understand these structures in a deep and principled way, we must reflect on how they are embedded within the continuous spaces described here. Those of you who want to explore this issue further should consult Appendix C. Those who are more interested in music analysis can instead focus on internalizing the two types of lattice described in this section. The important point is that it is typically sufficient to understand the discrete lattices, and to be able to construct the graph germane to a particular musical situation.

3.12 TWO MUSICAL GEOMETRIES

We have now developed two geometrical representations of the same musical facts—the circular pitch-class space of Chapter 2 and the higher dimensional spaces discussed in this chapter. These two representations are complementary, in the sense that relationships that are difficult to understand in one are often easier to understand in the other. Let's conclude by considering some of the strengths and weaknesses of the two models.

In some ways, the circular space of Chapter 2 is simpler and more convenient than its higher dimensional counterparts. In this chapter we've created individual spaces for chords of various sizes: the space of two-note chords is two-dimensional, the space of three-note chords is three-dimensional, and so on. But there is no analogous description of the space

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

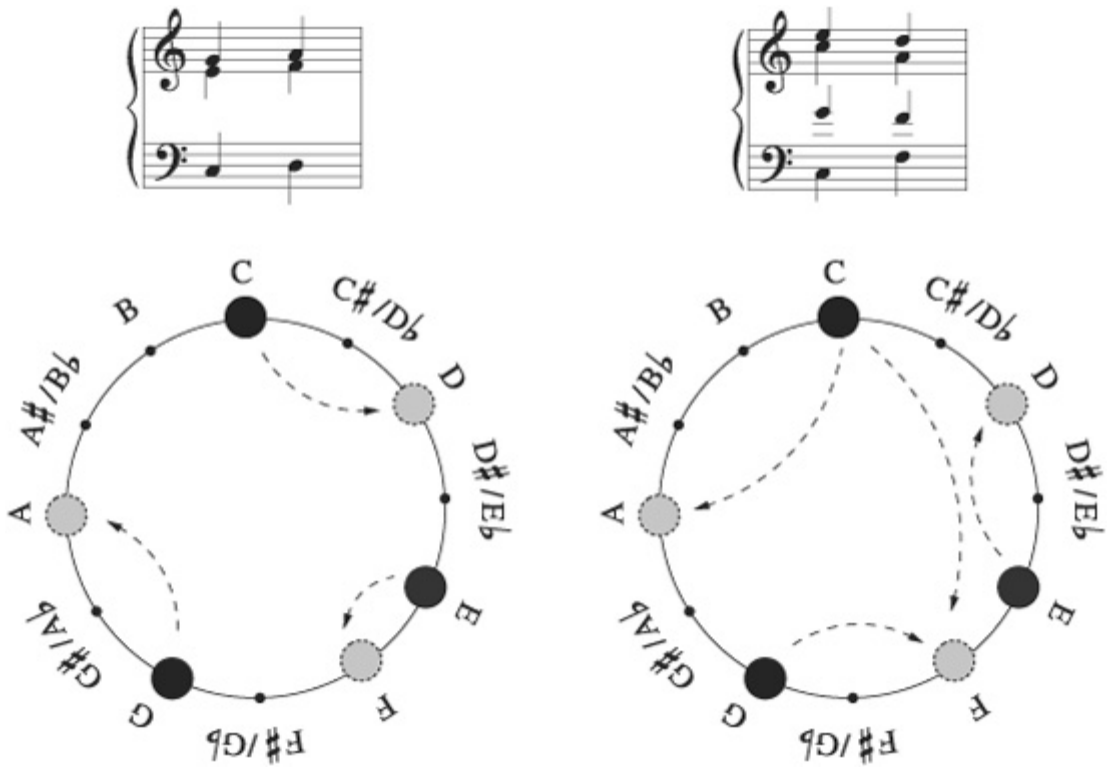
of *all* chords.[45] Furthermore, it can be very difficult to visualize relationships among larger chords, since a new dimension is required for each additional voice. By contrast, the circular representation uses a single circle to depict chords of any size, and there is no sense that two-note chords and three-note chords occupy fundamentally different worlds. Nor are there any special problems associated with large chords, as we can always add more points to the circle.

Moreover, the spaces discussed in this chapter do not provide a natural way to represent the similarity between chords with the same pitch-class content but different "doublings." For example, the collections {C, E, E, G} and {C, E, G, G} are represented by widely separated points in four-note chord space, yet they are musically quite similar; ordinarily, both would be considered C major chords. This problem is again ameliorated in the circular representation, which does not require us to distinguish the number of times each note appears: the chords {C, G}, {C, C, G} can both be modeled using the same two points on the circle, C and G. Voice leadings with "doublings" can be represented as different collections of paths between the same sets of points. Thus Figure 3.12.1 models the voice leadings (C, E, G)→(D, F, A) and (C, C, E, G)→(F, A, D, F) as two different ways to link the same configurations of black and gray points on the circle. Once again, the circular model seems more flexible than the higher dimensional alternative.

Figure 3.12.1 Two collections of paths linking the points {C, E, G} and {D, F, A}. On the left, the three-voice voice leading (C, E, G)→(D, F, A); on the right, the four-voice (C, C, E, G)→(F, A, D, F).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



The problem with the circle, however, is that it is easy to use but hard to understand. One could stare at circular pitch-class space for a long time without ever realizing that voice leadings can be decomposed into purely parallel and purely contrary components. And important voice-leading relationships can sometimes be hard to see. (Quick: which of its transpositions is the chord in Figure 3.12.2 closest to? Now look at Figure 3.11.8 and ask yourself the same question.) Nor does circular space make manifest structural relations such as those we uncovered in the *Allelujia Justus et Palma* or the Brahms C minor Piano Quartet. So at least in some cases, the circular model will obscure important features of musical structure.

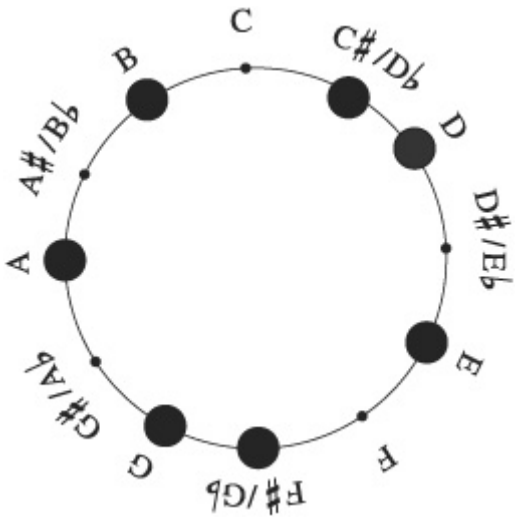
By contrast, higher dimensional chord spaces provide a powerful set of visual tools for thinking about music. We can describe the process of musical abstraction—the ignoring of octave and order information—as a matter of concentrating on a single "tile" of musical wallpaper. Abstract concepts like "individual T-relatedness" (or even "scalar transposition") can be translated into concrete visuospatial terms. Plotting music in the spaces can yield insight into its structure (§3.5). And though the higher dimensional spaces are unwieldy, lattices such as those in §3.11 can be used to convey a large amount of musical information very quickly, mapping a wealth of compositional possibilities in a

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

clear and extremely efficient manner. To use such lattices is to exploit the higher dimensional chord spaces described in this chapter, even if we do not explicitly represent the whole space from which they are drawn.

Figure 3.12.2 In circular pitch-class space, it is not immediately obvious which of this chord's transpositions it is closest to.



In all of these ways, then, the higher dimensional spaces provide useful complements to the simpler, circular model of Chapter 2. Ultimately, a deep understanding of music likely requires fluency with *both* models. (As the physicist Richard Feynman once put it, "every theoretical physicist who is any good knows six or seven different theoretical representations for exactly the same physics."[46]) By learning multiple ways to represent the same music, we can develop flexibility of mind and deepen our grasp of the underlying structures—even if, in the end, we prefer one representation to another. Readers who do not share my aesthetic appreciation for the higher dimensional chord spaces may therefore still be able to appreciate them as part of a program of music-theoretical calisthenics. In stretching our minds to understand these exotic structures, we will surely strengthen our appreciation for the richness latent in ordinary musical notation.

3.13 STUDY GUIDE

Readers who want to improve their understanding of musical geometry should practice three basic skills. First, representing voice leadings on the two-note Möbius strip of Figure 3.3.1. Second, representing single chords, as well as the pure contrary component of three-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

note voice leadings, on Figure 3.8.6. (It may also be useful to try representing pure contrary two-note voice leadings on Figure 3.7.2a.) And third, constructing the lattices describing voice-leading relationships among musically interesting objects—for instance, the most even two-note chords in the pentatonic and whole-tone scales, the most even three-note chords in the diatonic and whole-tone scales, the most even four-note chords in the diatonic and octatonic scales, and the most even five-note chords in the chromatic scale. (In constructing these lattices, use the various graphs in this chapter as guides, and consider the scale step to be a unit of distance.) After practicing these particular tasks, the more general geometrical principles should start to become clear. It may also be helpful to read the chapter once, spend some time with the questions in Appendix F (perhaps augmented by the more detailed discussion in Appendix B), and then return to the chapter a second time, repeating as necessary. In the end, passive reading is no substitute for a little bit of active, hands-on exploration. Just as nobody will ever learn to play the piano simply by reading about it, nobody will learn geometry unless they pick up a pencil and start working through some exercises.

Readers who are hungry to explore analytical applications can turn directly to Chapter 8, which is concerned with chromatic voice leading. Chapter 9 also uses the geometry of seven-dimensional scale space to investigate twentieth-century music, though it should be read after Chapter 4. The geometrical spaces described in this chapter also appear in Chapter 6, where they provide a framework for retelling the history of Western music. However, this chapter is probably best read after the remaining theoretical material in Part I.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

CHAPTER 4

Scales

Now we're ready to incorporate scales into our developing theoretical apparatus. We'll begin by modeling scales as "musical rulers" that allow us to measure distances in pitch and pitch-class space. We'll then investigate a number of familiar scales, including the pentatonic, diatonic, chromatic, and melodic minor. This leads to a discussion of the relation between modulation and voice leading. Finally, we investigate *scalar* and *interscalar* transpositions—concepts that are useful not just for understanding scalar music, but also for understanding voice leading more generally. The upshot is that there is an interesting duality, or complementarity, between chord and scale. Not only can chordal concepts (e.g. voice leading) help us understand scalar processes (e.g. modulation), but the reverse is also true; concepts like *scalar transposition* provide useful tools for understanding relationships between chords.[1]

4.1 A SCALE IS A RULER

Intuitively, the opening of *The Sound of Music's* "Do, Re, Mi" (Figure 4.1.1) involves three repetitions of the same musical pattern: when composing each phrase, Richard Rogers simply "did the same thing" at different pitch levels. (This, of course, is part of the song's cheery didacticism—it's supposed to teach kids music.) But we are hard pressed to describe this sense of sameness using the ideas developed so far. From the standpoint of Chapter 2, (C, D, E) and (D, E, F) are fundamentally different objects, since they are not related by transposition, inversion, or any other combination of OPTIC symmetries.

The difficulty, of course, is that we are thinking chromatically rather than diatonically. Instead of describing F as being one semitone above E, we should describe it as being one *scale step* above E. Once we start to think diatonically we realize that each phrase begins with two ascending scale steps, with the entire pattern moving up by step at each repetition. The moral is that a scale provides us with an alternative measure of musical distance, and that we must be careful to choose the right distance metric for the job. In

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

principle, any collection of pitches can be a scale: a scale's notes do not have to be very close to one another, nor are they required to repeat after every octave, nor is it necessary that scales have "first" or "tonic" notes (Figure 4.1.2). All the scale needs to do is to tell us how to move up and down by 1 unit, or *scale step*.[2]

Figure 4.1.1 The opening notes of the first phrases of "Do, Re, Mi," interpreted chromatically and diatonically. The numbers represent intervals as measured relative to the two scales.

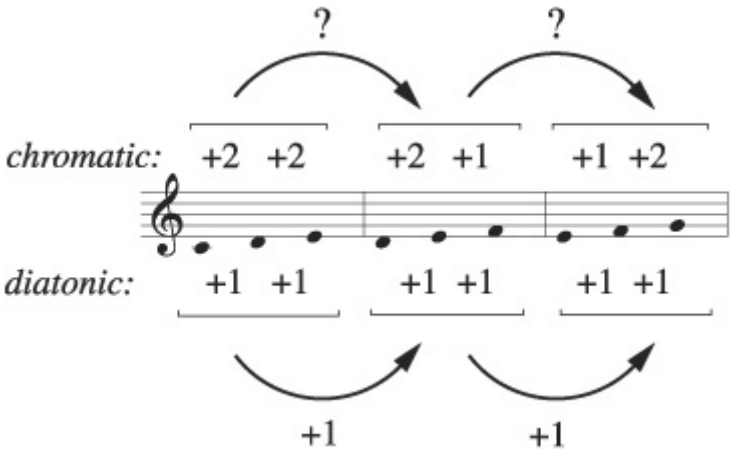
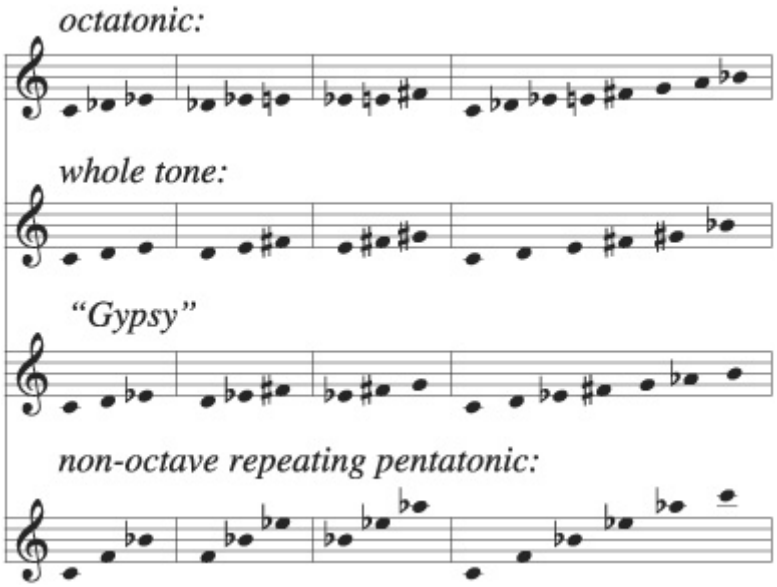


Figure 4.1.2 The "Do, Re, Mi" pattern in other scales.



Octave-repeating scales, which contain each of their pitches in every possible octave, can be represented as points in circular pitch-class space.[3] When depicting them geometrically we can choose whether the visual layout should reflect chromatic or scalar distance. (*Note:* throughout this chapter, the term "chromatic" refers to the continuous, log-frequency

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

measure of §2.1, rather than to distances along some chromatic scale.[4]) For example, Figure 4.1.3a uses chromatic distance, representing C and D as farther apart than E and F, while Figure 4.1.3b uses scalar distances, placing the seven notes around the circle so that steps all appear to be the same size. We can do something similar even when working with the higher dimensional chord spaces of the previous chapter. For example, Figure 4.1.4a draws the two-note Möbius strip, labeling only the dyads in the C diatonic scale. Points are connected by lines when they can be linked by voice leading in which both voices move by diatonic step: the roughly horizontal lines represent parallel motion within the scale; the roughly vertical lines represent contrary motion within the scale. The irregularity of the resulting grid reflects the fact that the scale's steps are not all the same chromatic size. Figure 4.1.4b redraws the graph using scalar distance, so that the grid is perfectly regular and all scale steps appear to be equally large.[5]

Figure 4.1.3 An octave-repeating scale is a circular arrangement of pitch classes. It can be drawn either using chromatic distance (*a*) or scalar distance (*b*).

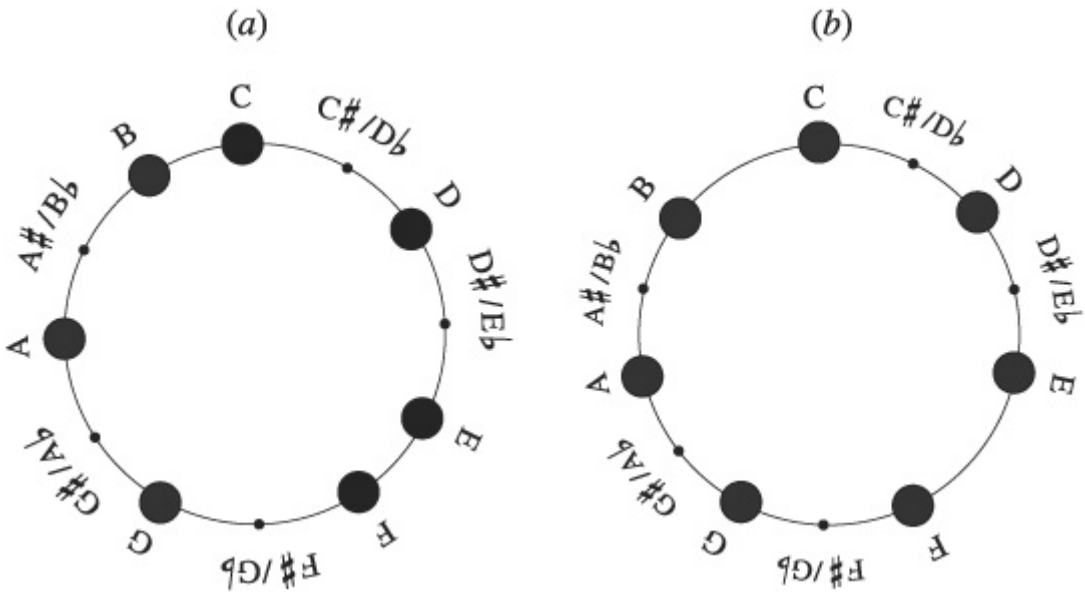
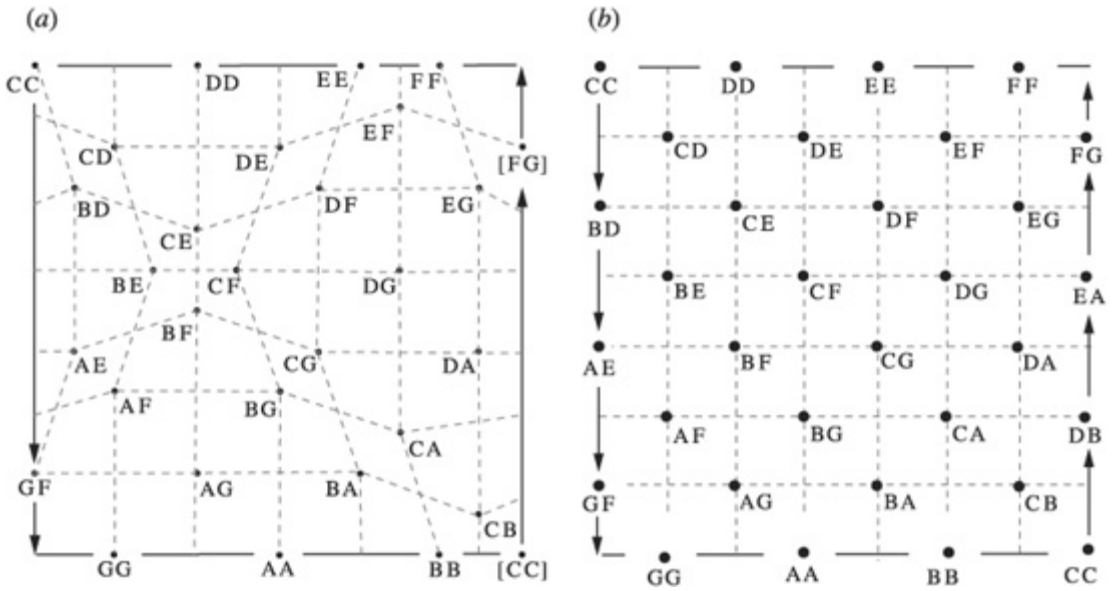


Figure 4.1.4 Two-dimensional chord space, drawn using chromatic distance (*a*) and scalar distance (*b*). Only the white notes are labeled.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



To my eye, Figure 4.1.4a has a three-dimensional quality, resembling a crumpled piece of paper or an aerial view of a hilly city, such as San Francisco. This impression of three-dimensionality testifies to the way our visual system has evolved to manipulate multiple kinds of distance. When we look down at San Francisco, we need to infer the *intrinsic* length of the city blocks from their *apparent* length on our retinas. Intrinsically the blocks are all (roughly) equal, since San Francisco is a grid city, but the blocks appear different, since some travel uphill. (When seen from above, these uphill blocks look shorter, as we lose the vertical dimension.) These two visual distances are closely analogous to the two musical distances we have been discussing. Just as the hot air balloonist needs to juggle multiple visual distances when looking down at San Francisco, so too does the musician need to realize that the line segments in Figure 4.1.4a have the same diatonic length, but different chromatic sizes. Remarkably, all of this is encapsulated in the crumpled appearance of Figure 4.1.4a—demonstrating once again that geometry can help us understand abstract music-theoretical relationships.

4.2 SCALE DEGREES, SCALAR TRANSPOSITION, AND SCALAR INVERSION

Since a scale provides a measure of musical distance, it also defines scale-specific notions of transposition, inversion, chord type, and set class. These allow us to adapt the ideas of the previous chapters to our new scalar perspective.

We begin by assigning numbers to the notes in the scale—arbitrarily selecting some note as scale degree 1, labeling the note immediately above it scale degree 2, the note two

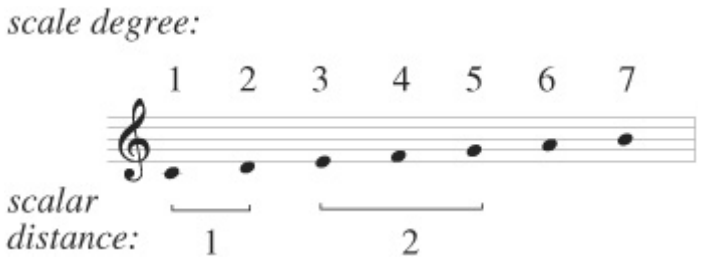
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

steps above it scale degree 3, and so on (Figure 4.2.1). (The choice of a first scale degree is a mere notational convenience and does not imply that this note is more significant than the others.) Scale degree numbers are analogous to numerical pitch-class labels, with one trivial difference: while music theorists label pitch classes starting from zero, they label scale degrees starting from one.[6] Note that while scale degree *numbers* are dependent on the arbitrary choice of a first scale degree, scalar distances are not—in C diatonic, for instance, the scalar distance between C and D is always one scale step, no matter what the first scale degree happens to be. Musicians refer to these distances in a somewhat confusing way, using the terms "a second," "a third," and "a fourth" to refer to scalar distances of one, two, and three steps, respectively.

Transposition and inversion are the two distance-preserving operations in chromatic space. We can define new operations of *scalar transposition* and *scalar inversion* that preserve scalar distances. To transpose relative to a scale, simply add a constant to each scale degree (Figure 4.2.2). (Here we use "scale degree arithmetic"—for an n-note scale, we add or subtract n until the result lies between 1 and n, inclusive.[7]) To invert relative to a scale, choose a fixed point that will remain unaffected by the inversion. Any note x scale steps above the fixed point gets sent to the note x scale steps *below* the fixed point, and vice versa. Scalar inversion can be represented algebraically by subtraction from a constant value: in C major, we invert (C, D, F) around the fixed point E by subtracting each scale degree from 6, transforming 1, 2, 4 into 5, 4, 2, or (G, F, D).[8] The mathematics is identical to that in §2.2, only using scale degree numbers rather than chromatic pitch-class labels.

Figure 4.2.1 Scale degrees and scalar distances. A scalar distance of one step is called a "second," a distance of two steps is "a third," and so on.



Relative to a scale, two chords belong to the same chord type (transpositional set class) if they are related by scalar transposition. Thus, relative to C harmonic minor, the chords {B, D, F} and {C, E♭, G} are both "triads," and hence transpositionally related, even

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

though one is diminished and the other is minor (Figure 4.2.3). Relative to C harmonic minor, {A♭, B, E♭} is *not* a triad, because A♭ and B are only *one* scale step apart. (Chromatically, of course, it is an A♭-minor triad and is transpositionally related to C minor.) This shows that the same chords can be transpositionally related relative to one scale, but not another.[9] Scalar "set classes" can be defined in the obvious way, as groups of chords related either by scalar transposition or scalar inversion. Thus {C, D, F} and {D, F, G} belong to the same set class relative to C harmonic minor, because they are related by scalar inversion around E♭.

Figure 4.2.2 (*a*) To transpose up by three scale steps, add 3 to each scale degree number. (*b*) To invert around E (scale degree 3), subtract each scale degree number from 6 (= 3 × 2). In both cases, use "scale degree arithmetic," adding or subtracting 7 until the result lies between 1 and 7.

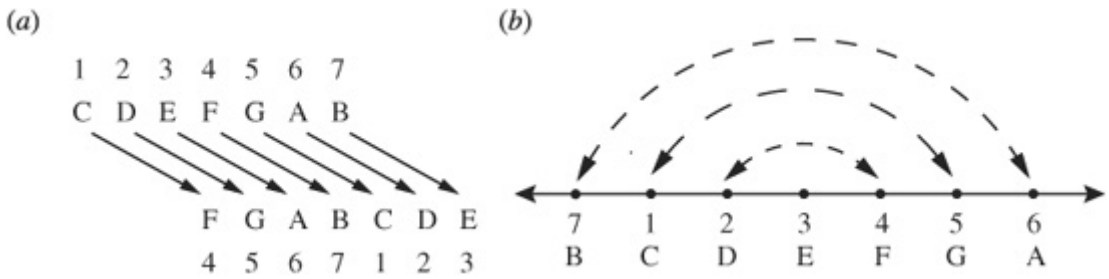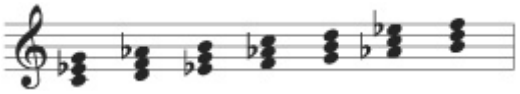


Figure 4.2.3 Relative to the C harmonic minor scale, these chords all belong to the same transpositional set class, and are therefore triads. {A♭, B, E♭} is not a member of this set class, and is not a triad, even though it is related to {C, E♭, G} by chromatic transposition.



It is even possible to assign scale degree numbers to notes that are not in the scale. Relative to C diatonic, the note C♯ can be considered scale degree 1.5, since it lies halfway between scale degrees 1 and 2 (Figure 4.1.3). Similarly, E𝄲 (E quarter-tone sharp) can be considered scale degree 3.5, since it is halfway between scale degrees 3 and 4. It follows
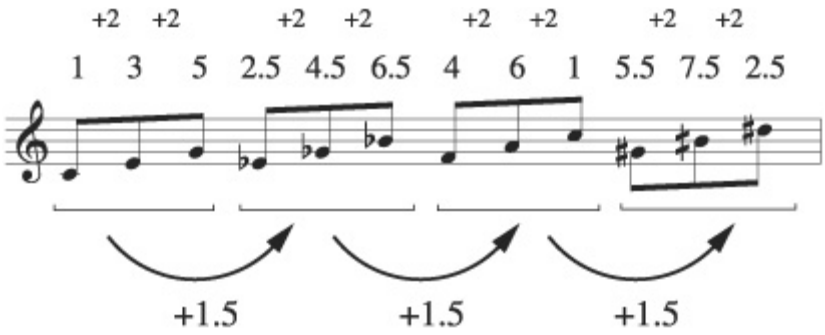
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

that scalar transposition and inversion can act on notes *outside the scale!* For example, we can shift (C, D, E) up by half a scale step to produce (C♯, D♯, E♯), or we can invert (C, D, D♯) around D♯ to produce (F, E, D♯).[10] Proceeding in this way, we can apply scalar transposition and inversion completely promiscuously—shifting *any* collection of notes by any fraction of a scale step.

This suggests the perverse musical strategy of writing music that uses the C diatonic scale to measure musical distance, while not privileging its notes in any way. For example, Figure 4.2.4 repeatedly transposes the C major triad by one and a half scale steps, producing {E♭, G♭, B♭}, {F, A, C} and {G♯, B♯, D♯}.[11] The resulting music is *guided* by the diatonic scale without emphasizing the white notes themselves. A brilliant music theorist, gifted with extremely sharp ears and a somewhat demented musical imagination, might even be able to discern the presence of the scale from the deformations it introduces as chords move through musical space. I stress that I am not recommending this as a practical possibility; my point is simply that a musical scale is very similar to what mathematicians call a *metric*, or a method of measuring distance. In principle, a scale can function in this way even in highly chromatic contexts such as Figure 4.2.4. This is just to repeat the point that a *scale* need not also be a *macroharmony*.

Figure 4.2.4 This music uses the C major scale to measure distance, while also containing notes foreign to the scale (compare Figure 4.1.1).



4.3 EVENNESS AND SCALAR TRANSPOSITION

Listeners are typically aware of both scalar and log-frequency distance at the same time. We know that (D, E, F) is in some sense "the same" as (C, D, E), but we also know that (E, F) is different from (D, E). Ordinary musical terminology adopts this twofold perspective,
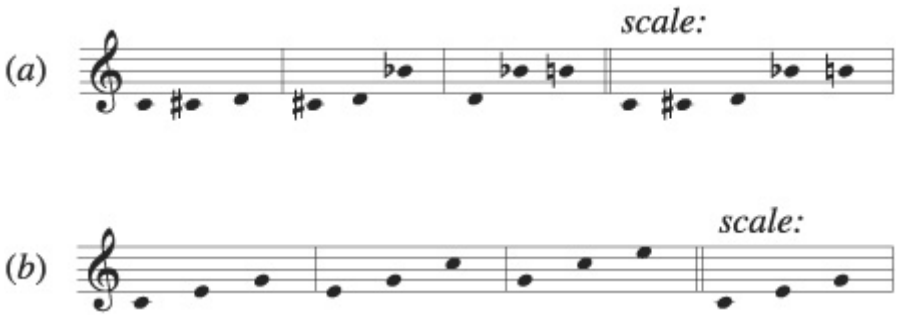
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

asserting that (E, F) and (D, E) are both "seconds" (one-step intervals), while adding the qualification that the first is "minor" and the second is "major."[12] Scalar music is interesting precisely because of this doubleness. When moving musical patterns along a scale, composers inevitably transform their material in subtle ways, creating a pleasing mixture of identity and difference.

For scalar music to be successful, these induced variations must be small—otherwise, listeners would not be able to treat (D, E, F) as being essentially similar to (C, D, E). Figure 4.3.1a presents an example of a scale that fails spectacularly in this regard. Scalar transposition transforms (C, C♯, D) (analogous to "Do, a deer") into (C♯, D, B♭) ("Re, a . . . *drop?!!?!?*"), creating an eight-semitone step that sounds nothing whatsoever like the one-semitone step between C and C♯. By contrast, Figure 4.3.1b presents a relatively even three-note scale, in which scalar transposition is completely successful and gestures retain their shape as they are transposed. Such passages are common in Western music, demonstrating that scalar terminology can be useful even when we are investigating objects that are not traditionally considered to be scales.

Figure 4.3.1 (*a*) An uneven scale, in which the step between D and B♭ is much larger than the others. (*b*) A small but nearly even scale, containing only the notes {C, E, G}.
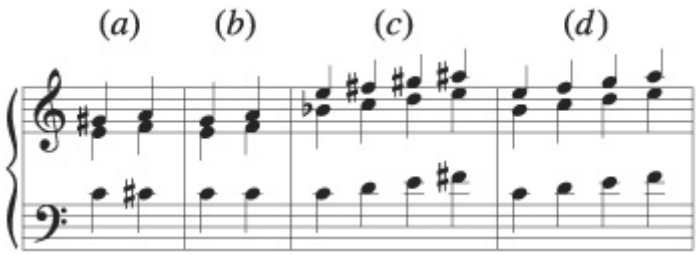


Clearly, scalar transposition will be precisely equal to chromatic transposition when the scale's notes are distributed perfectly evenly in pitch-class space—as in the case of the familiar whole-tone scale. When the scale is *nearly* even, then the two forms of transposition will be nearly but not exactly the same. In some ways this latter situation is preferable: scalar transposition along a perfectly even scale does not introduce any variations into the music and can become boring rather quickly. A bit of unevenness therefore adds musical interest by introducing a degree of variation. Interestingly, we can also make an analogous point about chords. When a chord divides the octave *perfectly*

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

evenly, efficient voice leading typically involves parallel motion in all voices and does not create the effect of independent melodic lines (Figure 4.3.2). A little bit of unevenness is preferable here as well, allowing us to escape parallelism by causing the voices to move unequally. We might therefore say that both scales and chords are subject to the *Goldilocks Principle*—not too much evenness, because that would be boring, and not too little, because that would lead to musical disorder.

Figure 4.3.2 (*a*) Chords that divide the octave completely evenly can be connected to their transpositions by efficient but parallel voice leading. (*b*) Slightly uneven chords can be connected to their transpositions by voice leading in which the voices move by different distances, creating the sense of counterpoint. (*c*) Along a completely even scale, scalar transposition introduces no variation into music. (*d*) A small amount of unevenness creates slight amounts of variation, which can often be desirable.



Near evenness therefore plays at least three fundamentally different roles in musical life: the nearly even collections can be linked to their transpositions by efficient voice leading, they include the acoustically consonant chords, and they also allow for scalar transpositions that introduce moderate but not overwhelming amounts of variation into musical textures. It is somewhat remarkable that these three separate considerations all point in the same musical direction. Intuitively, one would think that the requirements for constructing contrapuntal music had nothing in common with the requirements for constructing effective scales, and this in turn might lead to the expectation that composers had an enormous amount of freedom in creating alternative musical languages. The *overdetermination* of Western chords and scales leads to the very opposite conclusion. For insofar as composers are interested in harmonic consistency, acoustic consonance, *or* scalar transposition, they will necessarily find themselves gravitating toward the same familiar musical objects.

4.4 CONSTRUCTING COMMON SCALES

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Since scales typically function as macroharmonies, tonal composers have reason to be interested in scales containing many consonant intervals. And since the octave is the most consonant of the intervals, composers have reason to use scales that are maximally saturated with octaves—that is, scales in which each note has an octave both above it and below it. But this, of course, is just to say that the scale is *octave repeating*. Octave-repeating scales are therefore overdetermined in their own modest way: besides being easy to use and remember, they also contain as many octaves as they possibly can.

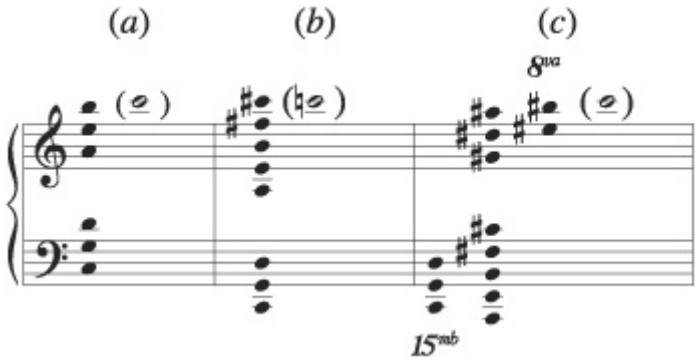
The second-most consonant interval, the acoustically pure perfect fifth, is just a hair's breadth larger than its familiar equal-tempered cousin. Suppose you would like to construct an octave-repeating scale that is maximally saturated with acoustically pure fifths—that is, you would like as many notes as possible to have fifths both above them and below them. It has been known for thousands of years that a finite scale can never have pure fifths above and below *each* of its notes; the best one can do is construct scales in which all but two notes have this property—in other words, "stacks of fifths" such as those in Figure 4.4.1. If the scale divides the octave nearly evenly, then the final note of the stack is *nearly* a perfect fifth away from the initial note.[13] Figure 4.4.1 identifies three salient possibilities: a stack of five perfect fifths falls approximately 0.9 semitones short of three octaves, a stack of seven perfect fifths exceeds four octaves by about 1.137 semitones, and a stack of twelve perfect fifths overshoots seven octaves by nearly 0.25 semitones.[14] We can therefore build nearly even scales containing four, six, and eleven perfect fifths, otherwise known as the pentatonic, diatonic, and chromatic scales. The pentatonic has a "near fifth" approximately equal to a minor sixth; the diatonic has a "near fifth" approximately equal to a tritone; and the chromatic scale has a "near fifth" a bit smaller than the perfect fifth. Different tuning systems will assign slightly different sizes to these intervals, depending on how exactly they compromise between the demands of evenness and acoustic purity. We will return to this issue momentarily.

Figure 4.4.1 (*a*) A stack of five perfect fifths falls 0.9 semitones short of three octaves. (*b*) A stack of seven perfect fifths overshoots four octaves by 1.14 semitones. (*c*) A stack of twelve fifths overshoots seven octaves by 0.25 semitones. (Note that the B♯ is not quite the same as C♮.) By eliminating the top note of each stack we can form a relatively even scale

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

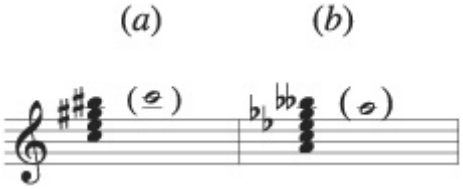
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

with one “imperfect fifth.”



Let us now play the same game with thirds. Suppose you would like to create a scale maximally saturated with acoustically pure major or minor thirds. Figure 4.4.2 shows that three acoustically pure major thirds, or four acoustically pure minor thirds, are about half a semitone away from an octave. To extend these stacks further would create very small melodic intervals that would be awkward to sing and play, and this is rarely done in Western music. Instead, musicians consider the cycle C-E-G♯-C to be closed, containing two pure thirds (C-E and E-G♯) and one “near major third” (G♯-C). (Modern equal temperament of course regularizes the thirds so that they are exactly the same size, with the resulting augmented triad dividing the octave into three precisely even parts.) Similarly, four minor thirds are considered to form a closed cycle A-C-E♭-G♭-A, consisting of three pure minor thirds and one “near minor third”—or in equal temperament, four completely equal intervals.

Figure 4.4.2 (*a*) A stack of four major thirds falls 0.41 semitones short of an octave. (*b*) A stack of five minor thirds overshoots an octave by 0.62 semitones.
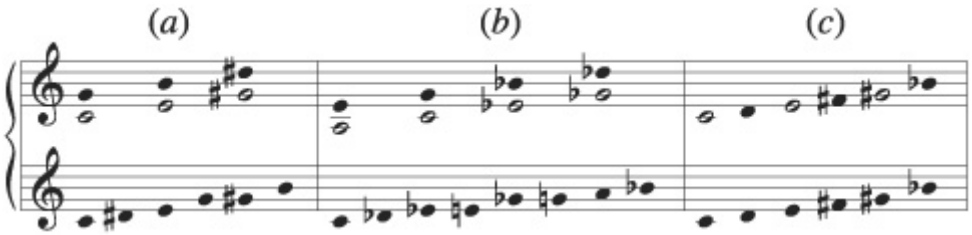


Unlike our stacks of perfect fifths, these thirds-cycles are too small to serve as satisfying macroharmonies. However, we can *combine* cycles to make larger collections. The hexatonic scale (Figure 4.4.3a) combines two augmented triads at a distance of a perfect fifth. This scale is maximally saturated with major thirds while also containing a large number of perfect fifths. However, it does not divide the octave particularly evenly, since its steps are (approximately) one and three semitones large. The octatonic scale (Figure

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

4.4.3b) is analogous to the hexatonic, but is built with minor-third cycles; it contains a large number of *minor* thirds and perfect fifths. Because its steps are one and two semitones large, it is considerably more even than the hexatonic scale. Hexatonic and octatonic are close cousins, being structurally similar and playing important roles in nineteenth- and twentieth-century music. A third possibility, the *whole-tone scale* (Figure 4.4.3c), combines two augmented triads at a distance of a major second; although it is not particularly saturated with consonances, it is perfectly even. In principle, we could use diminished seventh chords to construct a perfectly even eight-note scale, although it is not available in twelve-tone equal temperament.

Figure 4.4.3 (*a*) The hexatonic scale combines two augmented triads at a distance of a perfect fifth. (*b*) The octatonic scale combines two diminished sevenths at a distance of a perfect fifth. (*c*) The whole-tone scale combines augmented triads at a distance of a major second; it is perfectly even, but contains no perfect fifths.



We can also form two-octave stacks of major and minor thirds, as in Figure 4.4.4. The thirds can be arranged to produce four familiar scales: diatonic, “acoustic” (or melodic minor ascending), harmonic minor, and “harmonic major.”[15] The acoustic scale is so-called because it is approximately equal to the first seven pitch classes of the harmonic series; it is usually labeled relative to the ordering (C, D, E, F♯, G, A, B♭), known as the “C acoustic collection.”[16] (This is equivalent to either the mixolydian mode with raised fourth degree or the lydian with lowered seventh.) The harmonic scales are the only ones we have considered that are not inversionally symmetrical: the harmonic minor is ubiquitous in classical music, while its inversion, the so-called harmonic major scale, was investigated by nineteenth-century musicians such as Weitzmann and Rimsky-Korsakov.[17] It can be described as a major scale with lowered sixth degree or as a harmonic minor scale with raised third.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 4.4.4 Four scales can be represented as nearly even stacks of three major thirds and four minor thirds: diatonic (*a*), acoustic (*b*), harmonic minor (*c*), and harmonic major (*d*).

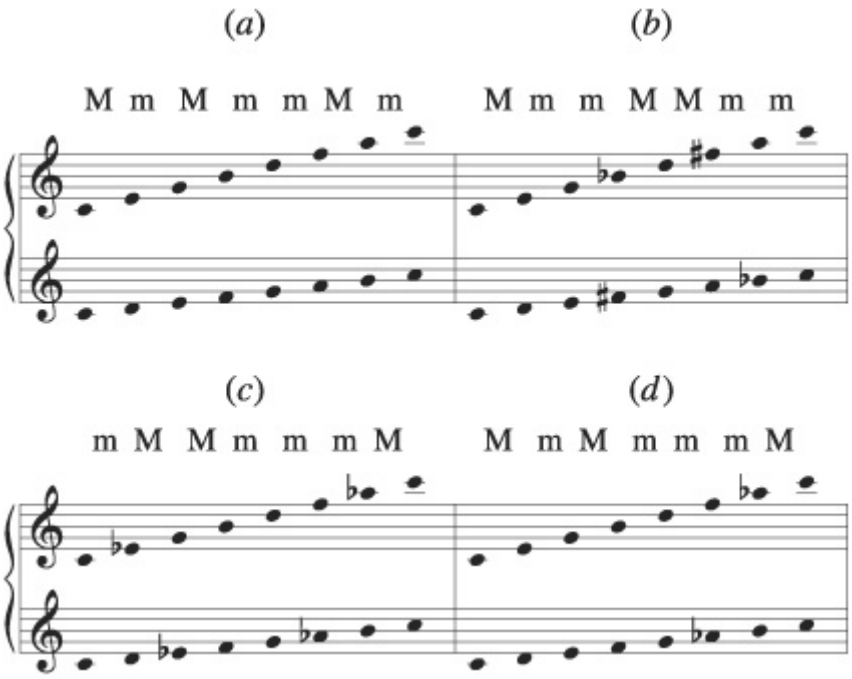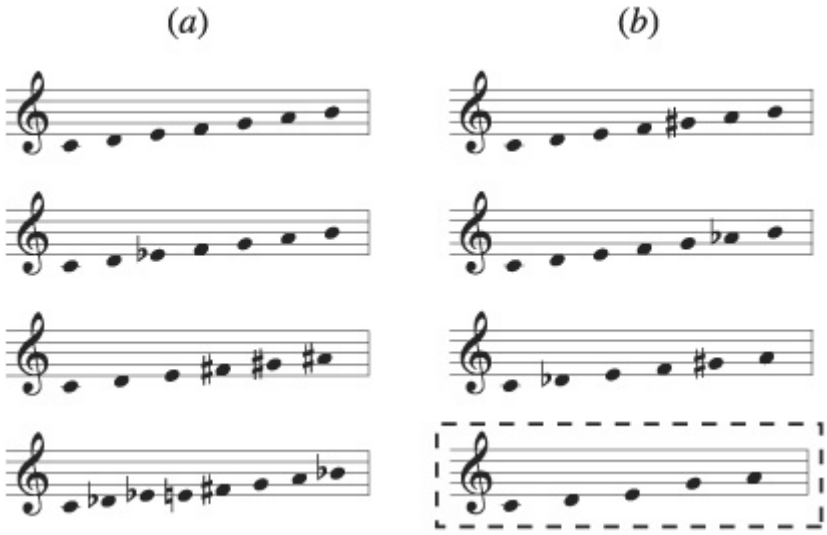


Figure 4.4.5 categorizes our eight scales based on their step sizes. The scales on the left—diatonic, acoustic, whole tone, and octatonic—have steps that are at most two semitones large, and thirds (two-step scalar intervals) that are either three or four semitones large.[18] These scales are important because they permit composers to import traditional compositional techniques to new scalar environments: by selecting adjacent scale tones, a composer can create melodies that are recognizably step-wise (i.e. whose notes are linked by one- or two-semitone intervals); and by selecting scalar "thirds" a composer can create chords that are recognizably "tertian" (i.e. whose adjacent notes are linked by three- or four-semitone intervals). The scales in the second group—harmonic minor, harmonic major, hexatonic, and pentatonic—have at least one step that is three semitones large, and hence lead to more exotic melodies. However, the first three scales have thirds that are three or four semitones large, and can again be used to construct "tertian" chords. By contrast, pentatonic thirds are four or five semitones large, which means that stacks of pentatonic thirds tend to sound like "fourth chords" (that is, chords whose notes are five *chromatic* steps apart). Figure 4.4.6 shows that twentieth-century tonal composers often used the pentatonic scale to create the impression of "fourthiness."

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 4.4.5 The scales in (*a*) have steps that are either one or two semitones large, and thirds that are three or four semitones large. Those in (*b*) have at least one step that is three semitones large. The top three scales in (*b*) have thirds that are three or four semitones large. The pentatonic scale is enclosed in a box because it is a subset of the diatonic.



In twelve-tone equal temperament, our scales have another useful property as well: they contain *every* chord that does not itself contain a "chromatic cluster" such as {C, C♯, D}.[19] The scales thus provide a technique for managing a greatly extended harmonic vocabulary that nevertheless stops short of atonality's extremes—that is, composers can construct harmonies by freely choosing notes from one of these scales, secure in the knowledge that they will never generate a dissonant chromatic cluster. Conversely, the scales provide a ready "reservoir" of melodic notes with which to accompany any cluster-free chord. Note in particular that "polychords," formed by superimposing two triads, can always be embedded within one of these scales. In this sense, they fit naturally with the extended tertian sonorities characteristic of twentieth-century tonal music.

A final word about tuning and temperament. Temperament enters the picture because of the need to balance the demands of regularity and acoustic purity. A Pythagorean twelve-tone chromatic scale contains acoustically pure fifths, but does not divide the octave precisely evenly. Conversely, the twelve-tone equal-tempered chromatic scale sacrifices purity in the name of evenness—with all its steps being precisely the same size, but its fifths being very slightly impure.[20] Beyond these two alternatives are a variety of other options, such as just intonation and mean-tone tuning, each enforcing a slightly

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.
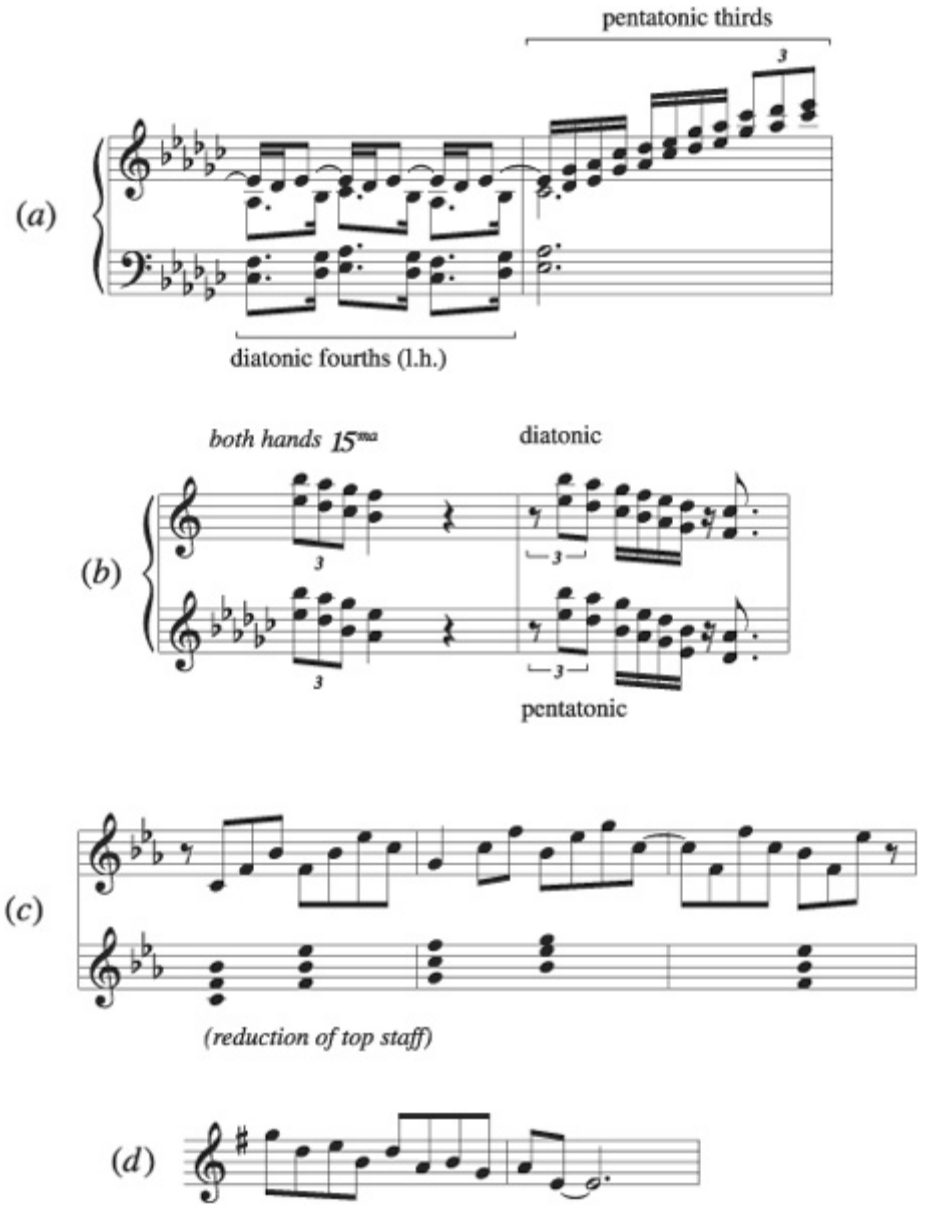
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

different compromise between evenness and euphony. The important point is that the scales we have been considering will be interesting no matter which intonational compromise we favor, simply because they divide the octave fairly evenly, while also containing a large number of consonances.[21] That so many of them can be played on the ordinary piano keyboard is testimony to the power and flexibility of the twelve-tone equal-tempered system.

Figure 4.4.6 Diatonic fourths and pentatonic thirds. In (*a*), from mm. 34–35 of Debussy's "La fille aux cheveux de lin," diatonic fourths give way to pentatonic thirds. In (*b*), from Ligeti's *Piano Concerto* (second movement, mm. 60–61) diatonic fifths are superimposed on pentatonic fourths (the inversions of pentatonic thirds). In (*c*), Herbie Hancock uses pentatonic thirds in the fifth chorus of "Eye of the Hurricane," from the album *Maiden Voyage*. (*d*) Pentatonic thirds at the end of the chorus of the Decemberists' "Here I Dreamt I Was an Architect."

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



4.5 MODULATION AND VOICE LEADING

Figure 4.5.1 presents the transition section from the first movement of Mozart's G major Piano Sonata, K. 283. The first three bars are sequential, transposing the same music up one step at each repetition, while the last three bars contain a second sequence, now transposed up two steps at a time. As the music repeats, the accidentals change: F♮ moves to F♯ in measure 4 and C♮ moves to C♯ in m. 6.
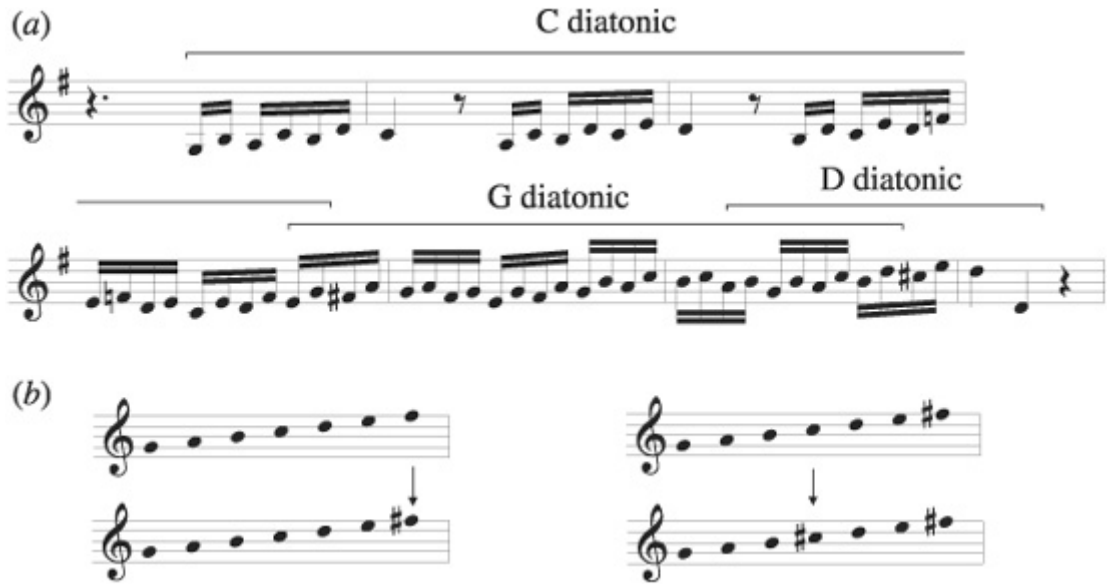
Suppose we ask a naive question: why say that Mozart's music changes scale? Why not say instead that Mozart uses a *nine*-note scale containing all the white notes plus F♯ and C♯ ? The answer lies in the music's sequential structure: we would like to say that the leaps (D, F), (E, G), and (F♯, A), are two-step scalar intervals, but there can be no scale in which

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

(D, F), (E, G), and (F♯, A) are all two steps apart.[22] Thus, if we wish to claim that the passage is a sequence—the same pattern repeated at different musical levels—we are required to postulate changes in the underlying scale.

Figure 4.5.1 (*a*) The transition from the first movement of Mozart's G major Piano Sonata, K. 283. (*b*) The passage involves two voice leadings between scales.



In fact, we are forced to postulate *specific voice leadings* between scales. This is because the sequence of scalar intervals will be preserved only if we assert that the F moves *up by semitone* to the F♯, and the C♮ moves *up by semitone* to C♯. To see why, look at the last six notes in the fourth bar of the example, D-F-E-G-F♯-A. Presumably, we would like to say that the music here consists in a repeating "up two, down one" pattern. But if this is right, then the F♮ and F♯ must occupy the same abstract position: the note F♮ is one step above E, the note F♯ is one step below G, and the distance E-G is two scale steps. The only way these facts are consistent is if the scale degree that used to be located at F has somehow moved to F♯—or in other words, if there has been a voice leading between scales.

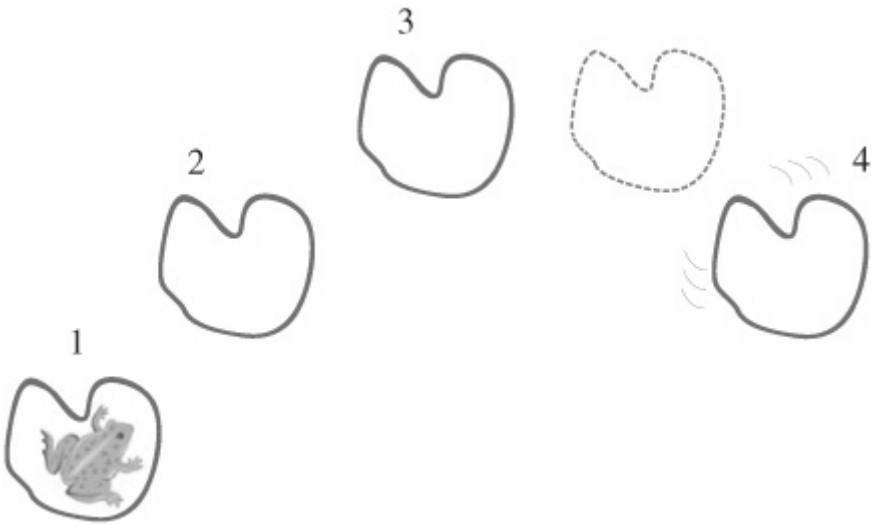
We conclude that a piece's *scalar* structure can determine specific voice leadings between its *macroharmonies*. It is our desire to analyze the music using scale steps, and in particular to do justice to its sequential structure, that requires us to postulate specific scalar voice leadings. I find it useful here to imagine a frog hopping in a regular way around a gently drifting circle of lily pads—say, forward by two lily pads, back by one lily pad, forward by two, back by one, and so on (Figure 4.5.2). As the frog hops, the fourth

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

lily pad moves into a new position, with the others remaining fixed. Clearly, to describe the frog's motion perspicuously, we need to measure in units of lily pads. (It would be somewhat odd to describe it as hopping forward by 10 centimeters, backward by 3 centimeters, and so on, oblivious to the fact that the size of its jumps is determined by the position of the pads!) But we also need to take into account the fact that the lily pads are themselves shifting: the phrase "lily pad #3" acts as a kind of variable that can refer to various locations in physical space.[23] In much the same way, we often want to analyze music using scale steps even while the scales themselves are changing. In these contexts a scale degree is a kind of abstract address ("the musical location one step above E and one step below G") that can point to a variety of different chromatic locations (such as F♭ or F♯).[24]

Figure 4.5.2 A frong, hopping along a circle of gently shifting lily pads.



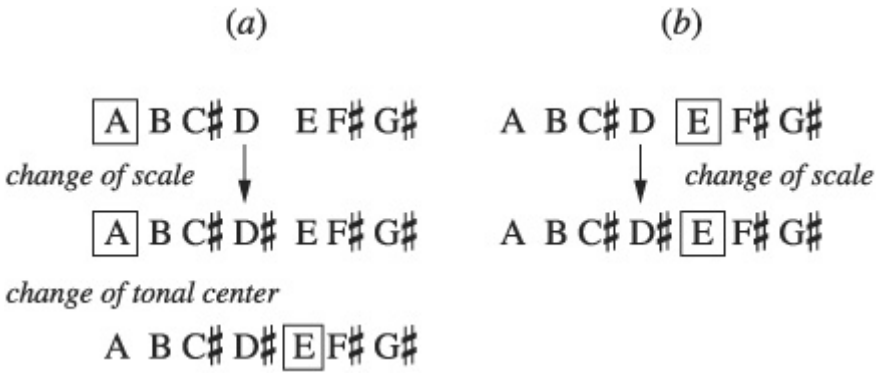
We can now see that traditional modulation is typically a two-stage business. Modulation is often initiated by voice leading between scales—the fourth degree gets raised by semitone, the leading tone gets lowered, and so on (Figure 4.5.3). This change of scale permits the introduction of the new key's V^7 chord, setting the stage for a shift in the tonal center. Thus, in the typical modulation from tonic to dominant, the raising of the fourth scale degree occurs *prior* to the conclusion of the V^7–I cadence in the new key. (In fact, it sometimes takes a while for our ears to be convinced that the scale shift represents a genuine change in tonic, rather than a passing melodic inflection.) In classical music, shifts between parallel keys (such as C major and C minor) are unique insofar as they do

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

not involve any change of tonic. But in twentieth-century tonality, we encounter a much wider range of tonic-preserving modulations, including all the shifts between parallel diatonic modes.

Figure 4.5.3 (*a*) Classical modulation as a two-stage process. Here a modulation from A major to E major, in which the scale shifts before the tonal center (represented by the boxed note). Thus we first change D to D♯ and then shift the tonic (boxed) note from A to E. (*b*) In twentieth-century tonality, one frequently finds a change of scale without a change of tonal center. For example, The Who's "I Can't Explain" moves from E mixolydian in the verse to E major in the chorus; here the tonal center stays fixed while D moves to D♯.



It is important to understand that the points I have been making do not depend in any way on notation. The question is about conceptualizing music in terms of scalar distances, and this does not require that the music be notated at all—in the case of our Mozart example, it merely requires that it be useful to conceptualize the distances F-A and F♯-A as *thirds*, and hence to infer that the note F has *moved to* F♯. Nevertheless, notation can often help us determine voice-leading relationships among scales. Suppose we imagine that the seven letter names A–G represent distinct musical voices, so that there is an "A" voice, a "B" voice, and so on. Accidentals can then be taken to indicate how the letter-voices move: thus when Mozart replaces F♮ with F♯, he signals that the "F voice" has moved up by semitone to F♯. In other words, standard musical notation faithfully reflects the fact that the first modulation in Figure 4.5.1 occurs by way of the voice leading (C, D, E, F, G, A, B) →(C, D, E, F♯, G, A, B), while the second occurs by way of (C, D, E, F♯, G, A, B)→(C♯, D, E, F♯, G, A, B).
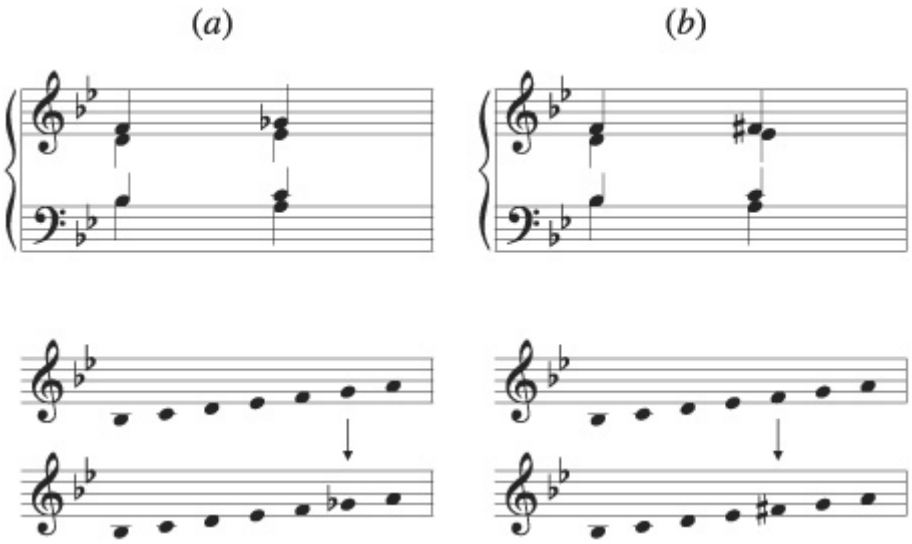
Of course, scalar voice leading can occur in nonmodulatory contexts as well. Consider, for example, the passages in Figure 4.5.4. In the first, the notation suggests that the G in a

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

B♭ major scale moves down to G♭, creating B♭ harmonic major, while in the second it suggests that F moves up to F♯, producing G harmonic minor. We can experience this difference without musical notation: for instance, an improvising musician might play the diminished chord in Figure 4.5.4a, anticipating a continuation whereby the G♭ moves downward to F, while on another occasion the same improviser might play the same chord anticipating a continuation where the F♯ rises to G. Our musical expectations are powerful enough that the same chord can even sound different in these two contexts. (Certainly, when imagining the passages in my head I experience them quite differently.) Here again, notation can clarify matters, even though the actual phenomena do not depend on notation itself.

Figure 4.5.4 Spelling often indicates a difference in musical function. Here, the chord {A, C, E♭, G♭} would typically lead back to a B♭ major triad, while {A, C, E♭, F♯} would lead to G minor. Spelling can be interpreted as indicating specific voice leadings between scales.



4.6 VOICE LEADING BETWEEN COMMON SCALES

In twelve-tone equal temperament, the four most even seven-note scales are the diatonic, acoustic, harmonic minor, and harmonic major. Consequently, their voice-leading relations are modeled by one of the lattices described in §3.11 and repeated here as Figure 4.6.1.[25] As mentioned earlier, the diatonic circle of fifths zigzags three-dimensionally through the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

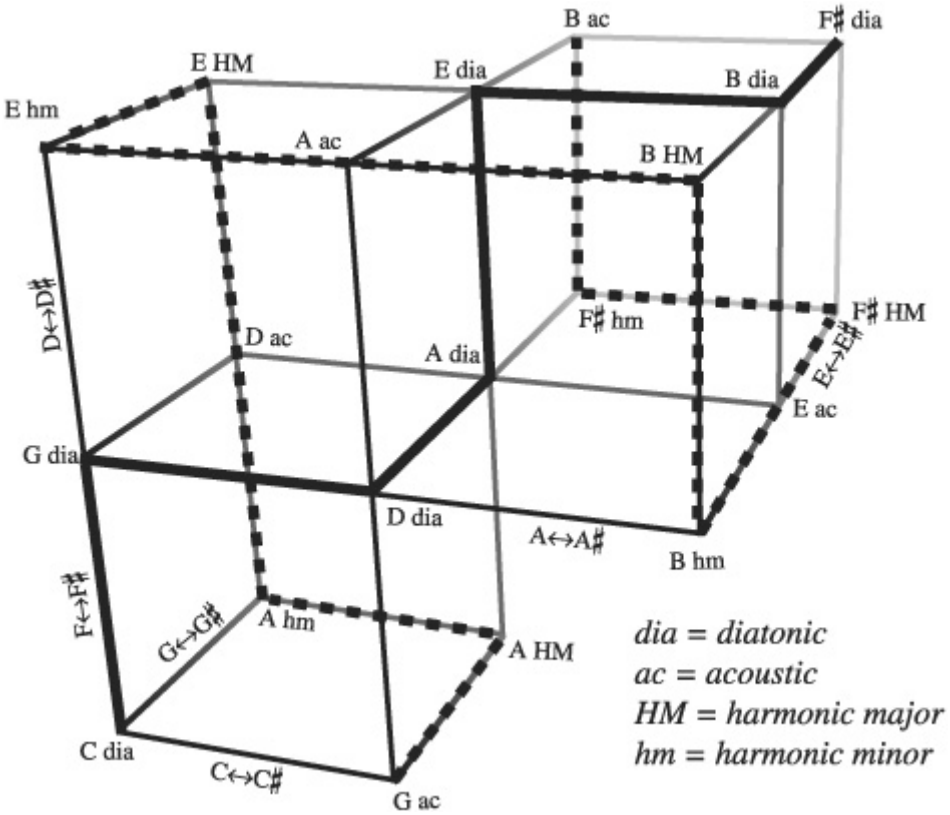
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

center of the figure, moving successively up, right, and into the paper. It turns out that the remaining scales are also linked by a chain of singlesemitone voice leadings. For instance, beginning with A harmonic major, we find

A harmonic major→A harmonic minor→D acoustic→E harmonic major→ E harmonic minor → A acoustic →. . . .

This is a second "circle of fifths" whose unit of sequential repetition is three scales long. These scales wind their way around the diatonic circle, somewhat in the manner of a double helix. However, the two strands are not quite the same shape: where the diatonic circle takes a right-angled turn after every step, the nondiatonic strand moves in a straight line through each acoustic collection.

Figure 4.6.1 Voice leadings among seven-note scales. The diatonic circle of fifths (dark solid line) starts at C diatonic (lower left front), while the nondiatonic circle (dashed line) starts at G acoustic (lower right front).
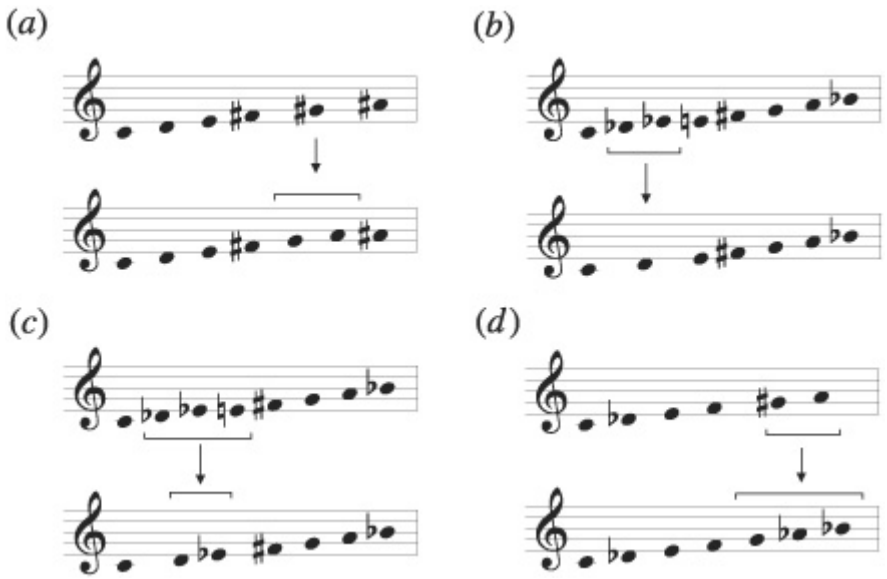


The harmonic and acoustic scales are closely related to the transpositionally symmetrical whole-tone, hexatonic, and octatonic scales. Figure 4.6.2 shows that we can change whole tone into acoustic by a voice leading in which a single pitch class "splits"

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

into its chromatic neighbors; because of the whole-tone scale's sixfold transpositional symmetry, there are six possible acoustic scales that can be reached in this way.[26] Similarly, octatonic can be transformed into acoustic by the reverse process—a transformation in which a major second "merges" into its central note. (Again, since the octatonic scale is transpositionally symmetrical, there are four ways to "merge" into an acoustic collection.) Octatonic can move to harmonic major or minor by a more complicated process of "merging": here, we fuse three notes spanning a minor third, producing the minor second in the center of the span—for instance, {C♯, D♯, E} becomes {D, E♭}, transforming C octatonic into G harmonic minor. (There are eight such possibilities for every octatonic scale, four leading to harmonic major and four leading to harmonic minor.) Finally, the hexatonic can be transformed into harmonic major or minor by the reverse process: here, a minor second such as {G♯, A} splits to become a three-note scale fragment whose outer notes span it, such as {G, A♭, B♭}. There are six such possibilities for every hexatonic scale, three leading to harmonic major and three leading to harmonic minor.

Figure 4.6.2 (*a*) To transform a whole-tone scale into an acoustic, "split" one note into its two chromatic neighbors. (*b*) To transform an octatonic scale into an acoustic, "merge" a major second into the note at its center. (*c*) To transform an octatonic scale into a harmonic major or minor, "merge" three notes spanning a minor third into the semitone they enclose. (*d*) To transform a hexatonic collection into a harmonic major or minor, "split" a semitone into a three-note scale-fragment enclosing it.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 4.6.3 describes these relationships using an inverted pyramid, with the three transpositionally symmetrical scales on the top line. The middle line contains the nondiatonic seven-note scales, while the diatonic scale is at the bottom. Scales are connected by line segments if they can be linked by particularly efficient voice leading: either a "split" or "merge" described in the preceding paragraph, or one of the single-semitone voice leadings represented on Figure 4.6.1. The numbers above each line indicate the presence of multiple possibilities for that particular kind of voice leading. Thus the notation "←6" on the line connecting whole tone to acoustic indicates that there are six different ways to "split" a note of the whole-tone scale, creating six different acoustic collections. By contrast, "1→" points in the opposite direction, because for any particular acoustic scale there is only one way to "merge" its notes to create a whole-tone collection.

The table in Figure 4.6.4 shows how the nondiatonic seven-note collections can be used to connect transpositionally symmetrical scales. Every scale shares six of its seven notes with the octatonic scale to its left, and five of six notes with the whole-tone or hexatonic scale above it. (For example, G acoustic and D♭ acoustic share six notes with C♯-D octatonic and five notes with D♭ whole tone.) In addition, each seven-note scale shares six of its notes with some diatonic collection. Figure 4.6.5 tries to convey the same information graphically: the central core features the full cycle of 36 seven-note nondiatonic scales, each linked to its neighbors by single-semitone voice leading. On the outside of this central circle runs the diatonic circle of fifths; another can be found within the central circle. (Here, the double-headed arrows connect scales sharing six of their seven notes: the A harmonic major collection, for example, shares six notes with A diatonic; while the next collection along the non-diatonic circle, A harmonic minor, shares six notes with C diatonic.) The labels along the central circle indicate the nearest octatonic, whole-tone, and hexatonic scales. This exotic and almost alchemistical graph provides a way to visualize all the voice leadings we have been discussing. I encourage you to verify that Figures 4.6.1, 4.6.3, and 4.6.5 provide different perspectives on the same basic relationships.

Figure 4.6.3 Voice leadings among familiar scales. The transpositionally symmetrical scales are at the top. The harmonic and acoustic scales mediate between these and the diatonic. The numbers indicate how many scales of each type can be connected by the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

voice leadings described in Figures 4.6.1 and 4.6.2. Thus the whole-tone scale can be connected to six different acoustic scales by a "split," while every acoustic scale can be connected to only one whole-tone scale by a "merge."

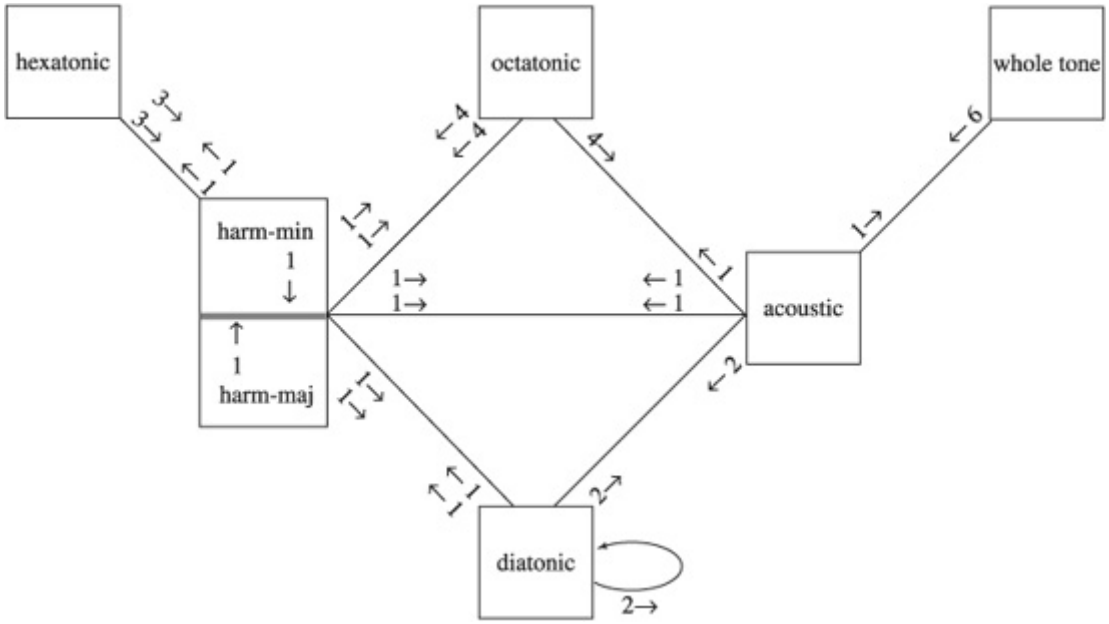


Figure 4.6.4 Each scale in the table shares six of its seven notes with the octatonic scale to its left, and contains five of the six notes in the whole-tone or hexatonic scale above it.

	C wt	D♭ wt	C-D♭ hex	C♯-D hex	D-E♭ hex	D♯-E hex
C-D♭ oct	C ac	E♭ ac	D♭ hm	B♭ hm	G hm	E hm
	F♯ ac	A ac	F HM	D HM	B HM	A♭ HM
C♯-D oct	E ac	D♭ ac	F hm	D hm	B hm	A♭ hm
	B♭ ac	G ac	A HM	F♯ HM	E♭ HM	C HM
D-E♭ oct	D ac	F ac	A hm	F♯ hm	E♭ hm	C hm
	A♭ ac	B ac	C♯ HM	B♭ HM	G HM	E HM

Figure 4.6.5 A geometrical representation of the voice leading between diatonic, acoustic, harmonic major/minor, whole-tone, hexatonic, and octatonic scales. Here "wt 1" and "wt 2" refer to the C♭ and D whole tone scales, respectively; "oct 1," "oct 2," and "oct 3" refer to the C♯-D, D-E♭, and D♯-E octatonic scales; and "hex 1," "hex 2," "hex 3," and "hex 4" refer to the C♯-D, D-E♭, D♯-E, and E-F hexatonic scales.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



4.7 TWO EXAMPLES

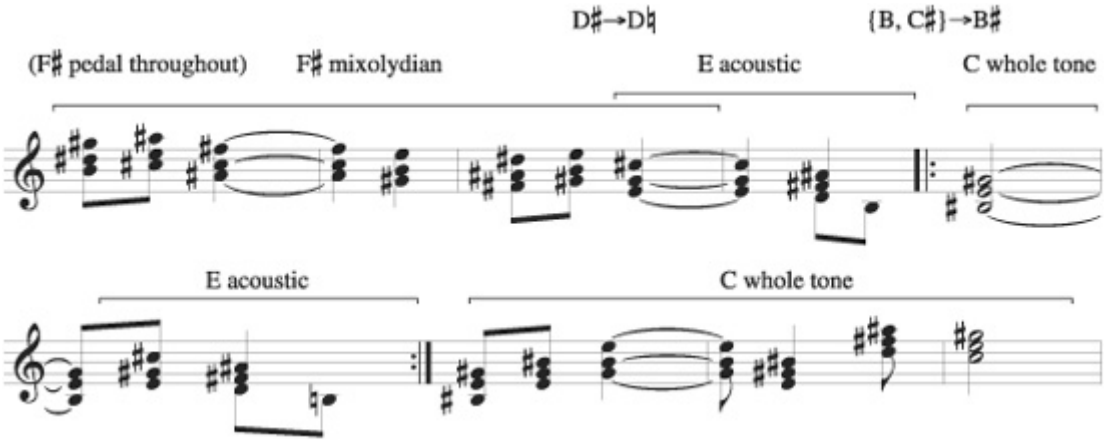
The previous section placed the acoustic and harmonic scales *between* the diatonic and the transpositionally symmetrical scales, since each shares six notes with some diatonic collection, while also being very close to some octatonic, whole-tone, and hexatonic collections. (This is represented by the fact that the harmonic and acoustic scales occupy the central line of the inverted pyramid in Figure 4.6.3.) And in fact, twentieth-century composers often use acoustic and harmonic collections to mediate between these different harmonic worlds. For example, at the end of the central section of "Mouvement," Debussy moves from F♯ mixolydian to C whole tone by way of the E acoustic scale (Figure 4.7.1). This acoustic collection is as close as possible to the other two, sharing five of the whole-tone scale's six notes, and six of the diatonic scale's seven notes. In this way, Debussy creates a smooth transition from familiar diatonic modality to the much more exotic world

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

of the whole-tone scale. This was one of his favorite compositional techniques, with similar progressions appearing repeatedly throughout his works.[27]

Figure 4.7.1 At the end of the B section of *Mouvement*, Debussy moves from F♯ mixolydian to the C whole tone by way of the F♯ mode of E acoustic.



For a more complicated example, consider "Sunlight Streaming in the Chamber," the first of Prokofiev's 1916 *Five Poems of Anna Akhmatova* (Figure 4.7.2). The music begins with a repeating figure that echoes the opening of Stravinsky's *Petrouchka*, composed just a few years earlier. I interpret the underlying harmony here as an E major triad, which can be connected to the following C major by semitonal voice leading. C major then initiates a stepwise descent through B minor, A major, G major, and F♯ major, each generating familiar scales. The music then shifts suddenly to C diatonic, moving toward a half cadence on an F♯ half-diminished seventh. The second half of the song begins by alternating between E minor (with added C) and A^9 chord (with G in the bass), vaguely suggesting ii–V in D major. The piece then shifts to D acoustic (A melodic minor) for six measures, ending with a turn to C-D♭ octa-tonic. This is the first appearance of the octatonic scale in the piece and it has a somewhat surprising effect, giving the impression of a question mark or raised eyebrow rather than an exclamation point.

Figure 4.7.3 graphs Prokofiev's scales using the two-dimensional scale lattice of Figure 3.11.8. The diatonic and acoustic collections form a compact region of the graph: the four diatonic collections (C, G, D, and A) are all adjacent on the zigzag of fifths, with the two acoustic collections (D and E) connected to these scales by singlesemitone voice leading.[28] Although the music does not always exploit these connections, there are a number of

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

transitions that do so. (For instance, the move from B melodic minor to A diatonic, or the progression from C diatonic to G diatonic to D diatonic.) Overall, there are six shifts involving single-semitone voice leading, five or six two-semitone shifts, one or two three-semitone shifts, and one four-semitone shift (Figure 4.7.4). It is interesting that the more dramatic shifts occur at the beginning and end of the song, with the smoother motions concentrated in the middle.

Figure 4.7.2 An outline of Prokofiev's Op. 27 No. 1.

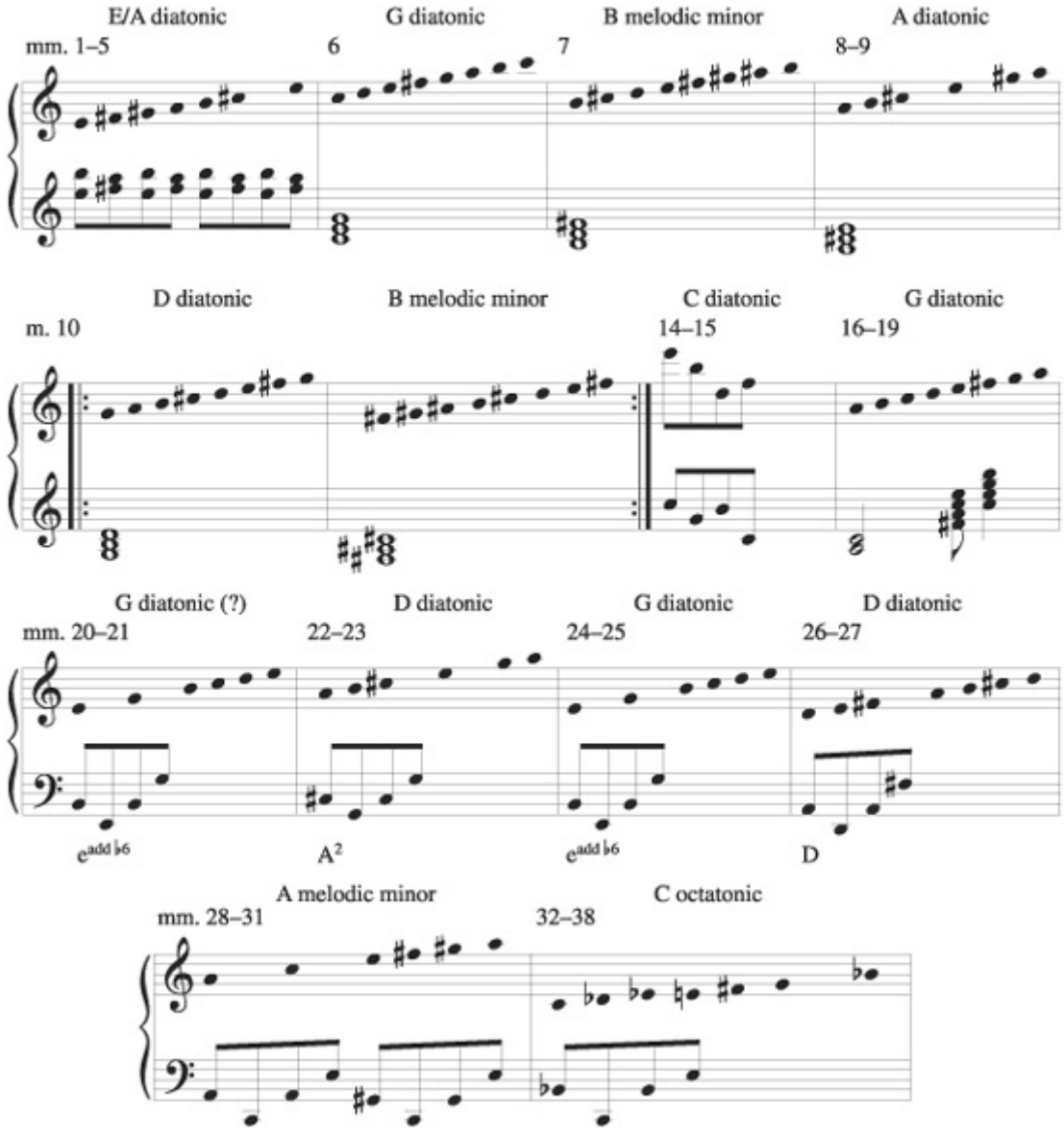


Figure 4.7.3 The scales in Prokofiev's piece form a connected region.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

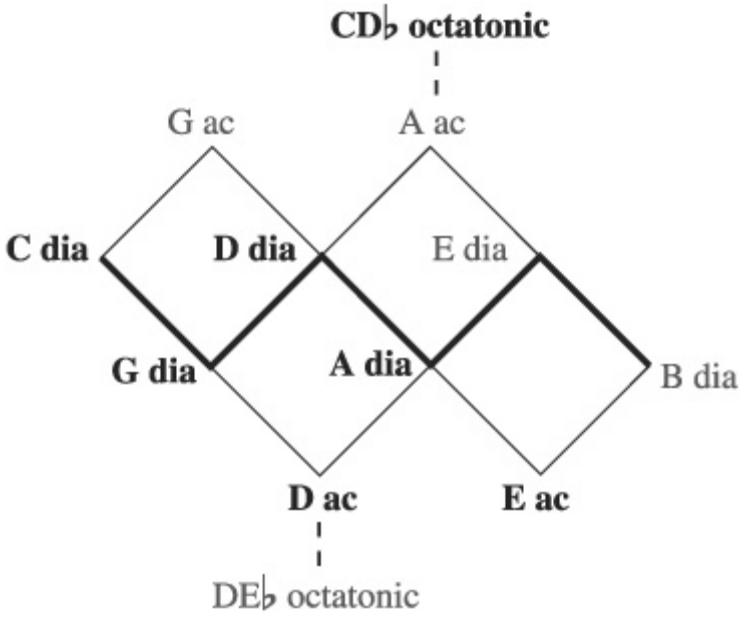


Figure 4.7.4 Prokofiev typically modulates by relatively efficient voice leading, with the most dramatic changes occurring at the beginning and end of the piece. Here, the numbers refer to the total number of semitonal shifts required to connect each scale to its successor: thus, the number "3" in the second line indicates that it takes 3 semitonal shifts (G→G♯, A→A♯, C→C♯) to change G diatonic into E acoustic.

E/A dia	2/3
G dia	3
E ac	1
A dia	1
D dia	2
E ac	2
D dia	2
E ac	2
C dia	1
G dia	1
D dia	1
G dia	1
D dia	2
D ac	4
CD♭ oct	

The final modulation deserves some comment. Had Prokofiev wanted to create a smooth transition from the antepenultimate D diatonic scale to the concluding CD♭ octatonic, he could have done so, since both scales share almost all their notes with A

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

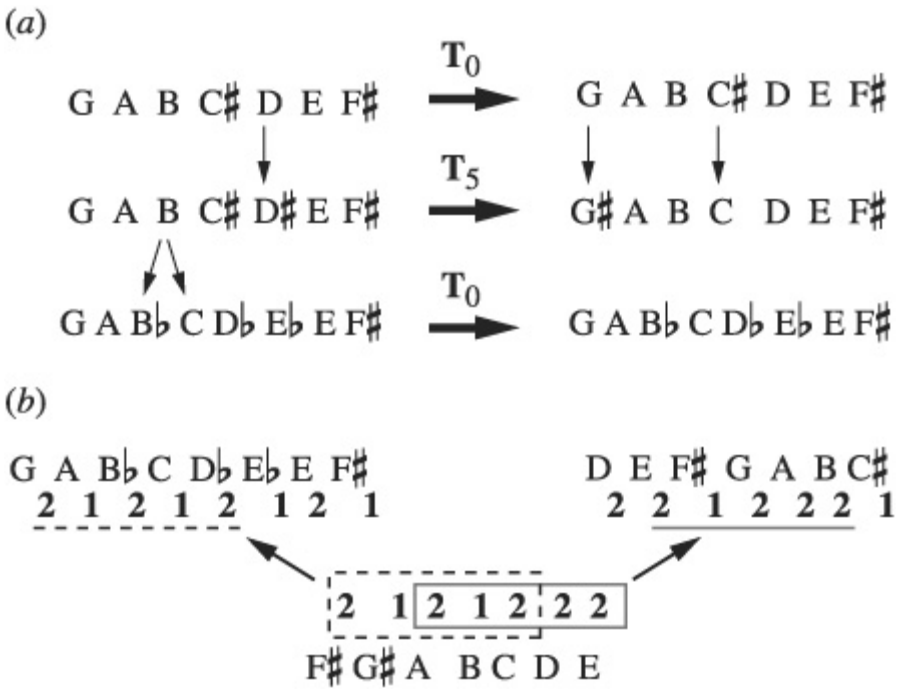
acoustic. (See Figure 4.7.3, where this is obvious.) However, the penultimate acoustic collection does *not* share a large number of notes with the final octatonic scale. (It does however share five notes with the preceding D diatonic.) Nevertheless, this D acoustic scale still has audible links to both the diatonic and octatonic scales, since it is "nearly transpositionally related" to both (§2.8). In other words, the mediation here is one of internal intervallic structure rather than particular common tones: Figure 4.7.5 shows that the acoustic scale shares the step-interval sequence 2-1-2-2-2 with the diatonic and the sequence 2-1-2-1-2 with the octatonic. Consequently, the piece articulates a gradual transition at the level of interval content, with the acoustic scale containing more tritones than the diatonic, but fewer than the octatonic, fewer fifths than the diatonic, but as many as the octatonic, and so on.[29]

Overall, Prokofiev's music suggests a somewhat loose and playful structure—the sense is of a relatively intuitive use of closely related scales, rather than a more logical or systematic exploration. Nevertheless, it is significant that he uses closely related scales that *can be* linked by efficient voice leading, and that he often modulates so as to highlight these connections. It is also relevant that his scalar vocabulary consists entirely of scales we have discussed, suggesting that our abstract, scale-theoretic considerations do indeed capture his actual concerns. Chapter 9 continues this line of argument, demonstrating that the techniques we have identified here are symptomatic of a much more general practice stretching from late nineteenth-century tonality to contemporary jazz and postminimalism.

Figure 4.7.5 In Prokofiev's piece the acoustic scale mediates between diatonic and octatonic in a somewhat abstract way. On the left side of (*a*) we see that the A acoustic scale can mediate between D diatonic and the C-D♭ octatonic, sharing almost all of its notes with both collections. However, the right side of (*a*) shows that Prokofiev presents the D acoustic scale instead. (*b*) When we consider intervallic structure, we see that the acoustic is similar to both diatonic and octatonic, sharing step patterns 2-1-2-1-2 with the octatonic and 2-1-2-2-2 with the diatonic.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.
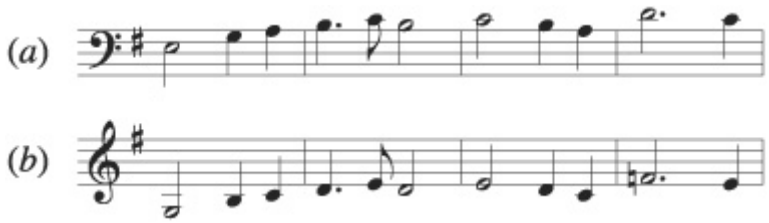
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



4.8 SCALAR AND INTERSCALAR TRANSPOSITION

Figure 4.8.1 shows two appearances of the subject of Shostakovich's Fugue in E minor, one in E aeolian, and the other in G mixolydian.[30] The two forms of the subject are related by a double process of transposition: a *scalar* transposition downward by diatonic step, from aeolian to mixolydian, combined with *chromatic* transposition upward by five semitones, from G diatonic to C diatonic (Figure 4.8.2). Though this combination of scalar and chromatic transposition can occasionally be found in earlier centuries, it plays a central role in twentieth-century tonal composition, particularly in the works of composers such as Debussy, Ravel, Stravinsky, Shostakovich, Steve Reich, and John Adams. By way of illustration, Figure 4.8.3 cites rehearsal 13 of Debussy's "Fêtes," where scalar transposition upward by two steps combines with chromatic transposition downward by three steps. The resulting voice leading shifts A♭ dorian into A♭ lydian, raising G♭, C♭, and D♭.

Figure 4.8.1 The subject in Shostakovich's Fugue in E minor, Op. 87 No. 4, as it appears at the opening of the piece and in m. 22.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 4.8.2 The two forms of the subject relate by scalar and chromatic transposition.
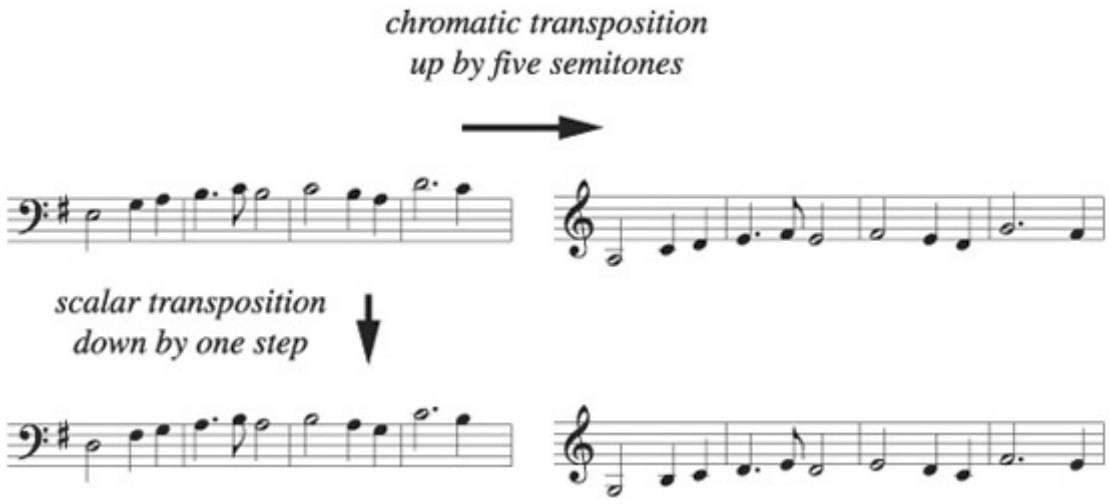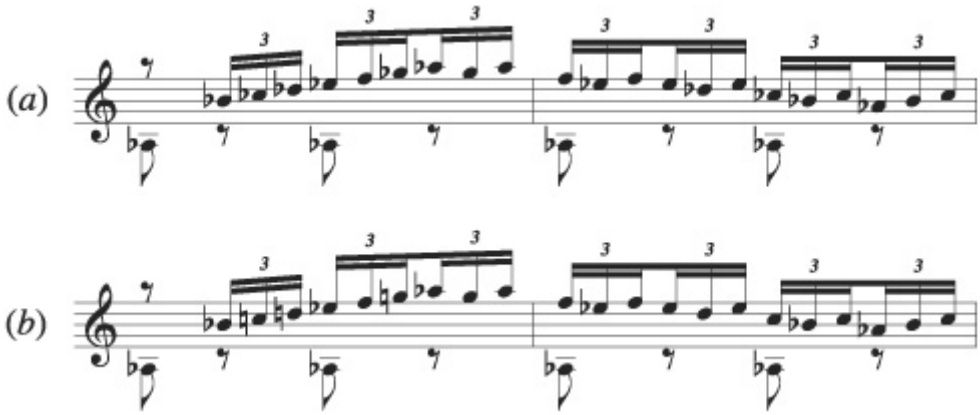


Figure 4.8.3 Rehearsal 13 of Debussy's "Fêtes" (*Nocturnes* II).



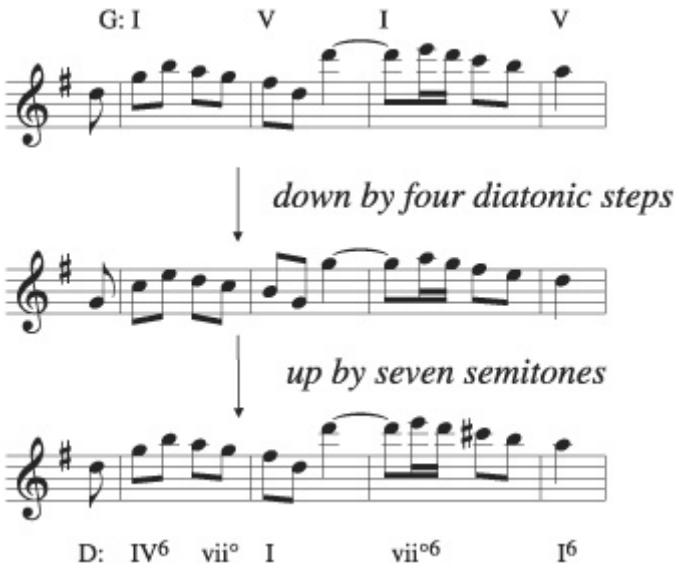
Note that these two kinds of transposition can sometimes combine to produce a simple change of accidentals, as in the opening of the Rondo from Clementi's Piano Sonata, Op. 25 No. 2 (Figure 4.8.4). The theme, which originally begins on scale degree 5 in G major, returns on scale degree 1 of D major, with C♯ replacing C♮. Though one could say that Clementi simply moved the note C♮ up by semitone, it is also possible to describe the relationship as combining two separate transpositions: scalar transposition *down* by four scale degrees followed by chromatic transposition *up* seven semitones. These two transpositions nearly cancel each other out, leaving behind the single-semitone shift C→C♯ as a residue. As we will see, there are a number of musical circumstances in which it is profitable to think in precisely this way.

Figure 4.8.4 The D-major theme in the Rondo from Clementi's Piano Sonata, Op. 25 No. 2, originally begins on scale degree 5 in G major, but returns in the transition with C♮

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

replaced by C♯. The two forms of the theme can be related by a combination of scalar and chromatic transposition.



If scalar transposition moves a musical pattern along a single scale, then a slightly more general process of "interscalar transposition" moves a pattern *from one scale to another.*[31] Figure 4.8.5 tracks the main motive of Bach's D minor invention as it moves from D harmonic minor scale to F major and D melodic minor ascending (G acoustic). Although the scale changes, each form of the motive uses precisely the same set of scalar intervals. In fact, if we label the notes of D harmonic minor and F major in the standard way, we can use the very same scale degree numbers to describe the first two passages in the figure. In this sense, we can say that they are related by *zero-step interscalar transposition*. Similarly, if we label the degrees of D melodic minor starting from D, then Figure 4.8.5a relates to (*c*) by *two-step* interscalar transposition—meaning that each note of Figure 4.8.5c is two scale degrees higher than the corresponding note of Figure 4.8.5a. Of course, the number of steps involved in an interscalar transposition depends on our arbitrary choice of the first scale degree. What is important is the fact that two passages are related by interscalar transposition, not the particular label we apply to the relationship.

Figure 4.8.5 Three forms of the motive in Bach's D minor two-part invention. Scale degrees are shown above each example.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
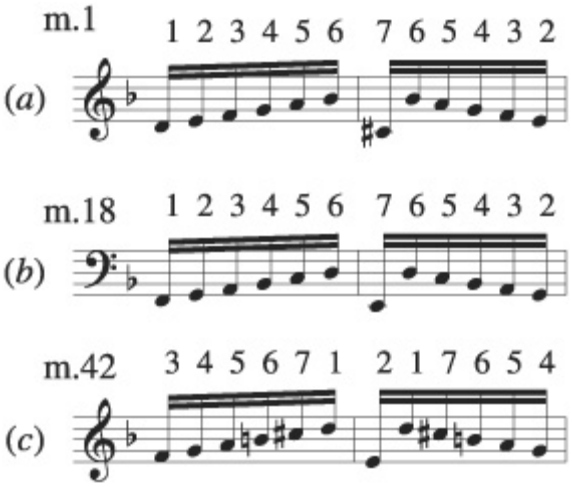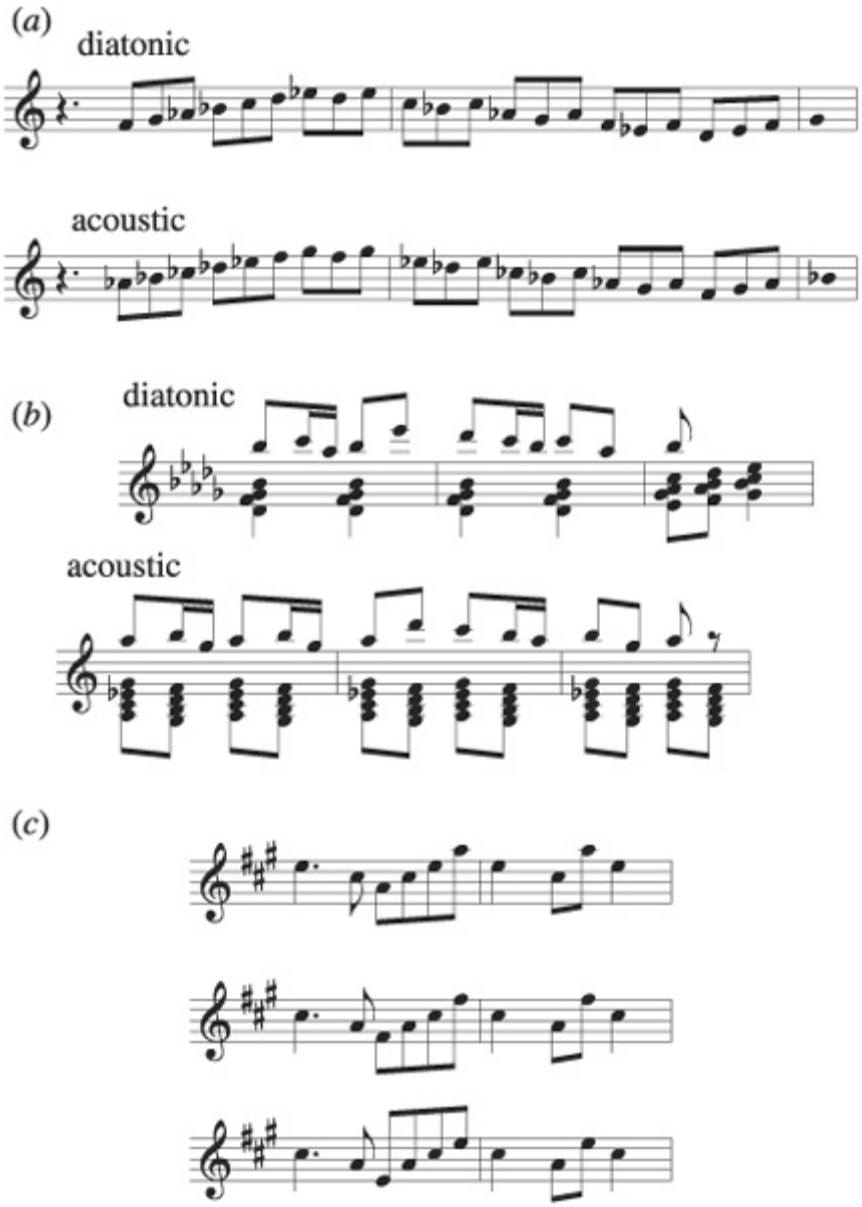


Figure 4.8.6 identifies interscalar transpositions in Debussy, Stravinsky, and Shostakovich. The first two examples involve interscalar transposition from the diatonic scale to the acoustic. The third is interesting insofar as its "scales" are simple triads. (This is the gimmick of Shostakovich's A major fugue; it is a "bugle fugue" that treats triads as if they were scales.) Thus in Figure 4.8.6c, the main theme originally appears in the A major triad (top line); when it returns in minor, at m. 21, it has moved to a new three-note scale, F♯ minor. This is interscalar transposition by zero steps (middle line). Eventually, in the stretto, the theme is transposed diatonically along the A major scale (bottom line). Thus the bottom two lines of Figure 4.8.6c are connected by descending-step interscalar transposition.[32]

Figure 4.8.6 Interscalar transposition in the opening of Debussy's "Fêtes" (*a*), Stravinsky's *Rite of Spring* (*b*), and Shostakovich's A major fugue (*c*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
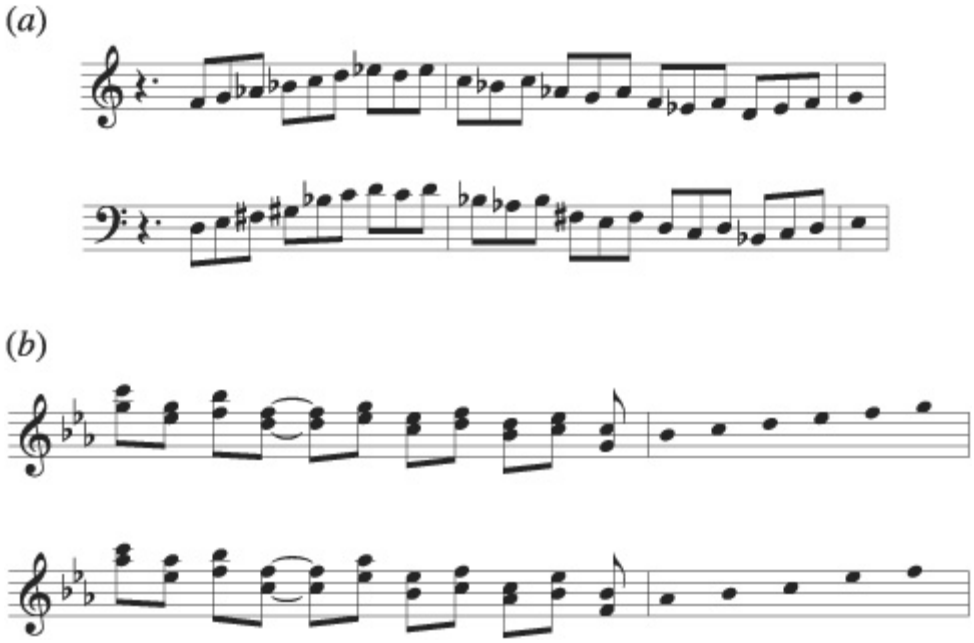


One terminological point: the previous examples describe what might be called "interscalar transpositions in pitch-class space," since they link octave-repeating scales with the same number of notes. There is an even more general musical transformation that relates scales of different sizes, which we can call "interscalar transposition in *pitch space*." Figure 4.8.7 presents examples from Debussy and Steve Reich. These transformations introduce octave-dependent changes into the music: for example, the D5 in the top line of Figure 4.8.7a gets mapped to C, whereas the D4 gets mapped to B♭. Consequently, we cannot speak of "what happens to the pitch class D"; instead, we need to speak about what happens to specific pitches in specific registers. By contrast, when the scales have the same number of notes, then all pitches in a pitch class are transformed in exactly the same way. (For example, the interscalar transposition in Figure 4.8.6a moves every D up by three semitones.) This in turn means that these interscalar transpositions can be modeled as

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

voice leadings between pitch-class sets. This is precisely why they will be important in what follows.

Figure 4.8.7 (*a*) "Fêtes" presents the same pattern of scalar intervals in the seven-note dorian mode and the six-note whole-tone scale. (*b*) Steve Reich's *Variations for Winds, Strings, and Keyboards* presents the same pattern of scalar intervals in the six-note diatonic hexachord and the five-note pentatonic scale.



4.9 INTERSCALAR TRANSPOSITION AND VOICE LEADING

In the final sections of this chapter I want to switch gears somewhat, exploring the connections between scale theory and voice leading. To prepare for this discussion we need to make a subtle conceptual shift: instead of thinking of chords as being embedded within larger scales, we will start to think of them as *scales unto themselves*. In other words, we will think of the C major chord as being a "scale" whose three scale degrees are C, E, and G, respectively. (This is just what we did in discussing Figure 4.8.6c, Shostakovich's A major fugue.) This will allow us to represent the voice leading (C, E, G)→(B, E, G) as an interscalar transposition that sends the first scale degree of the C major triad to the third scale degree of the E minor triad (C→B), the second scale degree to the first (E→E), and the third scale degree to the second (G→G). Once we start thinking in this way, we will see that there is a deep connection between the notion of an "interscalar transposition" and the general problem of identifying efficient voice leadings between chords.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

To understand why, notice that scalar and interscalar transpositions represent a very special kind of voice leading: not only are they free of voice crossings, but they remain so *no matter how their voices are distributed in pitch space*. (For this reason, I will say that they are "strongly crossing free.") For example, the first voice leading in Figure 4.9.1 has no crossings no matter how we change the register of its individual voices. By contrast, the second voice leading is only *weakly* crossing free, since transposing its lowest voice up by an octave creates a crossing between the bottom two voices. Given a strongly crossing-free voice leading, we can always arrange its voices so that each chord spans less than an octave, with ascending steps in one chord being sent to ascending steps in the other (Figure 4.9.2).[33] But this in turn implies that the voice leading is a scalar or interscalar transposition, since it sends any scalar interval in the first chord to the same scalar interval in the second.

Figure 4.9.1 The voice leading in (*a*) is strongly crossing free: no matter what octave its voices are in, there will never be crossing. The voice leading in (*b*) is crossing free but not strongly so, since a crossing is created when the lowest voice moves up by octave (*c*).
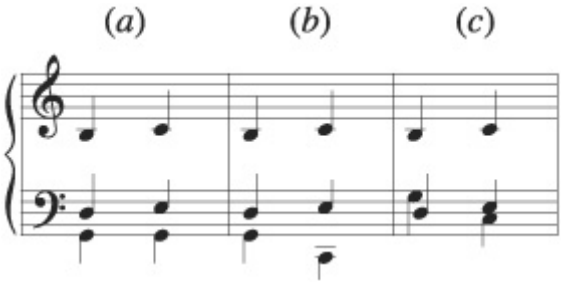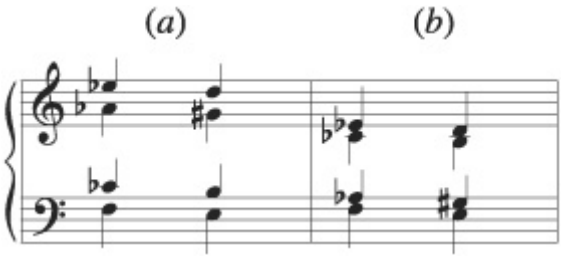


Figure 4.9.2 In a strongly crossing-free voice leading, the voices can be transposed by octave so that each chord is in "close registral position," spanning less than an octave. Ascending steps in one collection are sent to ascending steps in the other.
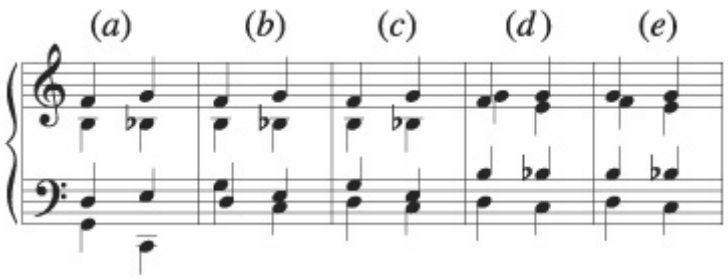


The connection to voice leading lies precisely here: in §2.7, we saw that, for any "reasonable" measure of voice-leading size, removing voice crossings never makes the voice leading larger. (See also Appendix A.) This means that we can take any voice leading

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

and, by repeatedly removing voice crossings and changing the octave in which voices appear, eventually produce a scalar or interscalar transposition. Since octave shifts don't affect the voice leading, and since removing crossings never makes it larger, the final voice leading is guaranteed to be at least as small as the original (Figure 4.9.3). Consequently, there is always a maximally efficient voice leading between any two chords that is a scalar or interscalar transposition.[34] It follows that the concept of interscalar transposition will be relevant wherever efficient voice leading is important.

Figure 4.9.3 Repeatedly octave-transposing and removing crossings will eventually produce a strongly crossing-free voice leading. (*a*) Transposing the bass voice up by octave produces a crossing (*b*). Switching the bass and tenor in the first chord removes the crossing (*c*). Transposing the tenor up by octave produces another crossing (*d*). Switching soprano and alto in the first chord removes this crossing (*e*), yielding a strongly crossing-free voice leading. Since removing crossings never makes a voice leading larger, the voice leading in (*e*) is at least as small as that in (*a*).



Let me illustrate with a few concrete examples. Figure 4.9.4 identifies the minimal voice leading between C major and A minor triads. If we consider each triad to be a scale, with root, third, and fifth being its first, second, and third scale degrees, then the voice leading is an interscalar transposition by one ascending step.[35] Similarly, the most efficient voice leading between the C major and C minor triads is an interscalar transposition by zero steps (root to root). The figure also shows that the most efficient voice leading between $C^{\emptyset 7}$ and $E\flat^{7}$ chords is an interscalar transposition by three steps (root to seventh), and the most efficient voice leading between F diatonic and G acoustic scales is an interscalar transposition by one descending step (scale degree one to scale degree seven).[36]

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Geometrically, interscalar transpositions can be represented as collections of crossing-free paths between two concentric circles, as in Figure 4.9.5. Appendix A shows that these voice leadings correspond to paths that do not "bounce off" the mirror boundaries of higher-dimensional chord space.

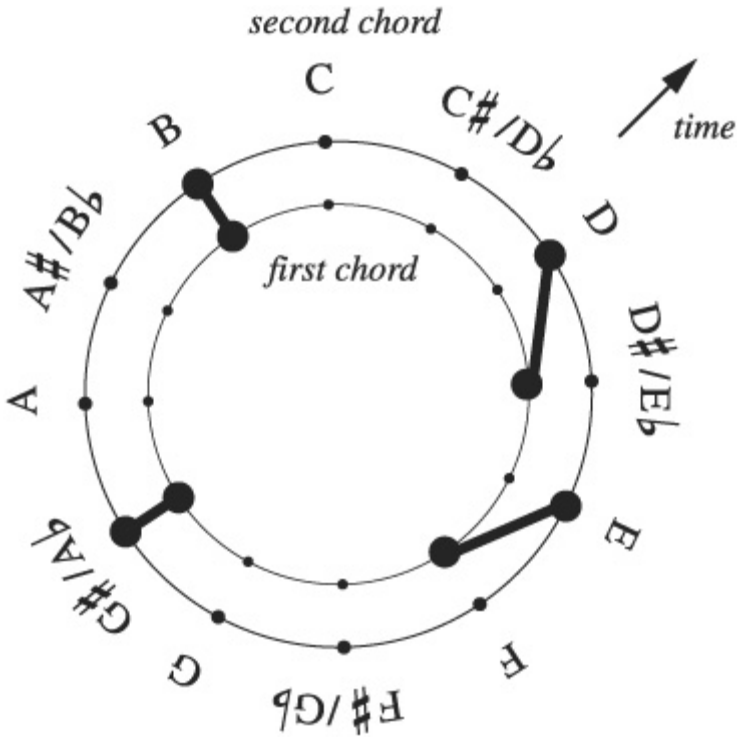
Figure 4.9.4 The minimal voice leading between C major and A minor triads (*a*), between C major and C minor triads (*b*), between C$^{\text{ø}7}$ and E♭^{7} chords (*c*), and between F diatonic and G acoustic scales (*d*). All are interscalar transpositions.
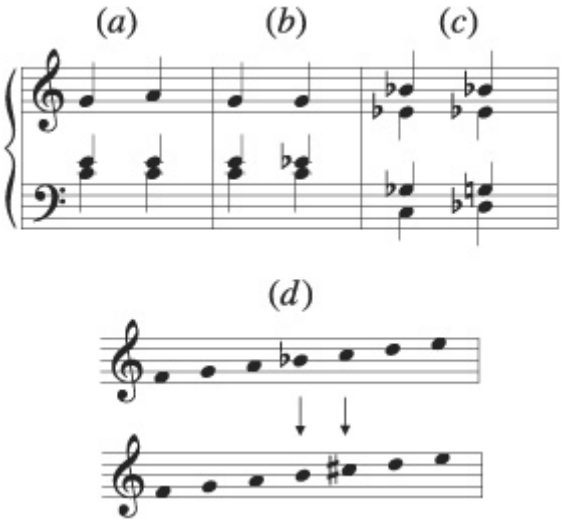


Figure 4.9.5 Strongly crossing-free voice leadings can be represented geometrically by non-intersecting paths connecting two concentric circles. Here the music progresses radially outward, from the inner circle to the outer. The voice leading (F, G♯, B, D♯)→(E, G♯, B, D) holds G♯ and B constant, moving F and D♯ down by semitone.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

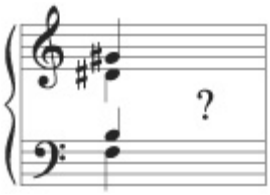


Throughout this book I have emphasized that Western music involves the simultaneous satisfaction of two independent constraints—a vertical constraint that requires chords to be structurally similar, and a horizontal constraint that dictates that they be connected by efficient voice leadings. Clearly, to satisfy these constraints, a composer must be able to *find* the efficient voice leadings between arbitrary chords. This need is all the more pressing when we reflect that composers regularly ask questions that require sorting through a large number of possibilities. Suppose, for example, you write the chord in Figure 4.9.6, and decide to move it to some nearby dominant seventh; to solve this problem, it is (in principle) necessary to search 288 different voice leadings to all twelve dominant sevenths. It is rather remarkable, therefore, that composers, theorists, and even beginning music students manage to find efficient voice leadings so quickly, and with so little apparent effort.

Figure 4.9.6 Suppose a composer decides to connect this F half-diminished seventh chord to some dominant seventh chord by maximally efficient voice leading. There are almost three hundred possible voice leadings to consider. Yet musicians manage to solve problems like this very quickly.
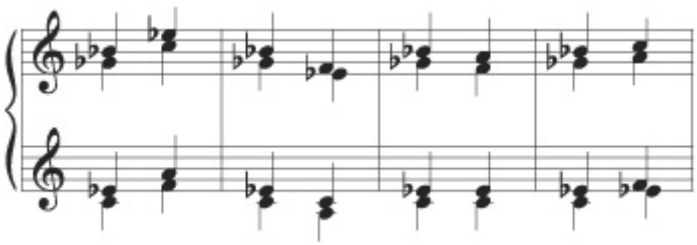
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Our discussion helps explain how this can be possible. Rather than searching *all* the different voice leadings between chords, musicians need consider only the small number of interscalar transpositions. For example, if you want to find a minimal voice leading between the C half-diminished and F dominant seventh chords, it is sufficient to consider the four interscalar transpositions shown in Figure 4.9.7—mapping the root of the half-diminished to the root, third, fifth, and seventh of the dominant seventh chord, respectively. This reduces the number of potential voice leadings by a factor of six, from twenty-four down to four. (For larger chords the reduction of effort is even more dramatic.) Thus when harmony teachers enjoin their students to avoid voice crossings, they are actually achieving two distinct aims: they are encouraging a kind of composition in which the voices remain registrally separate, and hence easy to distinguish aurally, while also drastically reducing the "search space" that students must consider when looking for efficient voice leadings. This last point, though rarely discussed, is arguably central to the whole enterprise of Western composition: for if "crossed" voice leadings could be smaller than their uncrossed counterparts, it would be enormously more difficult to combine harmonic consistency with efficient voice leading—simply because it would be very hard to sort through all the voice-leading possibilities.

Figure 4.9.7 In order to find the minimal voice leading between two chords, it is necessary to check only the interscalar transpositions between them. Here, the first interscalar transposition maps the root of $C^{ø7}$ to the root of F^{7}, the second maps root to third, the third maps root to fifth, and the fourth maps root to seventh. Having chosen a destination for the root, the rest of the voice leading is completely determined by the fact that it is an interscalar transposition.
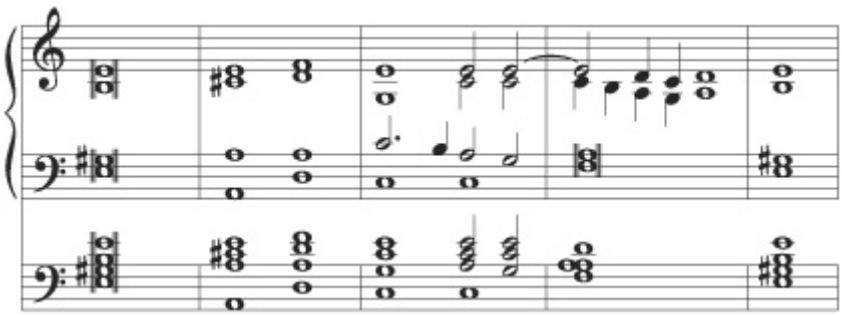


Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

It follows that there are both *perceptual* and *conceptual* reasons to expect that voice crossings should be infrequent, since crossings make life difficult for both listener and composer. And when we look at actual music we find that voice crossings are indeed rare, occurring only 5% of the time even in the most polyphonic of styles.[37] Furthermore, when they *do* occur it is often profitable to interpret them as embellishments of more basic, crossing-free paradigms. For example, the opening of Palestrina's motet "Adoramus Te," shown in Figure 4.9.8, can plausibly be said to be based on crossing-free voice leadings in the bottom staff—with the crossing not only serving to add melodic interest, but also allowing Palestrina to evade the parallel fifths and octaves that would otherwise occur.[38] (Notice that in this passage the top three voices typically articulate voice leadings between complete triads, with the bass adding doublings—a technique that we will explore later.) Chapters 6 and 7 will generalize this observation by showing that a large majority of voice leadings, in a large range of music, can be understood as embellishments of a few basic templates, all of which are strongly crossing free. This in turn suggests that composers really do privilege the crossing-free voice leadings, perhaps thinking of voice crossings as surface-level embellishments to be used only on special occasions.

Figure 4.9.8 The opening phrase of Palestrina's "Adoramus Te" can be interpreted as embellishing a fundamentally crossing-free template.



One word of caution: for simplicity, I have been considering voice leadings without doublings, in which the number of voices is equal to the number of notes in each of the two chords. However, it sometimes happens that a minimal voice leading maps multiple notes in the one chord to a single note in the other. For example, the first voice leading in Figure 4.9.9 is smaller than any of the four-voice voice leadings between the C and E major seventh chords. This, clearly, is not an interscalar transposition.[39] Fortunately, in many practical applications, we can ignore this complication, restricting our attention to

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

voice leadings without doublings. (Figure 4.9.10 demonstrates that, for nearly even three- and four-note chords, maximally efficient voice leadings are almost always doubling free.) In the grand scheme of things, this is quite fortunate, for otherwise the composer's task of finding efficient voice leadings would be considerably more difficult than it already is.

Figure 4.9.9 The five-voice voice leading on the left is smaller than any of the four-voice alternatives. Here, three voices move by one semitone, whereas the smallest four-voice alternative moves one voice by three semitones and one voice by one semitone. Voice leadings containing doublings cannot be identified using the techniques discussed in this chapter.
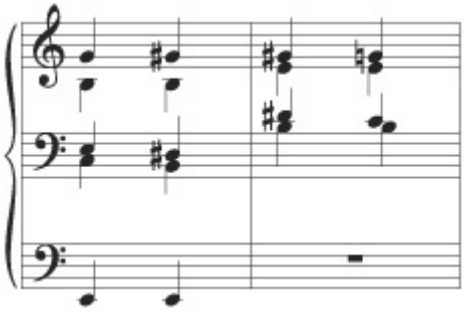


Figure 4.9.10 The probability that the minimal voice leading between two randomly chosen *n*-note chords will involve "doublings," as in Figure 4.9.9. The first column represents the likelihood of doublings when the chords are selected randomly from all pairs of *n*-note chords, while the second represents the likelihood when chords are restricted to the four most even *n*-note chords. Nearly even chords with four or fewer notes thus require doublings only rarely.

	All Chords	Nearly Even
2 notes	0%	**0%**
3 notes	10%	**1%**
4 notes	29%	**2%**
5 notes	48%	8%
6 notes	61%	19%
7 notes	69%	14%
8 notes	73%	11%

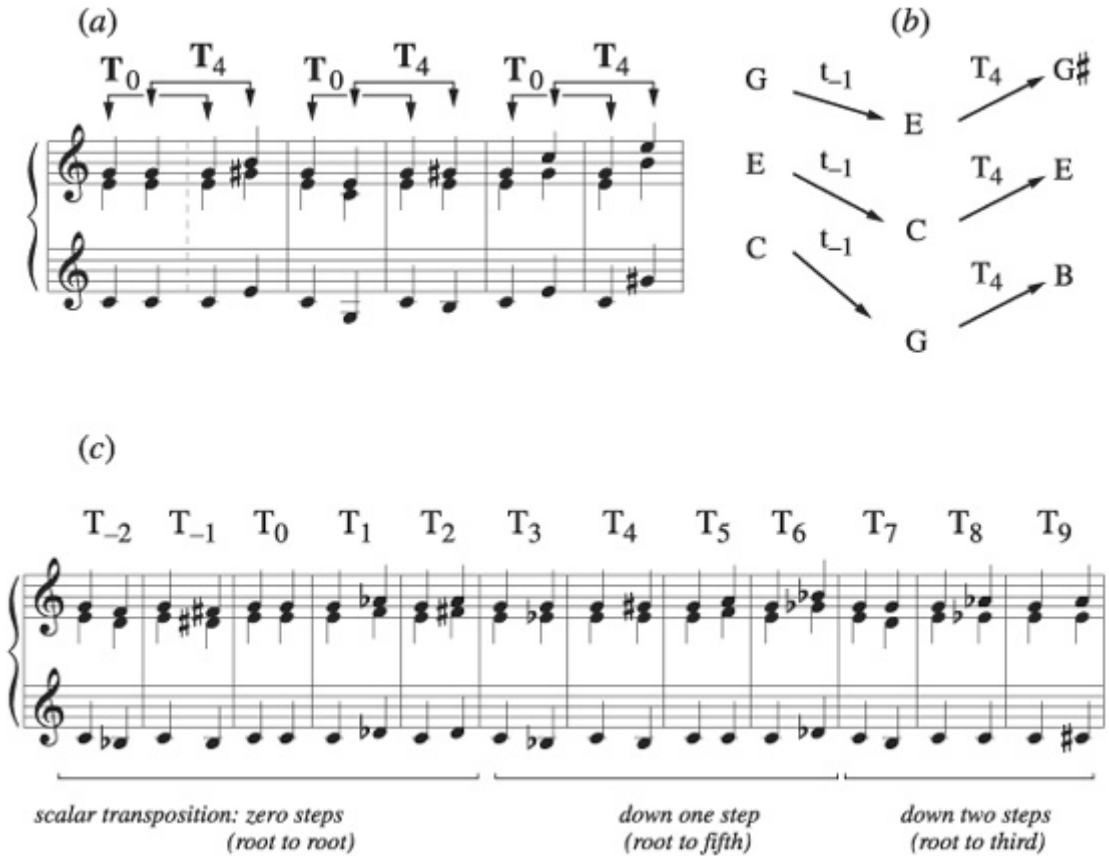
4.10 COMBINING INTERSCALAR AND CHROMATIC TRANSPOSITIONS

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Let me end by showing how scale theory can be used to model the intuitive knowledge possessed by sophisticated tonal composers. To begin, note that the interscalar transpositions linking any transpositions of the same two chord types will always be related by individual transposition. For instance, the scalar transpositions linking the C major triad to itself are individually transpositionally related to the interscalar transpositions linking the C major to the E major triad, and indeed to those connecting any other major triads (Figure 4.10.1). This is because transposition can never introduce voice crossings into a voice leading, even when applied individually.

Figure 4.10.1 (*a*) The scalar and/or interscalar transpositions between any transpositions of the same two chord types are always individually T-related. (*b*) The minimal voice leading between C and E major triads can therefore be analyzed as the combination of a one-step descending scalar transposition with a four-semitone ascending chromatic transposition. (*c*) More generally, the minimal (three-voice) voice leading between any major triads can be depicted as combining chromatic and scalar transposition. As the chromatic transposition increases, the descending scalar transposition increases as well, so that the two forms of transposition cancel out.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

From this it follows that we can decompose any strongly crossing-free voice leading from C to E major into two parts: a scalar transposition that moves each note down by some number of scale steps (relative to the C major triad) and a chromatic transposition that moves each note up by some number of semitones. This voice leading will be efficient when these two components nearly cancel out—as, for instance, when a one-step descending scalar transposition nearly neutralizes the four-semitone ascending chromatic transposition (Figure 4.10.1). In fact, we can produce a similar decomposition of any strongly crossing-free voice leading between *any* two major triads. Again, the voice leadings will be efficient when the two transpositions combine to leave each voice roughly where it was: thus in Figure 4.10.1c the descending scalar transposition increases as the ascending chromatic transposition does.

Figure 4.10.2 uses this idea to organize the efficient voice-leading possibilities between half-diminished and dominant seventh chords. Here, we represent voice leadings between half-diminished and dominant sevenths as combining an *interscalar* transposition (from C half-diminished to C dominant seventh) with a chromatic transposition to some other dominant seventh. Again, there is an inverse relationship between the two transpositions, so that one counteracts the other. Thus we see that *efficient voice leadings occur when scalar or interscalar transpositions neutralize the effects of chromatic transpositions*.

The upshot is that we can provide a surprising and nontrivial answer to what might otherwise seem like a hopeless question: what is it that a composer knows, when she knows all the most efficient voice leadings from one type of chord to another? For example, what does a composer know when she can easily identify the most efficient path between *any* particular half-diminished and dominant seventh chords? Our answer is: *she knows how to combine interscalar and chromatic transpositions*. That is, in the particular case of half-diminished and dominant sevenths, she knows how to combine the four templates in Figure 4.10.2a with the various chromatic transpositions. Thus a seemingly complicated musical skill—knowing *all* the most efficient voice leadings between two chord types—reduces to a much simpler kind of knowledge, knowing how to combine two familiar kinds of transposition.

In the second half of the book, I show how this idea allows us to understand chromatic music "from the inside," revealing some of the remarkable ways in which nineteenth-century composers explored the contrapuntal possibilities available to them. For now, let

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

me simply offer a few hints about what is to come. Figure 4.10.3 analyzes the first four resolutions of the half-diminished seventh chord in the prelude to Wagner's *Tristan*, showing that the first two involve interscalar transposition by zero steps, whereas the last two involve interscalar transposition by ascending step. Besides helping us understand these relationships, scale theory can prompt us to ask new questions: for example, we might find ourselves wondering whether Wagner uses the other two interscalar transpositions in his opera, or whether he ever substitutes one scalar transposition for another. Similarly, Figure 4.10.4 contains a series of progressions that resolve a seventh chord into a triad. Scale theory can show us that the first two voice leadings are closely related, since they map the root of the seventh chord to the root of the triad, whereas the third maps the root to the fifth.[40] Finally, consider the situation of a composer who wants to identify efficient (three-voice) voice leadings from major to minor triads. Having absorbed the ideas in this chapter, the composer will see that there are really just three basic possibilities: once the destination of the root is chosen, the rest of the interscalar transposition is determined. Each possibility gives rise to a family of voice leadings, all related by individual transposition and hence sharing the same basic voice-leading structure.

Figure 4.10.2 (*a*) The four interscalar transpositions from the C half-diminished seventh chord to the C dominant seventh. (*b*) The minimal voice leading between the C half-diminished and F dominant seventh chords combines the third of these interscalar transpositions with chromatic transposition upward by five semitones. (*c*) The minimal (four-voice) voice leading between any half-diminished and dominant seventh chords combines one of these interscalar transpositions with a chromatic transposition. Again, as the chromatic transposition increases, the descending interscalar transposition increases as well, so that the two forms of transposition cancel out.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

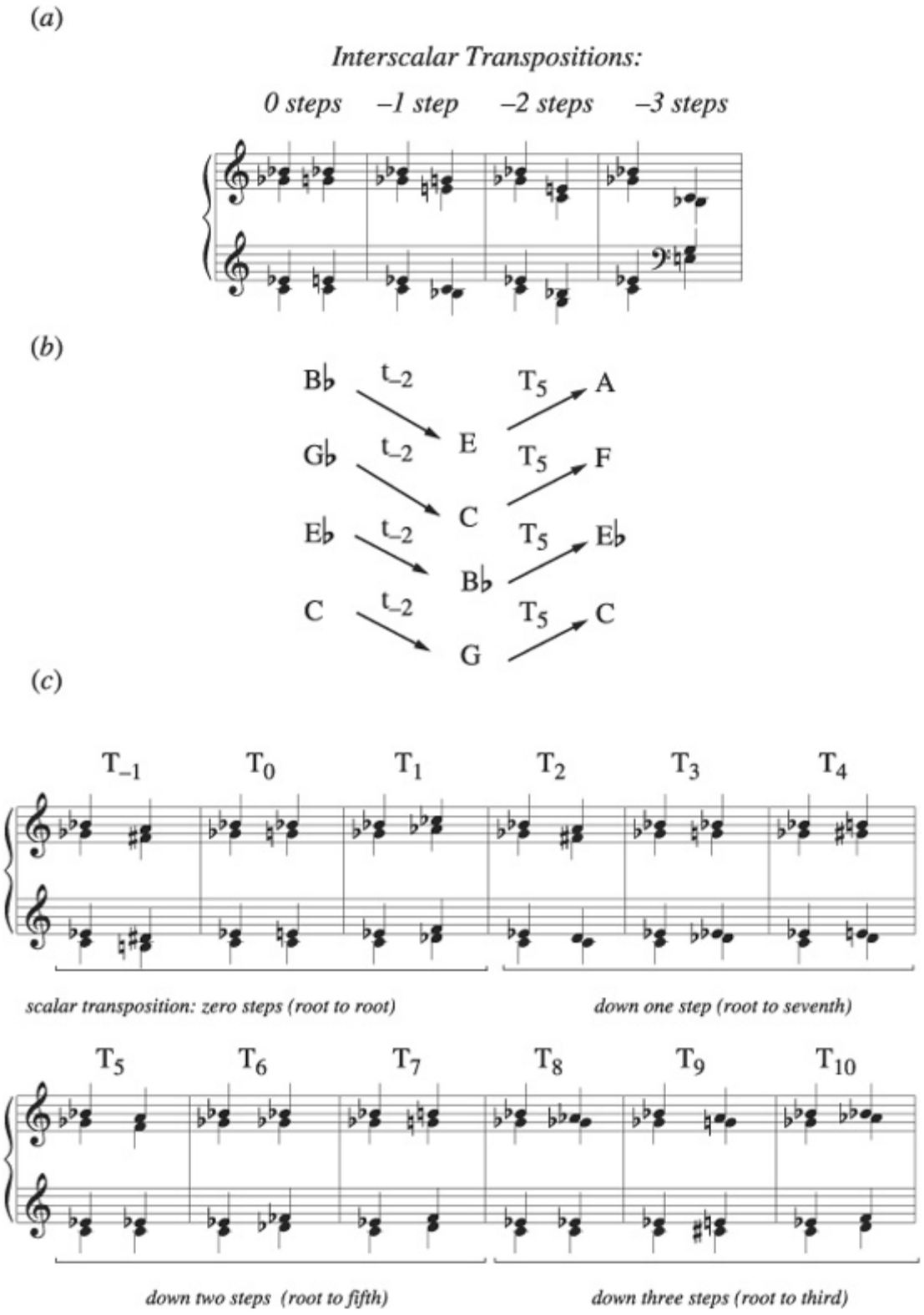


Figure 4.10.3 The first four resolutions of the half-diminished seventh chord in Wagner's *Tristan*. The first two map root to root, while the second two map root to third.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

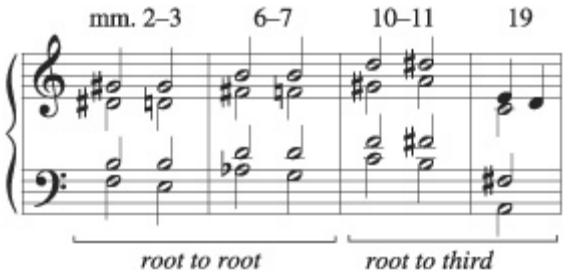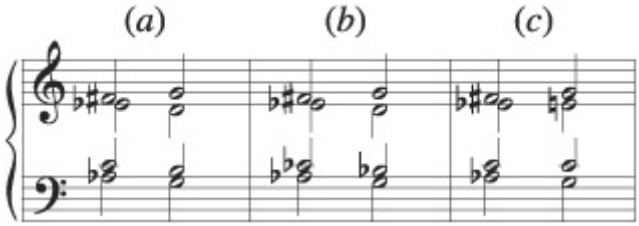


Figure 4.10.4 Three chromatic voice leadings. The first two map the root of the first chord to the root of the second, and can be considered interscalar transpositions from a seventh chord to a triad with doubled root. The last maps the root of the first chord to the fifth of the second.



The broader moral is that there is indeed a close connection between chord concepts and scale concepts. Fundamentally, a scale is a large chord, and a chord is just a small scale: both participate in efficient voice leadings, and both can be represented using the same basic geometries; composers develop musical motifs by transposing them along familiar chords, as if chords were just very small scales (Figs. 4.3.1b and 4.8.6c); and efficient voice leading frequently involves interscalar transposition or strongly crossing-free voice leadings. In fact, it even turns out that there is a close analogy between the idea of decomposing voice leadings into scalar and chromatic transpositions, and the idea of analyzing them into pure parallel and pure contrary components. Thus there are significant theoretical advantages to adopting a unified perspective that treats chords and scales similarly.

Those of you who are interested in pursuing this idea in greater technical detail should consult Appendix C, while those who are impatient for analysis can instead turn to Chapters 8 and 9. Everyone else is encouraged to proceed to Chapter 5 in an orderly fashion.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

CHAPTER 5

Macroharmony and Centricity

Having discussed harmony, counterpoint, and acoustic consonance, we'll now turn to macroharmony and centricity, the last of the five features. First, we'll explore the ways in which composers might combine harmonic and macroharmonic consistency. Then we'll develop two analytical tools for quantifying macroharmony: *pitch-class circulation graphs*, which record how fast a piece cycles through the pitch classes, and *global macroharmonic profiles*, which represent the relative proportion of large collections in a piece. We'll then introduce *pitch-class profiles* to describe both "local" centricity (or rootedness) and "global" centricity (or tonicity). Together, these tools amount to a "generalized theory of keys" allowing us to conceptualize the possibilities between complete atonality and traditional scale-based tonality.

5.1 MACROHARMONY

When we think about harmony, we automatically think about chords. In fact, we are so fixated on chords that we sometimes forget they tell only part of the story. To counteract this tendency, Figure 5.1.1 uses the same chords to construct two very different sequences: the first, containing twelve triads from the C diatonic scale followed by twelve triads from the F♯ diatonic scale, is placid and restful; the second, alternating between C and F♯ diatonic scales, is considerably more angular and energetic. The difference suggests that musical experience is strongly colored by what we have heard recently: after a number of white notes, a C diatonic triad will sound relatively consonant, while after a sequence of black notes, it will sound more jarring and out of place. It is as if previously heard notes linger in our memory, mixing with what we are currently hearing to create a harmonic penumbra—a "macroharmony" extending beyond the boundaries of the temporal instant.

In thinking about macroharmony we need to ask at least four questions:
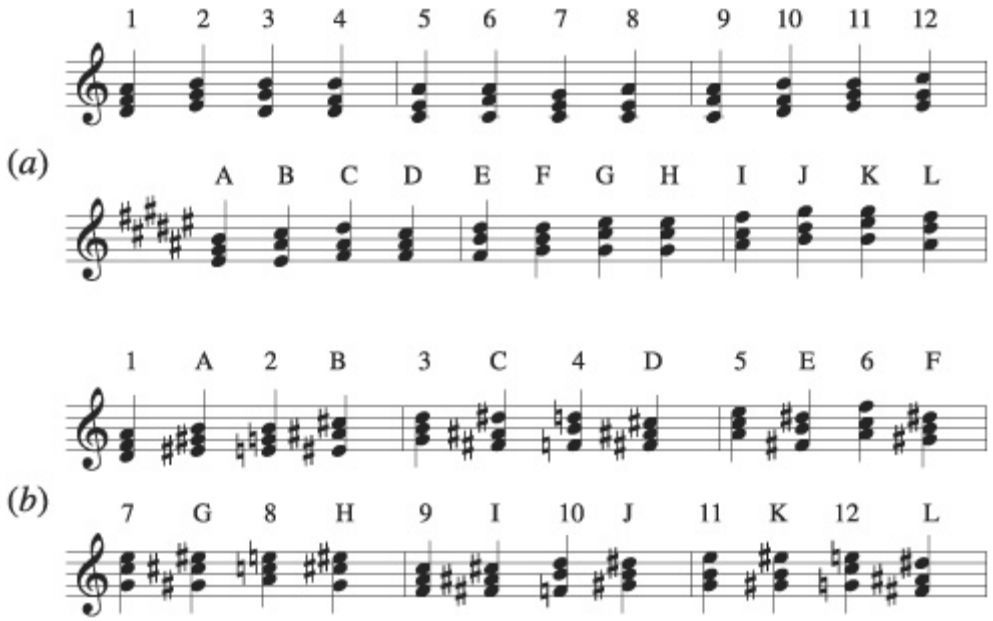
1. Does the music articulate identifiable macroharmonies other than the total chromatic?

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
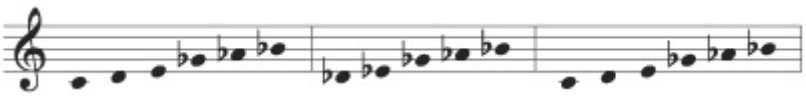
2. How fast do these macroharmonies change?
3. Are the various macroharmonies in the piece structurally similar—that is, related by transposition or nearly so?
4. Are the macroharmonies consonant or dissonant?

Figure 5.1.1 Two sequences containing the same chords, but in different order. The first sounds considerably less chromatic than the second.



From this point of view, our two sequences are very different. The first is both *macroharmonically consistent* (which is to say that it articulates a pair of diatonic collections, related by transposition) and *macroharmonically consonant* (since those collections are themselves relatively consonant). The second is macroharmonically consistent only in a trivial sense, since it cycles through the complete chromatic collection every two measures or so. More important, it is *macroharmonically dissonant*, because the chromatic scale is itself fairly dissonant. As a result, the second sequence sounds considerably less "tonal" than the first, even though they use exactly the same triads.

Figure 5.1.2 Each section of Debussy's "Voiles" uses a different scale. The switch from whole tone to pentatonic to whole tone moves from greater dissonance to greater consonance and back again.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.
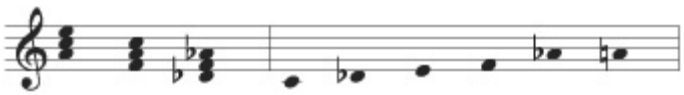
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Note that our questions echo those we might ask about ordinary harmonies: macroharmonies, like harmonies, can be consonant or dissonant, and sequences of macroharmonies can be harmonically consistent just as chord progressions can. Thus Figure 5.1.2 exhibits a *macroharmonic* transition from mild dissonance to greater consonance and back, somewhat akin to a classical V^7–I–V^7 progression. (To say this is just to repeat the basic point that traditional Western music uses similar techniques on different time scales.) In fact, in some twentieth-century music macroharmony becomes the primary bearer of harmonic significance. Mechanical, repetitive, or random chord changes can create a sonic "wash" in which changes in macro-harmony become more salient than the individual chords. Listening to Debussy's "Voiles," Stravinsky's "Dance of the Adolescents," or Reich's *Different Trains*, it is motion *between* macroharmonies that really strikes the ear. Harmonic states are here intermediate between chord and key, lasting longer and containing more notes than classical chords, but also moving faster than classical keys.

5.2 SMALL-GAP MACROHARMONY

Suppose we would like to write music that is harmonically consistent while also confining itself to a macroharmony with five to eight pitch classes. How should we go about doing this?

Figure 5.2.1 One way to combine harmonic and macroharmonic consistency is to choose a set of chords that together contain a relatively small number of notes. Here, the A minor, F major, and D♭ major triads contain the pitches of a hexatonic scale.



One strategy is just to choose a collection of similar chords that together contain no more than five to eight notes. For instance, we could chose A minor, F major, and D♭ major, which together form a six-note hexatonic scale (Figure 5.2.1). This works well until we get bored of our chords, at which point we notice—somewhat ruefully—that the macroharmony is quite limited. First, the *only* major or minor triads it contains are those with roots on F, A, and D♭. Second, it contains only a small number of chord types: if, for

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

example, we later decide to use stacks of fourths (such as F-B♭-E♭ or F-B-E), we must choose another macroharmony, since the hexatonic scale contains no fourth chords whatsoever. The problem, it turns out, is that the hexatonic scale has a number of three-semitone "gaps" between successive notes—or to coin a term, it is a "3-gap macroharmony." Because of this, there are no fourth chords in the collection, and no triads with roots other than F, A, or D♭.[1]
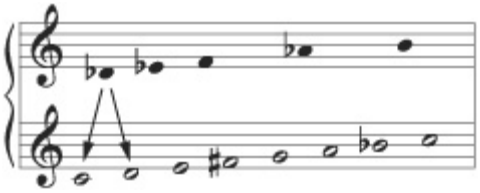
By contrast, a macroharmony whose steps are at most two semitones large (i.e. a "2-gap macroharmony") will always contain both a triad and a fourth chord above each of its notes. This is because harmonic terms like "triad" and "fourth chord" typically allow for some variation in interval size. For example, the term "triad" refers to a stack of three- or four-semitone intervals. Consequently, for each chord tone there are two semitonally adjacent options to choose from: given the root C, we can put the third at either E♭ or E♮, and given the third E♮, we can put the fifth at either G♮ or G♯. In a 2-gap macroharmony, one of these options will always be contained within the scale.

The principle here is related to what I call the "Fundamental Theorem of Jazz," which states that *you can never be more than a semitone wrong*.[2] The basic idea is that when trying to fit a particular chord into a 2-gap macroharmony, one only needs to shift its notes by semitone at most. This is illustrated in Figure 5.2.2. The connection to jazz lies in the fact that improvisers often make use of 2-gap macroharmonies; consequently, any note they play will either belong to the macroharmony or can be a chromatic neighbor to one of its notes. Practically speaking, this means that if one finds oneself accidentally playing a note outside the macroharmony, it can always be reinterpreted as a chromatic neighbor to an adjacent macroharmonic tone. In fact, with a 2-gap macroharmony, the improviser is free to shift either upward *or* downward by semitone. (If a particular note is not in a 2-gap macroharmony, then both of its chromatic neighbors are.[3]) This is quite useful, since out-of-macroharmony notes can be instantly corrected, without the improviser having to think about exactly how he or she has gone wrong.

Figure 5.2.2 In a 2-gap macroharmony, out-of-scale notes can be moved into the scale by semitone, in either direction.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Observe that there is a difference between "having small gaps" and the "near evenness" of Chapter 4. In a nearly even scale, scalar transposition resembles chromatic transposition, which means that we can take a harmony that is *inside* the scale and transpose it *along* the scale without distorting it much. In a 2-gap macroharmony, any note is at most one semitone away from some note in the macroharmony, which means that we can take *any sonority whatsoever* and "squeeze" it into the macroharmony without distorting it much. This difference is illustrated by Figure 5.2.3. Note, in particular, that a collection can be nearly even while still having large gaps (e.g. the major triad). Conversely, a collection can have reasonably small gaps while still being somewhat uneven. [4]

Figure 5.2.3 Near evenness and gaplessness. Chords can be transposed along a nearly even scale with minimal distortion (*a*). In a gapless scale (*b*), chords outside the scale can be "squeezed" into the scale with minimal distortion. Here, the E minor triad is squeezed into the C acoustic scale by shifting B down by semitone.
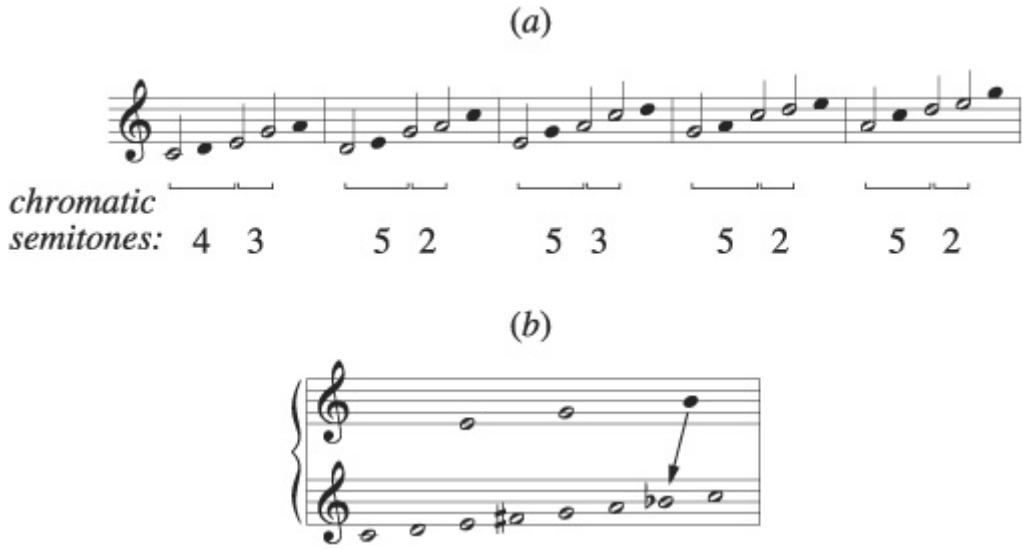


Figure 5.2.4 Set classes representing 2- and 3-gap macroharmonies.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Size	2-gap	3-gap
5 notes	—	pentatonic "dominant ninth" C-C♯-E♭-F♯-A C-C♯-E-F♯-A
6 notes	whole tone	*17 set classes*
7 notes	diatonic acoustic C-C♯-D-E-F♯-G♯-B♭	*19 set classes*
8 notes	*8 set classes*	*16 set classes*

Nevertheless, it is true that the most even collections generally have the smallest gaps, relative to other collections of that size. Figure 5.2.4 shows that the twelve-tone equal-tempered system has four five-note 3-gap macroharmonies, including the pentatonic scale, the "dominant ninth" chord, and the "diminished seventh plus one" chord. There is only one six-note 2-gap macroharmony (the whole-tone scale), while there are 17 six-note 3-gap macroharmonies. The three seven-note 2-gap macroharmonies are the diatonic, acoustic, and "whole tone plus one" scales. Finally, there are eight eight-note 2-gap collections, including the octatonic scale. Once again, we find familiar musical objects—such as the diatonic and acoustic scales—turning up in a variety of different theoretical contexts. And once again, we see that the goal of combining elementary tonal features (in this case harmonic and macroharmonic consistency) places nontrivial constraints on the composer. It is relatively easy to write music that exhibits harmonic or macroharmonic consistency, but more difficult to write music that exhibits both at once: we cannot simply choose harmonies and macroharmonies willy-nilly, mixing and matching them to our hearts' content.

5.3 PITCH-CLASS CIRCULATION

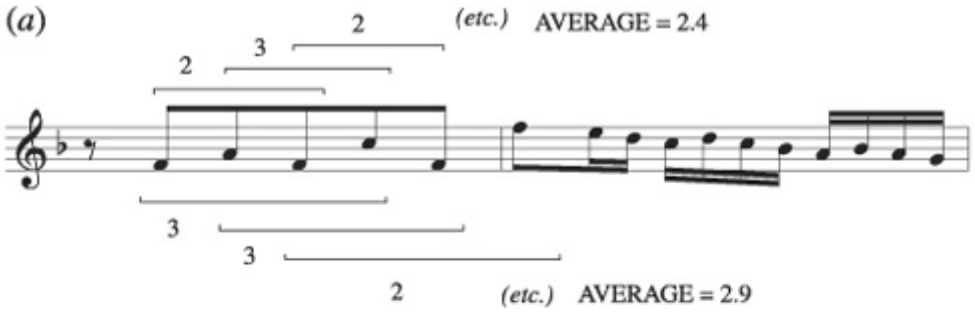
We'll now develop some tools for quantifying macroharmony, starting with graphs that represent how many pitch classes are used over various spans of musical time. The concept is easiest to explain by way of an example. Figure 5.3.1 shows that if we look at every three consecutive notes of the opening of Bach's F major two-part invention, we find on average 2.4 distinct pitch classes. Similarly, if we look at every four-note window of the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

music, we find on average 2.9 pitch classes. Figure 5.3.1c compiles this data into a *pitch-class circulation graph*, which shows how many pitch classes are found in windows of various sizes. Such graphs are very crude tools that do not tell us anything about the character of the macroharmonies; furthermore, they can be influenced by textural features independent of the music's underlying harmonic structure.[5] But by providing a rough picture of how fast a piece of music moves through the available notes, they can help us get a quantitative grip on how "chromatic" it is.

Figure 5.3.1 (*a*) The theme of Bach's F major two-part invention, along with the number of pitch classes in the first several three- and four-note windows. (*b*) A table listing the average number of pitch classes per window in the excerpt, for window sizes between 1 and 10 notes. (*c*) The same information expressed as a graph.



(*b*)

Window Size	Average Number of Pitch Classes
1	1
2	1.9
3	2.4
4	2.9
5	3.4
6	3.9
7	4.4
8	4.8
9	5.1
10	5.6

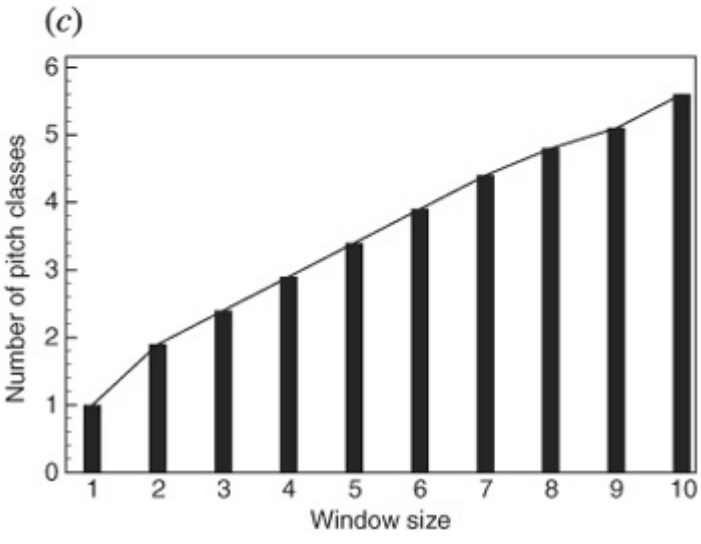


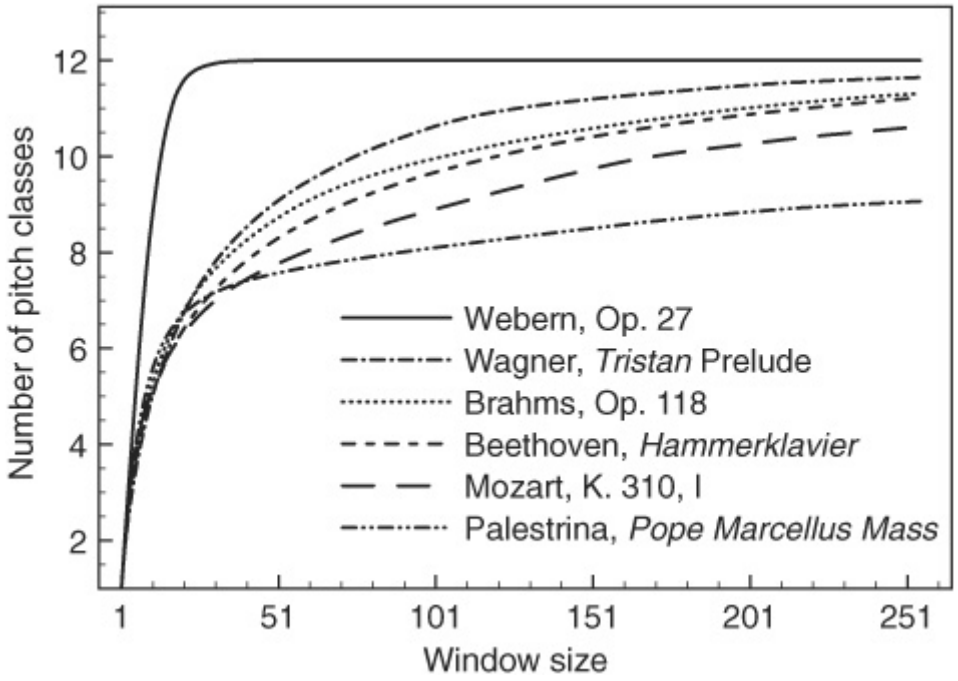
Figure 5.3.2 graphs the pitch-class circulation in a number of familiar pieces. Palestrina's *Pope Marcellus Mass* is at the bottom, while Webern's *Piano Variations*, Op. 27 is at the top. These two curves have a similar shape, rising quickly and flattening out rather sharply. The quick rise reflects the fact that, over short time scales, the two composers both tend to exhaust a particular collection of notes—the seven diatonic notes in Palestrina'

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

s case and the twelve chromatic notes in Webern's. The point at which the graphs level off tells us how large the macroharmony is, with Palestrina's leveling off below Webern's since the diatonic scale is smaller than the chromatic. The flattening itself indicates the relative absence of macroharmonic change: Webern's twelve-tone piece systematically cycles through the only twelve notes available to him, and as a result his graph reaches a completely horizontal plateau; Palestrina's occasional use of nondiatonic notes creates a very gradual slope over larger spans of time. Each piece is *macroharmonically static*, since any two of its mid-length segments will contain roughly the same pitch classes.

Figure 5.3.2 Pitch-class circulation in several well-known pieces.



Between these extremes lies most of the music of the classical tradition. Figure 5.3.2 shows that pieces by Mozart, Beethoven, Brahms, and Wagner exhibit a distinctive harmonic profile, with their graphs rising very quickly and then leveling off much more gradually. (In fact, a typical 10-note excerpt of classical music contains roughly half the pitch classes in a typical 100-note excerpt.[6]) The quick rise reflects the fact that the music again cycles through a collection of available notes, as in Palestrina and Webern. The more gradual flattening indicates that the macroharmonies are *themselves* changing, albeit at a much slower rate. Modulation here ensures a slow but steady supply of fresh pitch classes, causing the graph to level off much more gradually.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 5.3.2 is consistent with a truism of music history: that chromaticism gradually increased over time, beginning with modest explorations during the baroque and classical eras, thriving during the nineteenth century, and culminating in complete atonality. Our graph shows that from Palestrina through Wagner, later composers do indeed tend to utilize faster rates of pitch-class circulation, reflecting increasingly rapid modulations to increasingly distant tonal areas, increasing use of altered chords, and so on. (Of course, some of this is selection bias: including composers such as Gesualdo and Satie would complicate the matter considerably.) Furthermore, as we will see below, there is relatively little difference between the rate of pitch-class circulation in the highly chromatic tonal music of Max Reger and the fully atonal music of Schoenberg. In this sense, atonality does represent a relatively natural response to the saturated chromaticism of late nineteenth-century tonality. We will return to this thought below.

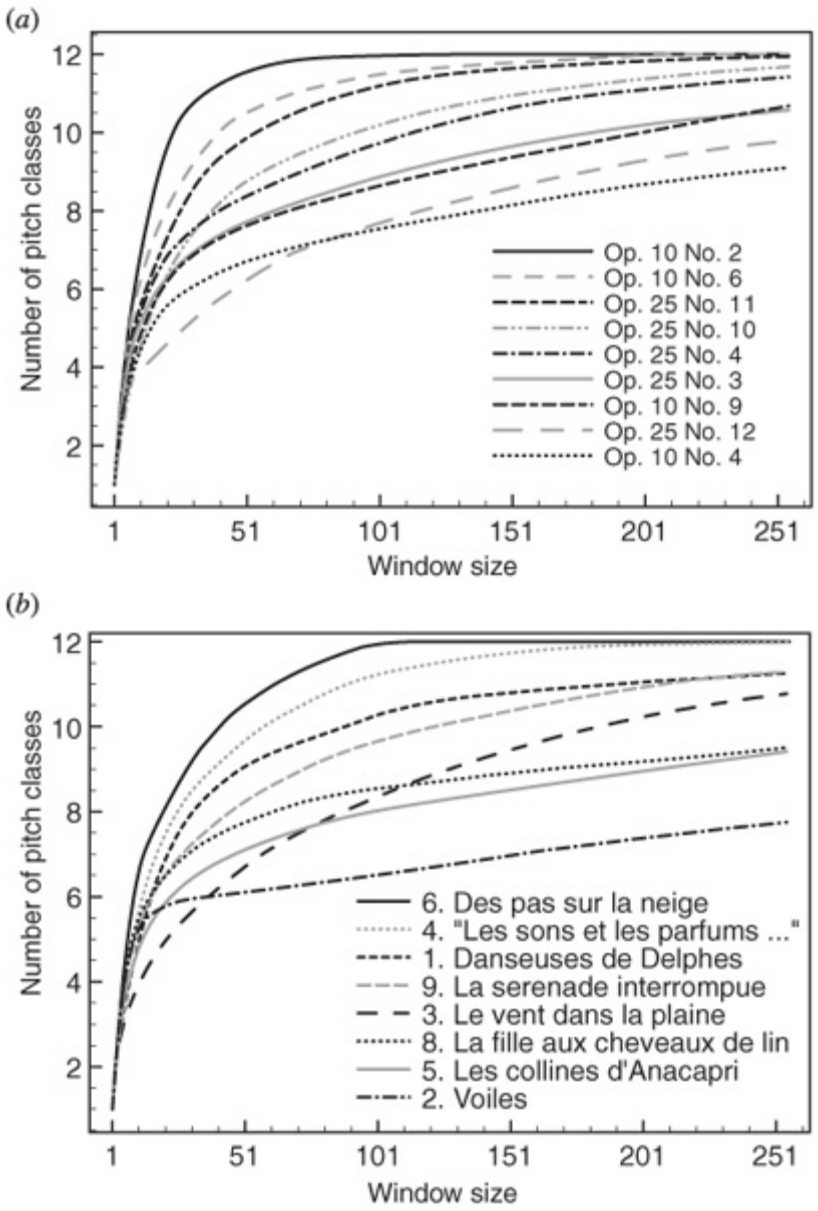
5.4 MODULATING THE RATE OF PITCH-CLASS CIRCULATION

In most Western music, the rate of pitch-class circulation is itself a harmonic variable to be manipulated. For example, Figure 5.4.1 graphs the rate of pitch-class circulation found in selected Chopin Etudes, the first book of Debussy's Preludes, the individual numbers of Stravinsky's *The Rite of Spring*, and a selection of Shostakovich's Op. 87 Preludes and Fugues. All four works display enormous variation, ranging from Palestrina-like sections that use just a few pitch classes, to highly chromatic sections in which all twelve notes are in play. This of course reflects a common technique for creating large-scale form: composers can generate long-term harmonic change by juxtaposing moments of relative calm—in which the music remains fixed in a small macroharmony—with passages of more rapid and aggressive chromaticism.

Figure 5.4.1 Pitch-class circulation in Chopin's *Etudes* (*a*) and Debussy's *Preludes* (*b*). Both collections cover an enormous range, comparable to that of the entire classical tradition (compare Figure 5.3.2).

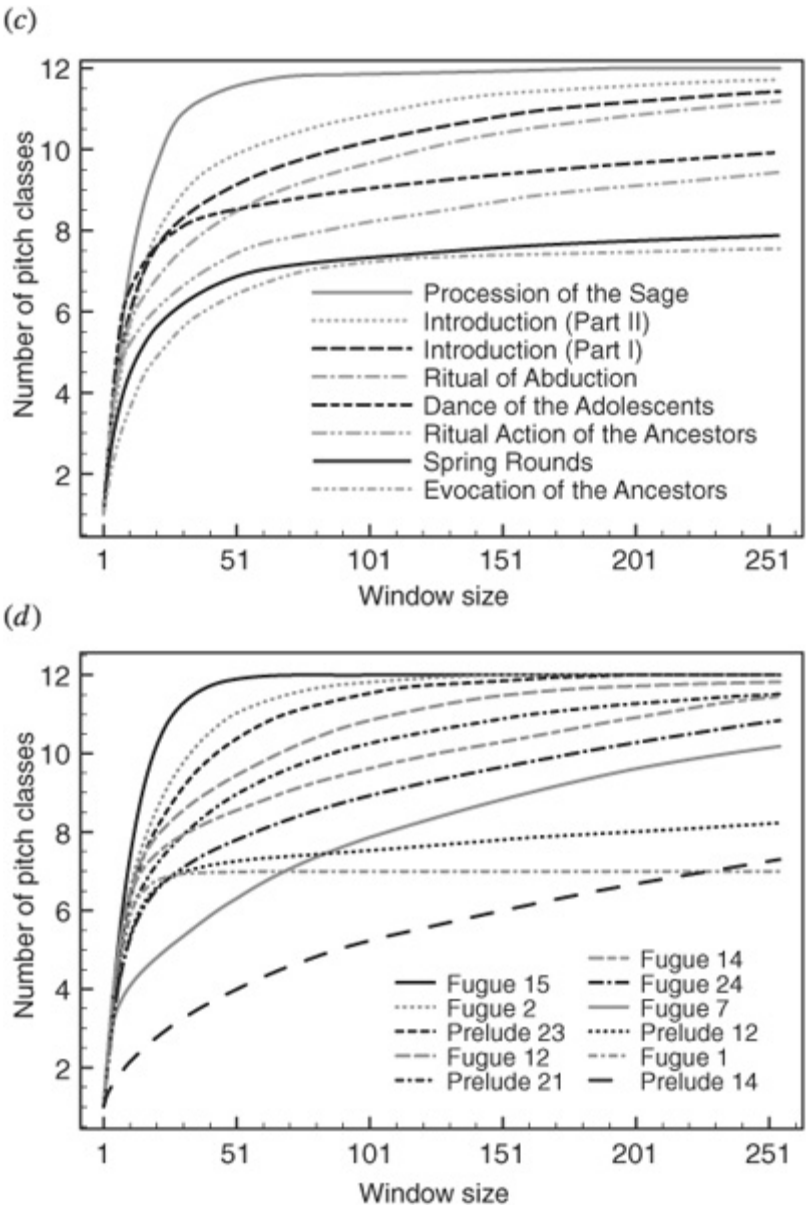
Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



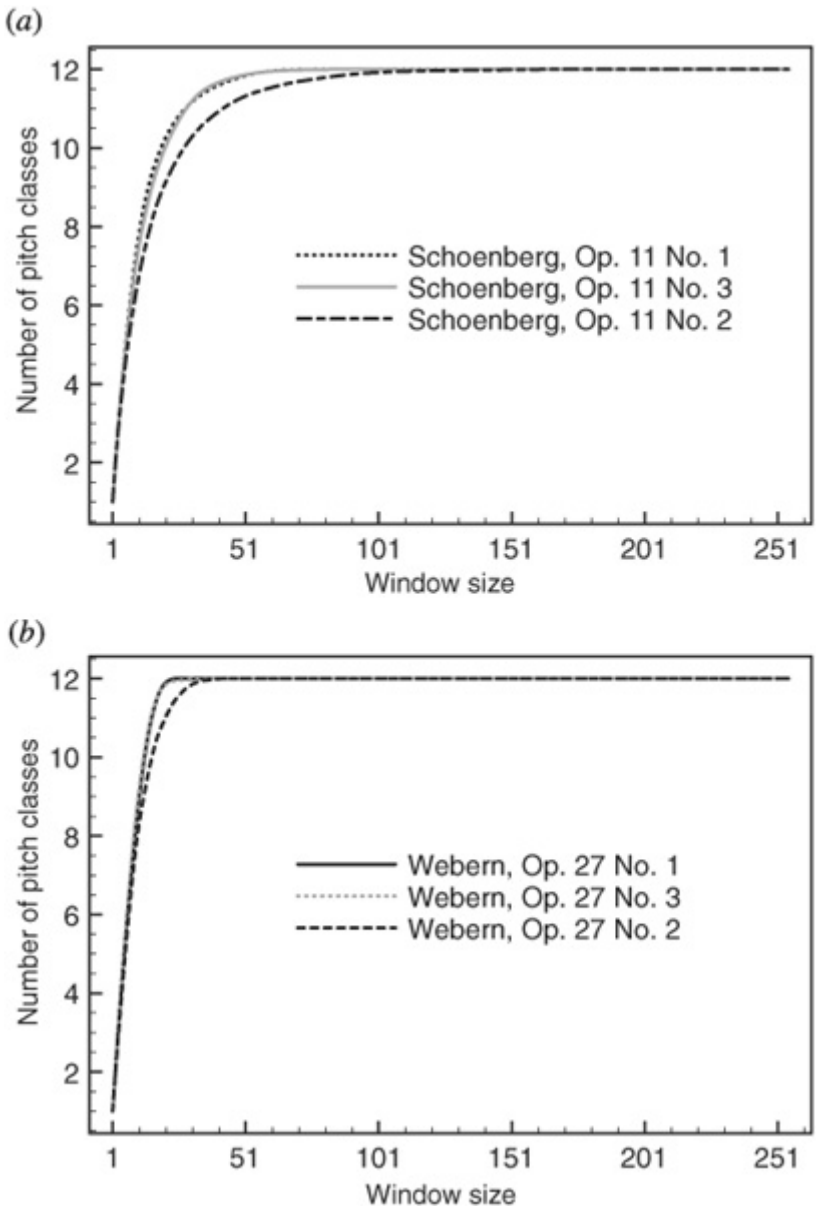
This diversity in the rate of pitch-class circulation stands in stark contrast to the homogeneity of much atonal music. Figure 5.4.2 graphs the pitch-class circulation in Schoenberg's Op. 11 piano pieces and Webern's Op. 27 variations. The outer movements of both pieces are almost indistinguishable from one another, while their middle movements are ever so slightly less chromatic. The difference between Figures 5.4.1 and 5.4.2 is truly remarkable: where Chopin, Debussy, Stravinsky, and Sho stakovich embraced a vast array of macroharmonic states, a diversity comparable to that of the *entire history of Western music*, Schoenberg and Webern restricted themselves to a much narrower region of musical space. This no doubt helps explain why some listeners find atonal music to be somewhat static. In writing music with a consistently fast rate of pitch-class circulation, atonal composers deprived themselves of one important tool for creating large-scale harmonic change.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 5.4.2 Pitch-class circulation in Schoenberg's Op. 11 and Webern's Op. 27 piano pieces. The individual movements of each piece are much more similar than in Figure 5.4.1

.



It is notable that historians have sometimes used *stylistic* categories to describe the macroharmonic diversity in the music of Debussy, Stravinsky, and Shostakovich. The low-circulation passages are said to evoke particular genres—the diatonicism of ancient music, the pentatonicism or exoticism of non-Western music, and so on—while the high-circulation passages are associated with modernism. In this way, composers favoring macroharmonic diversity are sometimes made out to be polyglots, stylistic magpies who borrow from many different sources. In some cases, this characterization is accompanied by an evaluative narrative, as if diatonicism represents a regressive, backward-looking

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

tendency, while thoroughgoing chromaticism represents a progressive or forward-looking attitude. Macroharmonic diversity is thus associated with stylistic mawkishness, testifying to an incomplete embrace of modernity.

Of course, it is true that many twentieth-century composers were interested in evoking a range of styles. Nevertheless, I think we should be careful to separate the issue of style from the purely harmonic effects that stylistic juxtaposition can create. Rather than seeing Debussy, Stravinsky, and Shostakovich as stopping short of a complete and consistent chromaticism, I would therefore prefer to describe them as embracing *modulation*—writing music that presents a wealth of different macroharmonic states and that changes the rate of pitch-class circulation to reinforce larger formal boundaries. (In fact, I suspect that critics have sometimes resorted to stylistic categories in part because we lack precise theoretical terms for talking about phenomena such as pitch-class circulation.) From this point of view, it is thoroughgoing chromaticism that is conservative, as it abandons macroharmonic change in favor of musical textures that are harmonically uniform in the large. In this respect it recalls the macroharmonic stasis of the earliest Western music.

5.5 MACROHARMONIC CONSISTENCY

Pitch-class circulation graphs show us *how fast* pitch classes are passing by, but do not provide any sense of what the macroharmonies actually are; as a result, they cannot distinguish between quickly modulating diatonic music and nondiatonic music in which all twelve pitch classes are constantly in play. This can be seen by comparing Schoenberg's Op. 11 No. 1 with John Coltrane's solo on "Giant Steps" (Figure 5.5.1). Both pieces have almost identical rates of pitchclass circulation, even though Schoe-nberg's music is intuitively "more chromatic" than Coltrane's.

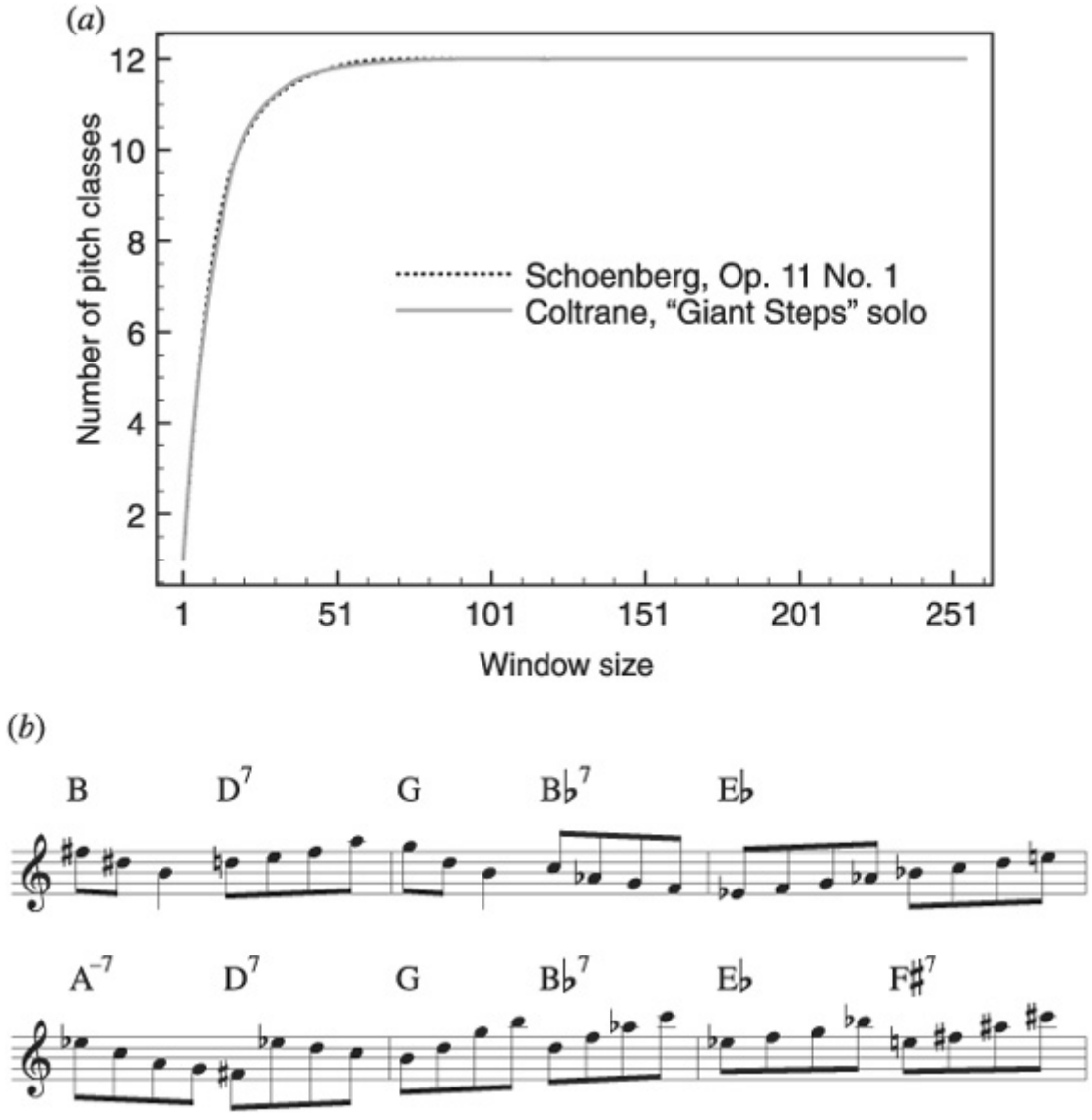
What we need are *global macroharmonic profiles* that identify the five- to eight-note macroharmonies used in a particular piece. Given an excerpt of music, we can exhaustively tabulate all the three-note chord types, four-note chord types, five-note chord types, and so on.[7] Figure 5.5.2 presents the six- and seven-note collections in Schoenberg and Coltrane's pieces. As we would expect, the graph of Coltrane's solo is highly peaked, reflecting the fact that it is saturated with diatonic scales and scale fragments. By contrast, Schoenberg's piece features a much more even distribution of chord types, not strongly emphasizing any

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

particular six- or seven-note collection. (On these graphs, the *x*-axis is labeled using Allen Forte's hard-to-decipher numerical labels for set classes; what is important here is just the overall difference in shape—Coltrane's graph is much more peaked than Schoenberg's.[8]) We can therefore say that Coltrane's solo is macroharmonically consistent in a way that Schoenberg's piece is not: Coltrane emphasizes one particular seven-note collection, while Schoenberg makes relatively indiscriminate use of almost all the available seven- and eight-note macroharmonies.[9] To be sure, Coltrane's music *modulates* very quickly, thus ensuring a high overall rate of pitch-class circulation; but unlike Schoenberg, the local structure of his music clearly articulates familiar scales.

Figure 5.5.1 Schoenberg's Op. 11 and John Coltrane's *Giant Steps* solo have identical rates of pitch-class circulation (*a*), even though Coltrane's solo is tonal (*b*).



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

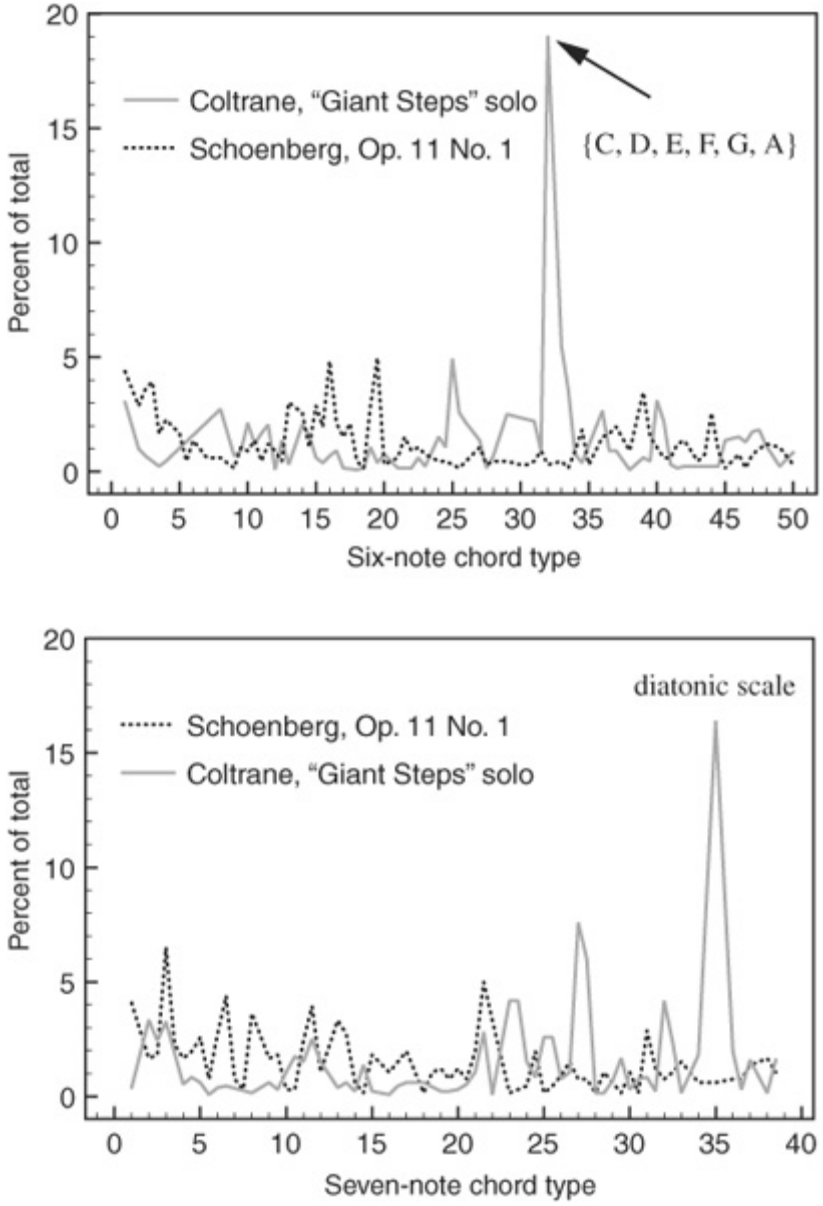
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

One can make a similar distinction among musical styles in which pitch classes circulate more slowly. Figure 5.5.3 contrasts Debussy's "La fille aux cheveux de lin" with Satie's "Theme of the Order," from *Sonneries de la Rose + Croix*. The opening of Debussy's piece very clearly articulates E♭ natural minor and G♭ acoustic scales, separated by a brief cadence on E♭ major. Satie's piece, although it uses relatively few accidentals, does so sporadically and without a clear system; as a result it is difficult to separate the music into regions exemplifying recurring macroharmonies. Yet both pieces have relatively low rates of pitch-class circulation, and are in this sense "not very chromatic."

Figure 5.5.2 The relative preponderance of different chord types in pieces by Schoenberg and Coltrane. Coltrane's graph is highly peaked at points representing familiar tonal collections. Schoenberg's piece does not emphasize any larger collections to the same extent. The *x*-axis uses a variant of Forte's (1973) set-class labels, where a half-integer $i + .5$ represents the inversion of chord *i*. Thus, the harmonic major scale, 32.5, is the inversion of the harmonic minor, 32.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Taken together, these four pieces demonstrate that the informal music-theoretical term "chromatic" involves the interaction of at least two independent variables: rate of pitch-class circulation and the degree of emphasis on particular macroharmonies. Figure 5.5.4 tries to represent the situation visually. Music such as Schoenberg's, which combines a high rate of pitch-class circulation with a low degree of macroharmonic consistency, will be heard as very chromatic; conversely, music such as Debussy's, with a low rate of pitch-class circulation and a high degree of macroharmonic consistency, will be heard as non-chromatic. Pieces such as "Giant Steps" or Satie's "Theme" are somewhat more difficult to classify, because the simple opposition between "chromatic" and "non-chromatic" breaks down: Coltrane's piece has faster pitch-class circulation than Satie's, but more clearly

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

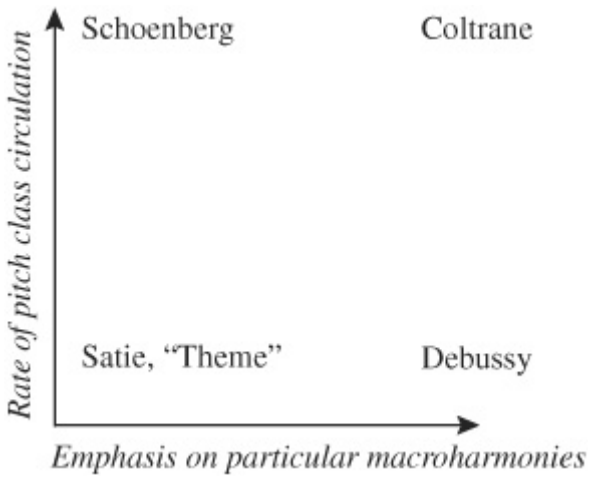
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

articulates specific macroharmonies; while Satie's circulates through the pitch classes more slowly, but without articulating identifiable macroharmonies. Each is somewhat chromatic in its own distinctive way.

Figure 5.5.3 Debussy's "La fille aux cheveux de lin" (*a*) and Satie's "Theme of the Order," from *Sonneries de la Rose + Croix* (*b*).
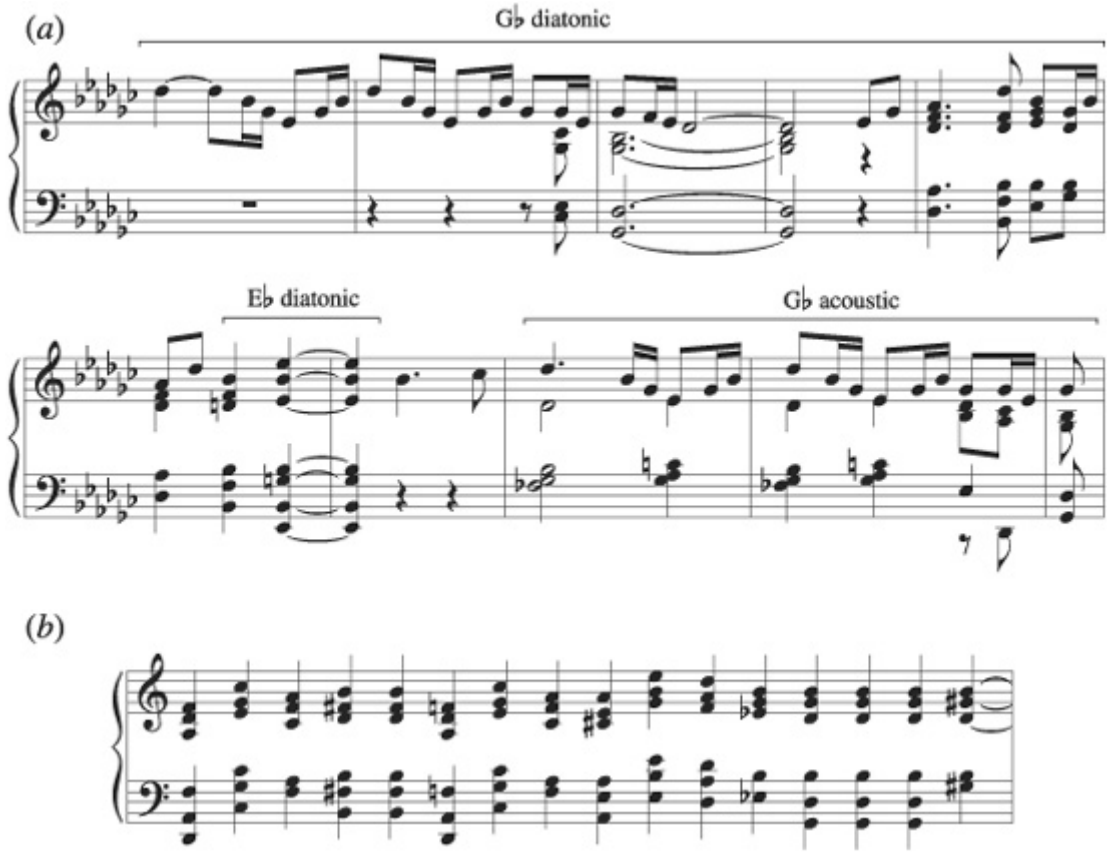


Figure 5.5.4 The rate of pitch-class circulation is independent of the degree of emphasis on particular macroharmonies. The four pieces we have been discussing stake out four different regions in the space of musical possibilities.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Global macroharmonic profiles can also be useful analytically. For example, Figure 5.5.5 identifies the most prominent seven- and eight-note collections in the opening four sections of *The Rite of Spring*. The introduction is relatively chromatic, superimposing multiple tonalities without emphasizing any familiar macroharmony. (The melodic minor scale is the most prominent seven-note collection, but it appears only 9% of the time.) The "Dance of the Adolescents" features a larger preponderance of harmonic and melodic minor scales—chiefly because its famous opening chord is a harmonic minor collection, while the final section emphasizes the A mode of F acoustic.[10] While neither of the opening sections involves much explicitly octatonic material, the "Ritual of Abduction" is highly (18%) octatonic. Finally, "Spring Rounds" is extremely diatonic (55%). Overall, these figures indicate that Stravinsky's piece exhibits a large-scale transition from the highly chromatic opening to the very diatonic "Spring Rounds," a transition mediated by familiar nondiatonic scales. This transition might be compared to those we encountered in Debussy and Prokofiev (§4.7), though spread out over a much longer span of time.

Figure 5.5.5 The most common large collections in the first four sections of the *Rite of Spring*. Each section has a distinct macroharmonic profile.

	Introduction	**Dance of the Adolescents**	**Ritual of Abduction**	**Spring Rounds**
Prevalent Seven-note Collections	melodic minor (9%)	harmonic minor (18%) melodic minor (16%) diatonic (8%)	*subsets of the octatonic*	diatonic (53%)
Prevalent Eight-note Collections	*Various chromatic collections:* (0, 1, 2, 3, 5, 6, 8, 9) (12%) (0, 1, 3, 4, 5, 6, 7, 8) (8%)	*supersets of the above*	octatonic (18%)	*supersets of the diatonic*

The Rite of Spring can be usefully contrasted with Shostakovich's Op. 87 Preludes and Fugues. Both pieces belong to the twentieth-century extended-tonal tradition, drawing on tonal techniques while also featuring moments of extreme chromaticism, superimposition of multiple scales, and other modernistic devices. However, where Stravinsky makes relatively frequent use of nondiatonic scales, the Preludes and Fugues are profoundly and almost stubbornly diatonic. Figure 5.5.6 shows that the diatonic scale is the most prominent macroharmony in almost every one of Shostakovich's 48 preludes and fugues,

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

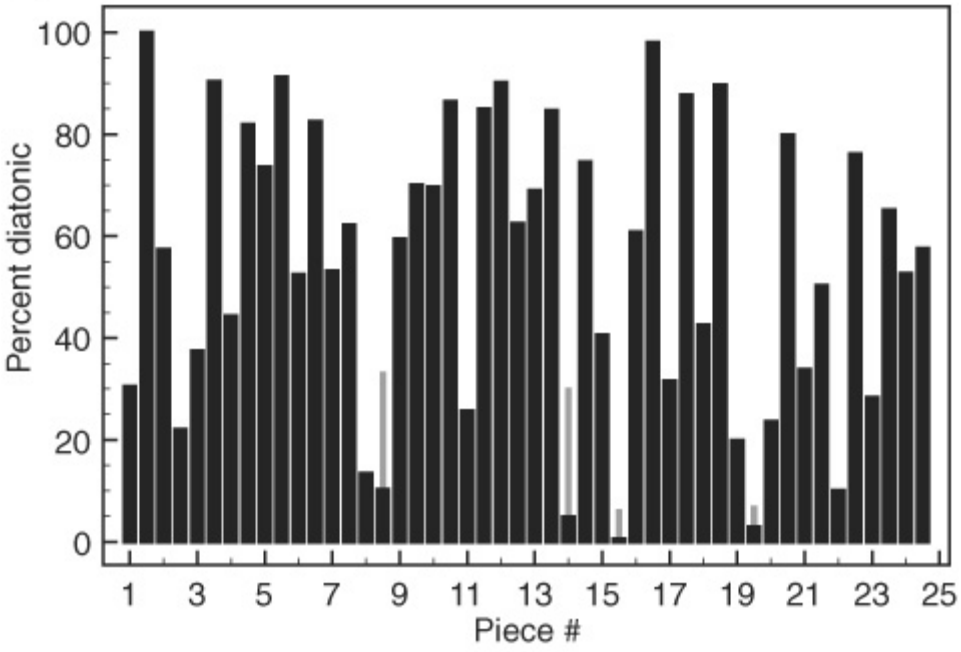
sometimes by an enormous degree.[11] (Indeed, in the first fugue it is the *only* macroharmony.) In fact, Shostakovich's minor-mode pieces are even more diatonic than Bach's—largely because Shostakovich eschews V–i progressions in favor of diatonic cadences that hearken back to a pretonal modality. In this respect, Shostakovich's pieces are more neo-classical than post-impressionist; like Copland, Piston, and many other mid-twentieth-century composers, the Preludes and Fugues tend to move between diatonicism, chromaticism, and polytonality without exploring nondiatonic scales. This is part of what endows the music with its distinctive austerity.

Figure 5.5.6 (*a*) The height of each bar represents the proportion of seven-note macroharmonies that are diatonic. The diatonic scale is the most common seven-note collection in all but four of Shostakovich's *Preludes and Fugues*, and in some pieces it is the only macroharmony. The most common non-diatonic collection is represented by a thin gray line; in only four movements does it rise above the black line. (*b*) Shostakovich's minor-key pieces are considerably more diatonic than Bach's minor-key Preludes and Fugues. In particular, Shostakovich rarely uses the harmonic and melodic minor scales, preferring the natural minor.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

(*a*)



(*b*)

	Bach	Shostakovich
Harmonic minor	25%	5%
Diatonic	20%	56%
Melodic minor	12%	3%

5.6 CENTRICITY

In many musical passages, a particular note is felt to be more prominent, important, or stable than the others—in other words, to be a tonal "center." The concept of centricity is complicated, in part because it encompasses two closely related phenomena: *rootedness*, which applies to individual chords, and *tonicity*, which refers to prominence over longer stretches of musical time. Rootedness and tonicity are music-theoretical cousins, sharing a number of physiognomic characteristics while being of distinct parentage. And just as the line between chord and macroharmony is sometimes blurry, so too is the distinction between rootedness and tonicity: in impressionist or minimalist music, it can be difficult to say whether the most important tone is a root, a tonic, or something in between. For this reason, it is useful to try to develop a unified approach to the two phenomena, without presupposing that there will always be a sharp line dividing them.

Elementary music theory teaches that the "root" of a tertian chord is the lowest note when the chord is arranged as an ascending chain of thirds: thus, E is the root of the

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

collection {E, G, B}, regardless of musical context. (I tell students that to find the root, they must first write the chord in *snowman form.*[12]) But this is an oversimplification: though the two progressions in Figure 5.6.1 use the same pitch classes, the first sounds like i-iv in E minor, while the second sounds more like V-I in C major. A more flexible theorist might therefore describe the chords in Figure 5.6.1b as G and C chords with "added sixths." This shows that a sophisticated notion of "root" cannot rely on mechanical rules like "the lowest note in a chain of thirds is always the root"; instead, we must sometimes make more delicate psychological judgments about the relative importance of notes. This issue arises all the time in twentieth-century music: it would be odd to describe Figure 5.6.2 as a series of 16 repetitions of the same chord, rather than a transition from a dissonant sonority (in which the notes D, E♭, and F♯ predominate) to a more consonant sonority featuring C, E, and G.

Figure 5.6.1 Two progressions that use the same pitch classes, but sound very different.
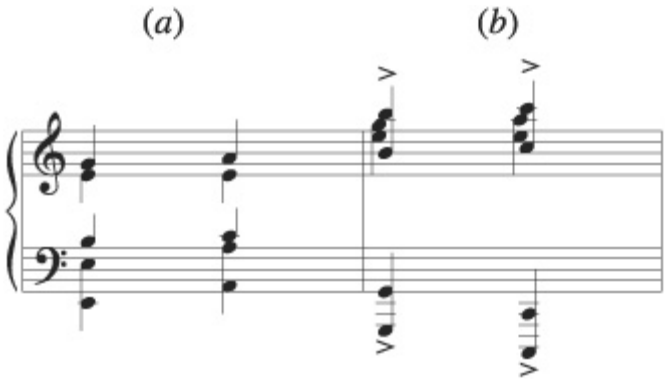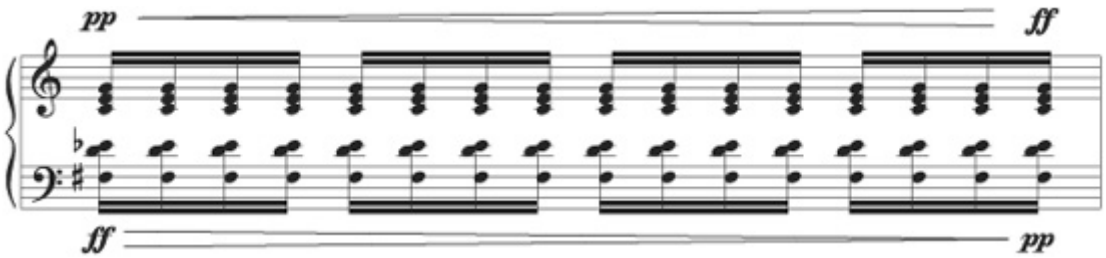


Figure 5.6.2 The same notes are attacked on every sixteenth note, but there is a palpable transformation over the course of the measure.



Contemporary theorists often use *pitch-class profiles* to represent differences of importance among the notes in a chord. These useful devices are bar graphs in which the *x*-axis represents pitch class and the *y*-axis represents a subjective assessment of prominence, stability, or importance.[13] The simplest conception of a chord is binary; notes can be either

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

inside the chord or outside of it, but no further differentiations are made. This approach can be modeled using two-tiered pitch-class profiles, with notes in the chord being assigned the value 1 and the rest being assigned 0 (Figure 5.6.3). The musical cases we have been considering require finer gradations of prominence. For example, Figure 5.6.4 shows how we might describe the progressions in Figure 5.6.1: the first profile assigns higher prominence to the pitch classes E and A, while the second assigns higher prominence to G and C. Figure 5.6.5 represents the music in. Figure 5.6.2, where there is a gradual shift from D, E♭, and F♯ to C, E, and G. In these cases, the notion of chord membership becomes fuzzy, admitting a range of values between 0 and 1.

Figure 5.6.3 The simplest model of a while the second assigns higher prominence to chord is a binary one: notes are either in the chord or outside of it.

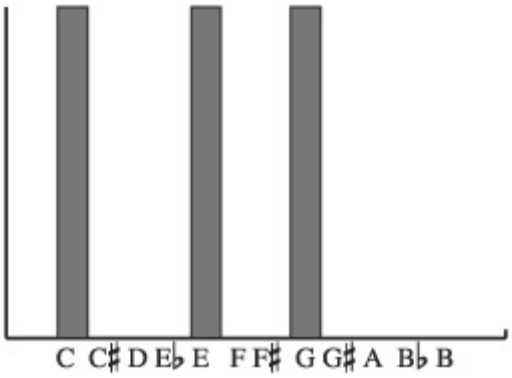
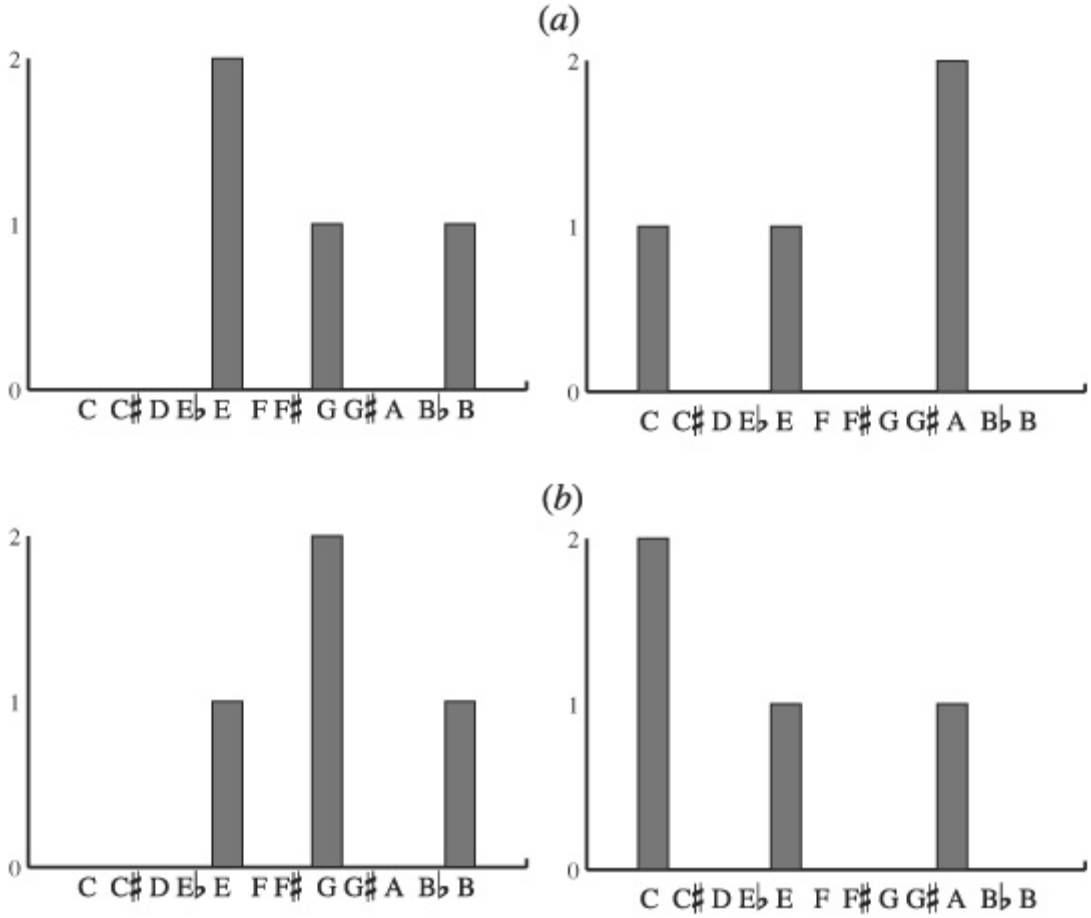


Figure 5.6.4 Pitch-class profiles can be used to reflect the fact that chord tones differ in terms of their importance. Here, (*a*) and (*b*) represent the two progressions in Figure 5.6.1.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



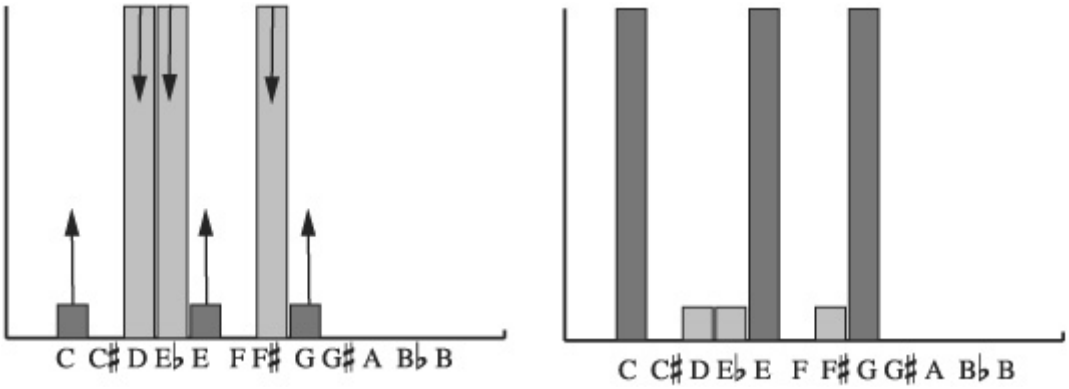
We can also use pitch-class profiles to represent global centricity, or tonicity.[14] Three-tiered profiles provide a simple but effective way to visualize musical modes: notes outside the macroharmony are assigned the value 0, non-centric notes inside the macroharmony are assigned the value 1, and centric notes are assigned the value 2.[15] Thus, Figure 5.6.6 represents C lydian, G ionian, C phrygian, and G locrian modes. In some theoretical contexts we may want to draw more fine-grained distinctions: for example, there is a palpable difference between an E phrygian mode in which the fifth scale degree is emphasized, and an E phrygian mode in which the fourth scale degree is important (Figure 5.6.7).[16] We could capture this with a four-tiered model that distinguishes tonal center, second-most important tone, within-macroharmony notes, and outside-macroharmony notes. Figure 5.6.8a goes even further by proposing a five-tiered pitch-class profile for the key of C major: here, C is most important, G the second-most important, E the third-most important, the remaining diatonic notes fourth-most important, and the black notes last. (This graph is strongly reminiscent of what Fred Lerdahl calls the "basic space" for C major, discussed in Appendix D.) In extreme cases, we may even want to use graphs such as that in Figure 5.6.8b, where we have continuous gradations of pitch-class stability. Here

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

it is impossible to draw sharp distinctions between notes “inside” the macroharmony and those outside it.

Figure 5.6.5 The music of Figure 5.6.2 can be represented as a continuous interpolation between these two graphs, with the more prominent notes becoming less so, and the less prominent notes becoming more so.



An interesting piece, in this regard, is the “Petit airs au bord du ruisseau” from Stravinsky’s *Histoire du soldat*. This playful bagatelle has an almost flirtatious relation to traditional scale-based tonality: the piece uses five fixed scale degrees (G-A-B-D-E) and two mobile degrees that appear in distinct flavors (F vs. F𝄯 and C vs. C𝄯). While some parts of the music clearly articulate distinct scales, others juxtapose different forms of the “mobile” scale degrees to create a mild form of polytonality (Figure 5.6.9). Interestingly, there is only one point in the piece where a single line sounds distinct forms of the mobile notes in direct succession; for the most part, it is as if the music were “locally scale-based” within each instrument. (This is precisely what leads me to consider F and F𝄯 to be two different forms of the same scale degree, rather than analyzing the music using a nine-note scale.) Figure 5.6.9b uses a pitch-class profile to represent the somewhat blurred tonality of this movement: here, the profile indicates that I consider A the central pitch; the “fixed” scale degrees are assigned the value 1, indicating their relative stability, while the mobile scale degrees are assigned lower values, indicating their status as fluctuating, “secondary” pitches. But as the graph shows, these notes are more significant than those that do not appear at all.

Figure 5.6.6 Pitch-class profiles for C lydian, G ionian, G locrian, and C phrygian (clockwise from upper left).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

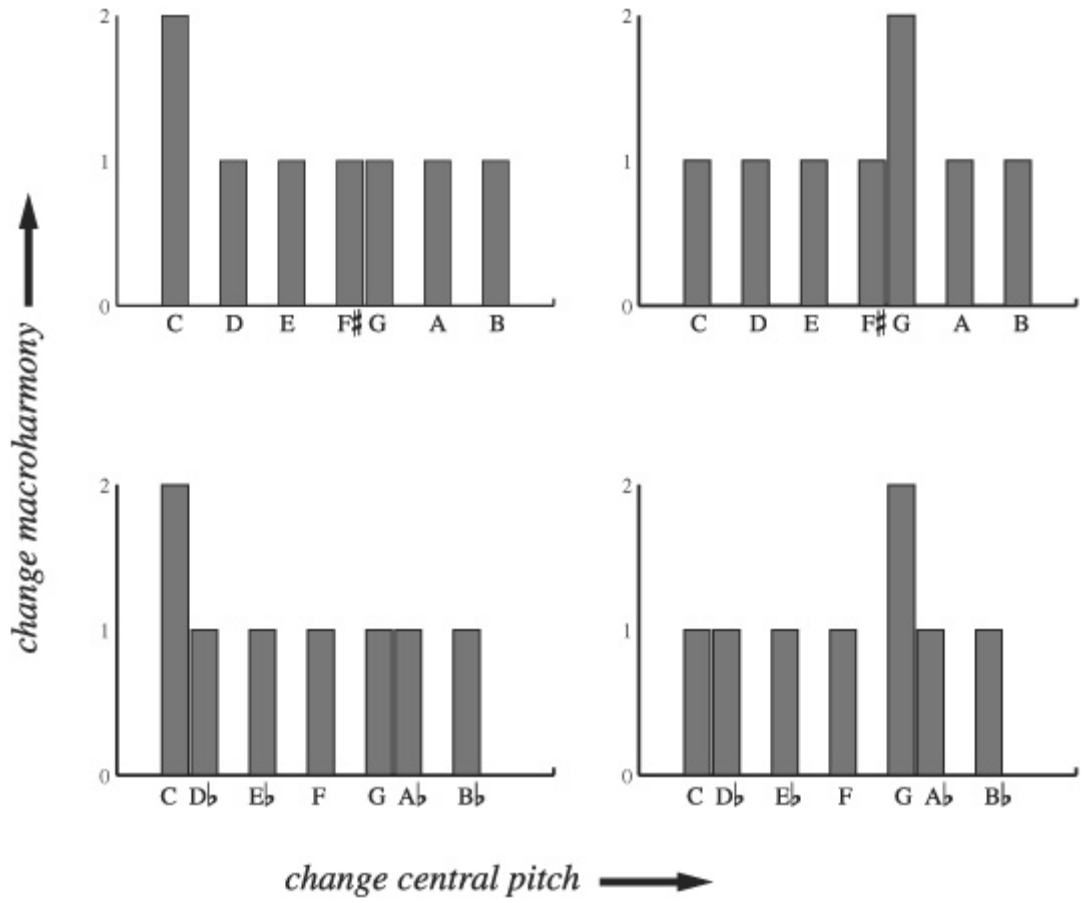
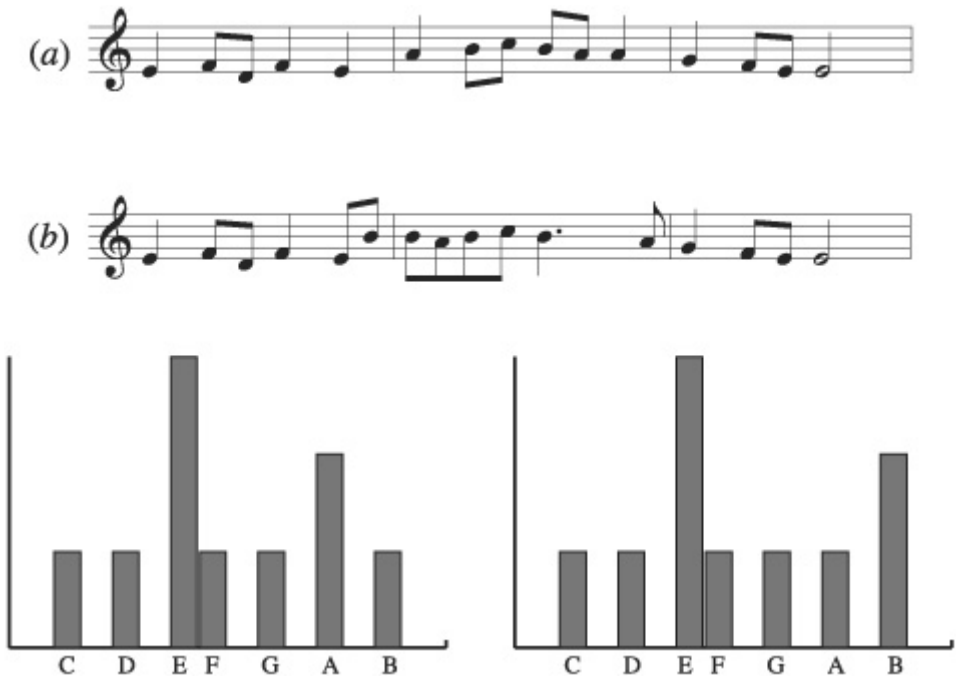


Figure 5.6.7 The melody in (*a*) emphasizes the notes E and A, while that in (*b*) emphasizes E and B. Though both might be said to be "in E phrygian," there is an important difference between them. We might represent this difference by using pitch-class profiles that identify the second-most important note in a mode.



Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 5.6.8 A five-tiered pitch-class profile representing the key of C major (*a*) and a profile that does not represent any common key (*b*).

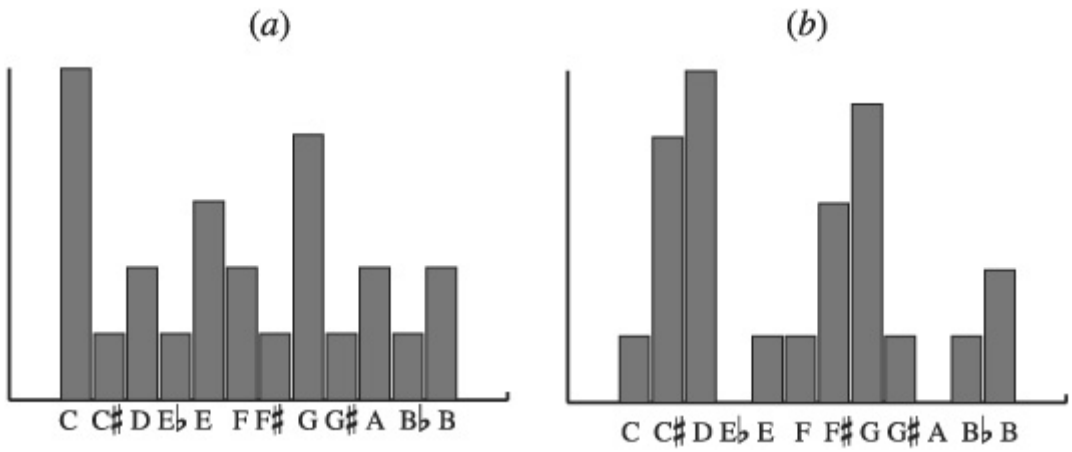
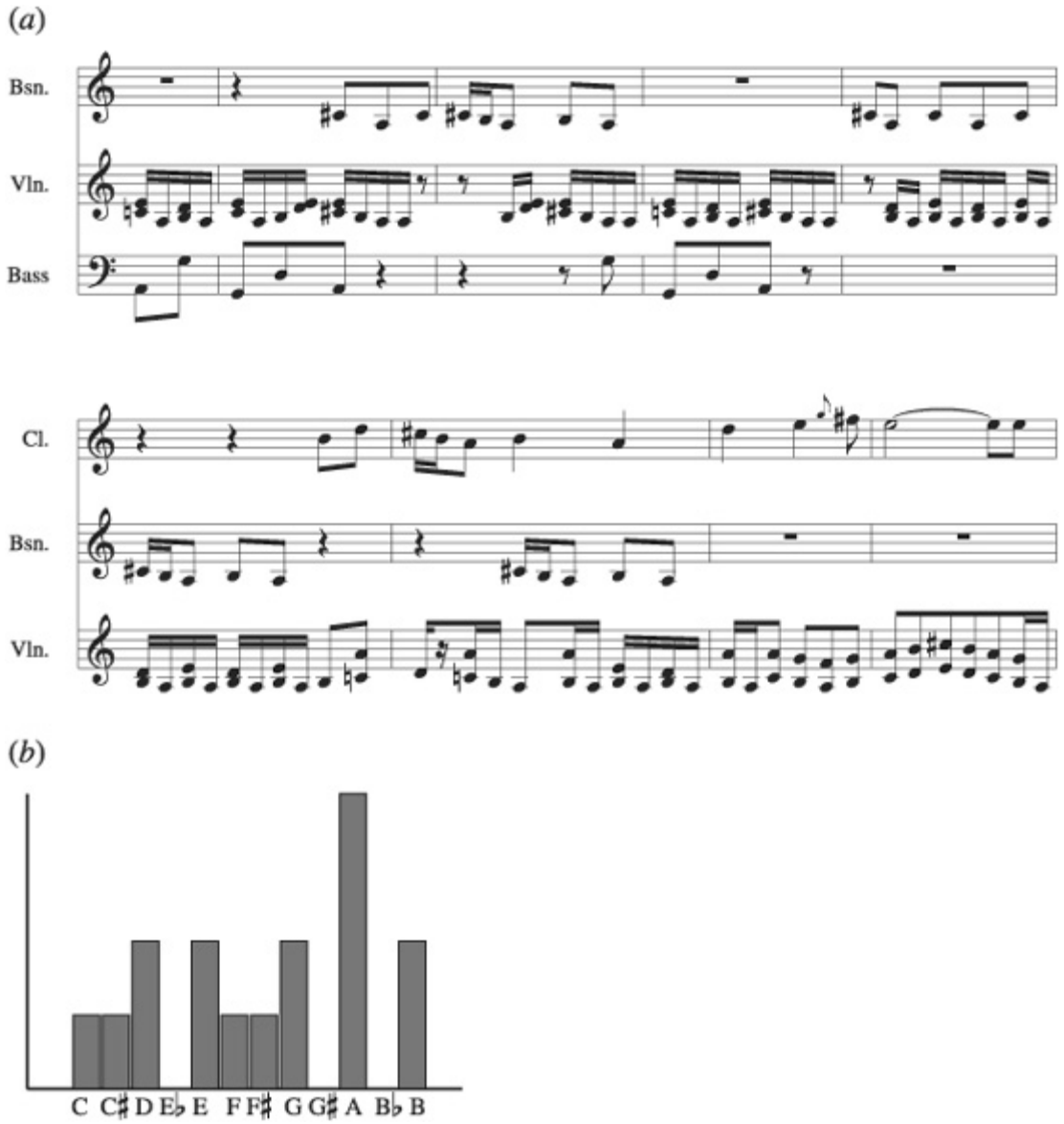


Figure 5.6.9 Stravinsky's "Petit airs," from *Histoire du soldat* uses five "fixed" pitches (G, A, B, D, E) and two "mobile" pitch classes (C/C♯, and F/F♯) (*a*). We can represent the music using a four-tiered pitch profile (*b*), in which the "mobile" pitch classes are assigned a lower weighting than the fixed pitch classes. Here, A is assigned the highest value, indicating that it functions as a tonic.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



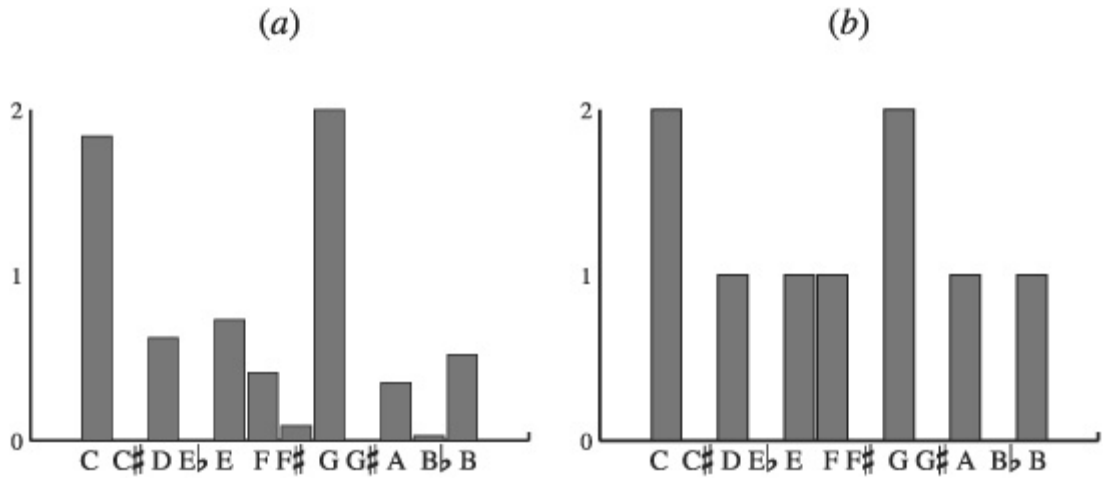
Global pitch-class profiles represent a psychological or conceptual phenomenon—the felt or imagined importance of the different pitch classes. However, the statistical distribution of notes in a piece often correlates reasonably well with the three-tiered pitch profiles we would intuitively describe them with.[17] (Terminological note: I distinguish *pitch-class distributions*, which measure the statistical frequencies of notes in a piece, from *pitch-class profiles*, which represent subjective assertions about psychological importance.) For example, Figure 5.6.10 shows the frequencies of the pitch classes in the first 23 measures of the first movement of Mozart's Jupiter Symphony. This statistical distribution resembles a three-tiered pitch-class profile in which both C and G are assigned the value 2, while the remaining diatonic notes are assigned the value 1—an arrangement that captures the intuitive sense that C and G are most stable in the key of C major, and that the diatonic tones are more stable than the nondiatonic tones. Note that we cannot recover *all* the information about pitch prominence simply by counting pitch classes, since G appears

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

more frequently than C. But this should not be surprising, as there are many ways of generating centricity besides note repetition. What is interesting is that a very crude count of the pitch classes provides a reasonable guide to the relative importance of the notes.

Figure 5.6.10 (*a*) The distribution of pitch classes in the opening phrase of Mozart's Jupiter Symphony. (*b*) A three-tiered pitch-class that matches Mozart's distribution reasonably well.



Indeed, pitch-class distributions can sometimes reveal interesting facts about the tonality of specific passages. For example, Figure 5.6.11 records the note frequencies of the three solos in "Freedom Jazz Dance," from the Miles Davis Group's *Miles Smiles*. The piece is an example of "modal jazz," in which the rhythm section plays a relatively static riff emphasizing B♭. Though the music is ostensibly in a single key, the pitch-class distributions are all very different, suggesting that the three soloists have different interpretations of the piece's "Bf-ness." Davis's distribution suggests B♭ dorian, with B♭, C, D♭, F, and G occurring most frequently, while Shorter's is more suggestive of B♭ mixolydian, emphasizing A♭, B♭, C, D, and F. Hancock's is considerably more chromatic than either of these, and the relative lack of A, C, and G♭ may suggest B♭ octatonic. In Chapter 9, we will see that some of these differences result from the fact that the soloists do, in fact, play different scales over the fixed B♭ ostinato. But scales are not the whole story. Even within a single scale, the musicians choose to emphasize different notes, creating very different shadings of the underlying B♭ tonality.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

Figure 5.6.11 Pitch-class distributions of the three solos in "Freedom Jazz Dance," from the album *Miles Smiles*. Each soloist emphasizes different pitches, suggesting different ways of conceiving of the key.

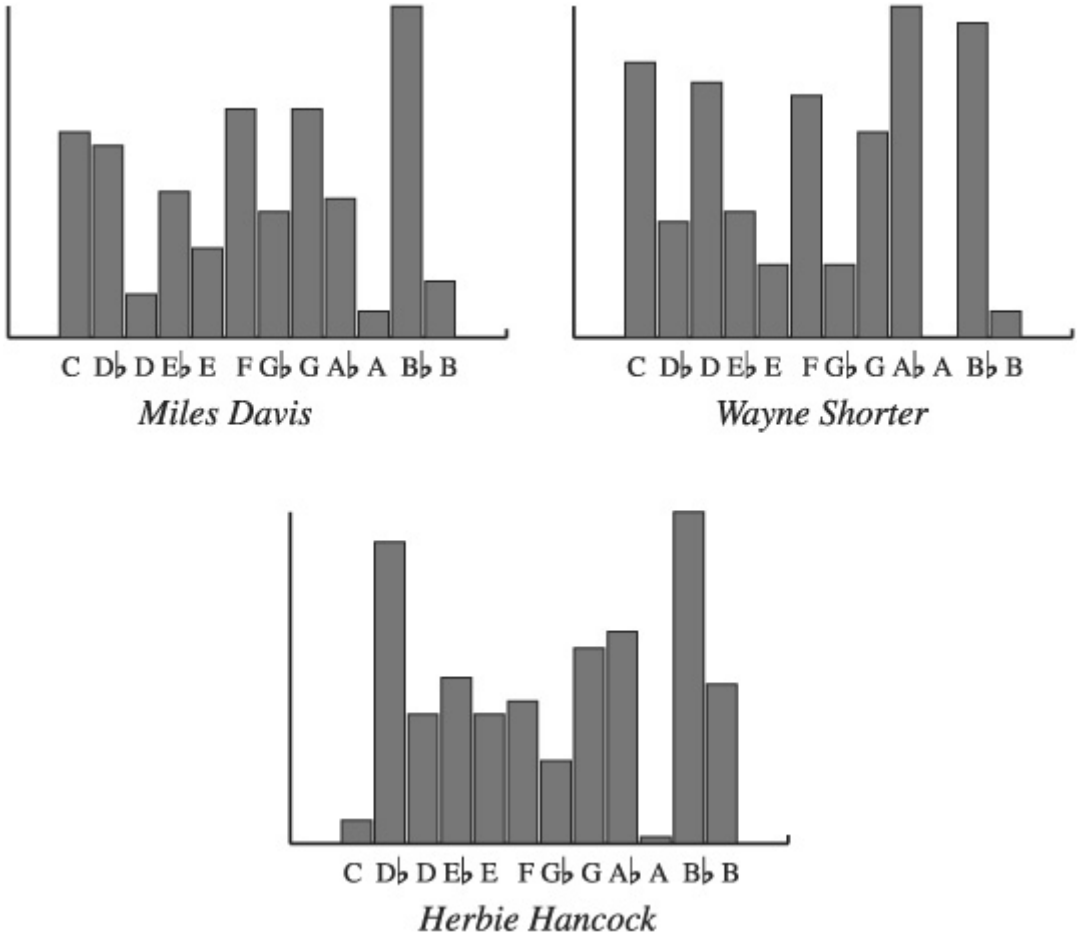
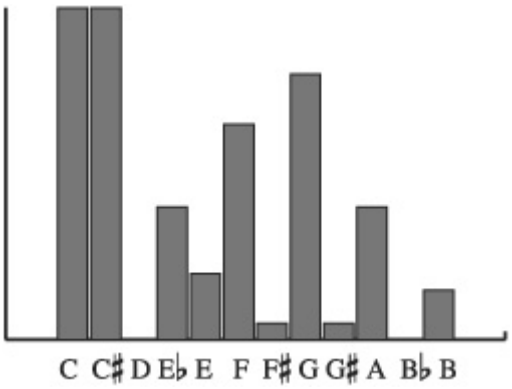


Figure 5.6.12 A hypothetical profile in which C and D♭ are equally important.



Compositionally, I find pitch-class profiles to be extremely useful devices. I like to ask: could I construct (or improvise) music that reflects Figure 5.6.8b? Could I write music that realizes Figure 5.6.12, in which both C and D♭ are perceived as equally important? The basic technique here is simply to make sure that the more prominent notes in the pitch-

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

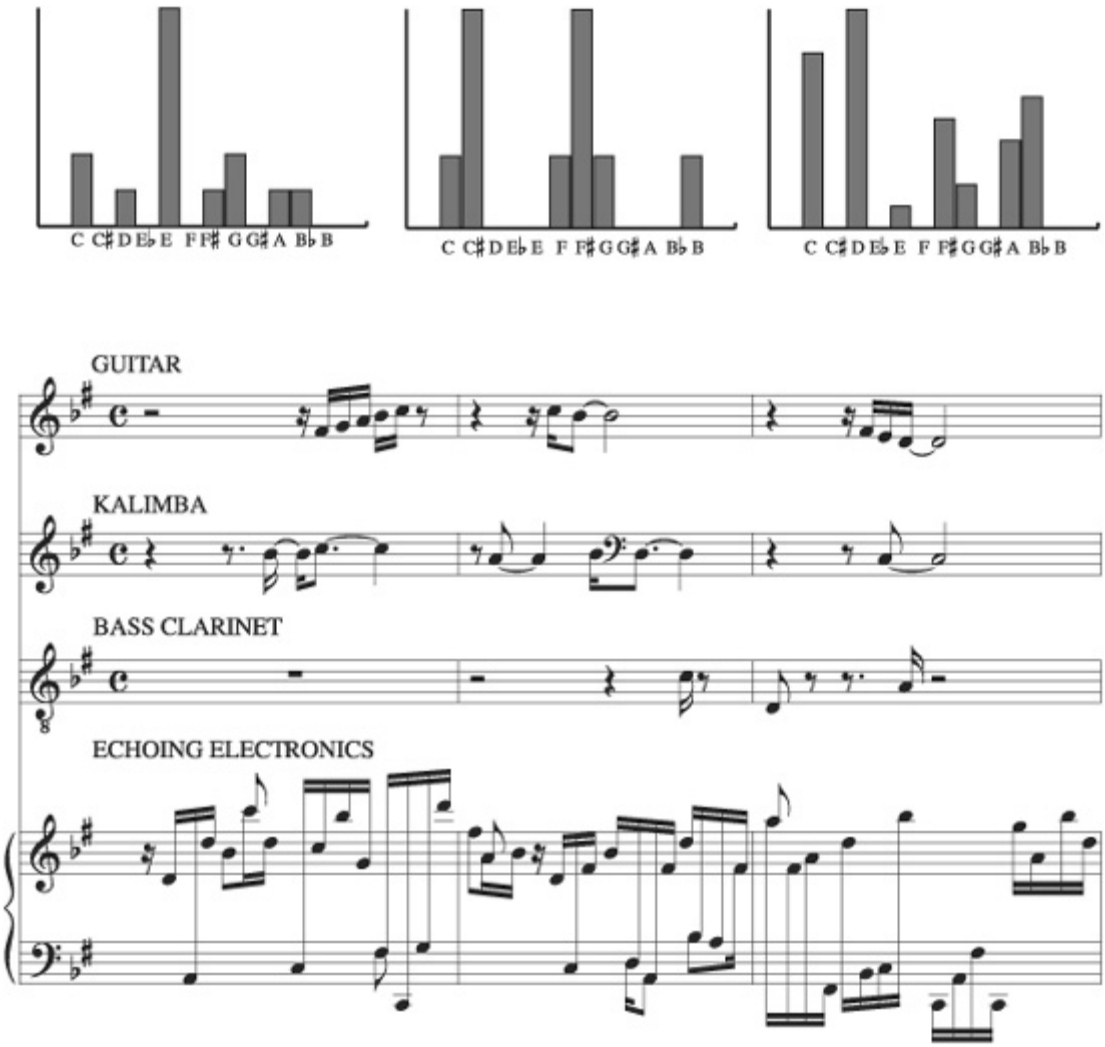
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

class profile are accented in various ways. Figure 5.6.13 provides an excerpt from a short computer-composed etude, in which pitch-class profiles were used as probability tables that determine the likelihood of each note's appearing. Another example, from a more conventional piece, is given in Figure 5.6.14. In composing this music, I began with the idea of "coloring" a single tonic note with a variety of secondary pitches. The pitch profiles in (*b*) provided an intuitive guide, allowing me to construct a sequence of improvisatory lines that gradually broadened the music's mac-roharmonic content. The result is a kind of blurred, burbling texture, in which the note A is a vey clear tonal center. (Note that the rhythmic language gradually congeals in concert with the harmonies, moving from indeterminate to determinate notation.) It seems to me that there is an enormous amount of unexplored musical territory here, representing the no-man's-land between traditional tonality and full-on atonality. Pitch-class profiles, by helping us visualize this territory, can help us imagine ways to explore it.

Figure 5.6.13 Three pitch-class profiles, along with a passage of computer-generated music embodying the third. Readers with access to the internet can hear a short computer-generated etude on the companion website that moves between the three profiles.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.
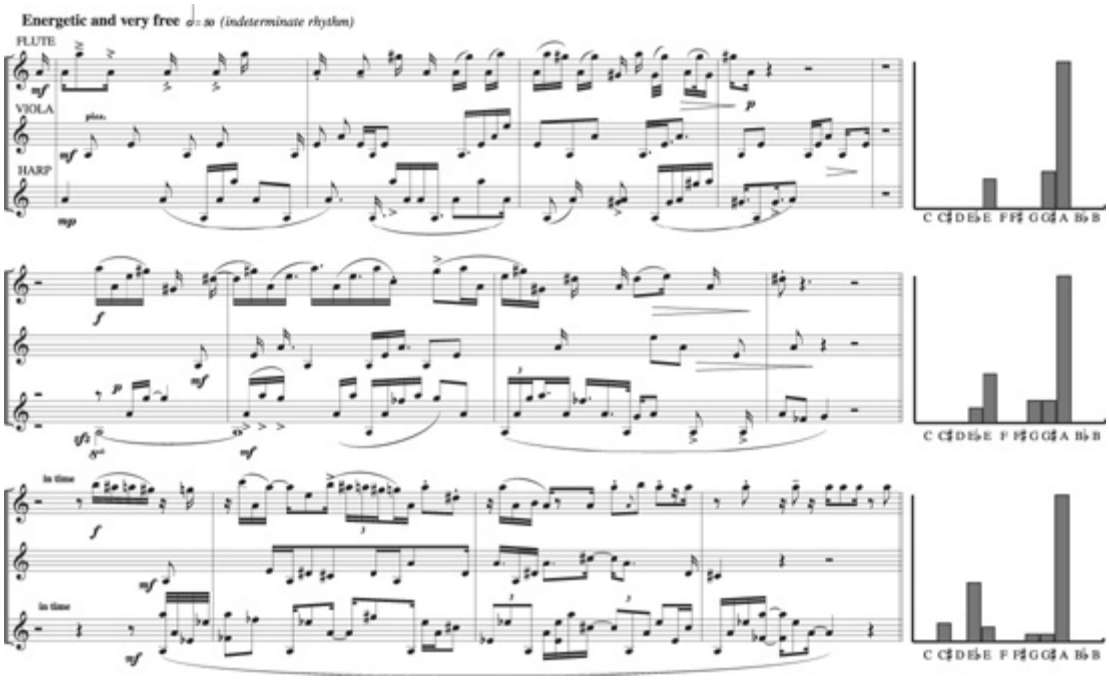


5.7 WHERE DOES CENTRICITY COME FROM?

Broadly speaking, theorists have explained centricity in two ways. *Internal* explanations assert that the structure of a group of notes is sufficient to pick one out as a tonal center, without any effort on the composer's part. *External* explanations focus on what composers do, asserting that composers make notes more prominent (or stable) by playing them more frequently, accenting them rhythmically or dynamically, placing them in registrally salient positions, and so on. Rather than being a property of collections considered abstractly, centricity is a property of collections as they are used in actual music.

Figure 5.6.14 The opening three phrases of my piece *Cathedral* (one phrase per line), along with the three pitch-class profiles that inspired them.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



Internalists typically cite principles such as the following:

I1. A note is more prominent if it is the lower note of one or more consonant intervals (perfect fifth, major third, or minor third) in the macroharmony.

I2. A note is more prominent if it does not form sharp dissonances (tritone, or minor second) with any note in the macroharmony.

Principle I1 suggests, for example, that C is the most prominent tone in the collection {C, E, G} and that the note B is poorly suited to be the tonic of the C diatonic scale, since there is no scale tone a perfect fifth above it. There is actually some psychological evidence in favor of this view: in an interesting series of experiments, Erkki Huovinen has shown that listeners, when asked to identify the tonic of a series of notes, generally prefer a note that has a perfect fifth above it.[18] Other theorists, such as Schoenberg and Ramon Fuller, have used these ideas to argue that ionian and aeolian are the most natural or appropriate modes of the diatonic scale.[19]

By contrast, external explanations assert that pitch classes become stable or prominent by

E1. Appearing more frequently;

E2. Being held for longer durations;

E3. Being accented dynamically;

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

E4. Being accented rhythmically;

E5. Being accented registrally (in other words, occurring as melodic high points and low points);

E6. Being the target of stepwise melodic motion, particularly stepwise contrary motion converging on a particular pitch class; and

E7. Being doubled at the octave, or paired with the note a fifth above.
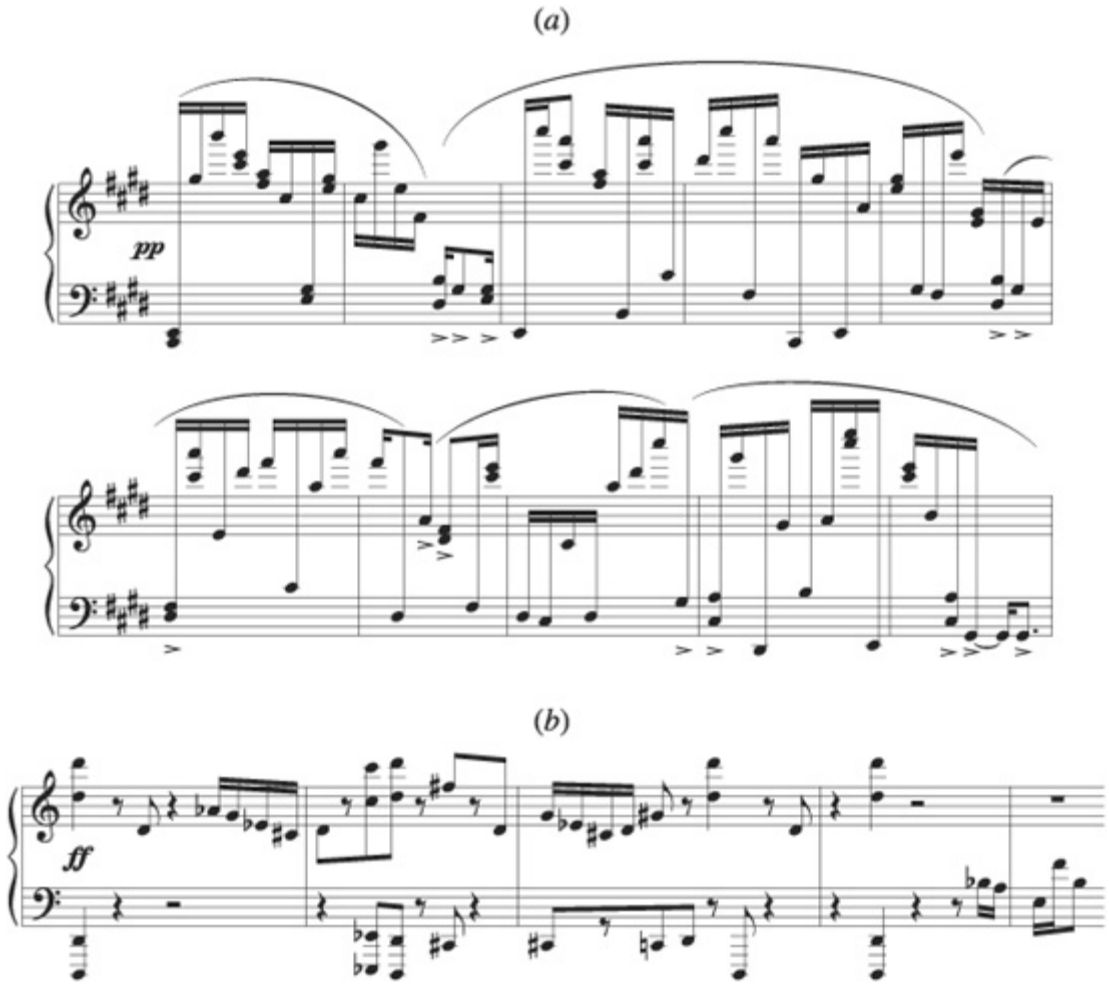
These external explanations all focus on notes as they are deployed compositionally.[20]

Note that the two approaches suggest two drastically different views about what is and is not musically possible. For suppose internal factors are relatively weak: in this case, composers can easily override "internal" centric tendencies, choosing to emphasize notes by a variety of external means. However, if internal factors are strong then composers are ill-advised to disregard them—for by attempting to emphasize something other than a collection's "natural" tonal center, they may create music that is unconvincing, inconsistent, or otherwise aesthetically deviant. It is interesting that these two views played a crucial role in early twentieth-century music history, with a number of prominent Germanic musicians (including Schoenberg and Schenker) inclined toward internalism, and a number of non-Germanic musicians (including Debussy) taking the opposite view.[21] The divergence between nonscalar atonality and scalar "extended tonality" can thus be traced, at least in part, to this difference of music-theoretical opinion: for Schoenberg and others, centricity was a matter of a natural law, and hence it was necessary to choose between traditional tonality and the wholesale abandonment of centricity. For composers in the French and Russian traditions, centricity was a compositional choice, and thus it was possible to contemplate musical styles that made use of new modes and scales.

Figure 5.7.1 (*a*) Diatonic music with no clear center. (*b*) Chromatic music with a clear center.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



My own sympathies lie very much with the external view: in most practical cases, I believe the internal contributions to centricity are relatively weak and can easily be overridden. Consequently it is entirely possible to write diatonic music that is acentric, or chromatic music that emphasizes a particular note (Figure 5.7.1). I do not consider either kind of music to be "unnatural."[22] Nor do I have any problem with any of the diatonic modes (or any other mode of any other scale): I am entirely convinced by the music of the Renaissance, of Debussy, Ravel, and Shostakovich, and of contemporary jazz and rock; I enjoy those numerous passages of twentieth-century music that make use of symmetrical scales while still asserting a tonal center; and as a composer I believe I can make virtually any note of virtually any collection sound like a tonic. (This last conviction has been reinforced by my experience with computers: by emphasizing particular notes through repetition, duration, loudness, and stepwise melodic motion, it is easy to create the effect of centricity in otherwise random sequences.) Any theorist who wants to argue against these convictions would have to fight an uphill battle: indeed, the very claim that the phrygian mode is deficient, or that centric music cannot use symmetrical scales, strikes me as evidence of a limited musical imagination.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

5.8 BEYOND "TONAL" AND "ATONAL"

I want to close this chapter—and by extension, the theoretical half of the book—by showing how we can use these ideas to orient ourselves relative to the broad spectrum of contemporary musical styles. I'll begin by contrasting two twentieth-century movements: the *chromatic tradition*, which rejects five- to eight-note macroharmonies in favor of the chromatic scale; and the *scalar tradition*, in which limited macroharmonies continue to play a significant role. I'll then suggest that our inquiry into the five basic components of tonality might help us envision new possibilities lying between these two extremes.

5.8.1 The Chromatic Tradition

A standard trope of music history asserts that the chromatic tradition originates in the extended tonality of post-Wagnerian chromaticism. According to this narrative, composers such as Strauss and Reger began to make heavy use of chromatic voice leading, to the point where familiar analytical concepts began to lose their purchase: pitch-class circulation increased, traditional scales became less important, and centricity became less obvious, with key-defining progressions (such as I–ii–V^7–I) gradually disappearing. Later composers, feeling that it was unreasonable to retain acoustic consonance while abandoning the other components of tonality, created atonality as we know it.

In my view, this narrative is essentially right: macroharmony was indeed the first casualty in the war on tonality, and highly chromatic "wandering tonality" does begin to approach atonality both in its rate of pitch-class circulation and in the absence of traditional macroharmonies.[23] Figure 5.8.1 shows that Reger's music sometimes circulates quickly through the twelve pitch classes while also abandoning clearly articulated macroharmonies. (The road to atonality, one might say, was paved with chromatic voice leading.) That said, there is still an important step from Reger to Schoenberg. For not only did Schoenberg "emancipate the dissonance," treating any possible combination of notes as a potential harmony, but he also took the much more radical step of rejecting harmonic consistency—in other words, the very idea that harmonies should be structurally similar to one another. By way of illustration, Figure 5.8.2 contrasts the distribution of three-note

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

chords in Schoenberg's Op. 11 No. 3 and Reger's Op. 58 No. 5: where Reger's graph is strongly peaked at the familiar major and minor triads—which together account for more than a third of the three-note chords in the piece—Schoenberg's graph is much flatter, suggesting a much more even distribution of chord types. In this respect, Schoenberg's music goes far beyond the simple rejection of tonality: by abandoning harmonic consistency, he took the radical step from "everything is permitted" to "everything is permitted *at all times*." Not only was he willing to use any collection of notes as a harmony, but he was also willing to use an enormous range of chord types within very short temporal spans.[24]

Figure 5.8.1 (*a*) Reger's tonal music sometimes circulates through the pitch classes almost as fast as Schoenberg's atonal music. (*b*) Some of Reger's pieces exhibit a very broad distribution of macroharmonies, comparable to that found in atonal music. Again, set-classes on the *x*-axis are labeled using Allen Forte's system, so that "15" corresponds to his set "7–15."

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

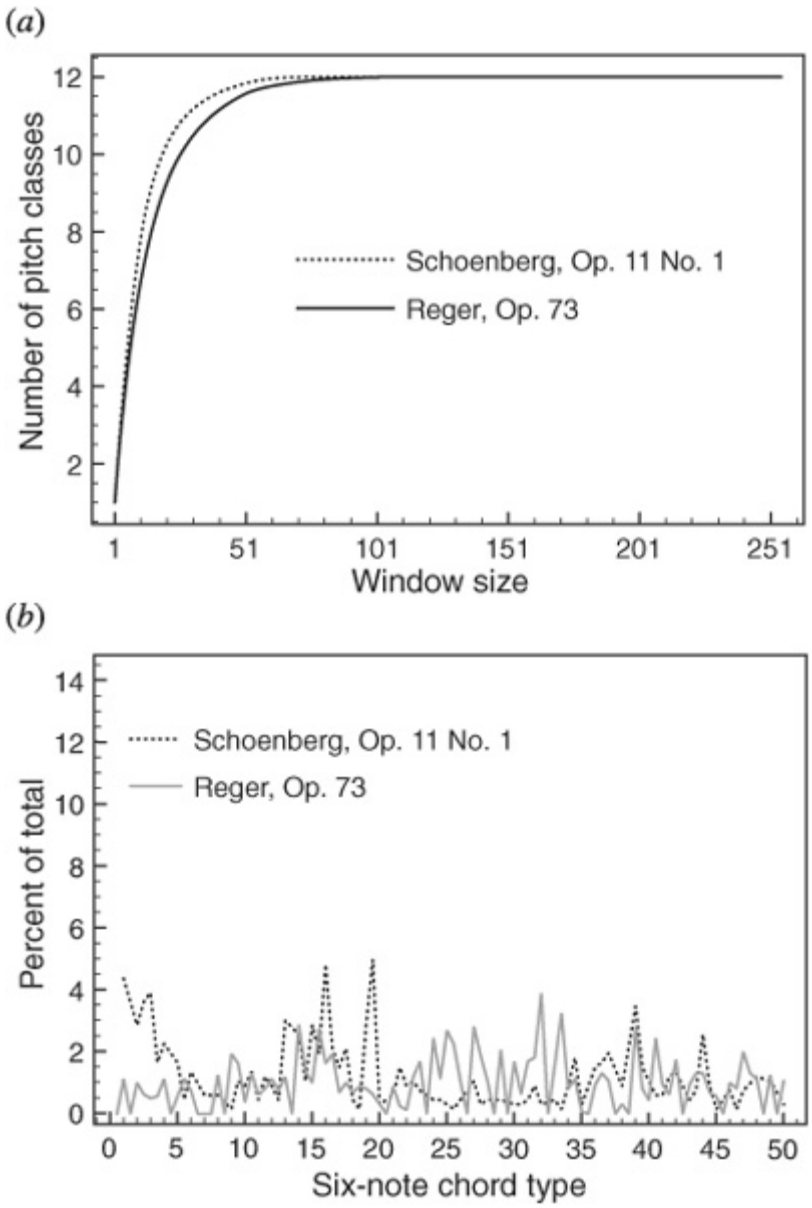
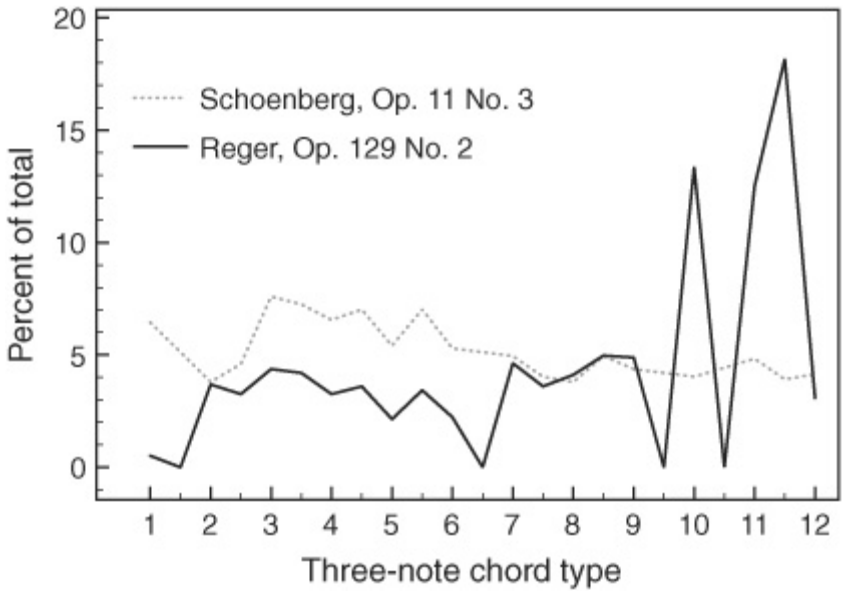


Figure 5.8.2 Reger's highly chromatic tonal music often exhibits more harmonic consistency than Schoenberg's atonal music. Here, the "spike" indicates that the music is saturated with major and minor triads.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.



It is worth asking whether this feature of Schoenberg's music might have been an *unintended byproduct* of the decision to abandon consonant chords. We have seen that harmonic consistency and conjunct melodic motion can be combined only under very special circumstances, typically involving acoustically consonant chords. It follows that composers who abandon consonance may put themselves in a difficult situation, sacrificing their ability to achieve the traditional two-dimensional coherence of Western music. Certainly, it is suggestive that the music of the second Viennese school often occupies two opposite poles: pointillistic textures that are harmonically consistent (as in the opening of Webern's *Concerto for Nine Instruments*), and conjunct passages that use a very wide range of harmonies (as in the opening of *Pierrot Lunaire*'s "Die Kreuze").[25] It seems possible that atonal composers did not explicitly *choose* to abandon harmonic consistency or conjunct melodic motion; instead, this choice may have been forced upon them by their prior rejection of acoustically consonant chords.

It is also clear that there are some specific ways in which atonal music resembles random music. Figure 5.8.3a shows that the pitch-class circulation graph of the first movement of Schoenberg's Op. 11 is nearly indistinguishable from that of an equally long series of random pitches. Figures 5.8.3b–c compare the distributions of harmonies in Schoenberg's piece with those in random sequences. Again, both atonality and random music contain a broad and relatively even distribution of chord types. Finally, Figure 5.8.3d contrasts the frequency of pitch classes in a Schoenberg movement with those in random sequences: the graphs are both relatively flat, reflecting the absence of clear points of tonal emphasis. (It is worth recalling that Schoenberg's later twelve-tone method was *explicitly*

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

designed to promote flat pitch-class profiles of this sort: as Schoenberg emphasized, doubling or repeating notes could potentially give rise to feelings of centricity, which would be reflected by unevenness in the pitch-class profile.[26]) Taken together, the figures suggest that there is some truth to a common ("naive") response to atonality—namely, "it sounds random." Statistically speaking, atonal music is often remarkably similar to random notes, and listeners perceive this fairly accurately.

Figure 5.8.3 (*a*) Pitch-class circulation in Schoenberg and in random notes. (*b*) Distribution of six-note macroharmonies in Schoenberg and in random notes. (*c*) Distribution of three-note harmonies in Schoenberg's Op. 11 No. 3 and in random notes.

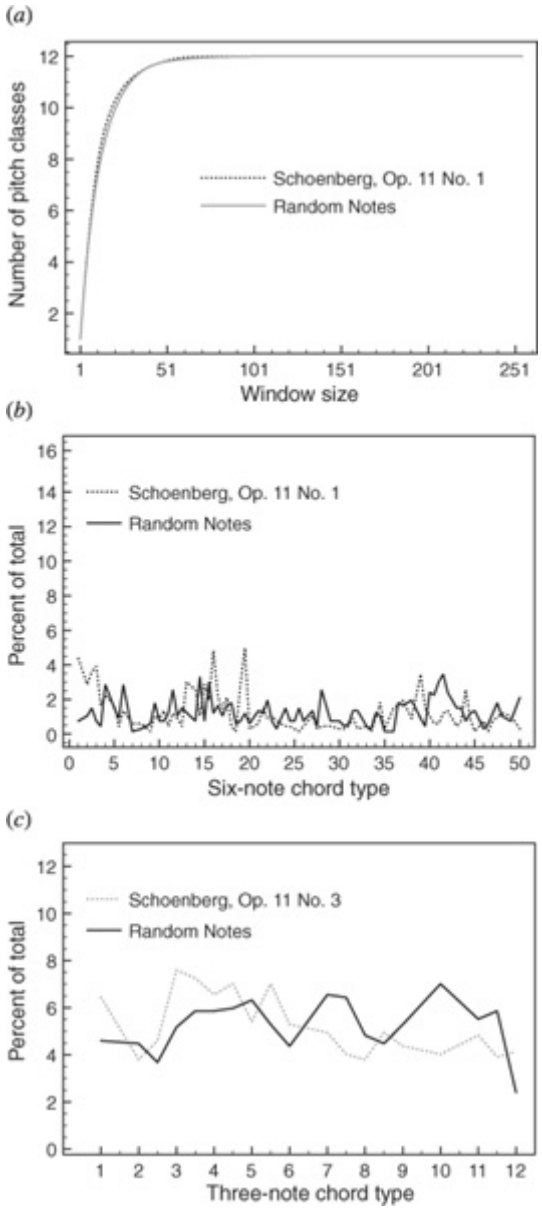
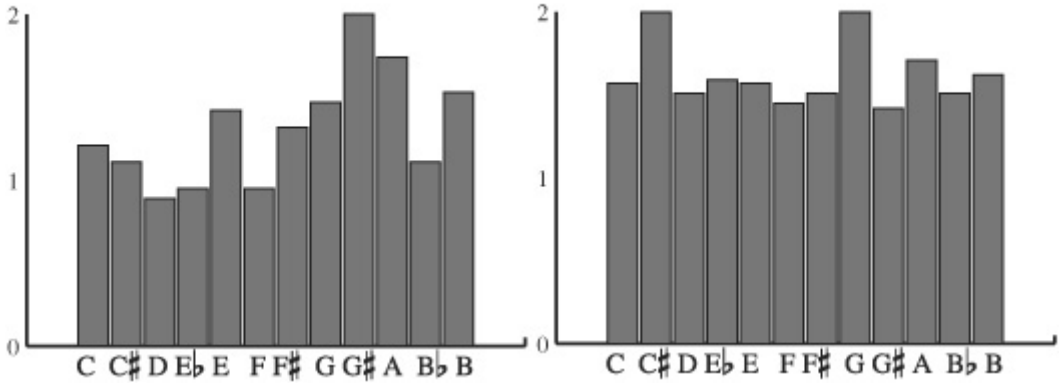


Figure 5.8.3 (*d*) Pitch-class distributions in Schoenberg's Op. 11 No. 1 (*left*) and in random notes (*right*).

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

(*d*)



Of course, in abandoning tonality, Schoenberg and other atonal composers attempted to substitute alternative methods of musical organization for those we have been considering—some of which, such as the twelve-tone method, involve the *order* of pitch sequences rather than their unordered pitch content. Discussions of atonal music often focus on the difficulties involved in perceiving this alternative organization: sympathetic observers sometimes explain atonality's unpopularity by comparing it to a language that is very difficult to learn, while more critical commentators, such as Fred Lerdahl and Diana Raffman, suggest that atonal music may be organized according to principles that are beyond *any* human perceptual understanding.[27] The preceding discussion suggests that we might want to reconsider this line of argument. For appreciating atonal music requires more than simply learning to appreciate alternative methods of musical organization; it also requires learning *not* to respond to those statistical features that the music shares with random sequences of notes. And insofar as a listener fundamentally dislikes the *sound* of random pitches, then it may not matter how atonal pieces are organized: after all, mere ordering does not typically convert unpleasant stimuli into pleasant ones. (Imagine someone causing you pain, or feeding you disgusting food, according to a perceptible and highly structured pattern!) If this is right, then atonal music is not so much analogous to a language that is hard to understand; instead, it is more like a taste that many people do not see the point of acquiring. To put it crudely: people dislike atonal music, not because they have a hard time understanding it, but *because they think it sounds bad*.[28]
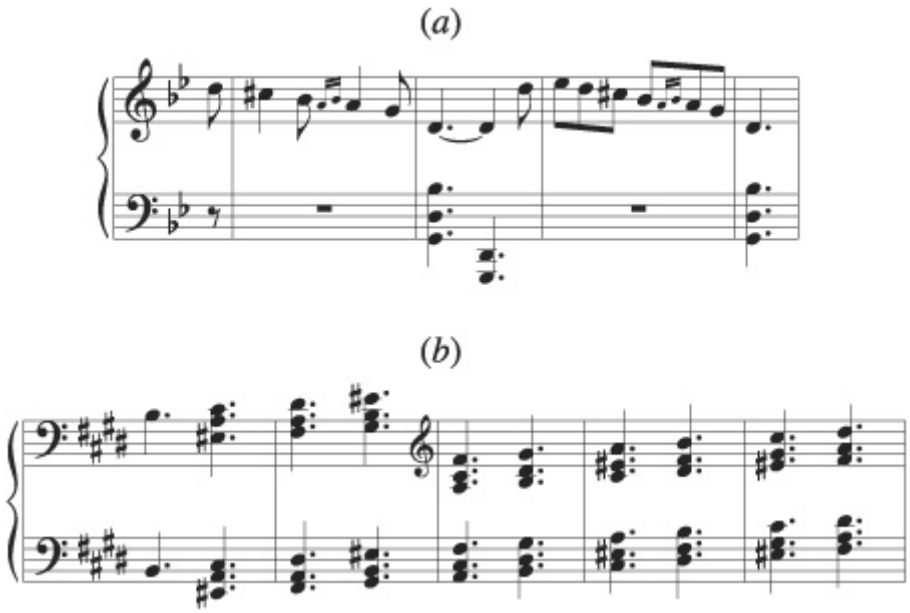
None of this implies that highly chromatic music actually *is* bad or otherwise aesthetically flawed: on the contrary, I think moments of extreme chromaticism (and even randomness) can often be artistically compelling. However, I do think that we might have

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

reason to resist the claim that fans of atonality constitute a cognitive elite, or that the enjoyment of atonal music is a straightforward function of unusual powers of musical understanding. For these claims tend to rely on the rather dubious suggestion that *if* people understood atonal music *then* they would like it—a claim that is no more likely to be true of Schoenberg and Babbitt than it is of heavy metal or polka. Instead, I think it is better to describe the aficionados of atonality as having managed to acquire a taste for highly chromatic musical textures: like the taste for clam chowder ice cream, this is one that people often do not care to cultivate. But that neither makes it worthy of approval nor condemnation—instead, it is just one of the myriad different specialized pleasures whose pursuit makes contemporary society so colorful.

Figure 5.8.4 Nondiatonic scales in nineteenth-century music. In (*a*), a fragment of the "Gypsy" scale (G-A-B♭-C♯-D-E♭-F♯) in Grieg's Lyric Piece "Gjetergutt" ("Shepherd's Boy"), Op. 54 No. 1. In (*b*), the acoustic scale in Liszt's "Angélus! Prière aux anges gardiens"), *Années de pèlerinage*, Year 3.



5.8.2 The Scalar Tradition

Diametrically opposed to atonality is the "scalar tradition" that makes extensive use of familiar scales and modes. This tradition encompasses at least six major twentieth-century movements—impressionism, neoclassicism, jazz, rock, minimalism/postmin-imalism, and neo-Romanticism—and a good deal of other music as well (including, to various extents, music of Scriabin, Stravinsky, Bartók, and Shostakovich). Of course, the scalar tradition is

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

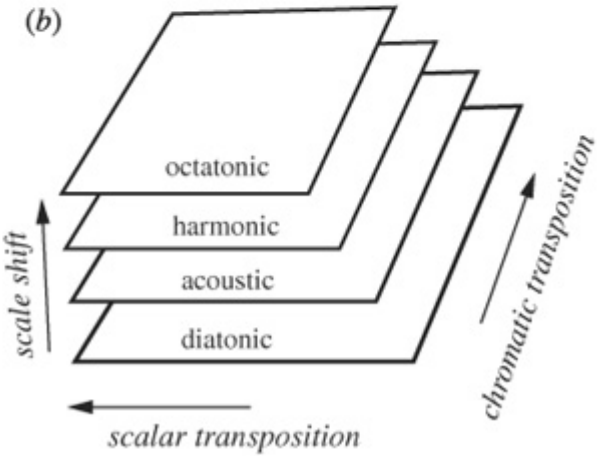
Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

itself diverse: some styles (including mid-century neoclassicism and contemporary rock) are predominantly diatonic, while others (including impressionism and jazz) make much greater use of nondiatonic scales; some composers write largely scalar music, while others juxtapose familiar scales with chromatic or even atonal passages. Nevertheless, we will see in Chapter 9 that there are enough commonalities among twentieth-century composers to justify talk of a scalar "common practice."

Figure 5.8.5 (*a*) The two-dimensional modulatory space of twentieth-century diatonic music. Horizontal motion corresponds to scalar transposition; vertical motion chromatic transposition. The figure's left and right edges are glued together, as are its top and bottom edges. (*b*) The three-dimensional space of twentieth-century scalar music. Each plane of the figure is analogous to (*a*); vertical motion changes the underlying scale.

(*a*)

C mixolydian	D aeolian	E locrian	F ionian	G dorian	A phrygian	B♭ lydian
B mixolydian	C♯ aeolian	D♯ locrian	E ionian	F♯ dorian	G♯ phrygian	A lydian
B♭ mixolydian	C aeolian	D locrian	E♭ ionian	F dorian	G phrygian	A♭ lydian
A mixolydian	B aeolian	C♯ locrian	D ionian	E dorian	F♯ phrygian	G lydian
A♭ mixolydian	B♭ aeolian	C locrian	D♭ ionian	E♭ dorian	F phrygian	G♭ lydian
G mixolydian	A aeolian	B locrian	C ionian	D dorian	E phrygian	F lydian
F♯ mixolydian	G♯ aeolian	A♯ locrian	B ionian	C♯ dorian	D♯ phrygian	E lydian
F mixolydian	G aeolian	A locrian	B♭ ionian	C dorian	D phrygian	E♭ lydian
E mixolydian	F♯ aeolian	G♯ locrian	A ionian	B dorian	C♯ phrygian	D lydian
E♭ mixolydian	F aeolian	G locrian	A♭ ionian	B♭ dorian	C phrygian	D♭ lydian
D mixolydian	E aeolian	F♯ locrian	G ionian	A dorian	B phrygian	C lydian
D♭ mixolydian	E♭ aeolian	F locrian	G♭ ionian	A♭ dorian	B♭ phrygian	C♭ lydian



Like the chromatic tradition, the scalar tradition has its origins in the late nineteenth century. Composers such as Chopin, Liszt, Mussorgsky, Rimsky-Korsakov, Grieg, and Fauré

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

began experimenting with the various modes of the diatonic scale, as well as with whole-tone, octatonic, and melodic and harmonic minor scales (Figure 5.8.4).[29] Historians have often conceptualized this tradition *extrinsically*—as reflecting "folk" influences external to the functionally harmonic tradition. But we can also understand the scalar tradition intrinsically, as representing an expansion and generalization of traditional compositional practices. In particular, twentieth-century scalar composers were the first to systematically combine three fundamental musical operations: change of tonal center, change of scale, and chromatic transposition. These operations are represented geometrically in Figure 5.8.5 . Here, each plane represents the combination of chromatic and scalar transposition, while motion between scales is represented by vertical motion between the planes.[30] The resulting modal system provides a truly vast range of different key areas (Figure 5.8.6): instead of seven modes or 24 major and minor keys, composers now have more than 450 different tonal areas to chose from!

Figure 5.8.6 Twentieth-century scalar composers potentially have a very wide range of scales and modes to work with.

Scale Type	Number of Scales	Number of Notes	Mode Types	Total Number of Modes
Pentatonic	12	5	5	60
Whole tone	2	6	1	12
Hexatonic	4	6	2	24
Diatonic	12	7	7	84
Acoustic	12	7	7	84
Harmonic Min.	12	7	7	84
Harmonic Maj.	12	7	7	84
Octatonic	3	8	2	24
Chromatic	1	12	1	12
TOTAL	70		39	468

Within the scalar tradition, we can distinguish chordal from "pandiatonic" (or perhaps "panscalar") approaches: chordal music preserves the two-tiered "chord within scale" organization of earlier Western music; while pandiatonic music effaces the role of chords as scales themselves begin to assert a more central harmonic role.[31] (As mentioned earlier, Debussy's "Voiles" is an early example of this second practice.) The distinction between these two approaches thus mirrors the distinction between the chromaticism of composers

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

like Reger and that of true atonality: Reger's music, like traditional tonality, preserves harmonic consistency, limiting itself to a small number of recognizable chord types, while pandiatonic music (like more radical atonality) abandons harmonic consistency altogether. This is illustrated graphically in Figure 5.8.7. Once again, we see that the binary opposition between "chromatic" and "nonchromatic" (or "tonal" and "atonal") blurs a number of more specific music-theoretical distinctions.

Returning now to a question broached earlier in this chapter: is it really true that Western music became more chromatic over the course of its history? Clearly, there is a sense in which it has: the progression from Palestrina to Mozart to Schubert to Wagner to Reger to Schoenberg exhibits a steady and quantifiable increase in chromaticism (Figure 5.3.2). What is left out of this story is that late nineteenth-century composers had already begun to rebel against this trend. This scalar tendency gathered steam throughout subsequent decades, culminating in an unprecedented explosion of diatonicism in the works of composers such as Shostakovich, Riley, Reich, Glass, Pärt, Górecki, Adams, Bryars, and ter Veldhuis, not to mention jazz, rock, and other popular forms of music. (The trend is also evident in the renewed appreciation for early music.) Thus, where early twentieth-century historians might have seen a relentless, ever-increasing drive toward chromaticism, twenty-first century musicians confront a very different environment, in which early twentieth-century scalar music seems as much a harbinger of things to come as a relic of the past.

Figure 5.8.7 In both the scalar and chromatic traditions, there are genres exhibiting harmonic consistency (top row) as well as genres in which harmonic consistency does not play as an important a role (bottom).

		Limited Macroharmony	
		YES	NO
Harmonic Consistency	YES	traditional scale-based tonality (Debussy, Jazz)	"triadic atonality" (Gesualdo, Reger)
	NO	pandiatonicism (*Voiles*)	complete atonality (Schoenberg, Babbitt)

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

5.8.3 Tonality Space

Of course, the chromatic and scalar traditions represent just a portion of the huge tapestry that is twentieth-century music. Figure 5.5.4 depicted twentieth-century music as lying on a two-dimensional continuum, whose axes represented the rate of pitch-class circulation and the explicitness of macroharmony. While the scalar tradition lies on the far right of the graph, and while complete atonality is in the lower left corner, there is a considerable amount of music that does not fit into either category. Many of Prokofiev's pieces, for example, make free use of accidentals without attempting to stay within a single scale for very long. Similarly, composers such as Stravinsky, Milhaud, Bartók, and Shostakovich explored polytonal textures in which familiar musical materials are superimposed, creating the effect of clashing keys.[32] Still others—such as the Messiaen of the *Quartet for the End of Time* or *Vingt regards sur l'enfant-Jésus*—wrote music that embraces many different techniques, being at different points traditionally tonal, polytonal, completely atonal, or several of these at once (Figure 5.8.8).
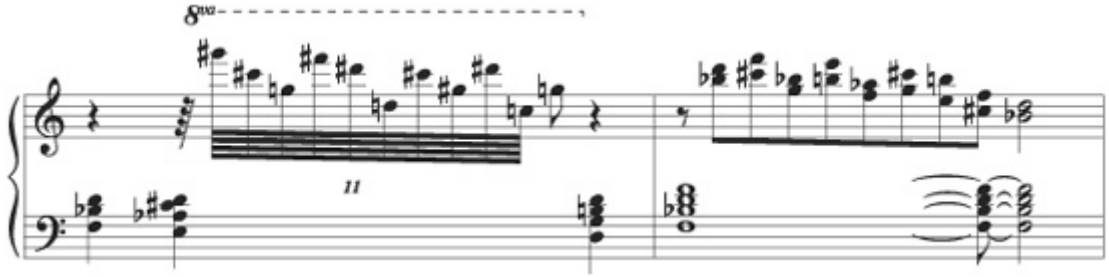
Among these myriad approaches, György Ligeti's "non-atonality" stands out as being particularly relevant to the present discussion. Though dissonant, his music often exhibits a harmonic consistency that contrasts with the more classical atonality of Schoenberg, Stockhausen, or Babbitt. This is in large part due to Ligeti's innovative use of chromatic clusters—and more specifically, to his "micropolyphonic" technique of moving multiple voices within a fixed chromatic collection. As I mentioned in Chapter 1, this "micropolyphony" constitutes a uniquely modern solution to the problem of combining conjunct melodic motion with harmonic consistency, one that exploits the *near-permutation symmetry* of very uneven chords (§1.3.1, §2.9). (Alternatively, we can say that Ligeti's music exploits the mirror boundaries of the higher-dimensional chord spaces, rather than taking advantage of their twists [§3.6]). I find it remarkable that there are just two general techniques for combining conjunct melodies with consistent harmonies, the first involving nearly even chords and underwriting the tonal procedures of the last five hundred years, the second involving clustered chords and featuring in the atonality of the 1960s. It is particularly interesting that this second solution arose almost *fifty years* after the first experiments with atonality—as if it took that long for composers to abandon the doomed project of writing traditional music with nontraditional sonorities, and to instead embrace

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

a more radical approach that exploits the distinctive virtues of nontraditional, clustered chords. In a real sense, this music is unclassifiable: atonal by virtue of its dissonance, but tonal in its aspiration to combine harmonic consistency with conjunct melodic motion.

Figure 5.8.8 Messiaen's "Première communion de la Vierge" (*Vingt regards sur l'enfant-Jésus* No. 11) combines an octatonic-infused tonality with atonal upper-register gestures.



One of my goals here has been to identify concepts that allow us to think about the full spectrum of musical possibilities, ranging from traditional tonality to outright atonality. From this point of view, the five basic components of tonality are useful insofar as they demarcate a collection of conceptual possibilities—as if each were a musical "vector" pointing into its own abstract dimension. Together, these (metaphorical) vectors span a region that could be called "tonality space." We can situate unfamiliar pieces or genres within this metaphorical space by asking questions such as those in Figure 5.8.9. The first set of questions concerns chords and melodies, and generalizes the traditional concerns of harmony and counterpoint; the second extends notions such as "mode," "key," and "tonic" to a broader range of macroharmonic possibilities. Collectively, they replace the opposition "tonal"/"atonal" with a much more fine-grained set of categories, more appropriate to the richness of contemporary musical practice.

Composers can use these questions to situate themselves within the world of musical possibilities. For some, this may be a matter of settling in a relatively small region of tonality space for the majority of their careers. (A strategy that is perfectly reasonable insofar as one has a strong preference for a very particular kind of music, or if one wants to appeal to audiences who do.) Others—following in the footsteps of Stravinsky, Bartók, Shostakovich, Messiaen, Ligeti, and John Adams—may prefer to compose music that traverses a broader range of tonality space, sometimes within the span of a single piece. Unimaginative critics may decry this as a matter of poly-stylistic or postmodern inconstancy, but it is perhaps better described as a desire to embrace the full range of

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.

Copyright © 2011. Oxford University Press, Incorporated. All rights reserved.

musical opportunities, an attitude encapsulated in Mahler's famous remark that composers create musical worlds "with all the technical means available."[33] For composers of a maximalist bent, tonality space is the field in which contemporary music operates, just as the 24 major and minor keys set the boundaries of eighteenth-century musical exploration.

Figure 5.8.9 Questions that help us situate music in tonality space.

LOCAL

1. Does the music use a small set of similar-sounding chords?
    - 1a. How consonant are these chords?
    - 1b. Do these chords progress freely or are certain progressions favored?
2. Does the music use efficient voice leading?
    - 2a. Does this voice leading take place in chromatic space or in another scale?
    - 2b. Does it exploit near symmetries, as discussed in Chapter 2?
    - 2c. Can it be usefully modeled with the geometrical spaces of Chapter 3?

GLOBAL

3. How fast do pitch classes circulate?
4. Does the music articulate clear macroharmonies?
    - 4a. Are these macroharmonies structurally similar to each other?
    - 4b. How consonant are the macroharmonies?
    - 4c. How fast do macroharmonies change?
    - 4d. Does the music exploit efficient voice leading between these macroharmonies?
    - 4e. Does the music exploit "polytonal" effects by juxtaposing multiple scales or macroharmonies, each retaining their own identity?
5. Is there a sense of tonal center?
    - 5a. Are there important secondary pitches?

To my mind there is a lot of fertile ground here. Having completed the theoretical portion of the book, my hope is that some of you will feel inspired to think about how to combine the five features—whether by making creative use of nearly symmetrical chords, exploring the voice-leading spaces of Chapter 3, manipulating interesting scales in novel ways, exploiting unusual pitch-class profiles, devising new combinations of scale, macroharmony, and centricity, or by doing something else entirely. Those who are uncertain about how to proceed may benefit from the analyses in Part II, which consider the ways in which previous composers have solved this problem. Besides deepening our appreciation for the music of the past, these analyses may suggest new directions to composers of the present.

Tymoczko, Dmitri. A Geometry of Music : Harmony and Counterpoint in the Extended Common Practice, Oxford University Press, Incorporated, 2011. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uvic/detail.action?docID=665450.
Created from uvic on 2026-11-09 23:10:51.